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01

PORTFOLIO MANAGEMENT

EXPECTED RETURN & RISK OF SINGLE

Question – 01

A stock costing ₹ 120 pays no dividends. The possible prices that the stock might sell for at the end of the year with the respective probabilities are :

Price	Probability
115	0.1
120	0.1
125	0.2
130	0.3
135	0.2
140	0.1

Required:

- Calculate the expected return.
- Calculate the Standard deviation of returns.

(SM TYK – 01)

Solution:**(i) Calculation of Return:**

$$\begin{aligned} \frac{115 - 120}{120} \times 100 &= -4.17\% \\ \frac{120 - 120}{120} \times 100 &= 0 \\ \frac{125 - 120}{120} \times 100 &= 4.17\% \\ \frac{130 - 120}{120} \times 100 &= 8.33\% \\ \frac{135 - 120}{120} \times 100 &= 12.50\% \\ \frac{140 - 120}{120} \times 100 &= 16.67\% \end{aligned}$$

(ii) Calculation of Standard Deviation of Return:

P	X	P (x̄)	(x - x̄)	(x - x̄) ² P
0.1	-4.17	-0.417	-11.253	12.663
0.1	0	0	-7.083	5.017
0.2	4.17	0.834	-2.913	1.697
0.3	8.33	2.499	1.247	0.466
0.2	12.50	2.500	5.417	5.869
0.1	16.67	1.667	9.587	9.191
	$\bar{x} =$	7.083%		34.90 (%) ²

$$\sigma_x = \sqrt{34.90\%} = 5.908\%$$



CORRELATION & RISK OF PORTFOLIO

Question – 02

The historical rates of return of two securities over the past ten years are given. Calculate the Covariance and the Correlation coefficient of the two securities:

Years:	1	2	3	4	5	6	7	8	9	10
Security 1 : (Return per cent)	12	8	7	14	16	15	18	20	16	22
Security 2: (Return per cent)	20	22	24	18	15	20	24	25	22	20

(SM TYK – 06)

Solution:

Year	R_1	Deviation $(R_1 - \bar{R}_1)$	Deviation $(R_1 - \bar{R}_1)^2$	R_2	Deviation $(R_2 - \bar{R}_2)$	Deviation $(R_2 - \bar{R}_2)^2$	Product of deviations
1	12	-2.8	7.84	20	-1	1	2.8
2	8	-6.8	46.24	22	1	1	-6.8
3	7	-7.8	60.84	24	3	9	-23.4
4	14	-0.8	0.64	18	-3	9	2.4
5	16	1.2	1.44	15	-6	36	-7.2
6	15	0.2	0.04	20	-1	1	-0.2
7	18	3.2	10.24	24	3	9	9.6
8	20	5.2	27.04	25	4	16	20.8
9	16	1.2	1.44	22	1	1	1.2
10	22	7.2	51.84	20	-1	1	-7.2
$R_1 = \frac{148}{10} = 14.8$		$\sum = 207.60$		$R_2 = \frac{210}{10} = 21$	$\sum = 84.00$		$\sum = -8$

Standard Deviation of Security 1

$$\sigma_1 = \sqrt{\frac{[R_1 - \bar{R}_1]^2}{N}}$$

$$\sigma_1 = \sqrt{\frac{207.60}{10}}$$

$$= \sqrt{20.76} = 4.56$$

Standard Deviation of Security 2

$$\sigma_2 = \sqrt{\frac{[R_2 - \bar{R}_2]^2}{N}}$$

$$\sigma_2 = \sqrt{\frac{84}{10}}$$

$$= \sqrt{8.46} = 2.90$$

$$\text{Covariance} = \frac{\sum_{i=1}^n [R_1 - \bar{R}_1][R_2 - \bar{R}_2]}{N} = -8/10 = -0.8$$

$$\begin{aligned} R_{xy} &= \frac{\text{COV}_{xy}}{\sigma_x \times \sigma_y} \\ &= \frac{-0.8}{4.56 \times 2.90} = -0.060 \end{aligned}$$

Question – 03

An investor has decided to invest ₹ 1,00,000 in the shares of two companies, namely, ABC and XYZ. The projection of returns from the shares of the two companies along with their probabilities are as follows:



Probabilities	ABC (%)	XYZ (%)
.20	12	16
.25	14	10
.25	-7	28
.30	28	-2

You are required to

- Comment on return and risk of investment in individual shares.
- Compare the risk and return of these two shares with a Portfolio of these shares in equal proportion.
- Construct minimum risk portfolio.

(SM TYK – 07 & RTP May – 2019)

Solution:

Calculation of ER, S.D. of Cov_{xy}

P	x	P(x)	(x - $\bar{x}$)	(x - $\bar{x}$) ² p	y	P(y)	(y - $\bar{y}$)	(y - $\bar{y}$) ² p	(x - $\bar{x}$)(y - $\bar{y}$)p
0.20	12	2.4	-0.55	0.0605	16	3.2	3.90	3.042	-0.429
0.25	14	3.5	1.45	0.5256	10	2.50	-2.10	1.1025	-0.761
0.25	-7	-1.75	-19.55	95.5506	28	7	15.90	63.20	-77.71
0.30	28	8.40	15.45	71.6107	-2	-0.60	-14.10	59.64	-65.95
	$\bar{x}$ =	12.55	Variance	167.75	$\bar{y}$	12.1	Variance	126.99	Cov _{xy} =
		%		(%) ²		0%			-144.85

$$\sigma_x = \sqrt{\text{Variance}}$$

$$= \sqrt{167.75} = 12.95\%$$

$$\sigma_y = \sqrt{126.99} = 11.27\%$$

(i) Return & Risk of Individual Stock

ABC Stock

$$ER = 12.55\%$$

$$\text{Risk} = 12.95\%$$

XYZ Stock

$$ER = 12.10\%$$

$$\text{Risk} = 11.27\%$$

(ii) Risk & Return of Portfolio

$$W_{ABC} = 50\%, \quad W_{xyz} = 50\%$$

$$ER_P = (ER_A \times W_A) + (ER_B \times W_B)$$

$$= (12.55 \times 0.5) + (12.10 \times 0.5)$$

$$= 12.325\%$$

$$R_{xy} = \frac{COV_{xy}}{\sigma_x \sigma_y} = \frac{-144.25}{12.95 \times 11.27} = -0.99$$

$$\sigma_p = \sqrt{\sigma_A^2 \times W_A^2 + \sigma_B^2 \times W_B^2 + 2 W_A W_B \times COV_{AB}}$$

$$= \sqrt{12.95^2 \times 0.5^2 + 11.27^2 \times 0.5^2 + 2 \times 0.5 \times 0.5 \times -144.24}$$

$$= 1.25\%$$



The Expected Return of portfolio is 12.325% and Standard Deviation is 1.25% hence portfolio results in the reduction of risk by the combination of two shares.

(iii) Minimum Variance Portfolio

$$W_A = \frac{\sigma_B^2 - \text{COV}_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\text{COV}_{AB}}$$

$$= \frac{(11.27)^2 - (-144.25)}{12.95^2 + 11.27^2 - (2 \times -144.25)}$$

$$= \frac{271.23}{583.21} = 0.46$$

$$W_B = 0.54$$

Question – 04

Ramesh has identified stocks of two companies A and B having good investment potential:

Following data is available for these stocks:

Year	A (Market Price per Share in ₹)	B (Market Price per Share in ₹)
2013	19.60	8.70
2014	18.75	12.80
2015	33.42	16.20
2016	42.64	18.25
2017	43.25	15.60
2018	44.60	13.25
2019	34.75	18.60

You are required to calculate:

- The Risk and Return by investing in Stock A and B
- The Risk and Return by investing in a portfolio of these Stocks if he invests in Stock A and B in proportion of 6 : 4.
- The better opportunity for investment

(Exam January – 2021) (10 Marks)

Solution:

	A				B				
Year	Market Price Per Share in ₹	Return (%)	Return - $\bar{A}$	Squared	Market Price Per Share in ₹	Return (%)	Return - B	Squared	(Return - A) × (Return - B)
2013	19.60				8.70				
2014	18.75	-4.34	-18.33	335.98	12.80	47.13	30.94	957.28	-567.13
2015	33.42	78.24	64.25	4128.06	16.20	26.56	10.37	107.53	666.27
2016	42.64	27.59	13.60	184.96	18.25	12.56	-3.54	12.53	-48.14
2017	43.25	1.43	-12.56	157.75	15.60	-14.52	-30.70	943.10	385.71
2018	44.60	3.12	-10.87	118.15	13.25	-15.06	-31.25	976.56	339.68
2019	34.75	-22.09	-36.08	1301.76	18.60	40.38	24.19	585.15	-872.77
Mean ($\bar{A}$)		83.95		6226.68	Mean ($\bar{B}$)	97.14		3582.17	-96.3718
		13.99	Variance 037.7814			16.19	Varianace	597.0291	Cov. = -16.0620



- (i) Return A = 13.99% and Risk (SD) = $\sqrt{1037.7814} = 32.2146$ and Return B = 16.19% and Risk (SD) = $\sqrt{597.0291} = 24.4342$
- (ii) Return of Portfolio = $0.60 \times 13.99\% + 0.40 \times 16.19\% = 14.87\%$
- Risk (Standard Deviation) of Portfolio = $[0.60^2 \times 1037.7814 + 0.40^2 \times 597.0291 + 2 \times 0.60 \times 0.40 \times (-16.0620)]^{1/2}$
- = $[373.6013 + 95.5247 - 7.7098]^{1/2} = 21.4806$
- (iii) On the basis of Return 'B' is preferable and on the basis of Risk 'Portfolio Investment' is preferable over the individual stocks.

Question – 05

Consider the following information on two stocks, X and Y.

Year	2016	2017
Return on X (%)	10	16
Return on Y (%)	12	18

You are required to calculate:

- (i) The expected return on a portfolio containing X and Y in the proportion of 40% and 60% respectively.
- (ii) The Standard Deviation of return from each of the two stocks.
- (iii) The Covariance of returns from the two stocks.
- (iv) The Correlation coefficient between the returns of the two stocks.

- (v) The risk of a portfolio containing X and Y in the proportion of 40% and 60%.

(SM TYK – 04 & Exam May – 2018) (10 Marks)

Solution:

Year	x	(x - $\bar{x}$)	(x - $\bar{x}$) ²	Y	(y - $\bar{y}$)	(y - $\bar{y}$) ²	(x - x)(y - y)
1	10	-3	9	12	-3	9	9
2	16	3	9	18	3	9	9
	26		18	30		18	18

$$\bar{x} = \frac{26}{2} = 13\%$$

$$\sigma_x = \sqrt{\frac{18}{2}} = 3\%$$

$$\bar{y} = \frac{30}{2} = 15\%$$

$$\sigma_y = \sqrt{\frac{18}{2}} = 3\%$$

$$\text{Cov}_{xy} = \frac{18}{2} = 9$$

$$\sigma_{xy} = \frac{9}{3 \times 3} = 1$$

$$\text{ER}_p = (13 \times 0.4) + (15 \times 0.6) = 14.2\%$$

$$\sigma_p = \sqrt{3^2 \times 0.4^2 + 3^2 \times 0.6^2 + 2 \times 3 \times 0.4 \times 3 \times 0.6 \times 1}$$

$$\sigma_p = 3\%$$

Question – 06

Mr. A is interested to invest ₹ 1,00,000 in the securities market. He selected two securities B and D for this purpose. The risk return profile of these securities are as follows :

Security	Risk (σ)	Expected Return (ER)
B	10%	12%
D	18%	20%

Co-efficient of correlation between B and D is 0.15.

You are required to calculate the portfolio return of the following portfolios of B and D to be considered by A for his investment.

- (i) 100 percent investment in B only;
- (ii) 50 percent of the fund in B and the rest 50 percent in D;
- (iii) 75 percent of the fund in B and the rest 25 percent in D; and
- (iv) 100 percent investment in D only.

Also indicate that which portfolio is best for him from risk as well as return point of view?

(SM TYK – 03)

Solution:

$$ER_p = ER_A \times W_A + ER_B \times W_B$$

$$\sigma_p = \sqrt{\sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times \sigma_A \times W_A \times \sigma_B \times W_B \times r_{AB}}$$

(i) 100 % in B Only

$$ER_p = 12\% \quad \sigma_p = 10\%$$

(ii) 50 % in B & 50% in D

$$ER_p = (12 \times 0.5) + (20 \times 0.5) = 16\%$$

$$\sigma_p =$$

$$\sqrt{10^2 \times 0.5^2 + 18^2 \times 0.5^2 + 2 \times 10 \times 0.5 \times 18 \times 0.5 \times 0.15} = 10.93\%$$

(iii) 75% in B & 25% in D

$$ER_p = (12 \times 0.75) + (20 \times 0.25) = 14\%$$

$$\sigma_p = \sqrt{10^2 \times 0.75^2 + 18^2 \times 0.25^2 + 2 \times 10 \times 0.75 \times 18 \times 0.25 \times 0.15} = 9.31\%$$

(iv) 100 % in D Only

$$ER_p = 20\% \quad \sigma_p = 18\%$$

From the point of view of return 100% in D is the best due to the highest return & from the point of view risk 75% in B & 25 % in D is the best due to the Lowest Risk.

Question – 07

Following information is available in respect of expected dividend, market price and market condition after one year.

Market condition	Probability	Market Price	Dividend per share
		₹	₹
Good	0.25	115	9
Normal	0.50	107	5
Bad	0.25	97	3

The existing market price of an equity share is ₹ 106 (F.V. ₹ 1), which is cum 10% bonus debenture of ₹ 6 each, per share. M/s. X Finance Company Ltd. had offered the buy-back of debentures at face value. Find out the expected return and variability of returns of the equity shares if buyback offer is accepted by the investor. And also advise-Whether to accept buy-back offer?

(SM TYK – 02)

Solution:

Calculation of return

$$\text{Good} = \frac{(115 - 100) + 9}{100} \times 100 = 24\%$$

$$\text{Normal} = \frac{(107 - 100) + 5}{100} \times 100 = 12\%$$



ER & SD

$$\sigma_x = \sqrt{72} = 8.485\%$$

Question – 08

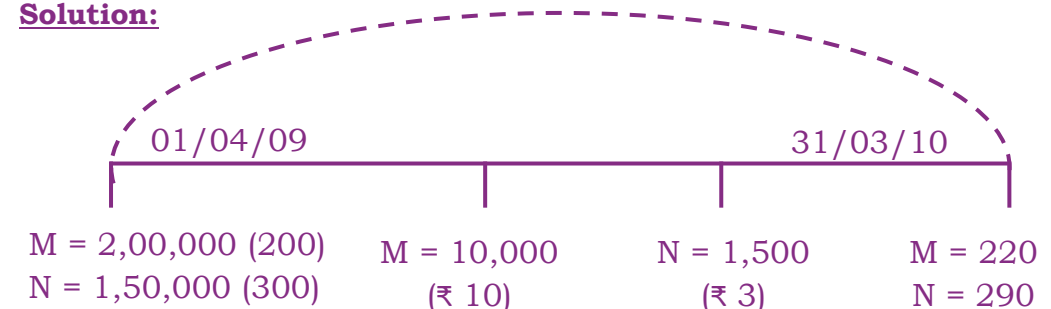
Name of Co.	No. of shares	Cost (₹)
M Ltd.	1,000 (₹ 100 each)	2,00,000
N Ltd.	500 (₹ 10 each)	1,50,000

On 1.4.2010, investment advisors indicate (a) that the dividends from M Ltd. and N Ltd. for the year ending 31.3.2011 are likely to be 20% and 35%, respectively and (b) that the probabilities of market quotations on 31.3.2011 are as below:

Probability factor	Price/share of M Ltd.	Price/share of N Ltd.
0.2	220	290
0.5	250	310
0.3	280	330

- (i) Calculate the average return from the portfolio for the year ended 31.3.2010;
- (ii) Calculate the expected average return from the portfolio for the year 2010-11; and
- (iii) Advise X Co. Ltd., of the comparative risk in the two investments by calculating the standard deviation in each case.

Solution:



Cost		Year End
M = 2,00,000	Dividend	M = 2,20,000
N = 1,50,000	10,000	N = 1,45,000
	1,500	
<u>3,50,000</u>	<u>11,500</u>	<u>3,65,000</u>

$$\text{Portfolio} = \frac{(3,65,000 - 3,50,000) + 11,500}{3,50,000} \times 100 = 7.57\%$$

Alternative

$$\text{Stock M} = \frac{(220 - 200) + 10}{200} \times 100 = 15\%$$

$$\text{Stock N} = \frac{(290 - 300) + 3}{300} \times 100 = -2.34\%$$

$$W_M = \frac{2,00,000}{3,50,000} \quad W_N = \frac{1,50,000}{3,50,000}$$

$$= 0.571 \quad = 0.429$$

$$\text{Portfolio Return} = (15 \times 0.571) + (-2.34 \times 0.429) = 7.56\%$$

(2) 2010 - 2011

	01/04/10	Dividend	31/03/11
M = 220		M = ₹ 20	M = (220 × 0.2) +
N = 290		N = ₹ 3.50	(250 × 0.5) + (280 ×
			0.3) = 253

$$ER_M = \frac{(253 - 220) + 20}{220} \times 100 = 24.10\%$$

$$ER_N = \frac{(312 - 290) + 3.50}{290} \times 100 = 8.79\%$$

$$W_M = \frac{2,20,000}{3,65,000} = 0.603$$

$$W_N = \frac{1,45,000}{3,65,000} = 0.397$$

$$ER_P = (24.10 \times 0.603) + (8.79 \times 0.397) = 18.02\%$$

(3) Standard Deviation

$$\text{M Ltd} = \frac{(220 - 220) + 20}{220} \times 100 = 9.09\%$$

$$= \frac{(250 - 220) + 20}{220} \times 100 = 22.73\%$$

$$= \frac{(280 - 220) + 20}{220} \times 100 = 36.36\%$$

$$\text{N Ltd} = \frac{(290 - 290) + 3.5}{290} \times 100 = 1.25\%$$

$$= \frac{(310 - 290) + 3.5}{290} \times 100 = 8.10\%$$

$$= \frac{(330 - 290) + 3.5}{290} \times 100 = 15\%$$

P	X	P (X)	(X - $\bar{X}$)	(X - $\bar{X}$) ² P	Y	P (Y)	(Y - $\bar{Y}$)	(Y - $\bar{Y}$) ² P
0.2	9.09	1.82	-	45.06	1.21	0.24	- 7.582	11.50
0.5	22.73	11.37	-1.37	0.94	8.10	4.04	- 0.692	0.24
0.3	36.36	10.91	12.26	45.09	15	4.5	6.208	11.56
	$\sum P (X) =$	24.10		91.09		8.792		23.30

$$\sigma_x = \sqrt{91.09} = 9.54\%$$

$$\sigma_y = \sqrt{23.30} = 4.83\%$$

Standard Deviation of M Ltd. is more than Standard Deviation of N Ltd.

Hence, stock of M Ltd. is more risky.

18.50%	20.00%	Under Valued	Buy
13.00%	13.00%	Correctly Valued	Hold



16.00%	17.00%	Under Valued	Buy
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Question – 09

Mr. A, a HNI invested on 1.4.2014 in certain equity shares as below:

Name of Co.	No. of shares	Cost (₹)
X Ltd.	1,00,000 (₹ 100 each)	2,00,00,000
Y Ltd.	50,000 (₹ 10 each)	1,50,00,000

In September, 2014, 10% dividend was paid out by X Ltd. and in October, 2014, 30% dividend paid out by Y Ltd. On 31.3.2015 market quotations showed a value of ₹ 220 and ₹ 290 per share for X Ltd. and Y Ltd. respectively.

On 1.4.2015, a technical analyst indicated as follows:

- (1) That the probabilities of dividends from X Ltd. and Y Ltd. for the year ending 31.3.2016 are as below:

Probability Factor	Dividend from X Ltd. (%)	Dividend from Y Ltd. (%)
0.2	10	15
0.3	15	20
0.5	20	35

- (2) That the probabilities of market quotations on 31.3.2016 are as below:

Probability factor	Price/share of M Ltd.	Price/share of N Ltd.
0.2	220	290
0.5	250	310
0.3	280	330

You are required to:

- (i) Calculate the average return from the portfolio for the year ended 31.3.2015;
- (ii) Calculate the expected average return from the portfolio for the year 2015-16; and

- (iii) Advise Mr. A, of the comparative risk in the two investments.
(RTP September – 2025)

Solution:

- (i) Average return from the portfolio for the year ended 31.3.2015

Calculation of return on Portfolio for 2014 – 15	(Calculation in ₹/share)		
	X Ltd.	Y Ltd.	
Dividend received during the year	10	3	
Capital gain/loss by 31.03.15			
Market value by 31.03.15	220	290	
Cost of investment	200	300	
Gain/loss	20	(-)10	
Yield	30	(-)7	
Cost	200	300	
% return	15%	(-)2.33%	
Weight in the portfolio	57.14	42.86	
Weighted average return			7.56%

- (ii) Average return from the portfolio for the year ended 2015-16 shall be calculated using the concept of joint probability as follows:

PORTFOLIO MANAGEMENT

X Ltd.

Path	Income from Dividend (₹)	Gain from Market Price (₹)	Total Yield (₹)	Joint Prob.	Exp. Yield (₹)
1	10	220 - 220 = 00	10	0.20 × 0.20 = 0.04	0.40
2	10	250 - 220 = 30	40	0.20 × 0.50 = 0.10	4.00
3	10	280 - 220 = 60	70	0.20 × 0.30 = 0.06	4.20
4	15	220 - 220 = 00	15	0.30 × 0.20 = 0.06	0.90
5	15	250 - 220 = 30	45	0.30 × 0.50 = 0.15	6.75
6	15	280 - 220 = 60	75	0.30 × 0.30 = 0.09	6.75
7	20	220 - 220 = 00	20	0.50 × 0.20 = 0.10	2.00
8	20	250 - 220 = 30	50	0.50 × 0.50 = 0.25	12.50
9	20	280 - 220 = 60	80	0.50 × 0.30 = 0.15	12.00
Expected Yield(₹)					49.50
Market Value on 01.04.2015(₹)					220
%Return					22.50

Y Ltd.

Path	Income from Dividend (₹)	Gain from Market Price (₹)	Total Yield (₹)	Joint Prob.	Exp. Yield (₹)
1	1.50	290 - 290 = 00	1.50	0.20 × 0.20 = 0.04	0.06
2	1.50	310 - 290 = 20	21.50	0.20 × 0.50 = 0.10	2.15
3	1.50	330 - 290 = 40	41.50	0.20 × 0.30 = 0.06	2.49
4	2.00	290 - 290 = 00	2.00	0.30 × 0.20 = 0.06	0.12
5	2.00	310 - 290 = 20	22.00	0.30 × 0.50 = 0.15	3.30
6	2.00	330 - 290 = 40	42.00	0.30 × 0.30 = 0.09	3.78
7	3.50	290 - 290 = 00	3.50	0.50 × 0.20 = 0.10	0.35
8	3.50	310 - 290 = 20	23.50	0.50 × 0.50 = 0.25	5.88
9	3.50	330 - 290 = 40	43.50	0.50 × 0.30 = 0.15	6.52
Expected Yield(₹)					24.65

Market Value on 01.04.2015(₹)	290
%Return	8.50

Weight in portfolio $(1,00,000 \times 220) : (50,000 \times 290) = 60.27:39.73$

Weighted average (Expected) return $(0.6027 \times 22.50 + 0.3973 \times 8.50) = 16.94\%$

(iii) To analyze the risk of each investment we need to calculate the Standard Deviation of each investment as follows:

X Ltd.

Path	Prob. (1)	Yield (₹)	Dev. $(P_x - P_{\bar{x}})$	Square of dev. (2)	(1)×(2)
1	0.04	10	-39.50	1560.25	62.41
2	0.10	40	-9.50	90.25	9.03
3	0.06	70	20.50	420.25	25.22
4	0.06	15	-34.50	1190.25	71.42
5	0.15	45	-4.50	20.25	3.04
6	0.09	75	25.50	650.25	58.52
7	0.10	20	-29.50	870.25	87.03
8	0.25	50	0.50	0.25	0.06
9	0.15	80	30.50	930.25	139.54
					$\sigma^2 M = 456.27$

Standard Deviation (σ_x) 21.36



Y Ltd.

Path	Prob. (1)	Yield (₹)	Dev. (Py - Pȳ)	Square of dev.(2)	(1) × (2)
1	0.04	1.50	-23.15	535.92	21.44
2	0.10	21.50	-3.15	9.92	0.99
3	0.06	41.50	16.85	283.92	17.04
4	0.06	2.00	-22.65	513.02	30.78
5	0.15	22.00	-2.65	7.02	1.05
6	0.09	42.00	17.35	301.02	27.09
7	0.10	3.50	-21.15	447.32	44.73
8	0.25	23.50	-1.15	1.32	0.33
9	0.15	43.50	18.85	355.32	53.30
					σ²N=196.75

Standard Deviation (σy) 14.03

Although expected return is higher in case of X Ltd. but it also has higher risk due to High S.D.

- (ii) Assuming perfect correlation, show whether it is preferable to invest 75% in A and 25% in C or to invest 100% in E
(SM TYK – 05)

Solution:

(i) Efficient Securities

- Security B & F are inefficient because Security A provides same return at lower level of risk.
- Security D is inefficient because security A provides higher return at the same level of risk
Hence efficient stocks are A, C & E.

(ii) 75% in A & 25% in C

$$ER_p = (8 \times 0.75) + (12 \times 0.25) = 9\%$$

$$SD_p = (4 \times 0.75) + (12 \times 0.25) = 6\%$$

100 % in E

ER = 9%

S.D = 5%

Investment in 100% in stock E is better than portfolio because Stock E provides same return at lower level risk.

Question – 11

Suppose that economy A is growing rapidly and you are managing a global equity fund and so far you have invested only in developed-country stocks only. Now you have decided to add stocks of economy A to your portfolio. The table below shows the expected rates of return, standard deviations, and correlation coefficients (all

EFFICIENT & OPTIMUM PORTFOLIO

Question – 10

Following is the data regarding six securities:

	A	B	C	D	E	F
Return (%)	8	8	12	4	9	8
Risk (Standard deviation)	4	5	12	4	5	6

- (i) Assuming three will have to be selected, state which ones will be picked.



estimates are for aggregate stock market of developed countries and stock market of Economy A).

	Developed Country Stocks	Stocks of Economy A
Expected rate of return (annualized percentage)	10	15
Risk [Annualized Standard Deviation (%)]	16	30
Correlation Coefficient (ρ)	0.30	

Assuming the risk-free interest rate to be 3%, you are required to determine:

- What percentage of your portfolio should you allocate to stocks of Economy A if you want to increase the expected rate of return on your portfolio by 0.5%?
- What will be the standard deviation of your portfolio assuming that stocks of Economy A are included in the portfolio as calculated above?
- Also show how well the Fund will be compensated for the risk undertaken due to inclusion of stocks of Economy A in the portfolio?

(SM TYK – 24)

Solution:

(1) Calculation of weights

$$ERP = ER_A \times W_A + ER_B \times W_B$$

$$10.5 = 10W_A + 15(1 - W_A)$$

$$10.5 = 10W_A + 15 - 15W_A$$

$$W_A = \frac{4.5}{5} = 0.9$$

$$W_B = 0.1$$

Developed Country = 90%

Economy A = 10%

(2) SD of Portfolio

$$\begin{aligned}\sigma_p &= \sqrt{\sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times \sigma_A \times W_A \times \sigma_B \times W_B \times r_{AB}} \\ &= \sqrt{16^2 \times 0.9^2 + 30^2 \times 0.1^2 + 2 \times 16 \times 0.9 \times 30 \times 0.1 \times 0.3} \\ &= 15.56\%\end{aligned}$$

(3) Sharpe ratio

$$SR = \frac{ER - RF}{\sigma}$$

$$\text{Develop country only} = \frac{10 - 3}{16} = 0.4375$$

$$\text{Portfolio} = \frac{10.5 - 3}{15.56} = 0.482$$

Sharpe ratio will improve.



Question – 12

Security	Standard Deviation (σ)	Weight (w)
R	20%	0.8
S	50%	0.2

Calculation upto two decimal place.

From the information given above, you are required to calculate Portfolio Standard Deviations, if –

- Securities returns are independent.
- Securities returns are perfectly negatively correlated.
- Securities returns are perfectly positively correlated.

(Exam May – 2025) (6 Marks)

Solution:

- Portfolio Variance and Standard Deviation of Portfolio if securities returns are independent:

$$\sigma_P^2 = x_1^2 \sigma_1^2 + x_2^2 \sigma_2^2$$

Accordingly,

$$\text{Variance } (\sigma_P^2) = (0.80)^2 (20)^2 + (0.20)^2 (50)^2$$

$$\text{Variance } (\sigma_P^2) = 256 + 100 = 356$$

$$\text{Standard Deviation } (\sigma_P) = \sqrt{356} = 18.87\%$$

- Portfolio Variance and Standard Deviation of Portfolio if securities returns are perfectly negatively related:

$$\sigma_P^2 = (x_1 \sigma_1 - x_2 \sigma_2)^2$$

$$\sigma_P = x_1 \sigma_1 - x_2 \sigma_2$$

Accordingly,

$$\text{Variance } (\sigma_P^2) = (0.80 \times 20 - 0.20 \times 50)^2$$

$$\text{Variance } (\sigma_P^2) = (6)^2 = 36$$

$$\text{Standard Deviation } (\sigma_P) = \sqrt{36} = 6\%$$

- Portfolio Variance and Standard Deviation of Portfolio if securities returns are perfectly positively related:

$$\sigma_P^2 = (x_1 \sigma_1 + x_2 \sigma_2)^2$$

$$\sigma_P = x_1 \sigma_1 + x_2 \sigma_2$$

Accordingly,

$$\text{Variance } (\sigma_P^2) = (0.80 \times 20 + 0.20 \times 50)^2$$

$$\text{Variance } (\sigma_P^2) = (26)^2 = 676$$

$$\text{Standard Deviation } (\sigma_P) = \sqrt{676} = 26\%$$

MINIMUM VARIANCE PORTFOLIO**Question – 13**

An investor has two portfolios known to be on minimum variance set for a population of three securities A, B and C having below mentioned weights:



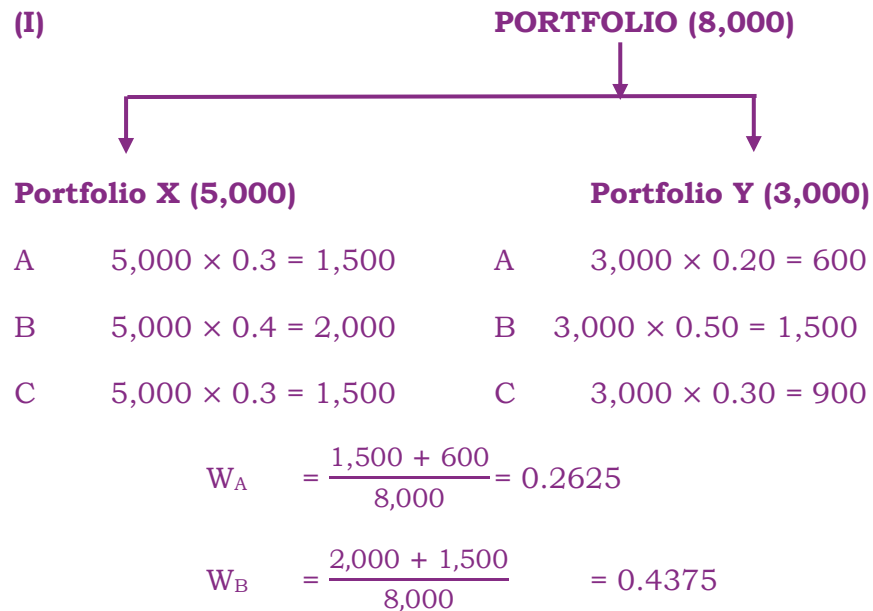
	WA	WB	WC
Portfolio X	0.30	0.40	0.30
Portfolio Y	0.20	0.50	0.30

It is supposed that there are no restrictions on short sales.

- (i) What would be the weight for each stock for a portfolio constructed by investing ₹ 5,000 in portfolio X and ₹ 3,000 in portfolio Y?
- (ii) Suppose the investor invests ₹ 4,000 out of ₹ 8,000 in security A. How he will allocate the balance between security B and C to ensure that his portfolio is on minimum variance set?

(SM TYK - 31)

Solution:



$$W_C = \frac{1,500 + 900}{8,000} = 0.30$$

(II) Calculation of W_B & W_C if $W_A = 0.5$

$$Y = a + bx$$

$$0.4 = a + 0.30b \quad \text{.....(i)}$$

$$0.5 = a + 0.20b \quad \text{.....(ii)}$$

$$-0.1 = 0.1b$$

$$b = \frac{-0.1}{0.1} = -1 \text{ Slope}$$

Put Slope in equation (i)

$$0.4 = a + 0.30 \times -1$$

$$0.4 = a - 0.30$$

$$a = 0.70$$

Equation

$$Y = 0.70 + -1x$$

$$\text{If } W_A = 0.5$$

$$W_B = 0.70 (-1 \times 0.5) = 0.2$$

$$W_C = 1 - 0.5 - 0.2 = 0.3$$

$$\text{Allocation} = A = 8,000 \times 0.5 = 4,000$$



$$B = 8,000 \times 0.2 = 1,600$$

$$C = 8,000 \times 0.3 = 2,400$$

Question – 14

An Investor is proposing to invest ₹ 10,000/- in two Portfolios A and B in the ratio of 3:2. The Portfolios have three securities each with following weights:

	W_x	W_y	W_z
Portfolio A	0.30	0.25	0.45
Portfolio B	0.20	0.45	0.35

You are required to

- Calculate the weight of each stock.
- Calculate the amount allocated to B and C if half of the funds are allocated to security X.

Solution:

1. Calculation of Weights

10,000 in 3:2	X	Y	Z
(6,000) Portfolio A	1,800	1,500	2,700
(4,000) Portfolio B	800	1,800	1,400
	<u>2,600</u>	<u>3,300</u>	<u>4,100</u>

$$W_x = 2600/10000 = 0.26$$

$$W_y = 3,300/10,000 = 0.33$$

$$W_z = 4,100/10,000 = 0.41$$

2. Calculation of W_y & W_z when W_x = 0.5 :

Using Critical Line:

$$Y = a + bx$$

$$0.25 = a + 0.30 \quad \dots\dots\dots(i)$$

$$0.45 = a + 0.20 \quad \dots\dots\dots(ii)$$

$$-0.20 = 0.1b$$

$$b = \frac{-0.20}{0.1} = -2$$

Putting b value in equation (i)

$$0.25 = a + (-2 \times 0.30)$$

$$a = 0.85$$

Equation:

$$y = 0.85 - 2x$$

$$W_x = 0.85 (-2 \times 0.5)$$

$$W_x = -0.15$$

$$W_z = 1 - (0.5 - 0.15) \\ = 0.65$$

Investment:

$$X = 10,000 \times 0.5 = 5,000$$

$$Y = 10,000 \times (-0.15) = (1,500) \text{ Short sale.}$$

$$Z = 10,000 \times 0.65 = 6,500$$



PART II: CAPITAL ASSET PROCING MODEL

CALCULATION OF BETA

Question – 15

The distribution of return of security 'F' and the market portfolio 'P' is given below:

Probability	Return %	
	F	P
0.30	30	-10
0.40	20	20
0.30	0	30

You are required to calculate the expected return of security 'F' and the market portfolio 'P', the covariance between the market portfolio and security and beta for the security.

(SM TYK – 09)

Solution:

Calculation of Cov_{xy}

p	x	p(x)	(x - $\bar{x}$)	(x - $\bar{x}$) ² p	y	P(y)	(y - $\bar{y}$)	(y - $\bar{y}$) ² p	(x - $\bar{x}$)(y - $\bar{y}$)p
0.30	30	9	13	50.7	-10	-3	-24	172.80	-93.60
0.40	20	8	3	3.60	20	8	6	14.40	7.20
0.30	0	0	-17	86.70	30	9	16	76.80	-81.60
	$\bar{x} =$	17%		141(%) ²	$\bar{y} =$	14%		264(%) ²	-168(%) ²

$$\sigma_x = \sqrt{141(\%)^2} = 11.87\%$$

$$\sigma_y = \sqrt{264(\%)^2} = 16.25\%$$

$$\beta = \frac{Cov_{xm}}{\sigma_m^2} = \frac{-168(\%)^2}{264(\%)^2} = -0.636$$

Question – 16

Given below is information of market rates of Returns and Data from two Companies A and B:

	Year 2007	Year 2008	Year 2009
Market (%)	12.0	11.0	9.0
Company A (%)	13.0	11.5	9.8
Company B (%)	11.0	10.5	9.5

You are required to determine the beta coefficients of the shares of Company A and Company B.

(SM TYK – 10)

Solution:

Beta of Portfolio in case of A & Market

Year	x	(x - $\bar{x}$)	(x - $\bar{x}$) ²	y	(y - $\bar{y}$)	(y - $\bar{y}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	13	1.57	2.46	12	1.33	1.77	2.09
2	11.5	0.07	0.0049	11	0.33	0.11	0.02
3	9.8	-1.63	2.66	9	-1.67	2.79	2.72
	34.3		5.12	32		4.67	4.83

$$\bar{x} = 11.43\%$$



$$\sigma_x = \sqrt{\frac{5.12}{3}} = 1.31\%$$

$$\bar{y} = 10.67\%$$

$$\sigma_y = \sqrt{\frac{4.67}{3}} = 1.25\%$$

$$\text{Cov}_{xy} = \frac{4.83}{3} = 1.61(\%)^2$$

$$B_p = \frac{1.31}{1.25} \times 0.983 = 1.03$$

$$r_{xm} = \frac{1.61}{1.31 \times 1.25} = 0.983$$

Beta of Portfolio in case of B & Market

Year	x	(x - $\bar{x}$)	(x - $\bar{x}$) ²	y	(y - $\bar{y}$)	(y - $\bar{y}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	11	0.67	0.45	12	1.33	1.77	0.89
2	10.5	0.17	0.03	11	0.33	0.11	0.06
3	9.5	-0.83	0.69	9	-1.67	2.79	1.39
	31		1.17	32		4.67	2.34

$$\bar{x} = 10.33\%$$

$$\sigma_x = \sqrt{\frac{1.17}{3}} = 0.62\%$$

$$\bar{y} = 10.67\%$$

$$\sigma_y = \sqrt{\frac{4.67}{3}} = 1.25\%$$

$$\text{Cov}_{xy} = \frac{2.34}{3} = 0.78(\%)^2$$

$$r_{xy} = \frac{0.78}{0.62 \times 1.25} = 1.006$$

$$B_p = \frac{0.62}{0.125} \times 1.006 = 0.50$$

Question - 17

The rates of return on the security of Company X and market portfolio for 10 periods are given below

Period	Return of Security X (%)	Return on market portfolio (%)
1	20	22
2	22	20
3	25	18
4	21	16
5	18	20
6	-5	8
7	17	-6
8	19	5
9	-7	6
10	20	11

- What is the beta of Security X?
- What is the characteristic line for Security X?

(SM TYK - 12)

Solution:

Year	x	(x - $\bar{x}$)	(x - $\bar{x}$) ²	y	(y - $\bar{y}$)	(Y - $\bar{Y}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	20	5	25	22	10	100	50



2	22	7	49	20	8	64	56
3	25	10	100	18	6	36	60
4	21	6	36	16	4	16	24
5	18	3	9	20	8	64	24
6	-5	-20	400	8	-4	16	80
7	17	2	4	-6	-18	324	-36
8	19	4	16	5	-7	49	-28
9	-7	-22	484	6	-6	36	132
10	20	5	25	11	-1	1	-5
x =	15%	$\sum(x - \bar{x})^2 =$	1148 (%) ²	y =120%		706(%) ²	357(%) ²

$$\sigma_x = \frac{\sqrt{1148}}{10} = 10.71\%$$

$$\sigma_y = \frac{\sqrt{706}}{10} = 8.402\%$$

$$\text{Cov}_{xy} = \frac{357\%}{10} = 35.7(\%)^2$$

$$(1) \text{ Beta} = \frac{\text{COV}_{xm}}{\sigma_m^2} = \frac{35.7}{(8.402)^2} = 0.505$$

Or,

$$\text{Beta} = \frac{\sigma_x}{\sigma_m} \times r_{xm}$$

$$r_{xm} = \frac{\text{COV}_{xm}}{\sigma_x \times \sigma_y} = \frac{35.7}{10.71 \times 8.402} = 0.397$$

$$= \frac{10.71}{8.402} \times 0.397 = 0.505$$

Characteristic Line (CL): Characteristic Line is used to calculate Beta (β) & it starts from Alpha (α).

Equation of Characteristic Line

$$\bar{y} = a + b\bar{x}$$

$$\bar{y} = \alpha + bR_m$$

(2) Characteristic Line

$$\bar{y} = \alpha + b\bar{x} \quad \bar{x} = \text{Market}$$

$$15 = \alpha + 0.505 \times 12$$

$$\bar{y} = \text{Stock}$$

$$15 = \alpha + 6.06$$

$$\alpha = 8.94$$

Equation of Characteristic Line

$$\bar{y} = 8.94 + 0.505x$$

Question – 18

The following information is available in respect of Security X

Equilibrium Return	15%
Market Return	15%
7% Treasury Bond Trading at	\$140
Covariance of Market Return and Security Return	225%
Coefficient of Correlation	0.75

You are required to determine the Standard Deviation of Market Return and Security Return.

(SM TYK – 19)

Solution:

First we shall compute the β of Security X



$$\text{Risk Free Rate} = \frac{\text{Coupon Payment}}{\text{Current Market Price}} = \frac{7}{140} = 5\%$$

Assuming equilibrium return to be equal to CAPM return then:

$$15\% = R_f + \beta_X(R_m - R_f)$$

$$15\% = 5\% + \beta_X(15\% - 5\%)$$

$$\beta_X = 1$$

or it can also be computed as follows:

$$\frac{R_m}{R_s} = \frac{15\%}{15\%} = 1$$

(i) Standard Deviation of Market Return

$$\beta_x = \frac{\text{Cov}_{X, M}}{\sigma_m^2} = \frac{225\%}{\sigma_m^2} = 1$$

$$\sigma_m^2 = 225$$

$$\sigma_m = \sqrt{225} = 15\%$$

(ii) Standard Deviation of Security Return

$$\begin{aligned} \beta_X &= \frac{\sigma_x}{\sigma_m} \times \sigma_{xm} \\ &= \frac{\sigma_x}{15} \times 0.75 = 1 \end{aligned}$$

$$\sigma_x = \frac{15}{0.75} = 20\%$$

PORTFOLIO BETA & BETA MANAGEMENT

Question – 19

A Portfolio Manager (PM) has the following four stocks in his portfolio:

Security	No. of Shares	Market Price per Share (₹)	β
VSL	10,000	50	0.9
CSL	5,000	20	1.0
SML	8,000	25	1.5
APL	2,000	200	1.2

Compute the following:

- Portfolio beta.
- If the PM seeks to reduce the beta to 0.8, how much risk free investment should he bring in?
- If the PM seeks to increase the beta to 1.2, how much risk free investment should he bring in?

(SM TYK – 36)

Solution:

(I) Portfolio Beta

Security	Share	Market Price	Amount	Weight	Beta	Weight $\times$ Beta
VSL	10,000	50	5,00,000	0.416	0.9	0.375
CSL	5,000	20	1,00,000	0.083	1.0	0.083
SML	8,000	25	2,00,000	0.3	1.5	0.250
APL	2,000	200	4,00,000	0.166	1.2	0.400
			12,00,000	0.333		1.108

(II) $B_T = 0.8$



PORTFOLIO MANAGEMENT

$$W_A = \frac{B_T}{B_P} = \frac{0.8}{1.108} = 0.722$$

$$\text{Portfolio Amount} = \frac{12,00,000}{0.722} = 16,62,050$$

$$R_f \text{ Amount} = 16,62,050 - 12,00,000 = 4,62,050$$

Revised Portfolio Beta

Stock	Amount	Weight	Beta	Weight × Beta
VSL	5,00,000	0.301	0.9	0.271
CSL	1,00,000	0.060	1.0	0.060
SML	2,00,000	0.120	1.5	0.180
APL	4,00,000	0.241	1.2	0.289
Risk Free Asset	4,62,050	0.278	0	0
	16,62,050			0.800

(III) $B_T = 1.2$

$$W_A = \frac{B_T}{B_P} = \frac{1.20}{1.108} = 1.083$$

$$\text{Portfolio Amount} = \frac{12,00,000}{1.083} = 11,08,030$$

$$R_f \text{ Amount} = 11,08,030 - 12,00,000 = -91,967$$

$$\text{Borrow at } R_f = ₹ 91,967$$

Revised Portfolio Beta

Stock	Amount	Weight	Beta	W × B
VSL	5,00,000	0.451	0.9	0.406
CSL	1,00,000	0.090	1	0.909
SML	2,00,000	0.181	1.5	0.272

APL	4,00,000	0.361	1.2	0.433
Risk Free Asset	-91,967	-0.083	0	0
	11,08,030			1.20

Question – 20

Mr. Tempest has the following portfolio of four shares:

Name	Beta	Investment Lac.
Oxy Rin Ltd.	0.45	0.80
Boxed Ltd.	0.35	1.50
Square Ltd.	1.15	2.25
Ellipse Ltd.	1.85	4.50

The risk free rate of return is 7% and the market rate of return is 14%.

Required.

- Determine the Portfolio Return.
- Calculate the Portfolio Beta.

(SM TYK – 16, RTP November – 2018)

Solution:

Portfolio Beta

$$\beta_P = \frac{(0.45 \times 0.80) + (0.35 \times 1.50) + (1.15 \times 2.25) + (1.85 \times 4.50)}{9.05}$$

$$= 1.3035$$

(i) Portfolio Return

$$E_{RP} = R_f + (R_m - R_f) \beta_P$$

$$= 7 + (14 - 7) 1.3035$$



$$= 16.1245\%$$

(ii) Portfolio Beta

$$\beta_P = 1.3035$$

Question – 21

Mr. Fed Up wants to invest an amount of ₹ 520 lakhs and had approached his Portfolio Manager. The Portfolio Manager had advised Mr. Fed Up to invest in the following manner:

Security	Moderate	Better	Good	Very Good	Best
Amount (in ₹ Lakhs)	60	80	100	120	160
Beta	0.5	1.00	0.80	1.20	1.50

You are required to advise Mr. Fed Up in regard to the following, using Capital Asset Pricing Methodology:

- Expected return on the portfolio, if the Government Securities are at 8% and the NIFTY is yielding 10%.
- Advisability of replacing Security 'Better' with NIFTY.

(SM TYK – 25)

Solution:

(1) Computation of Expected Return from Portfolio

Stock	Amount	Weight	Beta	Weight × Beta
Moderate	60	0.115	0.5	0.058
Better	80	0.154	1.00	0.154
Good	100	0.192	0.80	0.154
Very Good	120	0.231	1.20	0.277

PORTFOLIO MANAGEMENT

Best	160	0.308	1.50	0.462
	520			1.105

$$\begin{aligned} ER_P &= R_f + (R_m - R_f) B_P \\ &= 8 + (10 - 8) 1.105 \\ &= 10.21\% \end{aligned}$$

- (2)** As computed above the expected return from Better is 10% same as from Nifty, hence there will be no difference even if the replacement of security is made. The main logic behind this neutrality is that the beta of security 'Better' is 1 which clearly indicates that this security shall yield same return as market return.

Question – 22

M/s. Siri Ltd. Has a surplus amount of ₹ 3 crores to invest and has shortlisted the following equity shares:

Company	Beta
S Ltd.	1.6
K Ltd.	1
P Ltd.	-0.3
D Ltd.	2
C Ltd.	0.6

Required:

- If M/s. Siri Ltd. invests an equal amount in all securities, what is the beta of the portfolio?
- If M/s. Siri Ltd. invests 15% of its investment in S Ltd., 15% in P Ltd., 10% in C Ltd. and the balance in equal amount in the other two securities, what is the beta of the portfolio?



- (iii) If the expected return of market portfolio is 12% at a beta factor of 1.0, what will be the portfolios expected return in both the situations given above?
- (iv) If the Company changes its policy to invest in any 3 securities with a minimum of 20% in each of these 3 securities to diversify risk, you are requested to advise the company to have a right mix of securities to maximize the return in the following two scenarios and also calculate the expected return:

(1) Bull Phase: Expected Market returns 10%

(2) Bear Phase: Expected Market returns — 5%

(Exam Nov – 2022) (8 Marks)

Solution:

$$(i) \quad B_P = \frac{1.6 + 1 - 0.30 + 2 + 0.6}{5}$$

$$= 0.98$$

$$(ii) \quad B_P = (1.60 \times 0.15) + (1 \times 0.30) + (-0.3 \times 0.15) + (2 \times 0.30) + (0.6 \times 0.10)$$

$$= 1.155$$

- (iii) [Since R_f is not given hence we calculate ER_P as below]

$$ER_P = R_m \times B_P$$

$$\text{Situation I} = 12 \times 0.98 = 11.76\%$$

$$\text{Situation II} = 12 \times 1.155 = 13.86\%$$

- (iv) **Bull Phase**

In this situation portfolio Beta should be higher hence we invest 20% in S Ltd. 20% in K Ltd. & 60% in D Ltd.

$$B_P = (1.60 \times 0.20) + (1 \times 0.20) + (2 \times 0.60)$$

$$= 1.72$$

$$ER_P = 10 \times 1.72$$

$$= 17.20\%$$

Bearish Phase

In this situation portfolio beta should be lower hence we invest 20% in C Ltd., 20% in K Ltd. & 60% in P Ltd.

$$B_P = (0.6 \times 0.2) + (1 \times 0.2) + (-0.3 \times 0.6)$$

$$= 0.14$$

$$ER_P = -5 \times 0.14$$

$$= -0.7\%$$

Question – 23

Mr. A is having 1 lakh shares of K Ltd. The beta of the company is 1.40.

Mr. B a financial advisor has suggested having the following portfolio:

Security	Beta	% Holding
L	1.20	10
M	0.75	10
N	0.40	30
O	1.40	50
		100

Market Return is 12%



Risk free rate is 8%

You are required to calculate the following for the present investment and suggested portfolio:

- (i) What is the expected return based on CAPM and also
- (1) If the market goes up by 2.5%.
 - (2) If the market goes down by 2.5%.
 - (3) If the market is giving a negative return of 2.5%.
- (ii) If the probability of market giving negative return is more, please advise Mr. A whether to continue the holdings of M/s. K Ltd. or to buy the portfolio as per the suggestion of Mr. B. If so, why?

(RTP May – 2022)

Solution:

(i) Expected Return as per CAPM

Beta of K Ltd. = 1.40

$$B_p = (1.20 \times 0.10) + (0.75 \times 0.10) + (0.40 \times 0.30) + (1.40 \times 0.50)$$

$$= 1.015$$

$$E_R = R_f + \beta (R_m - R_f)$$

$$\text{K Ltd.} = 8 + 1.40 (12 - 8) = 13.6\%$$

$$\text{Portfolio} = 8 + 1.015 (12 - 8) = 12.06\%$$

(1) If market return goes up by 2.5%

$$\text{It means } R_m = [12 + 2.5] = 14.5\%$$

$$E_R \text{ K Ltd.} = 8 + (14.5 - 8) 1.40 = 17.1\%$$

$$\text{Portfolio} = 8 + (14.5 - 8) 1.015 = 14.6\%$$

(2) If market goes down by 2.5%

$$R_m = [12 - 2.5] = 9.5\%$$

$$E_R \text{ K Ltd.} = 8 + (9.5 - 8) 1.40 = 10.1\%$$

$$\text{Portfolio} = 8 + (9.5 - 8) 1.015 = 9.52\%$$

(3) $R_m = -2.5$

$$\text{K Ltd.} = 8 + (-2.5 - 8) 1.40 = -6.7\%$$

$$\text{Portfolio} = 8 + (-2.5 - 8) 1.015 = -2.66\%$$

- (ii) Since $B_p = 1.015$ is less than Beta of K Ltd. (1.40) hence it is better to invest in portfolio.

Question – 24

Following is the information related to return on shares of three different companies:

Years	A Ltd.	B Ltd.	C Ltd.
2018	2%	3%	5%
2019	6%	8%	7%
2020	13%	14%	15%
2021	7%	9%	11%

Required:



- (i) Construct maximum number of portfolio and its return, if each portfolio consists of any two Company's shares in proportion of 65% and 35% and suggest which portfolio provides highest return.
- (ii) Calculate portfolio return and beta (β), if Mr. X invests ₹ 65,000 in A Ltd. having beta (β) of 0.45; ₹ 20,000 in B Ltd. having beta (β) of 1.15 and ₹ 15,000 in C Ltd. having beta (β) of 1.8.

(Exam Nov – 2022) (8 Marks)

Solution:

Year	A Ltd.	B Ltd.	C Ltd.
2018	2	3	5
2019	6	8	7
2020	13	14	15
2021	7	9	11
	7%	8.5%	9.5%

(I) No. of Portfolio & Return

- (1) A & B (65:35)

$$ER_p = (0.65 \times 7) + (0.35 \times 8.5) = 7.525\%$$

- (2) B & A (65:35)

$$ER_p = (0.65 \times 8.5) + (0.35 \times 7) = 7.98\%$$

- (3) A & C (65:35)

$$ER_p = (0.65 \times 7) + (0.35 \times 9.5) = 7.88\%$$

- (4) C & A (65:35)

$$= (0.65 \times 9.5) + (0.35 \times 7) = 8.63\%$$

- (5) B & C (65:35)

$$= (0.65 \times 8.5) + (0.35 \times 9.5) = 8.85\%$$

- (6) C & B (65:35)

$$= (0.65 \times 9.5) + (0.35 \times 8.5) = 9.15\%$$

Since maximum return is under combination (6) i.e. 65% investment in C Ltd. & 35% investment in B Ltd. hence it should be opted.

(II) Portfolio Return & Beta.

$$ER_p = \frac{(65,000 \times 7) + (20,000 \times 8.5) + (15,000 \times 9.5)}{1,00,000} = 7.68\%$$

$$\beta_p = (0.65 \times 0.45) + (0.2 \times 1.15) + (0.15 \times 1.8) = 0.793$$

Question – 25

Your client is holding the following securities:

Particulars of Securities	Cost ₹	Dividends ₹	Market Price ₹	BETA
Equity shares:				
Co. X	8,000	800	8,200	0.8
Co. Y	10,000	800	10,500	0.7
Co. Z	16,000	800	22,000	0.5
PSU Bonds	34,000	3,400	32,300	0.2

Assuming a Risk-free rate of 15%, calculate:

- Expected rate of return in each, using the Capital Asset Pricing Model (CAPM).
- Simple Average return of the portfolio.

(SM TYK – 28)

Solution:

Calculation of expected return on market portfolio (R_m)

Investment	Cost (₹)	Dividends (₹)	Capital Gains (₹)
shares X	8,000	800	200
shares Y	10,000	800	500
shares Z	16,000	800	6,000
PSU Bonds	<u>34,000</u>	<u>3,400</u>	<u>-1700</u>
	<u>68,000</u>	<u>5,800</u>	<u>5,000</u>

$$R_m = \frac{5,800 + 5,000}{68,000} \times 100 = 15.88\%$$

Calculation of expected rate of return on individual security:

Security

Shares X	15 + 0.8 (15.88 – 15.0)	= 15.70%
Shares Y	15 + 0.7 (15.88 – 15.0)	= 15.62%
Shares Z	15 + 0.5 (15.88 – 15.0)	= 15.44%
PSU Bonds	15 + 0.2 (15.88 – 15.0)	= 15.18%

Calculation of the Average Return of the Portfolio:

$$= \frac{15.70 + 15.62 + 15.44 + 15.18}{4} = 15.49\%.$$

Question – 26

Your client is holding the following securities:

PORTFOLIO MANAGEMENT

Particulars of Securities	Cost ₹	Dividends/ Interest ₹	Market Price ₹	Beta
Equity Shares:				
Gold Ltd.	10,000	1,725	9,800	0.6
Silver Ltd.	15,000	1,000	16,200	0.8
Bronze Ltd.	14,000	700	20,000	0.6
GOI Bonds	36,000	3,600	34,500	0.01

Average return of the portfolio is 15.7%, calculate:

- Expected rate of return in each, using the Capital Asset Pricing Model (CAPM).
- Risk free rate of return

(SM TYK – 26)

Solution:

Expected rate of return on market portfolio

$$\frac{\text{Dividend Earned} + \text{Capital nappreciation}}{\text{Initial investment}} \times 100$$

$$= \frac{₹ 7,025 + ₹ 5,500}{₹ 7,025} \times 100 = 16.7\%$$

Risk Free Return

$$\text{Average of Betas} = \frac{0.6 + 0.8 + 0.6 + 0.01}{4} = \text{Average of Betas}^* = 0.50$$

Average return = Risk free return + Average Betas (Expected return of market – Risk free return)

$$15.7 = \text{Risk free return} + 0.50 (16.7 - \text{Risk free return})$$

$$\text{Risk free return} = 14.7\%$$



* Alternatively, it can also be calculated through Weighted Average Beta.

Expected Rate of Return for each security is

$$\text{Rate of Return} = R_f + B (R_m - R_f)$$

$$\text{Gold Ltd.} = 14.7 + 0.6 (16.7 - 14.7) = 15.90\%$$

$$\text{Silver Ltd.} = 14.7 + 0.8 (16.7 - 14.7) = 16.30\%$$

$$\text{Bronze Ltd.} = 14.7 + 0.6 (16.7 - 14.7) = 15.90\%$$

$$\text{GOI Bonds} = 14.7 + 0.01 (16.7 - 14.7) = 14.72\%$$

*Alternatively, it can also be computed by using Weighted Average Method.

Solution updated in May-2025 SM:

*Alternatively, it can also be computed as follows:

Particulars of Securities (1)	Cost ₹ (2)	Market Price ₹ (3)	Dividend ₹ (4)	Capital Gain ₹ (5)	Beta (6)	(3) × (6)
Gold Ltd.	10,000	9,800	1,725	-200	0.6	5,880
Silver Ltd.	15,000	16,200	1,000	1,200	0.8	12,960
Bronz Ltd.	14,000	20,000	700	6,000	0.6	12,000
GOI Bonds	36,000	34,500	3,600	-1,500	0.01	345
Total	75,000	80,500	7,025	5,500		31,185

$$\beta \text{ of the portfolio} = \frac{31,185}{80,500} = 0.387$$

Using GOI Bond we can compute Risk Free Rate of Return as follows:

$$R_f = \frac{(34,500 - 36,000) + 3,600}{36,000} = 0.0583 \text{ i.e. } 5.83\%$$

Now we can calculated Market Return (R_m) using average return of the portfolio as follows:

$$R_p = R_f + \beta (R_m - R_f)$$

$$15.7\% = 5.83\% + 0.387(R_m - 5.83\%)$$

Question – 27

A holds the following portfolio:

Share/Bond	Beta	Initial Price ₹	Dividends ₹	Market Price at end of Year ₹
Epsilon Ltd.	0.8	25	2	50
Sigma Ltd.	0.7	35	2	60
Omega Ltd.	0.5	45	2	135
GOI Bonds	0.01	1,000	140	1,005

Calculate:

- The expected rate of return of each security using Capital Asset Pricing Method (CAPM)
- The average return of his portfolio.

Risk-free return is 14%.

(SM TYK – 27)

Solution:

(i) Expected rate of return

	Total Investments	Dividends	Capital Gains
Epsilon Ltd.	25	2	25
Sigma Ltd.	35	2	25
Omega Ltd.	45	2	90
GOI Bonds	1,000	140	5
	<u>1,105</u>	<u>146</u>	<u>145</u>

$$\text{Expected Return on market portfolio} = \frac{146 + 145}{1105} = 26.33\%$$



CAPM $E(R_p) = R_F + \beta [E(R_M) - R_F]$

Epsilon Ltd.	$14 + 0.8[26.33 - 14] =$	$14 + 9.86$	$= 23.86\%$
Sigma Ltd.	$14 + 0.7[26.33 - 14] =$	$14 + 8.63$	$= 22.63\%$
Omega Ltd.	$14 + 0.5[26.33 - 14] =$	$14 + 6.17$	$= 20.17\%$
GOI Bonds	$14 + 0.01[26.33 - 14] =$	$14 + 0.12$	$= 14.12\%$

(ii) Average Return of Portfolio

$$\frac{23.86 + 22.63 + 20.17 + 14.12}{4} = \frac{80.78}{4} = 20.20\%$$

$$\text{Alternatively, } = \frac{0.8 + 0.7 + 0.5 + 0.01}{4} = \frac{2.01}{4} = 0.5025$$

$$14 + 0.5025 (26.33 - 14) = 14 + 6.20 = 20.20\%$$

Question – 28

Suppose one of your HNI clients is holding the following portfolio as per his risk appetite:

Particulars	Securities
Equity Shares:	
G Ltd.	1,000
S Ltd.	1,000
B Ltd.	500
PSU Bonds	20,000

The other data related to each of these securities is as follows:

Cost	Dividends/Interest	Market Price	Beta
₹	₹	₹	
10,000	1,725	9,800	0.6
15,000	1,000	16,200	0.8
28,000	1,400	28,300	0.6
1,800	180	1,725	0.10

Your client is interested in investing some more funds in Bonds issued by GOI.

- (1) Estimate the minimum rate of return that your client would expect from these Bonds keeping in view his risk appetite and assuming Market Return as 12.70%.
- (2) Analyze whether this portfolio has out-performed the market or not assuming Risk Free Rate of Return as 7%.

(MTP Nov – 2021)

Solution:

(1) Working Notes

Calculation of Return on each single security

	Cost ₹ (1)	No. of Secur- ities (2)	Total Cost (3) = (1)×(2)	Divide nd /Inte- rest	Cap- ital Gain	Tota l (4)	Total Income (5) = (2) × (4)	Bet a (6)	(6) × (3)
G Ltd.	10,000	1,000	1,00,00,000	1,725	-200	1,525	15,25,000	0.6	60,00,000
S Ltd.	15,000	1,000	1,50,00,000	1,000	1,200	2,200	22,00,000	0.8	1,20,00,000
B Ltd.	28,000	500	1,40,00,000	1,400	300	1,700	8,50,000	0.6	84,00,000
PSU Bonds	1,800	20,000	3,60,00,000	180	-75	105	21,00,000	0.10	36,00,000
Total			7,50,00,000				66,75,000		3,00,00,000

Rate of Return on earned on the Portfolio

$$\frac{\text{Dividend Earned + Capital Appreciation}}{\text{Initial Investment}} \times 100$$

$$= \frac{₹ 66,75,000}{₹ 7,50,00,000} \times 100 = 8.90\%$$

Weighted Average Beta of the Portfolio

$$\frac{3,00,00,000}{₹ 7,50,00,000} = 0.40$$



Expected Risk Free Rate of Return using CAPM

$$8.90\% = R_f + 0.40[12.7\% - R_f]$$

$$8.90\% = R_f + 5.08 - 0.40 R_f$$

$$3.82\% = 0.60 R_f$$

$$R_f = 6.37\%$$

Thus keeping in view the present risk appetite, the client would expect at least a return of 6.37% on Bonds.

(2) The expected return on the Portfolio using CAPM:

$$= 7\% + 0.40[12.7 - 7\%] = 9.28\%$$

Since the actual return is 8.90% which is quite lower than expected return considering the systematic risk borne by the investor and hence portfolio has not outperformed the market rather has underperformed.

Question – 29

Your client is holding the following securities:

Particulars of Securities	Cost	Dividend s/ Interest	Market price at the end of holding period	Beta
	₹	₹	₹	
Equity Shares:				
G Ltd.	20,000	1,450	19,600	0.6
S Ltd.	30,000	1,000	30,400	0.8
B Ltd.	28,000	1,400	32,000	0.6
GOI Bonds	72,000	5,060	71,980	0.01

Evaluate:

- Risk free rate of return.
- Expected rate of return of each security (except GOI Bond), using the Capital Asset Pricing Model (CAPM).

Note: (1) Use weighted average Beta in calculations.

(2) Round off calculations upto 3 decimal points.

(SM TYK, MTP April – 2024)

Solution:

Particulars of Securities	Cost ₹	Market Price	Capital gain	Dividend/ Interest
G Ltd.	20,000	19,600	-400	1,450
S Ltd.	30,000	30,400	400	1,000
B Ltd.	28,000	32,000	4,000	1,400
GOI Bonds	72,000	71,980	-20	5,060
Total	1,50,000	1,53,980	3,980	8,910

- Risk free return [Return on Govt. Security (GOI Bond)]

$$\frac{5,060 + (72,000 - 71,980)}{72,000} = 7\%$$

- Weighted Average of Beta

$$0.6 \times 19,600/1,53,980 + 0.8 \times 30,400/1,53,980 + 0.60 \times 32,000/1,53,980 + 0.01 \times 71,980/1,53,980$$

$$= 0.076 + 0.158 + 0.125 + 0.005 = 0.364$$

Average Return on Portfolio

$$(8,910 + 3,980) / 1,50,000 \times 100\% = 8.593\%$$



Market Return

$$8.593\% = 7\% + (R_m - 7\%) \times 0.364$$

$$R_m = 11.376\%$$

Expected Rate of Return for each security is

$$\text{Rate of Return} = R_f + \beta (R_m - R_f)$$

$$\text{G Ltd.} = 7.000\% + 0.6 (11.376\% - 7.000\%) = 9.626\%$$

$$\text{S Ltd.} = 7.000\% + 0.8 (11.376\% - 7.000\%) = 10.501\%$$

$$\text{B Ltd.} = 7.000\% + 0.6 (11.376\% - 7.000\%) = 9.626\%$$

Question – 30

On the basis of given information, Mr. XLY want to create a portfolio equally as risky as the market and is having ₹ 20,00,000 to invest.

Assets	Investment	Beta
Stock A	₹ 4,00,000	0.70
Stock B	₹ 5,00,000	1.10
Stock C	?	1.60
Debenture (D)	?	0

How do you recommend and interpret the risk scenario and investment in all the securities?

(MTP April – 2023)

Solution:

Since portfolio equally as risky as the market hence portfolio beta shall be “1”

$$1 = \frac{(4,00,000 \times 0.7) + (5,00,000 \times 1.10) + (x \times 1.60) + (y \times 0)}{20,00,000}$$

$$20,00,000 = 8,30,000 + 1.60x$$

$$x = \frac{20,00,000 - 8,30,000}{1.60}$$

$$= 7,31,200$$

$$\text{Investment in stock C} = 7,31,200$$

$$\text{Investment in Debenture D} = 3,68,800$$

ALPHA CALCULATION**Question – 31**

Assuming that shares of ABC Ltd. and XYZ Ltd. are correctly priced according to Capital Asset Pricing Model. The expected return from and Beta of these shares are as follows:

Share	Beta	Expected return
ABC	1.2	19.8%
XYZ	0.9	17.1%

You are required to derive Security Market Line.

(SM TYK – 20)



Solution:**Calculation of R_f & MRP**

$$19.8 = R_f + 1.2 \text{ MRP} \quad \dots\dots\dots(i)$$

$$17.1 = R_f + 0.9 \text{ MRP} \quad \dots\dots\dots(ii)$$

$$2.70 = 0.3 \text{ MRP}$$

$$\text{MRP} = 9$$

Put MRP in Equation.....(i)

$$19.8 = R_f + (1.2 \times 9)$$

$$R_f = 9$$

SML

$$\rightarrow ER = 9 + 9B$$

Question – 32

The expected returns and Beta of three stocks are given below

Stock	A	B	C
Expected Return (%)	20	13	17
Beta Factor	1.9	0.8	1.4

If the risk-free rate is 9% and the expected rate of return on the market portfolio is 14%, examine which of the above stocks are over, under or correctly valued in the market? What shall be the strategy?

(MTP March – 2024)

Solution:

Required Rate of Return is given by

$$\beta R_j = R_f + (R_m - R_f)$$

$$\text{For Stock A, } R_j = 9\% + 1.9 (14\% - 9\%) = 18.50\%$$

$$\text{Stock B, } R_j = 9\% + 0.8 (14\% - 9\%) = 13.00\%$$

$$\text{Stock C, } R_j = 9\% + 1.4 (14\% - 9\%) = 16.00\%$$

Required Return %	Expected Return %	Valuation	Decision
18.50%	20.00%	Under Valued	Buy
13.00%	13.00%	Correctly Valued	Hold
16.00%	17.00%	Under Valued	Buy

Question – 33

An investor holds two stocks A and B. An analyst prepared ex-ante probability distribution for the possible economic scenarios and the conditional returns for two stocks and the market index as shown below:

Economic Scenario	Probability	Conditional Returns %		
		A	B	Market
Growth	0.40	25	20	18
Stagnation	0.30	10	15	13
Recession	0.30	-5	-8	-3

The risk free rate during the next year is expected to be around 11%. Determine whether the investor should liquidate his holdings in stocks A and B or on the contrary make fresh investments in them. CAPM assumptions are holding true.

(SM TYK – 33)



Solution:

Calculation of ER_A , ER_M , σ_A , σ_M & β_A

P	A	P (A)	z	$(A - \bar{A})^2 P$	M	P (M)	$(M - \bar{M})$	$(M - \bar{M})^2 P$	$(A - \bar{A}) \times (M - \bar{M}) \times P$
0.40	25	10	13.5	72.9	18	7.2	7.8	24.336	42.32
0.30	10	3	-1.5	0.675	13	3.9	2.8	2.352	-1.26
0.30	-5	-1.5	-16.5	81.675	-3	-0.9	-	52.272	65.34
	$\bar{A} =$	11.5%	$\sigma_A^2 =$	115.25 (%) ²		10.2	$\sigma_M^2 =$	78.96(%) ²	106.2 (%) ²

$$\sigma_A = 12.46\%$$

$$\sigma_M = 8.88\%$$

$$\beta_A = \frac{COV_{AM}}{\sigma_M^2} = \frac{106.2}{78.96} = 1.345$$

Calculation of ER_A , ER_B , σ_B & β_A

P	A	P (B)	$(B - \bar{B})$	$(B - \bar{B})^2 P$	$(M - \bar{M})$	$(B - \bar{B}) \times (M - \bar{M}) \times P$
0.4	20	8	9.9	39.204	7.8	30.888
0.3	15	4.5	4.9	7.203	2.8	4.116
0.3	-8	-2.4	-18.1	98.283	-13.2	71.676
	$\bar{B} =$	10.1%	$\sigma_B^2 =$	144.69(%) ²		106.68(%) ²

$$\sigma_B^2 = 12.029\%$$

$$\beta_B = \frac{COV_{BM}}{\sigma_M^2} = \frac{106.68}{78.96} = 1.351$$

Calculation of K_e as per CAPM

(i) Required return for A

$$K_e = R_f + (R_m - R_f) \beta_A$$

$$= 11 + (10.2 - 11) 1.345$$

$$= 9.924\%$$

(ii) Required return for B

$$K_e = 11 + (10.2 - 11) 1.351$$

$$= 9.92\%$$

Calculation of Alpha $\alpha = ER - K_e$

$$A = \alpha_A = 11.5 - 9.924 = 1.576\%$$

$$B = \alpha_B = 10.1 - 9.92 = 0.18\%$$

Since α of both stock A & stock B is positive, they are underpriced. It is advised to make fresh investment in them.

Question – 34

Following is the data regarding Three Securities.

Stock	Expected Return (%)	Std. Deviation	Correlation with the Market Return
A	19%	2.50	0.840
B	13.50%	2.00	0.540
C	11.00%	0.80	0.975
Market Risk	-	1.20	-
Market Rate of Return	14.00%	-	-
Risk Free* Rate	9.00%	-	-

(i) Advise which of the above stocks are over, under or correctly valued in the market?

(ii) What will be strategy you would like to recommend?

(Exam November – 2024) (6 Marks)



Solution:

- (i) To decide which of this security is over, under or correctly valued we need to compute the Required Rate of Return of each Stock using CAPM for which Beta shall be calculated as follows:

$$\text{Beta of Stock A } (\beta_A) = \frac{0.840 \times 2.50}{1.20} = 1.75$$

$$\text{Beta of Stock B } (\beta_B) = \frac{0.540 \times 2.00}{1.20} = 0.90$$

$$\text{Beta of Stock C } (\beta_C) = \frac{0.975 \times 0.80}{1.20} = 0.65$$

Required rate of return is given by CAPM

$$R_j = R_f + \beta (R_m - R_f)$$

For Stock A $R_A = 9\% + 1.75 (14\% - 9\%) = 17.75\%$

Stock B $R_B = 9\% + 0.90 (14\% - 9\%) = 13.50\%$

Stock C $R_C = 9\% + 0.65 (14\% - 9\%) = 12.25\%$

Stock	Required rate of return %	Expected rate of return %	Valuation
A	17.75%	19.00%	Under Valued
B	13.50%	13.50%	Correctly Valued
C	12.25%	11.00%	Over Valued

- (ii) **Strategy**

Stock	Decision
A	Buy

B	Hold
C	Sell

Question – 35

XYZ Ltd. has substantial cash flow and until the surplus funds are utilized to meet the future capital expenditure, likely to happen after several months, are invested in a portfolio of short-term equity investments, details for which are given below:

Investment	No. of shares	Beta	Market price per share ₹	Expected dividend yield
I	60,000	1.16	4.29	19.50%
II	80,000	2.28	2.92	24.00%
III	1,00,000	0.90	2.17	17.50%
IV	1,25,000	1.50	3.14	26.00%

The current market return is 19% and the risk free rate is 11%.

Required to:

- Calculate the risk of XYZ's short-term investment portfolio relative to that of the market;
- Whether XYZ should change the composition of its portfolio.

(SM TYK – 22)

Solution:

(i) Calculation of Beta of Portfolio

Investment	No. of Shares	Market Price per Share ₹	Total	Weight	Beta	Weight × Beta
I	60,000	4.29	2,57,400	0.2339	1.16	0.27
II	80,000	2.92	2,33,600	0.2123	2.28	0.48



PORTFOLIO MANAGEMENT

III	1,00,000	2.17	2,17,000	0.1972	0.90	0.18
IV	1,25,000	3.14	3,92,500	0.3566	1.50	0.50
			11,00,500	1.0000	B _p	1.46

$$ER_p = (19.50 \times 0.2339) + (24 \times 0.2123) + (17.50 \times 0.1972) + (26 \times 0.3566)$$

$$= 22.38\%$$

Calculation of B_p assuming required rate of return is 22.38%

(जो 22.38% Return मिलना है उसके हिसाब से Beta कितना चाहिए)

$$K_e = R_f + (R_m - R_f) B_p$$

$$22.38 = 11 + (19 - 11) B_p$$

$$11.38 = 8 B_p$$

$$B_p = \frac{11.38}{8} = 1.42$$

Portfolio Beta (1.46) is more than market implicit Beta (1.42) hence portfolio is Risky in Comparison to market.

(ii) Calculation of required Rate of Return

	Ke	ER		
I	= 11 + (19 - 11) 1.16 = 20.28%	19.50%	Ke > ER	Sell
II	= 11 + (19 - 11) 2.28 = 29.24%	24%	Ke > ER	Sell
III	= 11 + (19 - 11) 0.9 = 18.20%	17.50%	Ke > ER	Sell
IV	= 11 + (19 - 11) 1.50 = 23%	26%	Ke < ER	Buy

Question – 36

The following information are available with respect of Krishna Ltd.

Year	Krishna Ltd. Average share price	Dividend per Share	Average Market Index	Dividend Yield	Return on Govt. Bonds
2012	₹ 245	₹ 20	2013	4%	7%
2013	253	22	2130	5%	6%
2014	310	25	2350	6%	6%
2015	330	30	2580	7%	6%

Compute beta value of the Krishna Ltd. at the end of 2015 and state your observation.

(SM TYK – 08)

Solution:

(1) Annual Return of Krishna Ltd. (x)

$$x = \frac{(P_1 - P_0) + D_1}{P_0} \times 100$$

$$2013 = \frac{(253 - 245) + 22}{245} \times 100 = 12.24\%$$

$$2014 = \frac{(310 - 253) + 25}{253} \times 100 = 32.41\%$$

$$2015 = \frac{(330 - 310) + 30}{310} \times 100 = 16.13\%$$

(2) Calculation of R_m (y)

$$2013 = \left(\frac{2,130 - 2,013}{2,013} \times 100 \right) + 5 = 10.81\%$$



$$2014 = \left(\frac{2,350 - 2,130}{2,130} \times 100 \right) + 6 = 16.33\%$$

$$2015 = \left(\frac{2,580 - 2,350}{2,350} \times 100 \right) + 7 = 16.79\%$$

$$R_f = \frac{6 + 6 + 6}{3} = 6\%$$

Calculation of Cov_{xy}

Year	x	(x - $\bar{x}$)	y	(y - $\bar{y}$)	(y - $\bar{y}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	12.24	-8.02	10.81	-3.83	14.67	30.717
2	32.41	12.15	16.33	1.69	2.86	20.533
3	16.13	-4.13	16.79	2.15	4.62	-8.879
$\Sigma x =$	60.78	$\Sigma y =$	43.93	$\Sigma (y - \bar{y})^2 =$	22.15	42.371(%) ²

$$\bar{x} = \frac{60.78}{3} = 20.26\%$$

$$\bar{y} = \frac{43.92}{3} = 14.64\%$$

$$\sigma_m^2 = \frac{22.15}{3} = 7.383(\%)^2$$

$$\text{Cov}_{xy} = \frac{42.371}{3} = 14.123(\%)^2$$

$$\beta = \frac{\text{Cov}_{xy}}{\sigma_m^2} = \frac{14.123}{7.383} = 1.91$$

$$\begin{aligned} K_e &= R_f + \beta (R_m - R_f) \\ &= 6 + 1.91(14.64 - 6) = 22.50\% \end{aligned}$$

Observation

$$2013 = K_e = 6 + (10.81 - 6) 1.91 = 15.19\% > ER = 12.24 \quad \text{Sell}$$

$$2014 = K_e = 6 + (16.33 - 6) 1.91 = 25.71\% < ER = 32.41 \quad \text{Buy}$$

$$2015 = K_e = 6 + (16.79 - 6) 1.91 = 26.61\% > ER = 16.13 \quad \text{Sell}$$

Question – 37

The following information is available with respect of AK Ltd.

Year	AK Ltd. Average share price	AK Ltd. Dividend per Share	Average Market Index	Dividend Yield	Return on Govt. Bonds
	₹	₹			
2022	242	20	1812	4%	6%
2023	279	25	1950	5%	5%
2024	305	30	2258	6%	4%
2025	322	35	2220	7%	5%

Compute beta value of the company as at the end of 2025. What is your observation?

(RTP Jan. – 2026)

Solution:

(a) Computation of Beta Value

Calculation of Returns

$$\text{Returns} = \frac{D_1 + (P_1 - P_0)}{P_0} \times 100$$

Year	Returns
2022-2023	$\frac{25 + (279 - 242)}{242} \times 100 = 25.62\%$
2023-2024	$\frac{30 + (305 - 279)}{279} \times 100 = 20.07\%$
2024-2025	$\frac{35 + (322 - 305)}{305} \times 100 = 17.05\%$

Calculation of returns from market Index



Year	Returns	Dividend Yield %	Total Return %
2022-2023	$\frac{1,950 - 1,812}{1,812} \times 100 = 7.62\%$	5%	12.62%
2023-2024	$\frac{2,258 - 1,950}{1,950} \times 100 = 15.79\%$	6%	21.79%
2024-2025	$\frac{2,220 - 2,258}{2,258} \times 100 = (-1.68\%)$	7%	5.32%

Computation of Beta

Year	X	Y	XY	Y ²
2022-2023	25.62	12.62	323.32	159.26
2023-2024	20.07	21.79	437.33	474.80
2024-2025	17.05	5.32	90.71	28.30
	$\Sigma X = 62.74$	$\Sigma Y = 39.73$	$\Sigma XY = 851.36$	$\Sigma Y^2 = 662.36$

$$\bar{X} = \frac{62.74}{3} = 20.91, \quad \bar{Y} = \frac{39.73}{3} = 13.24$$

$$\begin{aligned} \beta &= \frac{\Sigma XY - n\bar{X}\bar{Y}}{\Sigma Y^2 - n\bar{Y}^2} \\ &= \frac{851.36 - 3(20.91)(13.24)}{662.36 - 3(13.24)^2} \\ &= \frac{851.36 - 830.55}{662.36 - 525.89} = \frac{20.81}{136.47} = 0.15 \end{aligned}$$

- (b) **Observation:** Based on the beta value of the company and the analysis of returns, it can be concluded that changes in the return on share of AK Ltd. are less affected by changes in market returns.

Question – 38

Expected returns on two stocks for particular market returns are given in the following table:

Market Return	Aggressive	Defensive
7%	4%	9%
25%	40%	18%

You are required to calculate:

- The Betas of the two stocks.
- Expected return of each stock, if the market return is equally likely to be 7% or 25%.
- The Security Market Line (SML), if the risk free rate is 7.5% and market return is equally likely to be 7% or 25%.
- The Alphas of the two stocks.

(SM TYK – 13)

Solution:

(a) Beta

$$\text{Beta} = \frac{\text{Change in Stock Return}}{\text{Change in Market Return}}$$

$$A = \frac{40 - 4}{25 - 7} = 2$$

$$D = \frac{18 - 9}{25 - 7} = 0.5$$

(b) Expected Return

$$A = (4 \times 0.5) + (40 \times 0.5) = 22\%$$



$$= (9 \times 0.5) + (18 \times 0.5) = 13.5\%$$

(c) Security Market Line

$$R_m = (7 \times 0.5) + (25 \times 0.5)$$

$$= 16\%$$

$$SML = R_f + MRP \beta$$

$$= 7.5 + (16 - 7.5) \beta$$

$$= 7.5 + 8.5 \beta$$

(d) Alpha

$$Y = \alpha + bx$$

Security A

$$22 = \alpha + 2 \times 16$$

$$\alpha = -10$$

Security B

$$13.5 = \alpha + 0.5 \times 16$$

$$\alpha = 5.5$$

Question – 39

A Ltd. has an expected return of 22% and Standard deviation of 40%. B Ltd. has an expected return of 24% and Standard deviation of 38%. A Ltd. has a beta of 0.86 and B Ltd. a beta of 1.24. The correlation coefficient between the return of A Ltd. and B Ltd. is 0.72. The Standard deviation of the market return is 20%. Suggest:

- (i) Is investing in B Ltd. better than investing in A Ltd.?
- (ii) If you invest 30% in B Ltd. and 70% in A Ltd., what is your expected rate of return and portfolio Standard deviation?
- (iii) What is the market portfolios expected rate of return and how much is the risk-free rate?
- (iv) What is the beta of Portfolio if A Ltd.'s weight is 70% and B Ltd.'s weight is 30%?

(SM TYK – 21)

Solution:

	A	B
ER	22%	24%
SD	40%	38%
Beta	0.86	1.24

$$r_{AB} = 0.72$$

$$\sigma_m = 20\%$$

- (I) Yes Investment in B Ltd is better than A Ltd due to higher return & lower standard deviation.

$$(II) \quad ER_p \text{ \& S.D}_p \quad W_A = 0.70 \quad W_B = 0.30$$

$$ER_p = (22 \times 0.7) + (24 \times 0.3) = 22.60\%$$

$$\sigma_p = \sqrt{\sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times \sigma_A \times W_A \times \sigma_B \times r_{AB}}$$

$$= \sqrt{40^2 \times 0.7^2 + 38^2 \times 0.3^2 + 2 \times 40 \times 0.7 \times 38 \times 0.3 \times 0.72}$$

$$= 37.06\%$$

- (III) R_m & R_f



It is assumed that both securities are correctly priced

$$ER = R_f + MRPB$$

$$22 = R_f + 0.86 MRP \quad \dots\dots (i)$$

$$24 = R_f + 1.24 MRP \quad \dots\dots (ii)$$

$$-2 = -0.38 MRP$$

$$MRP = \frac{2}{0.38} = 5.26\%$$

Put MRP in equation (i)

$$22 = R_f + 0.86 \times 5.26$$

$$R_f = 17.46$$

$$MRP = R_m - R_f$$

$$5.26 = R_m - 17.46$$

$$R_m = 22.72\%$$

(IV) B_p

$$B_p = (0.7 \times 0.86) + (0.3 \times 1.24)$$

$$= 0.974$$

ASSETS PRICING

Question – 40

Amal Ltd. has been maintaining a growth rate of 12% in dividends. The company has paid dividend @ ₹ 3 per share. The rate of return on market portfolio is 15% and the risk-free rate of return in the

market has been observed as 10%. The beta co-efficient of the company's share is 1.2.

You are required to calculate the expected rate of return on the company's shares as per CAPM model and the equilibrium price per share by dividend growth model.

(SM TYK – 18)

Solution:

Capital Asset Pricing Model (CAPM) formula for calculation of expected rate of return is

$$ER = R_f + \beta (R_m - R_f)$$

$$ER = \text{Expected Return}$$

$$\beta = \text{Beta of Security}$$

$$R_m = \text{Market Return}$$

$$R_f = \text{Risk free Rate}$$

$$= 10 + [1.2 (15 - 10)]$$

$$= 10 + 1.2 (5)$$

$$= 10 + 6 = 16\% \text{ or } 0.16$$

Applying dividend growth mode for the calculation of per share equilibrium price:-

$$ER = \frac{D_1}{P_0} + g$$

$$\text{or } 0.16 = \frac{3(1.12)}{P_0} + 0.12 \quad \text{or} \quad 0.16 - 0.12 = \frac{3.36}{P_0}$$



$$\text{or } 0.04 P_0 = 3.36 \quad \text{or} \quad P_0 = \frac{3.36}{0.04} = ₹ 84$$

Therefore, equilibrium price per share will be ₹ 84.

Question – 41

An investor is holding 1,000 shares of Fatlass Company. Presently the rate of dividend being paid by the company is ₹ 2 per share and the share is being sold at ₹ 25 per share in the market. However, several factors are likely to be changed during the course of the year as indicated below:

	Existing	Revised
Risk free rate	12%	10%
Market risk premium	6%	4%
Beta value	1.4	1.25
Expected growth rate	5%	9%

In view of the above factors whether the investor should buy, hold or sell the shares? And why?

(SM TYK – 29)

Solution:

On the basis of existing and revised factors, rate of return and price of share is to be calculated.

Existing rate of return

$$= R_f + \text{Beta} (R_m - R_f) = 12\% + 1.4 (6\%) = 20.4\%$$

Revised rate of return

$$= 10\% + 1.25 (4\%) = 15\%$$

Price of share (original)

$$P_0 = \frac{D(1 + g)}{K_e - g} = \frac{2(1.05)}{.204 - .05} = \frac{2.10}{.154} = ₹ 13.63$$

Price of share (Revised)

$$P_0 = \frac{2(1.09)}{.15 - .09} = \frac{2.18}{.06} = ₹ 36.33$$

In case of existing market price of ₹ 25 per share, rate of return (20.4%) and possible equilibrium price of share at ₹ 13.63, this share needs to be sold because the share is overpriced (₹ 25 – 13.63) by ₹ 11.37. However, under the changed scenario where growth of dividend has been revised at 9% and the return though decreased at 15% but the possible price of share is to be at ₹ 36.33 and therefore, in order to expect price appreciation to ₹ 36.33 the investor should hold the shares, if other things remain the same.

Question – 42

An investor is holding 5,000 shares of X Ltd. Current year dividend rate is ₹ 3/ share. Market price of the share is ₹ 40 each. The investor is concerned about several factors which are likely to change during the next financial year as indicated below:

	Current Year	Next Year
Dividend paid/anticipated per share (₹)	3	2.5
Risk free rate	12%	10%
Market Risk Premium	5%	4%
Beta Value	1.3	1.4
Expected growth	9%	7%

In view of the above, advise whether the investor should buy, hold or sell the shares.

(SM TYK – 30)



Solution:

$$K_e = 12 + 5 \times 1.4 = 18.5\%$$

$$P_0 = \frac{3(1.09)}{0.185 - 0.09} = ₹ 34.42$$

As per existing situation, share is overpriced, hence investor should sell the shares.

Revised Situation

$$K_e = 10 + 4 \times 1.25 = 15.6\%$$

$$P_0 = \frac{2.5(1.07)}{0.156 - 0.07} = ₹ 31.10$$

In revised situation share is also overpriced, hence the investor should sell the share.

SYSTEMATIC & UNSYSTEMATIC RISK

Question – 43

A has portfolio having following features:

Security	β	Random Error σ_{ei}	Weight
L	1.60	7	0.25
M	1.15	11	0.30
N	1.40	3	0.25
K	1.00	9	0.20

You are required to find out the risk of the portfolio if the standard deviation of the market index (σ_m) is 18%.

(SM TYK – 37)

Solution:

$$\beta_p = (1.60 \times 0.25) + (1.15 \times 0.30) + (1.40 \times 0.25) + (1.00 \times 0.20) = 1.295$$

$$\begin{aligned} SR_p &= \sigma_P^2 \times \sigma_M^2 \\ &= 1.295^2 \times 324 \\ &= 543.36(\%)^2 \end{aligned}$$

$$\begin{aligned} \sigma_{ep}^2 &= \sigma_{eA}^2 W_A^2 + \sigma_{eB}^2 \times W_B^2 + \sigma_{eC}^2 W_C^2 + \sigma_{eD}^2 \times W_D^2 \\ &= 49 \times 0.25^2 + 121 \times 0.30^2 + 9 \times 0.25^2 + 81 \times 0.20^2 \\ &= 17.755 \end{aligned}$$

$$\begin{aligned} TR &= 543.36 + 17.755 \\ &= 561.115 \quad = 23.69\% \end{aligned}$$

Question – 44

A study by a Mutual fund has revealed the following data in respect of three securities:

Security	$\sigma(\%)$	Correlation with Index, P_m
A	20	0.60
B	18	0.95
C	12	0.75

The standard deviation of market portfolio (BSE Sensex) is observed to be 15%.



- (i) What is the sensitivity of returns of each stock with respect to the market?
- (ii) What are the covariance's among the various stocks?
- (iii) What would be the risk of portfolio consisting of all the three stocks equally?
- (iv) What is the beta of the portfolio consisting of equal investment in each stock?
- (v) What is the total, systematic and unsystematic risk of the portfolio in (iv)?

(SM TYK – 14)**Solution:****(i) Sensitivity**

$$\beta = \frac{\sigma_{X}}{\sigma_M} \times r_{XM}$$

$$A = \frac{20}{15} \times 0.60 = 0.80$$

$$B = \frac{18}{15} \times 0.95 = 1.14$$

$$C = \frac{12}{15} \times 0.75 = 0.60$$

(ii) Covariance

$$COV_{AB} = B_A \times B_B \sigma_M^2$$

$$COV_{AB} = 0.8 \times 1.14 \times 15^2 = 205.20 (\%)^2$$

$$COV_{AC} = 0.8 \times 0.6 \times 15^2 = 108 (\%)^2$$

$$COV_{BC} = 1.14 \times 0.6 \times 15^2 = 153.90 (\%)^2$$

(iii) Risk of Portfolio (Sharpe)

$$\begin{aligned} \sigma_p^2 &= \sigma_A^2 W_A^2 + \sigma_B^2 \times W_B^2 + \sigma_C^2 \times W_C^2 + 2 \times W_A \times W_B \times \\ &\quad COV_{AB} + 2 \times W_A \times W_C \times COV_{AC} + 2 \times W_B \times W_C \times COV_{BC} \\ &= 20^2 \left(\frac{1}{3}\right)^2 + 18^2 \times \left(\frac{1}{3}\right)^2 + 12^2 \times \left(\frac{1}{3}\right)^2 + 2 \times \frac{1}{3} \times \frac{1}{3} \times \\ &\quad 205.20 + 2 \times \frac{1}{3} \times \frac{1}{3} \times 108 + 2 \times \frac{1}{3} \times \frac{1}{3} \times 153.90 \\ &= 200.24 (\%)^2 \end{aligned}$$

$$\sigma_p = \sqrt{200.24} = 14.15\%$$

(iv) Beta of Portfolio

$$\begin{aligned} \sigma_p &= \frac{0.80 + 1.14 + 0.60}{3} \\ &= 0.8467 \end{aligned}$$

(v) TR, SR & UR of Portfolio

$$\begin{aligned} (i) \quad SR_p &= \beta_p^2 \times \sigma_M^2 \\ &= 0.8467^2 \times 15^2 \\ &= 161.30 (\%)^2 \end{aligned}$$

$$(ii) \quad TR_p = 200.24 (\%)^2$$



$$\begin{aligned} \text{(iii)} \quad UR_p &= TR_p - 8R_p \\ &= 200.24 - 161.30 \\ &= 38.94 (\%)^2 \end{aligned}$$

Question – 45

Following are the details of a portfolio consisting of three shares:

Share	Portfolio Weight	Beta	Expected Return in %	Total Variance
A	0.20	0.40	14	0.015
B	0.50	0.50	15	0.025
C	0.30	1.10	21	0.100

Standard Deviation of Market Portfolio Returns = 10%

You are given the following additional data:

Covariance (A, B) = 0.030

Covariance (A, C) = 0.020

Covariance (B, C) = 0.040

Calculate the following:

- The Portfolio Beta
- Residual variance of each of the three shares
- Portfolio variance using Sharpe Index Model
- Portfolio variance (on the basis of modern portfolio theory given by Markowitz)

Solution:

(I) Portfolio Beta

$$\begin{aligned} B_p &= (0.4 \times 0.2) + (0.5 \times 0.5) + (1.10 \times 0.30) \\ &= 0.66 \\ \sigma_m &= 10\% \\ \sigma_m &= 0.10 \\ \sigma_m^2 &= (0.10)^2 \\ &= 0.010 \end{aligned}$$

(II) Residual Variance

$$\begin{aligned} \text{(i)} \quad SR &= \beta_p^2 \times \sigma_M^2 \\ A &= 0.40^2 \times 0.010 = 0.0016 \\ B &= 0.50^2 \times 0.010 = 0.0025 \\ C &= 1.10^2 \times 0.010 = 0.0121 \\ \text{(ii)} \quad UR &= TR - SR \\ A &= 0.015 - 0.0016 = 0.0134 \\ B &= 0.025 - 0.0025 = 0.0225 \\ C &= 0.100 - 0.0121 = 0.0879 \end{aligned}$$

(III) Portfolio Variance [Sharpe]

$$\begin{aligned} SR_P &= \beta_p^2 \times \sigma_M^2 \\ &= 0.66^2 \times 0.010 = 0.004356 \end{aligned}$$

$$\begin{aligned} \sigma_{ep}^2 &= \sigma_{eA}^2 W_A^2 + \sigma_{eB}^2 \times W_B^2 + \sigma_{eC}^2 \times W_C^2 \\ &= 0.0134 \times 0.20^2 + 0.0225 \times 0.50^2 + 0.0879 \times 0.30^2 \\ &= 0.014072 \end{aligned}$$

TR_P, SR_P & UR_P

$$\begin{aligned} &= 0.004356 + 0.014072 \\ &= 0.018428 \end{aligned}$$

(IV) Portfolio Variance (Markowitz)

$$\begin{aligned} \sigma_p^2 &= \sigma_A^2 W_A^2 + \sigma_B^2 \times W_B^2 + \sigma_C^2 \times W_C^2 + 2 \times W_A \times W_B \times \\ &\text{Cov}_{AB} + 2 \times W_A \times W_C \times \text{Cov}_{AC} + 2 \times W_B \times W_C \times \text{Cov}_{BC} \\ &= 0.015 \times 0.20^2 + 0.025 \times 0.50^2 + 0.100 \times 0.30^2 + 2 \\ &\times 0.20 \times 0.50 \times 0.030 + 2 \times 0.20 \times 0.30 \times 0.020 + 2 \\ &\times 0.50 \times 0.30 \times 0.040 \\ &= 0.03625 \end{aligned}$$

Question – 46

Following are risk and return estimates for two stocks

Stock	Expected returns (%)	Beta	Specific SD of expected return (%)
A	14	0.8	35
B	18	1.2	45

The market index has a Standard Deviation (SD) of 25% and risk free rate on Treasury Bills is 6%.

You are required to calculate:

- The standard deviation of expected returns on A and B.
- Suppose a portfolio is to be constructed with the proportions of 25%, 40% and 35% in stock A, B and Treasury Bills respectively, what would be the expected return, standard deviation of expected return of the portfolio?

(Exam November – 2019) (8 Marks)

Solution:**(I) Standard Deviation**

$$SR = \beta^2 \times \sigma_M^2$$

$$A = 0.8^2 \times 25^2 = 400(\%)^2$$

$$B = 1.2^2 \times 25^2 = 900(\%)^2$$

$$UR$$

$$A = 1225(\%)^2$$

$$B = 2025(\%)^2$$



PORTFOLIO MANAGEMENT

Proportion of allocated fund	50%	50%	
------------------------------	-----	-----	--

You are required to

- Calculate the systematic and unsystematic risk for the companies stocks.
- Calculate portfolio risk.
- If the proportion of fund allocation is changed to 60:40 for TC Limited and PC Limited respectively, Advise whether it is preferable or not.

Calculation up to 3 decimal points.

(Exam September – 2025) (6 Marks)

Solution:

(i) SR & UR of Individual Stock

Systematic Risk

$$SR = \beta^2 \sigma_m^2$$

$$TC = 0.8^2 \times 2.10 = 1.3344$$

$$PC = 0.7^2 \times 2.10 = 1.029$$

Unsystematic Risk

$$UR = TR - SR$$

$$TC = 5.8 - 1.344 = 4.456$$

$$\begin{aligned} TR &= SR + UR \\ A &= 1225 + 400 = 1625 \\ B &= 2025 + 900 = 2925 \\ \sigma_A &= \sqrt{1625} = 40.31\% \\ \sigma_B &= \sqrt{2925} = 54.08\% \end{aligned}$$

(II) Expected return & σ_p

$$\begin{aligned} ER_p &= (14 \times 0.25) + (18 \times 0.40) + (6 \times 0.35) \\ &= 12.8\% \end{aligned}$$

$$\begin{aligned} \sigma_p^2 &= \sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times W_A \times W_B \times Cov_{AB} + 2 \times W_A \\ &\times W_C \times Cov_{AC} + 2 \times W_B \times W_C \times Cov_{BC} \\ &= 1625 \times 0.25^2 + 2925 \times 0.40^2 + 2 \times 0.25 \times 0.40 \times \\ &0.8 \times 1.2 \times 0.25^2 \\ &= 689.59 (\%)^2 \\ &= \sqrt{689.59} = 26.26\% \end{aligned}$$

Question – 47

The following details are given for TC Limited and PC Limited stocks and Nifty Index for a period of one year:

	TC Limited	PC Limited	Nifty Index
Average return	0.12	0.18	0.6
Variance of return	5.8	4.8	2.10
Beta (β)	0.8	0.7	



$$PC = 4.8 - 1.029 = 3.771$$

(ii) Portfolio Risk

$$COV_{AB} = \beta_A \beta_B \sigma_m^2$$

$$= 0.8 \times 0.7 \times 2.10 = 1.176$$

$$\sigma_p^2 = \sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times W_A \times W_B \times COV_{AB}$$

$$= 5.8 \times 0.50^2 + 4.8 \times 0.50^2 + 2 \times 0.5 \times 0.5 \times 1.176$$

$$= 3.238$$

(iii) If Proportion of Fund is 60:40**Existing Portfolio**

$$R_P = (12 \times 0.5) + (18 \times 0.5) = 15\%$$

$$\sigma_p^2 = 3.2387$$

Revised Portfolio

$$R_P = (12 \times 0.6) + (18 \times 0.4) = 14.4\%$$

$$\sigma_p^2 = 5.8 \times 0.60^2 + 4.8 \times 0.40^2 + 2 \times 0.60 \times 0.40 \times 1.176$$

$$= 3.420$$

It is not preferable to invest 60:40 because it provides lower return & higher risk.

Question – 48

Mr. Abhishek is interested in investing ₹ 2,00,000 for which he is considering following three alternatives:

- (i) Invest ₹ 2,00,000 in Mutual Fund X (MFX)
- (ii) Invest ₹ 2,00,000 in Mutual Fund Y (MFY)
- (iii) Invest ₹ 1,20,000 in Mutual Fund X (MFX) and ₹ 80,000 in Mutual Fund Y (MFY)

Average annual return earned by MFX and MFY is 15% and 14% respectively. Risk free rate of return is 10% and market rate of return is 12%.

Covariance of returns of MFX, MFY and market portfolio Mix are as follow:

	MFX	MFY	Mix
MFX	4.800	4.300	3.370
MFY	4.300	4.250	2.800
Mix	3.370	2.800	3.100

You are required to calculate:

- (i) Variance of return from MFX, MFY and market return,
- (ii) Portfolio return, beta, portfolio variance and portfolio standard deviation,
- (iii) Expected return, systematic risk and unsystematic risk; and
- (iv) Sharpe ratio, Treynor ratio and Alpha of MFX, MFY and Portfolio Mix

(SM TYK – 17)



Solution:

(i) Variance of X, Y & Market

Variance of X

(r between x & x is 1)

$$r_{xx} = \text{Cov}_{xx} / \sigma_x$$

$$1 = \frac{4.800}{\sigma_x^2}$$

$$\sigma_x^2 = 4.800$$

$$\sigma_y^2 = 4.250$$

$$\sigma_m^2 = 3.100$$

(ii) ER_p , β , σ_p^2 , & σ_p

$$ER_p = (15 \times 0.6) + (14 \times 0.4) = 14.6\%$$

Beta of Portfolio

$$\beta_x = \frac{\text{COV}_{xm}}{\sigma_m^2} = \frac{3.370}{3.100} = 1.087$$

$$\beta_y = \frac{\text{COV}_{ym}}{\sigma_m^2} = \frac{2.800}{3.100} = 0.903$$

$$\beta_p = (1.087 \times 0.6) + (0.903 \times 0.4) = 1.0134$$

$$\begin{aligned} \sigma_p^2 &= \sigma_x^2 \times w_x + \sigma_y^2 \times w_y + 2 \times w_x \times w_y \times \text{Cov}_{xy} \\ &= (4.800 \times (0.6)^2 + (4.250 \times (0.4)^2) + 2 \times 0.6 \times 0.4 \times 4.300 \end{aligned}$$

$$= 4.472 (\%)^2$$

$$\sigma_p = 2.1147\%$$

(iii) ER as per CAPM, SR & UR:

$$ER = R_f + (R_m - R_f) \beta$$

$$X = 10 + (12 - 10) 1.087 = 12.17\%$$

$$Y = 10 + (12 - 10) 0.903 = 11.806\%$$

$$ER_p = 10 + (12 - 10) 1.0134 = 12.027\%$$

$$SR = B^2 \sigma_m^2$$

$$X = (1.087)^2 \times 3.100 = 3.6628 (\%)^2$$

$$Y = (0.903)^2 \times 3.100 = 2.5278 (\%)^2$$

$$SR_p = (1.0134)^2 \times 3.100 = 3.1836 (\%)^2$$

$$UR = TR - SR$$

$$X = 4.800 - 3.6628 = 1.1372 (\%)^2$$

$$Y = 4.250 - 2.5278 = 1.7222 (\%)^2$$

$$Z = 4.472 - 3.1836 = 1.289 (\%)^2$$

(iv) (a) Sharpe Ratio $= \frac{ER - R_f}{S.D.}$

$$X = \frac{15 - 10}{2.19} = 2.28$$

$$Y = \frac{14 - 10}{2.06} = 1.94$$

$$\text{Portfolio} = \frac{14.6 - 10}{2.11} = 2.18$$

(b) **Treynor Ratio** $= \frac{ER - R_f}{\beta}$

$$X = \frac{15 - 10}{1.087} = 4.60$$

$$Y = \frac{14 - 10}{0.903} = 4.43$$

$$\text{Portfolio} = \frac{14.6 - 10}{1.0134} = 4.54$$

$$\text{Alpha} = ER - K_e$$

$$X = 15 - 12.174 = 2.826\%$$

$$Y = 14 - 11.806 = 2.194\%$$

$$\text{Portfolio} = 14.6 - 12.027 = 2.573\%$$

Question – 49

The returns on stock A and market portfolio for a period of 6 years are as follows:

Year	Return on A (%)	Return on market portfolio (%)
1	12	8
2	15	12
3	11	11
4	2	-4
5	10	9.5
6	-12	-2

You are required to determine:

(i) Characteristic line for stock A

(ii) The systematic and unsystematic risk of stock A.

(SM TYK – 11)

Solution:

Characteristic line is given by

$$\alpha + \beta_{Rm}$$

$$\beta_i = \frac{\sum xy - n \bar{x} \bar{y}}{\sum x^2 - n (\bar{x})^2}$$

$$\alpha_i = \bar{y} - \beta \bar{x}$$

Return on A (Y)	Return of market (X)	xy	x ²	(x - $\bar{x}$)	(x - $\bar{x}$) ²	(y - $\bar{y}$)	(y - $\bar{y}$) ²
12	8	96	64	2.25	5.06	5.67	32.15
15	12	180	144	6.25	39.06	8.67	75.17
11	11	121	121	5.25	27.56	4.67	21.81
2	-4	-8	16	-	95.06	-4.33	18.75
10	9.5	95	90.25	9.75	14.06	3.67	13.47
<u>-12</u>	<u>-2</u>	<u>24</u>	<u>4</u>	<u>3.75</u>	<u>60.06</u>	<u>-</u>	<u>335.99</u>
38	34.5	508	439.25	-7.75	240.86	18.33	497.34

$$\bar{y} = 38/6 = 6.33$$

$$\bar{x} = 34.5/6 = 5.75$$

$$\beta = \frac{\sum xy - n \bar{x} \bar{y}}{\sum x^2 - n (\bar{x})^2} = \frac{508 - 6 (5.75)(6.33)}{439.25 - 6(5.75)^2} = \frac{508 - 218.385}{439.25 - 198.375}$$



$$= \frac{289.615}{240.875} = 1.202$$

$$\alpha = \bar{y} - \beta \bar{x} = 6.33 - 1.202 (5.75) = -0.58$$

Hence the characteristic line is $-0.58 + 1.202 (R_m)$

$$\text{Total Risk of Market} = \sigma_m^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{240.86}{6} = 40.14\%$$

$$\text{Total Risk of Stock} = \frac{497.34}{6} = 82.89 (\%)$$

$$\text{Systematic Risk} = \beta^2 \sigma_2 = (1.202)^2 \times 40.14 = 57.99(\%)$$

$$\begin{aligned} \text{Unsystematic Risk is} &= \text{Total Risk} - \text{Systematic Risk} \\ &= 82.89 - 57.99 = 24.90(\%) \end{aligned}$$

PART III: ARBITRAGE PRICING THEORY

Question – 50

Mr. Nirmal kumar has categorized all the available stock in the market into the following types:

- (i) Small cap growth stocks
- (ii) Small cap value stocks
- (iii) Large cap growth stocks
- (iv) Large cap value stocks

Mr. Nirmal Kumar also estimated the weights of the above categories of stocks in the market index. Further, the sensitivity of returns on these categories of stocks to the three important factor are estimated to be:

Category of Stocks	Weight in the Market Index	Factor I (Beta)	Factor II (Book Price)	Factor III (Inflation)
Small cap growth	25%	0.80	1.39	1.35
Small cap value	10%	0.90	0.75	1.25
Large cap growth	50%	1.165	2.75	8.65
Large cap value	15%	0.85	2.05	6.75
Risk Premium		6.85%	-3.5%	0.65%

The rate of return on treasury bonds is 4.5%

Required:

- (a) Using Arbitrage Pricing Theory, determine the expected return on the market index.
- (b) Using Capital Asset Pricing Model (CAPM), determine the expected return on the market index.
- (c) Mr. Nirmal Kumar wants to construct a portfolio constituting only the 'small cap value' and 'large cap growth' stocks. If the target beta for the desired portfolio is 1, determine the composition of his portfolio.

(SM TYK – 40)

Solution:

(a) ER of Market as per APT

$$\begin{aligned} \text{SCG} &= 4.5 + (6.85 \times 0.8) + (-3.5 \times 1.39) + (0.65 \times 1.35) \\ &= 5.9925\% \end{aligned}$$

$$\text{SCV} = 8.8525\%$$

$$\text{LCG} = 8.477\%$$



$$LCV = 7.535\%$$

Expected return of market

$$= (5.9925 \times 0.25) + (8.8525 \times 0.10) + (8.477 \times 0.50) + (7.535 \times 0.15)$$

$$= 7.7525\%$$

(b) ER as per CAPM

$$B_p = (0.8 \times 0.25) + (0.9 \times 0.10) + (1.165 \times 0.50) + (0.85 \times 0.15)$$

$$= 1$$

$$ER = RF + MRP B$$

$$= 4.5 + 6.85 \times 1$$

$$= 11.35\%$$

(c) Calculation of weight

$$\beta = 0.9 \times W_A + 1.165 (1 - W_A)$$

$$1 = 0.9 W_A + 1.165 - 1.165 W_A$$

$$0.265 W_A = 0.165$$

$$W_A = 0.62$$

$$W_B = 1 - W_A = 1 - 0.62 = 0.38$$

$$\% \text{ of investment in small cap value} = 62\%$$

$$\% \text{ of investment in large cap growth} = 38\%$$

Question – 51

Mr. Kapoor owns a portfolio with the following characteristics:

	Security X	Security Y	Risk Free Security
Factor 1 sensitivity	0.75	1.50	0
Factor 2 sensitivity	0.60	1.10	0
Expected Return	15%	20%	10%

It is assumed that security returns are generated by a two factor model.

- If Mr. Kapoor has ₹ 1,00,000 to invest and sells short ₹ 50,000 of security Y and purchases ₹ 1,50,000 of security X, what is the sensitivity of Mr. Kapoor's portfolio to the two factors?
- If Mr. Kapoor borrows ₹ 1,00,000 at the risk free rate and invests the amount he borrows along with the original amount of ₹ 1,00,000 in security X and Y in the same proportion as described in part (i), what is the sensitivity of the portfolio to the two factors?
- What is the expected return premium of factor 2?

(SM TYK – 15)

Solution:

(i) Beta of Portfolio

Calculation of Weights

Own fund	= 1,00,000
Short sell (Y)	= 50,000
Investment in X	= 1,50,000
W_x	$= \frac{1,50,000}{1,00,000} = 1.5$



$$W_y = \frac{-50,000}{1,00,000} = -0.5$$

Factor 1

$$B_p = (0.75 \times 1.5) + (1.5 \times -0.5) = 0.375$$

Factor 2

$$B_p = (0.60 \times 1.5) + (1.10 \times -0.5) = 0.35$$

(ii) Beta of Portfolio

$$\text{Own fund} = ₹ 1,00,000$$

$$R_f = \frac{₹ 1,00,000}{₹ 2,00,000}$$

$$\text{Investment in X} = 2,00,000 \times 1.5 = 3,00,000$$

$$\text{Short sell in Y} = 2,00,000 \times -0.5 = -1,00,000$$

$$W_x = \frac{3,00,000}{1,00,000} = 3$$

$$W_y = \frac{-100,000}{1,00,000} = -1$$

$$R_f = \frac{-1,00,000}{1,00,000} = -1$$

Factor 1

$$B_p = (0.75 \times 3) + (1.5 \times -1) = 0.75$$

Factor 2

$$B_p = (0.6 \times 3) + (1.10 \times -1) = 0.70$$

(ii) Calculation of FRP₂

$$5 = 0.75 \text{ FRP}_1 + 0.60 \text{ FRP}_2 \quad \dots\dots\dots(1)$$

$$10 = 1.50 \text{ FRP}_1 + 1.10 \text{ FRP}_2 \quad \dots\dots\dots (2)$$

Multiply equation (1) with 1.5 & equation (2) with 0.75

$$7.5 = 1.125 \text{ FRP}_1 + 0.9 \text{ FRP}_2$$

$$7.5 = 1.125 \text{ FRP}_1 + 0.825 \text{ FRP}_2$$

$$0 = \frac{0.075 \text{ FRP}_2}{0.075}$$

$$\text{FRP}_2 = \frac{0}{0.075} = 0$$

Question – 52

Mr. Tamarind intends to invest in equity shares of a company the value of which depends upon various parameters as mentioned below:

Factor	Beta	Expected Value in %	Actual Value in %
GNP	1.20	7.70	7.70
Inflation	1.75	5.50	7.00
Interest rate	1.30	7.75	9.00
Stock market index	1.70	10.00	12.00
Industrial production	1.00	7.00	7.50

If the risk free rate of interest be 9.25%, how much is the return of the share under Arbitrage Pricing Theory?

(SM TYK – 38)

Solution:

Return as per Arbitrage Pricing Theory

Factor	Actual value	Expected value	Diff	Beta	Diff × B
GNP	7.70	7.70	0.00	1.20	0.00
Inflation	7.00	5.50	1.50	1.75	2.63
Int. rate	9.00	7.75	1.25	1.30	1.63
Stock Mrk index	12.00	10.00	2.00	1.70	3.40
Ind. Production	7.50	7.00	0.50	1.00	0.50

Risk free rate in %					8.16
Return under APT					9.25
					17.41

Question – 53

An investor has categorized all the available stock in the market into the following types and the estimated weights of the categories of stocks in the market index are given below. Further, the sensitivity of returns of these categories of stocks to two factors Inflation and Stock market are also given below:

Category	Weight in Market Index	Factor 1 (Inflation)			Factor 2 (Stock Market)		
		Beta 1	Expected Value in %	Actual Value in %	Beta 2	Expected Value in %	Actual Value in %
Small Cap	20%	1.20	6.70	6.70	0.80	10.00	10.50
Medium Cap	30%	1.75	4.50	6.00	0.90	7.00	8.00
Large Cap	15%	1.30	6.75	8.00	1.165	9.00	10.00
Flexi Cap	35%	1.70	7.00	6.50	0.85	8.85	9.75

Risk Free Rate of Interest is 7.50%.

Round off to 2 decimal.

You are required to calculate:

- Expected return on the market index for both the factors.
- Expected return on the market index under Arbitrage Pricing Theory (Existing Scenario).
- Expected return on the market index under Arbitrage Pricing Theory if the composition of the Portfolio is changed to 25% equally in all four categories.
- Which alternative (Existing or Changed) will be more profitable?

(Exam November – 2023) (8 Marks)

Solution:

(i) Expected Return on Market

Factor 1

$$= (6.70 \times 0.20) + (4.50 \times 0.30) + (6.75 \times 0.15) + (7 \times 0.35)$$

$$= 6.15\%$$

Factor 2

$$= (10 \times 0.20) + (7 \times 0.30) + (9 \times 0.15) + (8.85 \times 0.35)$$

$$= 8.55\%$$

(ii) Expected Return as per APT

$$ER = R_f + FRP_1 B_1 + FRP_2 B_2$$

$$\text{Small cap} = 7.5 + (0 \times 1.20) + (0.5 \times 0.80) = 7.90\%$$

$$\text{Medium cap} = 7.5 + (1.5 \times 1.75) + (1 \times 0.90) = 11.03\%$$

$$\text{Large Cap} = 7.5 + (1.25 \times 1.30) + (1 \times 1.165) = 10.29\%$$

$$\text{Flexi Cap} = 7.5 + (-0.5 \times 1.70) + (0.9 \times 0.85) = 7.42\%$$

Expected return of market

$$(7.90 \times 0.20) + (11.03 \times 0.3) + (10.29 \times 0.15) + (7.42 \times 0.35) = 9.03\%$$

(iii) Expected Return (Equal Weights)



$$ER = (7.90 \times 0.25) + (11.03 \times 0.25) + (10.29 \times 0.25) + (7.42 \times 0.25)$$

$$= 9.16\%$$

(iv) **Equal weights (Portfolio 2) is better due to higher return.**

PORTFOLIO REBELANCING

Question – 54

Ms. Sunidhi is working with an MNC at Mumbai. She is well versant with the portfolio management techniques and wants to test one of the techniques on an equity fund she has constructed and compare the gains and losses from the technique with those from a passive buy and hold strategy. The fund consists of equities only and the ending NAVs of the fund he constructed for the last 10 months are given below:

Month Ending	NAV (₹/unit)	Month Ending	NAV (₹/unit)
December 2008	40.00	May 2009	37.00
January 2009	25.00	June 2009	42.00
February 2009	36.00	July 2009	43.00
March 2009	32.00	August 2009	50.00
April 2009	38.00	September 2009	52.00

Assume Sunidhi had invested a notional amount of ₹ 2 lakhs equally in the equity fund and a conservative portfolio (of bonds) in the beginning of December 2008 and the total portfolio was being rebalanced each time the NAV of the fund increased or decreased by 15%.

You are required to determine the value of the portfolio for each level of NAV following the Constant Ratio Plan.

Solution:

Constant Ratio Plan

NAV	Bonds	Equity	Total	Rebalancing	Balance (units)
40	100,000	1,00,000	2,00,000	-	2,500
25	100,000	62,500	1,62,500	-	2,500
	81,250	81,250	1,62,500	Buy 750 unit	3,250
36	81,250	1,17,000	1,98,250	-	3,250
	99,125	99,125	1,98,250	Sell 496.53 unit	2,753.47
32	99,125	88,111.04	1,87,236.04	-	2,753.47
38	99,125	1,04,631.86	2,03,756.86	-	2,753.47
	1,01,878.43	1,01,878.43	2,03,756.86	Sell 72.46 Unit	2,681.01
37	1,01,878.43	99,197.37	2,01,075.80	-	2,681.01
42	1,01,878.43	1,12,602.42	2,14,480.85	-	2,681.01
43	1,01,878.43	1,15,283.43	2,17,161.86	-	2,681.01
50	1,01,878.43	1,34,050.50	2,35,928.93	-	2,681.01
	1,01,878.43	1,17,964.47	2,35,928.93	Sell 321.72 Unit	2,359.29
52	1,17,964.47	1,22,683.08	2,40,647.54	-	2,359.29

Buy & Hold (December – 2008)

Bonds = 100000

$$\text{Equity} = \frac{1,00,000}{40} = 2,500 \text{ Units}$$

Sep 2009

$$\text{Bond} = 1,00,000$$

$$\text{Equity } (2,500 \times 52) = 1,30,000$$

$$\underline{\underline{2,30,000}}$$



Gain due to adoption of Constant Ratio $(2,40,647.54 - 2,30,000) = 10,647.54$

Question – 55

Indira has a fund of ₹ 3 lacs which she wants to invest in share market with rebalancing target after every 10 days to start with for a period of one month from now. The present NIFTY is 5326. The minimum NIFTY within a month can at most be 4793.4. She wants to know as to how she should rebalance her portfolio under the following situations, according to the theory of Constant Proportion Portfolio Insurance Policy, using "2" as the multiplier:

(1) Immediately to start with.

(2) 10 days later-being the 1st day of rebalancing if NIFTY falls to 5122.96.

(3) 10 days further from the above date if the NIFTY touches 5539.04.

For the sake of simplicity, assume that the value of her equity component will change in tandem with that of the NIFTY and the risk free securities in which she is going to invest will have no Beta.

Solution:

Floor

$$\% \text{ Fall in Nifty} = \frac{5,326 - 4,793.40}{5,326} \times 100$$

$$= 10\%$$

$$\text{Floor} = 3,00,000 \times 90\% = 2,70,000$$

Today Investment

$$E = M (A - F)$$

$$= 2 (3,00,000 - 2,70,000) = 60,000$$

$$\text{Bonds} = 3,00,000 - 60,000 = 2,40,000$$

After 10 days

$$\text{Equity} = \left(\frac{60,000}{5326} \times 5,122.96 \right) = 57,712.65$$

$$\text{Bonds} = 2,40,000$$

$$A = 2,97,712.65$$

Rebalancing

$$\text{Equity} = 2 (2,97,712.65 - 2,70,000) = 55,425.31$$

$$\text{Bonds} = 2,97,712.65 - 55,425.31 = 2,42,287.34$$

Sell equity & buy bonds $(57,712.65 - 55,425.31)$ ₹ 2,287.34

Next 10 days

$$\text{Equity} = \frac{55,425.31}{5,122.96} \times 5,539.04 = 59,926.88$$

$$\text{Bonds} = 2,42,287.34$$

$$A = 3,02,214.22$$

Rebalancing

$$\text{Equity} = 2 (3,02,214.22 - 2,70,000) = 64,428.44$$



$$\text{Bonds} = (3,02,214.22 - 64,428.44) = 2,37,785.78$$

Buy equity & sell bonds $(64,428.44 - 59,926.88)$ ₹ 4501.56

Question – 56

Indira has a fund of ₹ 3 lacs which she wants to invest in share market with rebalancing target after every 10 days to start with for a period of one month from now. She has 3 close friends who have advised following different strategies:

- (i) Buy and Hold strategy
- (ii) Constant Ratio
- (iii) CPPI

Suppose she immediately starts with investment in Bonds (non-fluctuating) and Equity and decides to rebalance her portfolio after each 10 days and to invest in Nifty as equity component changes in tandem with that of Nifty. Further, Bond has no Beta.

As on date (i.e. beginning of month) Nifty is 5326 and minimum Nifty within a month can be most be 4793.40. If she chooses CPPI she will use “2” as the multiplier. If she chooses Constant Ratio plan she will maintain the ratio of 60:40 in Equity and Bonds respectively. Further, portfolio will be rebalanced each time Nifty is changed by 5% as compared to the previous Nifty.

You are required to evaluate Portfolio Position of Indira under each of the Strategies suggested by her friends and highlight the course of action to be taken if in the coming month after a gap of 10 days Nifty happened:

- (1) 10 days later-being the 1st day of rebalancing if NIFTY falls to 5122.96.

- (2) 10 days further from the above date if the NIFTY touches 5539.04.

(MTP March – 2024)**Solution:****(i) Buy & Hold**

$$\text{Fund} = ₹ 3,00,000$$

$$= \% \text{ fall in nifty} = \frac{5,326 - 4,193.40}{5,326} \times 100$$

$$= 10\%$$

$$\text{Floor} = 3,00,000 \times 90\% = 2,70,000$$

$$\text{Bonds} = 2,70,000$$

$$\text{Equity} = 30,000$$

After 10 days

$$\text{Equity} = \frac{30,000}{5,326} \times 5,122.96$$

$$= 28,856$$

$$\text{Bonds} = 2,70,000$$

$$= 2,98,856$$

After 10 days

$$\begin{aligned}\text{Equity} &= \frac{28,856}{5,122.96} \times 5,539.64 \\ &= 31,199.65\end{aligned}$$

$$\begin{aligned}\text{Bonds} &= 2,70,000 \\ &\underline{\hspace{1.5cm}} \\ &= 3,01,199.65\end{aligned}$$

(ii) Constant Ratio

$$\text{Fund} = 3,00,000$$

$$\text{Equity (60\%)} = 1,80,000$$

$$\text{Bond (40\%)} = 1,20,000$$

After 10 days

$$\begin{aligned}\text{Equity} &= \frac{1,80,000}{5326} \times 5,122.96 \\ &= 1,73,138\end{aligned}$$

$$\begin{aligned}\text{Bond} &= 1,20,000 \\ &\underline{\hspace{1.5cm}} \\ &= 2,93,138\end{aligned}$$

$$\text{Change in Nifty} = \frac{5,122.96 - 5,326}{5,326} \times 100 = 3.81\%$$

Since Nifty changed by less than 5% hence no rebalancing.

After next 10 days

$$\begin{aligned}\text{Equity} &= \frac{1,73,138}{5,122.96} \times 5,539.04 \\ &= 1,87,200\end{aligned}$$

$$\begin{aligned}\text{Bond} &= 1,20,000 \\ &\underline{\hspace{1.5cm}} \\ &= 3,07,200\end{aligned}$$

$$\text{Change in nifty} = \frac{5,539.04 - 5,122.96}{5,122.96} \times 100 = 8.12\%$$

Rebalancing

$$\text{Equity (3,07,200} \times 60\%) = 1,84,320$$

$$\text{Bond (3,07,200} \times 40\%) = 1,22,880$$

$$\underline{\hspace{1.5cm}} = 3,07,200$$

Sell equity and buy bonds $(1,87,200 - 1,84,320) = ₹ 2,880$

(iii) CPPI**Today**

$$\text{Today fund} = 3,00,000$$

$$\text{Floor} = 2,70,000$$

$$\text{Equity} = M (A - E)$$

$$\text{Equity} = 2 (3,00,000 - 27,000) = 60,000$$



PORTFOLIO MANAGEMENT

$$\text{Bonds} = (3,00,000 - 60,000) = 2,40,000$$

After 10 days

$$\text{Equity} = \left(\frac{60,000}{5,326} \times 5,122.96 \right) = 57,713$$

$$\text{Bond} = 2,40,000$$

$$A = 2,97,713$$

No Rebalancing

Next 10 days

$$\text{Equity} = \left(\frac{57,713}{5,122.96} \times 5,539.40 \right) = 62,400$$

$$\text{Bond} = 2,40,000$$

$$3,02,400$$

Rebalancing

$$\text{Equity} = 2 (3,02,400 - 2,70,000) = 64,800$$

$$\text{Bonds} = (3,02,400 - 64,800) = 2,37,600$$

$$\text{Buy equity and sell bonds } (64,800 - 62,400) = ₹ 2,400$$

EFFICIENT MARKET HYPOTHESIS

Question – 57

The closing value of Sensex for the month of October, 2007 is given below:

Date Closing	Sensex Value
01/10/2007	2800
03/10/2007	2780
04/10/2007	2795
05/10/2007	2830
08/10/2007	2760
09/10/2007	2790
10/10/2007	2880
11/10/2007	2960
12/10/2007	2990
15/10/2007	3200
16/10/2007	3300
17/10/2007	3450
19/10/2007	3360
22/10/2007	3290
23/10/2007	3360
24/10/2007	3340
25/10/2007	3290
29/10/2007	3240
30/10/2007	3140
31/10/2007	3260

You are required to test the weak form of efficient market hypothesis by applying the run test at 5% and 10% level of significance.

Following value can be used :

Value of t at 5% is 2.101 at 18 degrees of freedom

Value of t at 10% is 1.734 at 18 degrees of freedom

Solution:

Sensex Value	Sign	
2800		
2780	–	1
2795	+	2



2830	+	
2760	-	3
2790	+	4
2880	+	
2960	+	
2990	+	
3200	+	
3300	+	
3450	+	
3360	-	5
3290	-	
3360	+	6
3340	-	7
3290	-	
3240	-	
3140	-	
3260	+	8

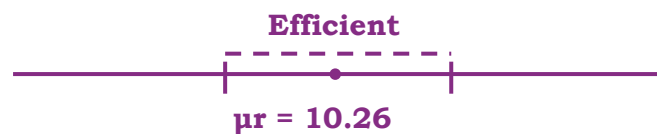
No. of sign change (r) = 8

n_1 (No. of plus) = 11

n_2 (No. of minus) = 8

$$\mu_r = \frac{2n_1n_2}{n_1 + n_2} + 1$$

$$= \frac{2 \times 11 \times 8}{11 + 8} + 1 = 10.26$$



Standard Deviation

$$\text{or} = \sqrt{\frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}}$$

$$= \sqrt{\frac{(2 \times 11 \times 8)(2 \times 11 \times 8 - 11 - 8)}{(11 + 8)^2 (11 + 8 - 1)}}$$

$$= 2.06 \quad \text{or}$$

$$\text{or} = \sqrt{\frac{(\mu_r - 1)(\mu_r - 2)}{n_1 + n_2 - 1}}$$

$$= \sqrt{\frac{(10.26 - 1)(10.26 - 2)}{11 + 8 - 1}}$$

$$= 2.06$$

Upper Limit = $\mu_r + \sigma_t$ (t depends on two factors)

Degree of Freedom (df) = $n_1 + n_2 - 1$

$$= 11 + 8 - 1 = 18$$

Level of significant (%)

5%

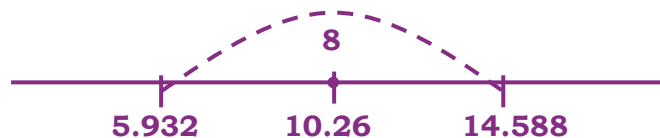
Upper Limit = $10.26 + 2.06 \times 2.101$

$$= 14.588$$

Lower Limit = $10.26 - 2.06 \times 2.101$

$$= 5.932$$





Since actual run (8) is in the range, hence market is weak form of efficient market.

10%

Upper Limit = $10.26 + 2.06 \times 1.734$
= 13.832

Lower Limit = $10.26 - 2.06 \times 1.734$
= 6.688

Since actual run (8) is in the range, hence market is weak form of efficient market.

Question – 58

Mr. X is of the opinion that market has recently shown the Weak Form of Market Efficiency. In order to test the validity of his impression he has collected the following data relating to the movement of the SENSEX for the last 20 days.

Days	Open	High	Low	Close
1	33470.94	33513.79	33438.03	33453.99
2	33453.64	33478.11	33427.82	33434.83
3	33414.06	33440.29	33397.65	33431.93
4	33434.94	33446.18	33377.78	33383.41
5	33372.92	33380.27	33352.12	33370.93
6	33375.85	33389.49	33331.42	33340.75
7	33340.89	33340.89	33310.95	33330.98
8	33326.84	33340.91	33306.17	33335.08
9	33307.16	33328.22	33296.43	33301.97
10	33298.64	33318.60	33254.28	33259.03
11	33260.04	33228.85	33241.66	33251.53
12	33255.92	33289.46	33249.46	33285.89

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13	33288.86	33535.67	33255.98	33329.28
14	33335.00	33346.21	33276.72	33284.17
15	33293.83	33310.86	33278.54	33298.78
16	33300.02	33337.79	33300.02	33325.38
17	33323.36	33356.34	33322.44	33329.95
18	33322.81	33345.98	33317.44	33319.67
19	33317.51	33321.18	33294.19	33302.32
20	33290.86	33324.96	33279.62	33319.61

You are required:

To test the **Weak Form of Market Efficiency** using Auto-Correlation test, taking time lag of 10 days.

(Exam January – 2021) (8 Marks)

Solution:

Days	x	(x - $\bar{x}$)	(x - $\bar{x}$) ²	y	(y - $\bar{y}$)	(y - $\bar{y}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	-19.16	2.50	6.25	34.36	26.80	718.24	67
2	-2.90	18.76	351.94	43.59	35.83	1283.79	672.17
3	-48.52	-26.86	721.46	-45.11	-52.67	2774.13	1414.71
4	-12.48	9.18	84.27	14.61	7.05	49.70	64.71
5	-30.18	-8.52	72.59	26.60	19.04	362.52	-162.22
6	-9.77	11.89	141.37	4.57	-2.99	8.94	-35.55
7	4.10	25.76	663.58	-10.28	-17.84	318.27	-459.55
8	-33.11	-11.45	131.10	-17.35	-24.91	620.51	285.21
9	-42.94	-21.28	452.84	17.29	9.73	94.67	-207.05
	-194.96		2625.4006	68.08		6230.7686	1637.98

$$\bar{x} = \frac{-194.96}{9} = -21.66$$

$$\sigma_x = 17.079$$

$$\bar{y} = \frac{68.08}{9} = 7.56$$



$$\sigma_y = 26.3117$$

$$\text{Cov}_{xy} = \frac{1637.98}{9} = 181.997$$

$$r_{xy} = \frac{\text{COV}_{xy}}{\sigma_x \times \sigma_y} = \frac{181.97}{17.079 \times 26.3117} = 0.405$$

Since correlation is not zero hence market is not weak form of efficient.

SHARPE OPTIMIZATION MODEL

Question – 59

Ramesh wants to invest in stock market. He has got the following information about individual securities:

Security	Expected Return	Beta	σ^2_{cl}
A	15	1.5	40
B	12	2	20
C	10	2.5	30
D	09	1	10
E	08	1.2	20
F	14	1.5	30

Market index variance is 10 percent and the risk free rate of return is 7%. What should be the optimum assuming no short sales?

(SM TYK – 35)

Solution:

Step 1: Calculate treynor's ratio & give rank highest to lowest

$$\text{Treynor's ratio} = \frac{E_R - R_f}{\beta}$$

Stock	$\frac{E_R - R_f}{\beta}$	Rank
A	5.33	I
B	2.50	III
C	1.20	V
D	2.00	IV
E	0.833	VI
F	4.667	II

Stock	TP	Bi	σ^2_{e1}	$\frac{\beta^2}{\sigma^2_{e1}}$	$\left(\frac{E_R - R_f}{\sigma^2_{e1}}\right)$	Cum .6	Completion 5	
A	5.33	1.5	40	0.056	0.30	0.30	0.056	CA = 1.923
F	4.667	1.5	30	0.075	0.35	0.65	0.131	CF = 2.814
B	2.50	2	20	0.20	0.50	1.15	0.331	CB = 2.668
D	2.00	1	10	0.10	0.20	1.35	0.431	CD = 2.542
C	1.20	2.5	30	0.208	0.25	1.60	0.639	CC = 2.165
E	0.833	1.2	20	0.072	0.06	1.66	0.711	CE = 2.047

Step 2: Calculation of Cutoff Point

$$C = \frac{\sigma_M^2 \times \sum \left(\frac{E_R - R_f}{\sigma_{e_i}^2} \right) B_i}{1 + \left(\sigma_M^2 \times \sum \frac{B_i^2}{\sigma_{e_i}^2} \right)}$$

$$CA = \frac{10 \times 0.30}{1 + (10 \times 0.056)} = 1.923$$

$$CF = \frac{10 \times 0.65}{1 + (10 \times 0.131)} = 2.814$$

$$\text{Highest cut off} = 2.814$$

Step 3: Select the stock having Treynor's Ratio is more than highest cut off Stock A & F should be selected because Treynor's Ratio is more than 2.814



Step 4: Calculation of weights of stock A & stock F

$$Z_i = \frac{B_i}{\sigma_{e_i}^2} \left(\frac{E_R - R_f}{B_i} - \text{Cut off} \right)$$

$$Z_A = \frac{1.5}{40} (5.33 - 2.814) = 0.0943$$

$$Z_F = \frac{1.5}{30} (4.67 - 2.814) = 0.0928$$

$$W_A = \frac{0.0943}{0.0943 + 0.0928} = 0.504 \text{ [50.40\%]}$$

$$W_f = 1 - 0.504 = 0.496 \text{ [49.60\%]}$$

Question – 60

Mrs. SRS is your HNI Client and wants to invest in stock market. She has got the following information about individual securities and wants to select the securities to form an optimal portfolio from amongst these securities:

Security	Expected Return (%)	Unsystematic Risk (%)	Beta
A	5	25	0.5
B	25	20	2.5
C	15	10	1.0
D	10	10	1.5
E	20	18	1.8

Market Index Variance is 25% and the Risk Free Rate of Return is 7%.

Required:

Based on this information help Mrs. SRS to:

- Prepare ranked table using Treynor's Ratio.
- Calculate Cut-off Point.
- Identify the securities to be included in optimal portfolio.

(Exam Jan. – 2026)

Solution:

(i) Ranked table using Treynor's Ratio

Security	R_i	β_i	$R_i - R_f$	$\beta_i / R_i - R_f$	Ranking
A	5	0.5	-2	-4	5
B	25	2.50	18	7.20	3
C	15	1.00	8	8.00	1
D	10	1.50	3	2.00	4
E	20	1.80	13	7.22	2

(ii) Calculation of Cut-off point

Security	$R_i - R_f$	β_i	σ_{ei}^2	$\frac{(R_i - R_f) \times \beta_i}{\sigma_{ei}^2}$	$\sum_{e=1}^N \frac{(R_i - R_f)}{\sigma_{ei}^2}$	$\frac{\beta_i^2}{\sigma_{ei}^2}$	$\sum_{e=1}^N \frac{\beta_i^2}{\sigma_{ei}^2}$	C_i
C	8	1.0	10	0.80	0.80	0.10	0.10	5.71
E	13	1.80	18	1.30	2.10	0.18	0.28	6.56
B	18	2.50	20	2.25	4.35	0.313	0.593	6.87
D	3	1.50	10	0.45	4.80	0.225	0.818	5.59
A	-2	0.50	25	-0.04	4.76	0.01	0.828	5.48

$$CC = 25 \times 0.80 / [1 + (25 \times 0.10)] = 5.71$$

$$CE = 25 \times 2.10 / [1 + (25 \times 0.28)] = 6.56$$

$$CB = 25 \times 4.35 / [1 + (25 \times 0.593)] = 6.87$$

$$CD = 25 \times 4.80 / [1 + (25 \times 0.818)] = 5.59$$

$$CA = 25 \times 4.76 / [1 + (25 \times 0.828)] = 5.48$$

Cut-off Point: 6.87

- (iii)** The stock whose excess-return to risk ratio (Treynor ratio in (i) above) is above the cut-off point of 6.87 are selected and whose ratios are below are rejected. Hence Security C, E & B are selected in the optimal portfolio while remaining securities i.e D & A are rejected as they are placed below the cut-off point.

MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

You are a financial analyst at a prominent investment firm tasked with empirically verifying the weak form of Efficient Market Theory (EMT) for the XYZ Stock Index, a collection of diverse stocks. You decide to conduct three different tests to assess whether the stock market follows the principles of the weak form of EMT.

Test 1

For the past five years, you collect daily price changes of the stocks in the XYZ Stock Index. You calculate autocorrelation coefficients for different lag periods and analyze whether past price changes exhibit any significant correlation with future price changes. The results indicate that most autocorrelation coefficients are close to zero and statistically insignificant, suggesting those past price changes do not predict future price changes.

Test 2

You further investigate the randomness of price changes in the XYZ Stock Index. Analyzing the sequence of daily price changes, you count the number of runs where price changes are consistently positive or negative. Upon comparing the observed number of runs with the expected number based on randomness, you find that they align closely, supporting the idea that price changes follow a random pattern.

Test 3

To examine the efficacy of trading strategies based on historical price trends, you implement a simple trading rule for the XYZ Stock Index. The rule involves buying when the price crosses a certain moving average % threshold and selling when it crosses another % threshold. Over a period of testing, you compute the returns generated by the trading strategy. The results reveal that the returns are not

consistently better than random chance, implying that past price trends do not reliably predict future price movements.

Conclusion:

After conducting the three tests the evidence supports the weak form of Efficient Market Theory for the XYZ Stock Index. The lack of correlation, the randomness of price change patterns, and the inability of the trading strategy to consistently outperform random chance suggests that past price trends do not reliably predict future price movements.

Based on the above information answer the following questions:

- I. Test 2 is called as:
 - (a) Serial Correlation test.
 - (b) Run test.
 - (c) Filter Rules test.
 - (d) Variance Ratio test.

- II. In the filter rule test, filter rules do not apply if:
 - (a) the behavior of stock price changes is predictable.
 - (b) the behavior of stock price changes is dependent on past trends.
 - (c) the behavior of stock price changes is correlated.
 - (d) the behavior of stock price changes is random.

- III. Test 1 is called as:
 - (a) Serial Correlation test.
 - (b) Filter Rules test.
 - (c) Run test.
 - (d) Variance Ratio test.

- IV. Test 3 is called as:
 - (a) Serial Correlation test.
 - (b) Filter Rules test.
 - (c) Run test.



PORTFOLIO MANAGEMENT

(d) Variance Ratio test.

V. Result of your studies support the_____

- (a) Semi strong EMH Theory
- (b) Semi EMH theory
- (c) Random walk Theory
- (c) Markowitz Theory

(RTP May – 2024)

Answer: Case Scenario – 01

- I. (a) Run Test.
- II. (d) The behavior of stock price changes is random.
- III. (a) Serial Correlation Test.
- IV. (b) Filter Rules Test.
- V. (C) Random walk Theory

Case Scenario – 02

Two friend Mr. A and Mr. N were discussing about the risks of market. While Mr. A is sort of risk averse, Mr. N is an aggressive investor and believes in taking risk.

Mr. N said we cannot diversify the market risk at all, and he quoted the Modern Portfolio Approach. Both friends analyze the market data for the few months and came out with expected returns on two stocks for a particular market.

Market Return	Aggressive	Defensive
7%	4%	9%
25%	40%	18%

Based on above scenario, answer the following questions:

- I. The Beta of Defensive stock is.....
 - (a) 2
 - (b) 0.5
 - (c) 4
 - (d) 1
- II. If the market return is equally likely to be 7% or 25% then expected return of Aggressive stock shall be.....

- (a) 18%
- (b) 13.50%
- (c) 22%
- (d) 11%

III. The Alpha of the Defensive stocks is.....

- (a) -10%
- (b) 22%
- (c) 5.50%
- (d) 12%

IV. The Modern Portfolio Theory was propounded by

- (a) William Sharpe
- (b) Black Scholes
- (c) Stephen Ross
- (d) Harry Markowitz

V. As per Capital Market Line (CML) Theory the Portfolios lying on the CML over the market portfolio are called

- (a) Lending Portfolios
- (b) Borrowing Portfolios
- (c) Diversified Portfolios
- (d) Risk- Free Portfolios

(RTP November – 2024)

Answer: Case Scenario – 02

- I. (b) 0.5
- II. (c) 22%
- III. (c) 5.50%
- IV. (d) Harry Markowitz
- V. (c) Diversified Portfolios

Case Scenario – 03

Mr. Abhishek is interested in investing ₹ 2,00,000 for which he is considering following three alternatives:

- (i) Invest ₹ 2,00,000 in Mutual Fund X (MFX)



PORTFOLIO MANAGEMENT

- (ii) Invest ₹ 2,00,000 in Mutual Fund Y (MFY)
(iii) Invest ₹ 1,20,000 in Mutual Fund X (MFX) and ₹ 80,000 in Mutual Fund Y (MFY)

Average annual return earned by MFX and MFY is 15% and 14% respectively. Risk free rate of return is 10% and market rate of return is 12%.

Covariance of returns of MFX, MFY and market portfolio Mix are as follow:

	MFX	MFY	Mix
MFX	4.800	4.300	3.370
MFY	4.300	4.250	2.800
Mix	3.370	2.800	3.100

- I.** Variance of Market
(a) 4.800 (b) 3.370
(c) 4.250 (d) 3.100
- II.** Portfolio standard Deviation
(a) 2.115 (b) 3.100
(c) 2.800 (d) 1.725
- III.** Systematic Risk of Portfolio
(a) 3.663 (b) 2.528
(c) 2.800 (d) 3.181
- IV.** Based on standard deviation, the optimum investment for Mr A would be.....
(a) Portfolio
(b) All investment in MFX
(c) All investment in MFY
(d) Both MFY and mix are indifference
- V.** Standard Deviation of MFX

- (a) 2.191 (b) 2.071
(c) 1.761 (d) 1.433

- VI.** Portfolio Return would be
(a) 13.50% (b) 14.60%
(c) 15.50% (d) 18.60%

Answer: Case Scenario – 03

- I. (d) 3.100**
II. (a) 2.115
III. (d) 3.181
IV. (c) All investment in MFY
V. (a) 2.191
VI. (b) 14.60%

Case Scenario – 04

Stock	ER	Beta	Specific Risk	TR
A	18%	1.2	5%	?
B	10%	0.5	4%	?
C	?	1.4	7%	22%

Assuming CAPM holding good.

- I.** Total Risk (SD) of Stock A
(A) 55.25 (B) 15.25
(C) 18.56 (D) 9.45
- II.** Total Risk (SD) of Stock B
(A) 8.45 (B) 76.24
(C) 10.45 (D) 14.35
- III.** Expected Return of Stock C
(A) 15 (B) 14.90



(C) 8.45

(D) 20.28

Answer: Case Scenario – 04**I. (C) 18.56****II. (A) 8.45****III. (D) 20.28****Case Scenario – 05**

An investor is evaluating two companies, A Ltd. and B Ltd., to determine which company is a better investment. Both companies have different expected returns, standard deviations, and betas.

- A Ltd. has an expected return of 22% and a standard deviation (risk) of 40%. Its Beta is 0.86.
- B Ltd. has an expected return of 24%, a standard deviation of 38%, and a Beta of 1.24.

The correlation coefficient between the returns of A Ltd. and B Ltd. is 0.72, and the standard deviation of the market return is 20%.

From the information given above, choose the correct answer to the following questions:

I. The portfolio's expected rate of return, if the investor invests 30% in B Ltd. and 70% in A Ltd. shall be.....

- (a) 22.00% (b) 22.60%
(c) 23.40% (d) 24.00%

II. The standard deviation of the investor's portfolio if 30% is invested in B Ltd. and 70% in A Ltd. shall be approximately.....

- (a) 40% (b) 39%
(c) 37% (d) 35%

III. The covariance between returns of Stock of A Ltd. and B Ltd. shall be approximately.....

- (a) 0.0230 (b) 0.1094
(c) 0.7200 (d) 0.2189

IV. Based on the data given the Risk Free Rate of Return shall be approximately

- (a) 17.48% (b) 18.00%
(c) 20.00% (d) 22.74%

V. Based on the data given the Market Rate of Return shall be approximately.....

- (a) 17.48% (b) 18.00%
(c) 20.00% (d) 22.74%

Answer: Case Scenario – 05**I. (b) 22.60%****II. (c) 37%****III. (b) 0.1094****IV. (a) 17.48%****V. (d) 22.74%****Case Scenario – 06**

Equity Researchers have estimated the rate of return for Stock A, Stock B and Market Portfolio under each state of the economy is as under –

Economy	Probability	Return on Stock A (%)	Return on Stock B (%)	Market Portfolio (%)
Boom	0.3	16	19	18
Normal	0.4	14	16	15
Recession	0.3	-9	-7	-8



The risk-free rate of return is expected to be 8%. The covariance between Stock A and the Market Portfolio is 122.70, while the covariance between Stock B and the Market Portfolio is 125.40. Assume that the CAPM framework is valid in this market.

From the information given above, choose the correct answer to the question No. 1 to 5:

(EXAM MAY–2025)

1. What is the expected rate of return (percentage) for Stocks A and B ?
 (A) 7.70% and 9.00%
 (B) 10% and 9.00%
 (C) 7.70% and 10%
 (D) 13.1% and 14.20%
2. What will be the variance of market portfolio ?
 (A) 38.4
 (B) -4.8
 (C) 126.8
 (D) 125.4
3. What will be the beta of Stock A and Stock B respectively ?
 (A) 0.9542 and 1.00
 (B) 0.9862 and 1.24
 (C) 0.9785 and 1.00
 (D) 0.9785 and 1.24
4. Required rate of return of Stock A is _____ and Stock B is _____.
 (A) 8.9785% and 9%
 (B) 9% and 8.978%
 (C) 8.9875% and 9%
 (D) 8.9785 and 10.4%
5. Based on calculated Alpha of Stock A and Stock B in the above case scenario, which of the following statement is correct for purchase or sale of Stock A and Stock B ?
 (A) Stock A has a positive alpha, and Stock B has a negative alpha. This indicates that Stock A is

underpriced and Stock B is overpriced. Therefore, purchase Stock A and sell Stock B.

- (B) Both Stock A and Stock B have positive alpha values, suggesting they are underpriced. Therefore, purchase both stocks.
- (C) Stock A has a negative alpha, while Stock B has a positive alpha. This means Stock A is overpriced and Stock B is underpriced. Therefore, sell Stock A and purchase Stock B.
- (D) Both Stock A and Stock B have negative alpha values, indicating they are overpriced. Therefore, sell both stocks

Answer: Case Scenario – 06

1. (C) 7.70% and 10%
2. (D) 125.4
3. (C) 0.9785 and 1.00
4. (A) 8.9785% and 9%
5. (C) Stock A has a negative alpha, while Stock B has a positive alpha.

This means Stock A is overpriced and Stock B is underpriced. Therefore, sell Stock A and purchase Stock B.

Case Scenario – 07

You are submitted the following information in respect of Mr. Z's portfolio:

Share/Bond	Cost (₹)	Dividend/Interest (₹)	Market Price (₹)	Beta
A Ltd.	40,000	4,000	41,000	0.60
B Ltd.	50,000	5,000	52,500	0.80
C Ltd.	80,000	3,000	1,10,000	0.60
GOI Bonds	1,70,000	17,000	1,61,500	0.01

Average Return of the portfolio is 15.70% per annum.



Note: Calculate upto two decimal points.

From the information given above, choose the correct answer to the following Question No. 7 to 9:

- I.** Expected return on market portfolio (R_m) is –
(A) 14.38% (B) 15.34%
(C) 15.88% (D) 16.32%
- II.** Risk-free rate of return (as per simple average) is –
(A) 15.42% (B) 15.52%
(C) 15.62% (D) 15.72%
- III.** Expected rate of return on shares of C Ltd. is –
(A) 12.38% (B) 13.54%
(C) 14.12% (D) 15.74%

(EXAM JANUARY – 2026)

Answer Case Scenario – 07

- I. (C) 15.88%**
II. (B) 15.52%
III. (D) 15.74%

Case Scenario – 08

An investor is evaluating two companies, X Ltd. and Y Ltd., to determine which company is a better investment. Both companies have different expected returns, standard deviations, and betas.

- X Ltd. has an expected return of 22% and a standard deviation (risk) of 40%. Its Beta is 0.86.
- Y Ltd. has an expected return of 24%, a standard deviation of 38%, and a Beta of 1.24.

The correlation coefficient between the returns of X Ltd. and Y Ltd. is 0.72, and the standard deviation of the market return is 20%.

From the information given above, choose the correct answer to the following questions:

- I.** Which company is expected to be more volatile in absolute terms?
(a) X Ltd.
(b) Y Ltd.
(c) Both have equal volatility
(d) Depends on correlation
- II.** The correlation coefficient of 0.72 between X Ltd. and Y Ltd. implies that.....
(a) the stocks move in opposite directions.
(b) the stocks are perfectly correlated.
(c) there is strong positive co-movement.
(d) there is no diversification benefit.
- III.** If the correlation between X Ltd. and Y Ltd. were zero, the portfolio risk would.....
(a) Increase
(b) Decrease
(c) Remain unchanged
(d) Become zero
- IV.** Which type of risk can be reduced by forming a portfolio of X Ltd. and Y Ltd.?
(a) Political risk
(b) Market risk
(c) Unsystematic risk
(d) Interest rate risk
- V.** If market volatility increases but betas remain unchanged, the expected return as per CAPM will.....
(a) Increase
(b) Decrease



- (c) Remain unchanged
- (d) Become zero

Answer Case Scenario – 08

- I. (a) X Ltd.
- II. (c) there is strong positive co-movement.
- III. (b) Decrease
- IV. (c) Unsystematic risk
- V. (c) Remain unchanged



Student Notes



Student Notes



02

MUTUAL FUND

(1) NET ASSET VALUE PER UNIT

Question – 01

Cinderella Mutual Fund has the following assets in Scheme Rudolf at the close of business on 31st March, 2014.

Company	No. of Shares	Market Price per Share
Nairobi Ltd.	25,000	₹ 20
Dakar Ltd.	35,000	₹ 300
Senegal Ltd.	29,000	₹ 380
Cairo Ltd.	40,000	₹ 500

The total number of units of Scheme Rudolf are 10 lacs. The Scheme Rudolf has accrued expenses of ₹ 2,50,000 and other liabilities of ₹ 2,00,000. Calculate the NAV per unit of the Scheme Rudolf.

(SM TYK – 07 & MTP April – 2021)

Solution:

Shares	No. of Shares	Price	Amount (₹)
Nairobi Ltd,	25,000	20.00	5,00,000
Dakar Ltd.	35,000	300.00	1,05,00,000
Senegal Ltd.	29,000	380.00	1,10,20,000
Cairo Ltd.	40,000	500.00	2,00,000

	4,20,20,000
Less: Accrued Expenses	2,50,000
Other Liabilities	2,00,000
Total Value	4,15,70,000
No. of Units	10,00,000
NAV per Unit (4,15,70,000/10,00,000)	41.57

Question – 02

A Mutual Fund Co. has the following assets under it on the close of business as on:

Company	No. of Shares	1 st February 2012 Market price per share ₹	2 nd February 2012 Market price per share ₹
L Ltd.	20,000	20.00	20.50
M Ltd.	30,000	312.40	360.00
N Ltd.	20,000	361.20	383.10
P Ltd.	60,000	505.10	503.90

Total No. of Units 6,00,000

- Calculate Net Assets Value (NAV) of the Fund.
- Following information is given:

Assuming one Mr. A, submits a cheque of ₹ 30,00,000 to the Mutual Fund and the Fund manager of this company purchases 8,000 shares of M Ltd; and the balance amount is



held in Bank. In such a case, what would be the position of the Fund?

- (iii) Find new NAV of the Fund as on 2nd February 2012.

(SM TYK – 08)

Solution:

- (i) NAV [1st Feb 2012]**

			Total
L Ltd.	20,000	20.00	4,00,000
M Ltd.	30,000	312.40	93,72,000
N Ltd.	20,000	361.20	72,24,000
P Ltd.	60,000	505.10	3,03,06,000
Net Asset			4,73,02,000
÷ No of units			6,00,000
NAV			78.84

$$\text{No. of new units} = \frac{30,00,000}{78.84} = 38,051.75 \text{ units}$$

- (ii) NAV [1st Feb 2012]**

Net Assets	4,73,02,000
M Ltd. [8,000 × 312.40]	24,99,200
(+) Bank [30,00,000 – 24,99,200]	5,00,800
Net Assets	5,03,02,000
÷ No. of units (6,00,000 + 38,051.75)	6,38,051.75
NAV	78.84

- (iii) On 2nd February 2012, the NAV of fund will be as follows:**

Shares	No. of Shares	Price	Amount (₹)
L Ltd.	20,000	20.50	4,10,000
M Ltd.	38,000	360.00	1,36,80,000

N Ltd.	20,000	383.10	76,62,000
P Ltd.	60,000	503.90	3,02,34,000
Cash			5,00,800
			<u>5,24,86,800</u>

$$\text{NAV as on 2nd February 2012} = \frac{₹ 5,24,86,800}{6,38,053} = ₹ 82.26 \text{ per unit}$$

Question – 03

The unit price of Equity Linked Savings Scheme (ELSS) of a mutual fund is ₹ 10/-. The public offer price (POP) of the unit is ₹ 10.204 and the redemption price is ₹ 9.80.

Calculate:

- (i) Front-end Load

- (ii) Back end Load

(SM TYK – 02 & Exam May – 2018) (5 Marks)

Solution:

- (i) Front End Load**

$$\frac{10.204 - 10}{10} \times 100 = 2.04\%$$

- (ii) Back End Load**

$$\frac{10 - 9.80}{10} \times 100 = 2\%$$

Question – 04

A mutual fund made an issue of 10,00,000 units of ₹ 10 each on January 01, 2008. No entry load was charged. It made the following investments:

MUTUAL FUND

Particulars	₹
50,000 Equity shares of ₹ 100 each @ ₹ 160	80,00,000
7% Government Securities	8,00,000
9% Debentures (Unlisted)	5,00,000
10% Debentures (Listed)	5,00,000
	<u>98,00,000</u>

During the year, dividends of ₹ 12,00,000 were received on equity shares. Interest on all types of debt securities was received as and when due. At the end of the year equity shares and 10% debentures are quoted at 175% and 90% respectively of face value. Other investments are at par.

Find out the Net Asset Value (NAV) per unit given that operating expenses paid during the year amounted to ₹ 5,00,000. Also find out the NAV, if the Mutual fund had distributed a dividend of ₹ 0.80 per unit during the year to the unit holders.

(SM TYK – 19)

Solution:

Working Note 1 Closing Cash

Opening cash	2,00,000
(+) Dividend Received	12,00,000
(+) Interest Received (56,000 + 45,000 + 50,000)	1,51,000
(-) Expenses Paid	5,00,000
Closing Cash	<u>10,51,000</u>

Calculation of NAV

Closing Cash (W.N.1)	10,51,000
ESC [50,000 × 175]	87,50,000
10% debentures [5,00,000 × 90%]	4,50,000
7% Govt. Securities	8,00,000

9% debentures	5,00,000
Net Assets	1,15,51,000
÷ No. of units	10,00,000
NAV	11.551

If Mutual Fund Distributed Dividend

Alternative 1	
Net Assets	1,15,51,000
(-) Dividend Paid (10,00,000 × 0.8)	8,00,000
Net Assets	1,07,51,000
(÷) No. of Units	10,00,000
NAV	₹ 10.75

Alternative 2	
Revised NAV (11.551 – 0.8)	₹ 10.75

Question – 05

A mutual fund made an issue of New Fund Offer (NFO) on 01/01/2021 of 10.00 Lakh Units of ₹ 10 each. No entry load was charged. It made the following investments:

Particulars	(₹)
25,000 Equity Shares of XYZ Ltd., ₹ 100 each @ ₹ 320	80,00,000
5% Government Securities	4,00,000
10% NCDs Unlisted	5,00,000
8% Listed Debentures	10,00,000

During the year, dividends of ₹ 8.00 lakhs were received on equity shares. Interest on all types of debt securities were received. On 31st December 2021 equity shares were appreciated by 15% while listed debentures were quoted at 20% premium. XYZ Ltd., on 15th December 2021 in its AGM declared the interim dividend of 10% and bonus shares at 1:10 with the record date of 28th December 2021.



- (i) Find out the NAV per unit as on 31st December given that the operating expenses paid during the year amounting to ₹ 3,00,000.
- (ii) Find out the NAV, if the MF had distributed a dividend of, ₹ 0.50 per unit during the year to the investors.
- (iii) If you are the investor, find out what is the annualized return you have got.

(Exam May – 2022) (8 Marks)

Solution:

Working Note 1 Closing Cash

Opening Cash [10,00,000 × 10 – 99,00,000]	1,00,000
(+) Dividend Received	
Final Dividend	8,00,000
Interim Dividend [25,000 × 100 × 10%]	2,50,000
Interest received [20,000 + 50,000 + 80,000]	1,50,000
(-) Expense Paid	3,00,000
Closing Cash	10,00,000

(I) Calculation of NAV

Closing Cash (W.N.1)	10,00,000
Equity $\left[25,000 + \left(25,000 \times \frac{1}{10} \right) \right] \times [320 \times 115\%]$	1,01,20,000
5% Govt. Securities	4,00,000
10% NCD	5,00,000
8% Listed Debenture (10,00,000 × 120%)	12,00,000
Total Assets	1,32,20,000
÷ No of Units	10,00,000
NAV	13.22

(II) If Mutual Fund distributed dividend 0.5 per unit

$$\text{NAV} = (13.22 - 0.5) = 12.72$$

(III) Annual Return

$$(i) \quad \text{Return} = \frac{13.22 - 10}{10} \times 100 = 32.20\%$$

$$(ii) \quad \text{Return} = \frac{(12.72 - 10) + 0.5}{10} \times 100 = 32.20\%$$

Question – 06

Based on the following information, determine the NAV of a regular income scheme on per unit basis:

Particulars	₹ Crores
Listed shares at Cost (ex-dividend)	20
Cash in hand	1.23
Bonds and debentures at cost	4.3
Of these, bonds not listed and quoted	1
Other fixed interest securities at cost	4.5
Dividend accrued	0.8
Amount payable on shares	6.32
Expenditure accrued	0.75
Number of units (₹ 10 face value)	20 lacs
Current realizable value of fixed income securities of face value of ₹ 100	106.5
The listed shares were purchased when Index was	1,000
Present index is	2300
Value of listed bonds and debentures at NAV date	8

There has been a diminution of 20% in unlisted bonds and debentures. Other fixed interest securities are at cost.

(SM TYK – 20 & MTP April – 2022)

Solution:

Calculation of NAV	₹ in Crore
Listed shares $\left[\frac{20}{1,000} \times 2,300 \right]$	46.00
Cash in hand	1.23
Unlisted bonds (1 × 0.8)	0.80
Listed bonds	8.00
Other fixed income securities	4.50
Dividend accrued	0.80
Expenditure accrued	(0.75)
Amount payable on shares	(6.32)
Net assets	54.26
÷ Number of unit	0.20
NAV	271.30

Question – 07

The following particulars relating to Vishnu Fund Schemes:

Particulars	Value ₹ in Crores
1. Investment in Shares (at cost)	
a. Pharmaceuticals companies	79
b. Construction Industries	31
c. Service Sector Companies	56
d. IT Companies	34
E. Real Estate Companies	10
2. Investment in Bonds (Fixed Income)	
a. Listed Bonds (8000, 14% Bonds of ₹ 15,000 each)	12
b. Unlisted Bonds	7
3. No. of Units outstanding (crores)	4.2
4. Expenses Payable	3.5
5. Cash and Cash equivalents	1.5
6. Market expectations on listed bonds	8.842%

Particulars relating to each sector are as follows:

Sector	Index on Purchase Date	Index on Valuation Date
Pharmaceutical companies	260	465
Construction Industries	210	450
Service Sector Companies	275	480
IT Companies	240	495
Real Estate Companies	255	410

The fund has incurred the following expenses:

Consultancy and Management fees	₹ 480 Lakhs
Office Expenses	₹ 150 Lakhs
Advertisement Expenses	₹ 38 Lakhs

You are required to calculate the following:

- Net Asset Value of the fund
- Net Asset Value per unit
- If the period of consideration is 2 years, and the fund has distributed ₹ 3 per unit per year as cash dividend, ascertain the Net return (Annualized).
- Ascertain the Expenses ratio.

(Exam May – 2019) (5 Marks)

Solution:

(i) Calculation of Net Asset Value

Equity	
Pharmaceutical companies $\left(\frac{79}{260} \times 465 \right)$	141.29
Construction Industries $\left(\frac{31}{210} \times 450 \right)$	66.43
Service Sector Companies $\left(\frac{56}{275} \times 480 \right)$	97.75
IT Companies $\left(\frac{34}{240} \times 495 \right)$	70.13

Real Estate Companies $\left(\frac{10}{255} \times 410\right)$	16.08
Bonds	
Listed Bonds $\left[\frac{12 \times 14\%}{8.842\%}\right]$	19.00
Unlisted Bonds	7.00
Cash	1.50
Expenses Payable	(3.50)
Net Asset	415.68

(ii) NAV per Unit

$$\text{Net Assets} = 415.68$$

$$\div \text{No of unit} = 4.20$$

$$\text{NAV} = 98.97$$

(iii) Net Return

NAV before 2 Years

$$\text{Shares } [79 + 31 + 56 + 34 + 10] = 210$$

$$\text{Bonds } [12 + 7] = 19$$

$$\text{Net Assets} = 229$$

$$\div \text{No. of Units} = 4.2$$

$$\text{NAV} = 54.52$$

Holding Period Return (HPR)

$$= \frac{(P_1 - P_0) + \text{Income}}{P_0} \times 100$$

$$= \frac{(98.97 - 54.52) + 6}{54.52} \times 100$$

$$= 92.53\%$$

$$\text{Annual Return} = 92.53 \times \frac{1}{2} = 46.27\%$$

It is assumed that there is no opening Cash.

(iv) Expenses Ratio

$$\text{Expenses Ratio} = \frac{\text{Expense Per Unit}}{\text{Average NAV per Unit}}$$

Expenses per Unit

$$\text{Expenses } [480 + 150 + 38] = 6.68 \text{ Cr.}$$

$$\div \text{No. of Units} = 4.20 \text{ Cr.}$$

$$\text{Expenses per Unit} = 1.59$$

$$\text{Average NAV} \frac{54.52 + 98.97}{2} = 76.75$$

$$= \frac{1.59}{76.75} \times 100$$

$$= 2.07\%$$

(2) RETURN CALCULATION**Question – 08**

A mutual fund that had a net asset value of ₹ 20 at the beginning of month - made income and capital gain distribution of ₹ 0.0375 and ₹ 0.03 per share respectively during the month, and then ended the month with a net asset value of ₹ 20.06. Calculate monthly return.

(SM TYK – 03)**Solution:**

$$\text{Monthly Return} = \frac{(20.06 - 20) + 0.0375 + 0.03}{20} \times 100$$

$$= 0.6375\% \text{ p.m.}$$

$$\text{Or,} \quad = 7.65\% \text{ p.a.}$$

Question – 09

An investor purchased 300 units of a Mutual Fund at ₹ 12.25 per unit on 31st December, 2009. As on 31st December, 2010 he has received ₹ 1.25 as dividend and ₹ 1.00 as capital gains distribution per unit.

Required:

- (i) The return on the investment if the NAV as on 31st December, 2010 is ₹ 13.00.
- (ii) The return on the investment as on 31st December, 2010 if all dividends and capital gains distributions are reinvested into additional units of the fund at ₹ 12.50 per unit.

(SM TYK – 04)**Solution:****(1) Annual Return**

$$= \frac{(13 - 12.25) + 1.25 + 1.00}{12.25} \times 100$$

$$= 24.49\%$$

(2) Dividend Reinvestment Plan

2009		2010
300	1.25 + 1.00 = 2.25	NAV @ 13
@ 12.25	2.25 × 300 = 675	354 × 13
= 3,675		= 4,602
	$\frac{675}{12.50} = 54 \text{ Units}$	

$$\text{Investment} = 300 \text{ units} \times 12.25 = 3,675$$

Dividend & Capital Gain Income

$$(1.25 + 1) \times 300 \text{ units} = 675$$

$$\text{Additional units} = \frac{675}{12.50} = 54 \text{ units}$$

$$\text{Ending value} = [354 \text{ units} \times 13] = ₹ 4,602$$

$$\text{Return} = \frac{4,602 - 3,675}{3,675} \times 100 = 25.22\%$$



Question – 10

Mr. DK has 1,000 units of AM Mutual Fund. NAV of it is ₹ 17.50 per unit and ₹ 18.90 per unit at the beginning and at the end of the year respectively. The Mutual Fund has given two options:

Option I: Pay ₹ 1.50 per unit as dividend and ₹ 1.20 per unit as a capital gain, or

Option II: Reinvest these distributions at an average NAV of ₹ 17.30 per unit.

You are required to compute the holding period return percentage for both the options and select which option is preferable.

Calculation up to 2 decimal points.

(Exam September – 2025)

Solution:

Holding Period Return

Option – I:

$$\text{HPR} = \frac{(18.90 - 17.50) + 1.20 + 1.50}{17.50} \times 100 = 23.43\%$$

Option – II:

$$\text{Purchase} = 1,000 \text{ units} \times 17.50 = ₹ 17,500$$

$$\begin{aligned} \text{Dividend \& Capital Gain} &= (1.20 + 1.50) \times 1,000 \text{ units} \\ &= ₹ 2,700 \end{aligned}$$

$$\text{Reinvestment} = \frac{₹ 2,700}{17.30} = 156.07 \text{ units}$$

$$\text{Total Units} = 1,000 + 156.07 = 1,156.07 \text{ units}$$

$$\begin{aligned} \text{HPR} &= \frac{(1,156.07 \times 18.90) - 17,500}{17,500} \times 100 \\ &= 24.86\% \end{aligned}$$

Option II is better due to higher return.

Question – 11

The following information is extracted from Steady Mutual Fund's Scheme:

- Asset Value at the beginning of the month - ₹ 65.78
- Annualized return - 15 %
- Distributions made in the nature of Income & Capital gain (per unit respectively). - ₹ 0.50 and ₹ 0.32

You are required to:

- (i) Calculate the month end net asset value of the mutual fund scheme (limit your answers to two decimals).
- (ii) Provide a brief comment on the month end NAV.

(SM TYK – 06)

Solution:

(i) Calculation of NAV at the end of month

$$\text{Annualized Return} = \text{HPR} \times \frac{12}{1}$$



$$15 = \text{HPR} \times \frac{12}{1}$$

$$\text{HPR} = 15 \times \frac{1}{12}$$

$$= 1.25\%$$

$$1.25\% = \frac{(x - 65.78) + 0.50 + 0.32}{65.78}$$

$$0.82 = x - 65.78 + 0.82$$

$$x = 65.78$$

(ii) There is no change in NAV.

Question – 12

Mr. A has invested in three Mutual Fund Schemes as per details below:

Particulars	MF A	MF B	MF C
Date of investment	01/12/2009	01/01/2010	01/03/2010
Amount of investment	₹ 9	₹ 1,00,000	₹ 50,000
Net Assets Value (NAV) at entry date	₹ 10.50	₹ 10	₹ 10
Dividend received upto 31/03/2010	₹ 950	₹ 1,500	Nil
NAV as at 31/03/2010	₹ 10.40	₹ 10.10	₹ 9.80

Required:

What is the effective yield on per annum basis in respect of each of the three schemes to Mr. A upto 31/03/2010?

(SM TYK – 10)

Solution:

Effective Yield

		MFA	MFB	MFC
(a)	Investment	50,000	1,00,000	50,000
(b)	NAV	10.50	10	10
(c)	No of units $\left(\frac{a}{b}\right)$	4,761.90	10,000	5,000
(d)	NAV (31.03.2010)	10.40	10.10	9.80
(e)	Ending value (c × d)	49,524	1,01,000	49,000
(f)	Dividend	950	1500	0
	$\text{HPR} = \frac{(e-a) + f}{a} \times 100$	0.95%	2.5%	-2%
	No. of days	121 days	90 days	31 days
	$\text{E.Y.} = \text{HPR} \times \frac{365}{N}$	2.87%	10.14%	-23.55%

Question – 13

Mr. Y has invested in the three mutual funds (MF) as per the following details:

Particulars	Scheme X	Scheme Y	Scheme Z
Amount of Investment (₹)	2,00,000	4,00,000	2,00,000
NAV at the time of purchase (₹)	10.30	10.10	10
Dividend Received up to 31.03.2018 (₹)	6,000	0	5,000



NAV as on 31.03.2018 (₹)	10.25	10	10.20
Effective yield per annum as on 31.03.2018 (percent)	9.66	-11.66	24.15

Assume 1 Year = 365 days

Mr. Y has misplaced the documents of his investment. Help him in finding the date of his original investment after ascertaining the following:

- Number of units in each scheme;
- Total NAV;
- Total Yield; and
- Number of days investment held.

(SM TYK – 12, MTP Oct – 2020 & Exam May – 2018) (8 Marks)

Solution:

(i) Number of Units

$$\text{Mutual Fund X} = \frac{2,00,000}{10.30} = 19,417.47$$

$$\text{Mutual Fund Y} = \frac{4,00,000}{10.10} = 39,603.96$$

$$\text{Mutual Fund Z} = \frac{2,00,000}{10} = 20,000$$

(ii) Total NAV

$$\text{Mutual Fund X} = [19,417.47 \times 10.25] = 1,99,029$$

$$\text{Mutual Fund Y} = [39,603.96 \times 10] = 3,96,040$$

$$\text{Mutual Fund Z} = [20,000 \times 10.20] = 2,04,000$$

$$= 7,99,069$$

(iii) Total Yield

$$= \frac{(7,99,029 - 8,00,000) + 11,000}{8,00,000} \times 100 = 1.25\%$$

(iv) Number of Days

Mutual Fund X

$$\text{HPR} = \frac{(1,99,019 - 2,00,000) + 6,000}{2,00,000} \times 100 = 2.51\%$$

$$\text{E.Y.} = \text{HPR} \times \frac{365}{n}$$

$$9.66 = 2.51 \times \frac{365}{n}$$

$$n = \frac{2.51 \times 365}{9.66} = 95 \text{ days}$$

Mutual Fund Y

$$\text{HPR} = \frac{(3,96,040 - 4,00,000) + 0}{4,00,000} \times 100 = -0.99\%$$

$$\text{E.Y.} = -0.99 \times \frac{365}{n}$$

$$-11.66 = -0.99 \times \frac{365}{n}$$

$$n = \frac{-0.99 \times 365}{-11.66} = 31 \text{ days}$$



Mutual Fund Z

$$\text{HPR} = \frac{(2,04,000 - 2,00,000) + 5,000}{2,00,000} \times 100 = 4.5\%$$

$$\text{E.Y.} = \text{HPR} \times \frac{365}{n}$$

$$24.15 = 4.5 \times \frac{365}{n}$$

$$n = \frac{4.5 \times 365}{24.15} = 68 \text{ days}$$

Date of Investment [31.03.2018]

$$\begin{aligned} \text{Mutual Fund X} &= 31.03.2018 - 95 \text{ days} \\ &= 26.12.2017 \end{aligned}$$

$$\begin{aligned} \text{Mutual Fund Y} &= 31.03.2018 - 31 \text{ days} \\ &= 28.02.2018 \end{aligned}$$

$$\begin{aligned} \text{Mutual Fund Z} &= 31.03.2018 - 68 \text{ days} \\ &= 22.01.2018 \end{aligned}$$

Question – 14

Mr. D had invested in three mutual funds (MF) as per the following details:

Particulars	MF 'A'	MF 'B'	MF 'C'
Amount of Investment	2,00,000	5,00,000	4,00,000
NAV at the time of purchase	10.00	25.00	20.00
Dividend Yield up to 31.03.2022	3%	5%	4%
NAV as on 31.03.2022	10.50	22.80	20.80
Annualized Yield as on 31.03.2022	9.733%	-11.185%	15%

Assume 1 Year = 365 Days.

Mr. D has misplaced the documents of his investments.

You are required to help Mr. D to find out the following:

- Number of units allotted in each scheme,
- Value of his investments as on 31.03.2022,
- Holding period of his investments in number of days as on 31.03.2022
- Dates of original investments
- Total Return on investments,
- Assuming past performance of all three schemes will continue for next one year, what action the investor should take? What will be the expected return for the next one year after the above action?
- Will your answer as above point no. (vi) changes if the Mutual fund charges exit load of 5% if the investment is redeemed within one year? If so, advise the investor what and when the action to be taken to optimize the returns.

(Exam May – 2022) (8 Marks)

Solution:

(i) No. of Units

$$A = \frac{2,00,000}{10} = 20,000$$

$$B = \frac{5,00,000}{25} = 20,000$$

$$C = \frac{4,00,000}{20} = 20,000$$

(ii) Value of investment as on 31st March 2022

$$A = [20,000 \times 10.50] = 2,10,000$$

$$B = [20,000 \times 22.80] = 4,56,000$$

$$C = [20,000 \times 20.80] = 4,16,000$$

$$= 10,82,000$$

(iii) No. of days

	MFA	MFB	MFC
No. of unit	20,000	20,000	20,000
NAV (31/03/2022)	10.50	22.80	20.80
Investment (31/03/2022)	2,10,000	4,56,000	4,16,000
Dividend	6,000	25,000	16,000
HPR	8%	-3.8%	8%
E.Y.	9.733%	-11.185%	15%
E.Y. = $HPR \times \frac{365}{n}$	300 days	124 days	195 days

(iv) Dates of Original Investment

$$\text{Mutual Fund A} = 31/03/2022 - 300 \text{ days} = 04/06/2021$$

$$\text{Mutual Fund B} = 31/03/2022 - 124 \text{ days} = 27/11/2021$$

$$\text{Mutual Fund C} = 31/03/2022 - 195 \text{ days} = 17/09/2021$$

(v) Total Return

$$\text{Total Return} = \frac{(10,82,000 - 11,00,000) + 47,000}{11,00,000} \times 100$$

$$= 2.64\%$$

(vi) Unit of Mutual Fund A & Mutual Fund B should be redeemed & invested in Mutual Fund C.

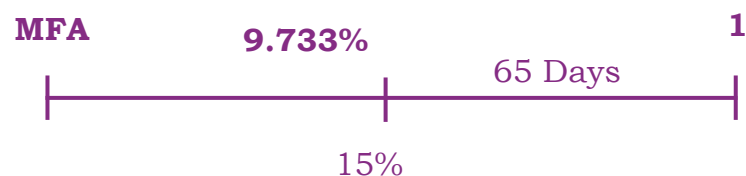
$$\text{Expected Return of next year} = 15\% \text{ p.a.}$$

(vii)



$$\text{Extra Return} = 15 - (-11.85) = 26.185\%$$

$$\text{Extra Return for 241 days} = 26.185 \times \frac{241}{365} = 17.29\%$$



$$\text{Extra Return} = 15 - 9.733 = 5.267\%$$

$$\text{Extra Return for 65 days} = 5.267 \times \frac{65}{365} = 0.94\%$$

If Mutual Fund charges exit load 5% then of Mutual Fund B should be redeemed now & unit of Mutual Fund A should be redeemed after 65 days.

Question – 15

Sun Moon Mutual Fund (Approved Mutual Fund) sponsored open-ended equity-oriented scheme “Chanakya Opportunity Fund”. There were three plans viz. ‘A’ – Dividend Re-investment Plan, ‘B’ – Bonus Plan & ‘C’ – Growth Plan.

At the time of Initial Public Offer on 1.4.1999, Mr. Anand, Mr. Bacchan & Mrs. Charu, three investors invested ₹ 1,00,000 each & chosen ‘B’, ‘C’ & ‘A’ Plan respectively.

The History of the Fund is as follows:

Date	Dividend%	Bonus Ratio	Net Asset Value per Unit (F.V. ₹ 10)		
			Plan A	Plan B	Plan C
28.07.2003	20	5 : 4	30.70	31.40	33.42
31.03.2004	70		58.42	31.05	70.05
31.10.2007	40		42.18	25.02	56.15
15.03.2008	25	1 : 3	46.45	29.10	64.28
31.03.2008			42.45	20.05	60.12
24.03.2009	40	1 : 4	48.10	19.95	72.40
31.07.2009			53.75	22.98	82.07

On 31st July all three investors redeemed all the balance units.

Calculate annual rate of return to each of the investors.

Consider:

1. Long-term Capital Gain is exempt from Income tax.
2. Short-term Capital Gain is subject to 10% Income tax.
3. Security Transaction Tax 0.2 per cent only on sale/redemption of units.
4. Ignore Education Cass.

(SM TYK – 17)

Solution:

Plan A: Dividend Re-Investment Plan

Date	Dividend Rate	Dividend Amount	NAV	Additional Unit	Balance Unit
01/04/1999	-	-	10.00	-	10,000
28/07/2003	20%	20,000	30.70	651.47	10,651.47
31/03/2004	70%	74,560	58.42	1,276.28	11,927.75
30/10/2007	40%	47,711	42.18	1131.13	13,058.88
15/03/2008	25%	32,647	46.45	702.85	13,761.73
24/03/2009	40%	55,047	48.10	1144.43	14,906.16

Redemption (31/07/2009)

Redeemable Value (14,906.16 × 53.75) 8,01,206

(-) STT @ 0.2 % 1,602

Net Amount Received 7,99,604

(-) Short Term Capital Gain Tax

$$1,144.43 (53.64^* - 48.10) \times 10\% \quad 634$$

$$\text{Net Amount} \quad 7,98,970$$

$$*53.64 = 53.75 - 0.2\%$$

$$\text{HPR} = \frac{7,98,970 - 1,00,000}{1,00,000} \times 100 = 698.97\%$$

$$\text{Annual Return} = 698.97 \times \frac{12}{124} = 67.64\%$$

Plan B: Bonus

	Bonus Units	Balance
01/04/1999	---	10,000
31/03/2004	$10,000 \times \frac{5}{4} = 12,500$	22,500
31/03/2008	$22,500 \times \frac{1}{3} = 7,500$	30,000
24/02/2009	$30,000 \times \frac{1}{4} = 7,500$	37,500

Redemption on 31/7/2009

$$\text{Sales } (37,500 \times 22.98) = 8,61,750$$

$$(-) \text{ STT @ } 0.2\% = 1,724$$

$$= 8,60,026$$

(-) Short Term Capital Gain

$$7,500 (22.93 - 19.95) \times 10\% = 2,235$$

$$= 8,57,791$$

$$\text{HPR} = \frac{8,57,791 - 1,00,000}{1,00,000} \times 100 = 757.79\%$$

$$\text{Annual Return} = 757.79 \times \frac{12}{124} = 73.33\%$$

Plan C: Growth Plan

$$\text{Sales Consideration } (10,000 \times 82.07) \quad 8,20,700$$

$$(-) \text{ STT @ } 0.2\% \quad 1,641$$

$$8,19,059$$

$$\text{HPR} = \frac{8,19,059 - 1,00,000}{1,00,000} \times 100 = 719.06\%$$

$$\text{Annual Return} = 719.06 \times \frac{12}{124} = 69.59\%$$

Question – 16

A mutual fund company introduces two schemes i.e. Dividend plan (Plan-D) and Bonus plan (Plan-B). The face value of the unit is ₹ 10. On 1-4-2005 Mr. K invested ₹ 2,00,000 each in Plan-D and Plan-B when the NAV was ₹ 38.20 and ₹ 35.60 respectively. Both the plans matured on 31-3-2010.

Particulars of dividend and bonus declared over the period are as follows:

Date	Dividend	Bonus Ratio	Net Asset Value (₹)	
			Plan D	Plan B
30-09-2005	10	1:5	39.10	35.60
30-06-2006			41.15	36.25
31-03-2007	15		44.20	33.10
15-09-2008	13	1:8	45.05	37.25
30-10-2008			42.70	38.30
27-03-2009	16		44.80	39.10
11-04-2009		1:10	40.25	38.90
31-03-2010			40.40	39.70

What is the effective yield per annum in respect of the above two plans?

(SM TYK – 18)

Solution:

Dividend Reinvestment Plan

Date	Dividend Rate	Dividend Amt	NAV	Units	Balance Units
01/04/2005	-	-	-	-	5,235.60
30/09/2005	10	5236	39.10	133.90	5,369.50
31/03/2007	15	8054	44.20	182.22	5,551.72
15/09/2008	13	7217	45.05	160.20	5,711.92
27/03/2009	16	9139	44.80	204	5,915.92

Investment value 31/03/2010

$$5,915.92 \times 40.40 = 2,39,003$$

Annual Return

$$\text{HPR} = \frac{2,39,003 - 2,00,000}{2,00,000} \times 100 = 19.50\%$$

$$\text{Annual Return} = 19.50 \times \frac{1}{5} = 3.90\%$$

Effective Yield

Alternative 1

$$2,00,000 (1 + r)^5 = 2,39,003$$

$$(1 + r)^5 = \frac{2,39,003}{2,00,000}$$

$$1 + r = (1.195)^{1/5}$$

$$r = 0.0363 - 1$$

$$= 0.0363 \text{ or } 3.63\%$$

Alternative 2

IRR Method (NPV Method)

IRR is a rate where NPV = 0

2%

$$\frac{2,39,003}{(1.02)^5} - 2,00,000 = 16,472$$

5%

$$\frac{2,39,003}{(1.05)^5} - 2,00,000 = -12,735$$

Interpolation Method (NPV Method)

$$2 \% \quad 16,472$$

$$5 \% \quad - 12,735$$

$$3 \% \quad 29,207$$



$$EY = 2 + \left(\frac{3}{29,207} \times 16,472 \right)$$

$$= 3.69 \% \text{ P.a.}$$

Bonus Plan

$$\text{No of Units} = \frac{2,00,000}{35.60} = 5,617.98$$

Date	Pre Bonus	Bonus Ratio	Bonus Unit	Balance
30/06/2006	5,617.98	1:5	$5617.98 \times \frac{1}{5} = 1123.60$	6,741.58
30/10/2008	6,741.58	1:8	$6741.58 \times \frac{1}{8} = 842.69$	7,584.28
11/04/2009	7,584.28	1:10	$7584.28 \times \frac{1}{10} = 758.428$	8,342.708

$$\begin{aligned} \text{Investment Value on 31/03/2010} &= (8,342.708 \times 39.70) \\ &= 3,31,205.5 \end{aligned}$$

$$HPR = \frac{3,31,205 - 2,00,000}{2,00,000} \times 100 = 65.60\%$$

$$\text{Annual Return} = 65.60 \times \frac{1}{5} = 13.12\%$$

Alternative

Effective Yield

$$EY = 2,00,000 (1 + r)^5 = 3,31,205$$

$$r = \left[\left(\frac{3,31,205}{2,00,000} \right)^{\frac{1}{5}} - 1 \right] \times 100$$

$$= 10.61\%$$

Question – 17

A mutual fund company introduces two schemes i.e. Dividend plan (Plan-D) and Bonus plan (Plan-B). The face value of the unit is ₹ 10. On 1-4-2018 Mr. K invested ₹ 2,00,000 each in Plan-D and Plan-B when the NAV was ₹ 38.20 and ₹ 35.60 respectively. Both the plans matured on 31-3-2023.

Particulars of dividend and bonus declared over the period are as follows:

Date	Dividend %	Bonus Ratio	Net Asset Value (₹)	
			Plan D	Plan B
30-09-2018	10	---	39.10	35.60
30-06-2019	---	1:5	41.15	36.25
31-03-2020	15	---	44.20	33.10
15-09-2021	13	---	45.05	37.25
30-10-2021	---	1:8	42.70	38.30
27-03-2022	16	---	44.80	39.10
11-04-2022	---	1:10	40.25	38.90
31-03-2023	---	---	40.40	39.70

Evaluate the Effective Yield Per Annum in respect of the above two plans.

Note:

- Use following PV Factors: PVIF (2%,5) = 0.9057, PVIF (4%,5) = 0.8219, PVIF (8%,5) = 0.6806, PVIF (13%,5) = 0.5428
- Round off calculations upto 2 decimal points.

(MTP April – 2024)

Solution:

Plan – D

$$\text{Unit acquired} = \frac{2,00,000}{38.20} = 5,235.60$$

Date	Units Held	Dividend		Reinvest ment Rate	New Units	Total Units
		%	Amount			
01.04.2018				39.10	133.90	5235.60
30.09.2018	5235.60	10	5235.60	44.20	182.22	5369.50
31.03.2020	5369.50	15	8054.25	45.05	160.20	5551.72
15.09.2021	5551.72	13	7217.24	44.80	204.00	5711.92
27.03.2022	5711.92	16	9139.07			5915.92
31.03.2023	Maturity (₹ 40.40 × 5915.92) Value					₹ 2,39,003.17
	Less: Cost of Acquisition					₹ 2,00,000.00
	Total Gain					₹ 39,003.17

$$\therefore \text{Approximate Effective Yield} = \frac{\text{₹ } 39,003.17}{\text{₹ } 2,00,000} \times \frac{1}{5} \times 100 = 3.90\%$$

Now more accurate effective yield can be computed by using the IRR method as follows:

$$\text{NPV at 4\%} = - \text{₹ } 2,00,000 + \text{₹ } 1,96,436.71 = - \text{₹ } 3,563.29$$

$$\text{NPV at 2\%} = - \text{₹ } 2,00,000 + \text{₹ } 2,16,465.17 = \text{₹ } 16,465.17$$

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR} - \text{NPV at HR}} (\text{HR} - \text{LR})$$

$$= 2\% + \frac{16465.17}{16465.17 - 3563.29} (4\% - 2\%) = 3.64\%$$

Plan – B

Date	Particulars	Calculation Working	No. of Units	NAV (₹)
01.04.2018	Investment	₹		35.60
30.06.2019	Bonus	2,00,000/35.60 =	5617.98	36.25
30.10.2021	"	5617.98/5 =	1123.60	38.30
11.04.2022	"	6741.58/8 =	6741.58	38.90
		842.70		
31.03.2023	Maturity Value	7584.28/10 =	7584.28	3,31,205.59
	Less: Investment Gain	8342.71 × ₹ 39.70 =	758.43	2,00,000.00
			8342.71	1,31,205.59

$$\therefore \text{Approximate Effective Yield} = \frac{\text{₹ } 1,31,205.59}{\text{₹ } 2,00,000} \times \frac{1}{5} \times 100 = 13.12\%$$

Now more accurate effective yield can be computed by using the IRR method as follows:

$$\text{NPV at 13\%} = - \text{₹ } 2,00,000 + \text{₹ } 1,79,778.39 = - \text{₹ } 20,221.61$$

$$\text{NPV at 8\%} = - \text{₹ } 2,00,000 + \text{₹ } 2,25,418.52 = \text{₹ } 25,418.52$$

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR} - \text{NPV at HR}} (\text{HR} - \text{LR})$$

$$= 8\% + \frac{25,418.52}{25,418.52 - (-20,221.61)} (13\% - 8\%) = 10.78\%$$

Question – 18

During the year 2017 an investor invested in a mutual fund. The capital gain and dividend for the year was ₹ 3.00 per unit, which were re-invested at the year end NAV of ₹ 23.75. The investor had

total units of 26,750 as at the end of the year. The NAV had appreciated by 18.75% during the year and there was an entry load of ₹ 0.05 at the time when the investment was made.

The investor lost his records and wants to find out the amount of investment made and the entry load in the mutual fund.

(Exam November – 2018) (5 Marks)

Solution:

NAV at the Beginning of the Year

$$\text{NAV} = \frac{23.75}{1.1875} = ₹ 20$$

Number of units

Let assume number of unit at beginning of the year be x

$$= x + \frac{3x}{23.75} = 26,750$$

$$= \frac{23.75x + 3x}{23.75} = 26,750$$

$$x = 26,750 \times \frac{23.75}{23.75 + 3} = 23,750 \text{ Units}$$

$$\begin{aligned} \text{Investment Amount} &= 23,750 \text{ Units} \times 20.50 \\ &= 4,76,187.50 \end{aligned}$$

$$\begin{aligned} \text{Entry Load} &= 23,750 \text{ Units} \times 0.50 \\ &= 1,187.50 \end{aligned}$$

Question – 19

In 2024, Mr. Raj, an investor made a lump sum investment in an equity mutual fund that had an entry load of ₹ 0.05 per unit. By the end of the year, the NAV appreciated by 13.60%. Additionally, the fund declared a total capital gain and dividend of ₹ 5.00 per unit, which were reinvested at a year-end NAV of ₹ 25. As a result, the investor held 15,000 units at year – end.

The fund also charges an exit load of 1% if redeemed within 1 year. The investor is in the 20% tax bracket. Inflation rate during the year is 4.50%.

You are to required to

- (i) Calculate the number of units purchased by Mr. Raj at the beginning of the investment.
- (ii) Calculate the NAV per unit and the total investment amount made by Mr. Raj at the beginning of the year.
- (iii) Appraise the return percentage and the real return percentage, if Mr. Raj decided to exit the investment at the end of the year.

Calculation up to 2 decimal points.

(Exam September – 2025) (7 Marks)

Solution:

- (i) Let X be the number of units purchased at the beginning of the year, then

$$X + \frac{5.00}{25.00} \times X = 15,000$$

X = 12,500 units

Thus, 12,500 units was purchased by Mr. Raj at the beginning of investment.

(ii) Let NAV per unit in the beginning of the year is N then

$$N \times 1.1360 = 25, N = ₹ 22.01$$

Thus, NAV per unit at the beginning of the year is ₹ 22.01 and total investment made by Mr. Raj is:

$$= (₹ 22.01 + ₹ 0.05) \times 12,500 = ₹ 2,75,750$$

(iii) If Mr. Raj decided to exit then Return and Real Return in percentage terms shall be computed as follows:

NAV at the end	₹ 25
No. of Units	15,000
Amount Redeemable	₹ 3,75,000
Less: Exit Load	₹ 3,750
Amount Received on Redemption of Units	₹ 3,71,250
Total Amount Realized	₹ 3,71,250
Less: Initial Investment	₹ 2,75,750
Return	₹ 95,500
Less: Tax @ 20%	₹ 19,100
Net Return	₹ 76,400
Return in Percentage Terms (₹ 76,400/₹ 2,75,750)	27.71%
Real Return in Percentage Terms (27.71%/1.045)	26.52%
Or Real Return $\frac{1 + 0.2771}{1 + 0.045} - 1 =$	22.21%

Question – 20

M/S. Corpus an AMC, on 1.04.2015 has floated two schemes viz. Dividend Plan and Bonus Plan. Mr. X, an investor has invested in both the schemes. The following details (except the issue price) are available:

Date	Dividend (%)	Bonus Ratio	NAV	
			Dividend Plan	Bonus Plan
01-04-2015			?	?
31-12-2016		1:4 (One unit on 4 units held)	47	40
31-03-2017	12		48	42
31-03-2018	10		50	39
31-12-2018			46	43
31-03-2019	15	1:5 (One unit on 5 units held)	45	42
31-03-2020	-		49	44

Additional details

Investment (₹)	₹ 9,20,000	₹ 10,00,000
Average Profit (₹)	₹ 27,748.60	
Average Yield (%)		6.40

You are required to calculate the issue price of both the schemes as on 01.04.2015.

(Exam November – 2020) (10 Marks)

Solution:

1. Dividend Plan

Particulars	Amount
Investment (01/04/15)	9,20,000
(+) Profit (27,748.60 × 5)	1,38,743

Investment (31/3/2020)	10,58,743
÷ NAV (31/3/2020)	49
No. of units	21607 units

Date	Dividend Per Unit	NAV	Post Dividend Units	Working	Pre Dividend Unit
31/3/2019	1.50	45	21,607	$\frac{21,607 \times 1.5}{45 + 1.5} = 697$	20,910
31/3/2018	1.00	50	20,910	$20,910 \times \frac{1}{51} = 410$	20,500
31/3/2017	1.20	48	20,500	$20,500 \times \frac{1.20}{49.20} = 500$	20,000

$$\text{Issued Price} = \frac{9,20,000}{20,000 \text{ Units}} = ₹ 46$$

2. Bonus Plan

Investment 01/04/2015	10,00,000
Profit $[10,00,000 \times 6.4\%] \times 5$	3,20,000
Investment	13,20,000
÷ NAV (31/03/2020)	44
No. of units (31/03/2020)	30,000 units

Date	Post Bonus Unit	Bonus Unit	Pre Bonus
31/3/2018	30,000	$30,000 \times \frac{1}{6} = 5,000$	25,000

31/12/2016	25,000	$25,000 \times \frac{1}{5} = 5,000$	20,000
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$$\text{NAV (1/4/2015)} = \frac{₹ 10,00,000}{20,000} = ₹ 50 \text{ Issue Price}$$

Question – 21

M/s. Wealth Builders, an Asset Management Company (AMC), launched a dividend bonus scheme on 1st April 2019. The fund demonstrated strong performance over the years.

Key events are as follows:

- On 30th September 2021, the fund declared a bonus of 1 : 4 (one bonus unit for every four existing units held).
- On 30th September 2023, a second bonus of 2 : 5 (two bonus units for every five existing units held) was declared.

Ms. Investor made a lump-sum investment of ₹ 25 lakhs in the scheme at its inception and remained invested throughout. As of 31st March 2025, her investment has generated an average annual yield of 16.8%.

The Net Asset Value (NAV) of the scheme on various dates is provided below:

Particulars	30.09.2021	30.09.2023	31.03.2025
NAV (in ₹)	78	88	110

Required:

Determine the opening NAV per unit as on 1st April 2019 for Ms. Investor's holding.

("Round off all intermediate and final calculations to two decimal places.")

(Exam Jan – 2026)

Solution:

(a)	Average Yield	0.168
(b)	Investment	₹ 25,00,000
(c)	Gain over a period of 6 years (a*b*6)	₹ 25,20,000
(d)	Market Value as on 31.03.2025 (b + c)	₹ 50,20,000
(e)	NAV as on 31.03.2023	110
(f)	Total units as on 31.03.2025 (d/e)	45,636.36
(g)	No of units as on 31.03.2023 Pre bonus = 45,636.36*5/ (5 + 2)	32,597.40
(h)	No of units as on 31.09.2019 Pre bonus = 32,597.40*4/ (4 + 1)	26,077.92

Issue Price as on 01.04.2019

Investment ₹ 25,00,000/ Units purchased 26,077.92 (b/h) = ₹ 95.87

Alternatively, it can also be computed as follows:

Units on 01.04.2019	X
Units after bonus on 30.09.2021 (1:4)	1.25X
Units after bonus on 30.09.2023 (2:5)	1.75X
Average yield	0.168
Investment	₹ 25,00,000
Gain for 5 years (25,00,000 × 0.168 × 6)	₹ 25,20,000

Total Value (₹ 25,00,000 + ₹ 25,20,000) ₹ 50,20,000

Where, $1.75 X \times ₹ 110 = ₹ 50,20,000$

Therefore, $X = 26,077.92$ units

Issue Price on 01.04.2019:

= ₹ 25,00,000 / 26,077.92 units = ₹ 95.87

Alternatively, it can also be computed as follows:

Average Yield = 16.80%

Investment = ₹ 25,00,000

Gain over a period of 6 years = ₹ 25,00,000*0.168*6 = ₹ 25,20,000

Thus, Maturity Value on 31.03.25 shall be ₹ 50,20,000

No. of units = $\frac{50,20,000}{110} = 45,636.36$

Now let B be the NAV on 01.04.19 then

Units acquired on 01.04.19 = $\frac{50,20,000}{B}$

Units acquired on 30.09.21 = $\frac{25,00,000}{B} \times \frac{1}{4} = \frac{6,25,000}{B}$

Units acquired on 31.12.23 = $\frac{31,25,000}{B} \times \frac{2}{5} = \frac{12,50,000}{B}$

Thus, total units can be shown as follows:

$\left[\frac{25,00,000}{B} + \frac{6,25,000}{B} + \frac{12,50,000}{B} \right] = 45,636.36$

$$B = ₹ 95.87$$

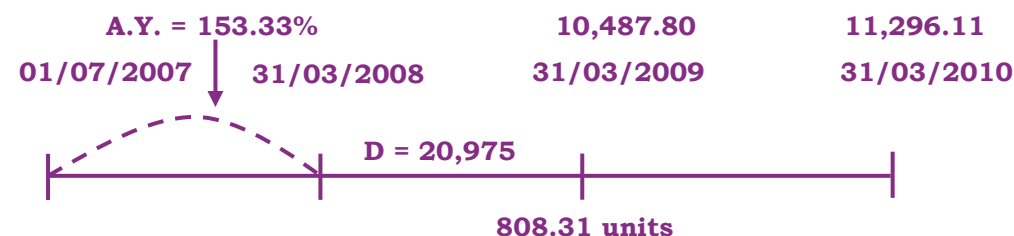
Thus, the issue price of units under bonus plan shall be ₹ 95.87.

Question – 22

Mr. X on 01.07.2007, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31.03.2008, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31.03.2009, 20% dividend was given. On 31.03.2010, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%. What are the NAVs as on 31.03.2008, 31.3.2009 and 31.03.2010?

(SM TYK – 13)

Solution:



10,000 Units @ ₹
10 = ₹ 1,00,000

31/3/2008

$$\text{Annualized Yield} = \text{HPR} \times \frac{12}{n}$$

$$153.33 = \text{HPR} \times \frac{12}{9}$$

$$\text{HPR} = 153.33 \times 9/12$$

$$= 115\%$$

$$\text{HPR} = \frac{\text{Closing Value} - \text{Opening Value}}{\text{Opening Value}}$$

$$115\% = \frac{x - 1,00,000}{1,00,000}$$

$$1,15,000 = x - 1,00,000$$

$$x = 2,15,000$$

$$\text{Dividend Amount} = ₹ 10,000$$

$$\text{Pre Dividend Investment Value} = 2,05,000$$

$$\text{No. of units pre dividend} = 10,000 \text{ Units}$$

$$\text{NAV (31/3/2008)} = \frac{2,05,000}{10,000} = 20.50$$

$$\text{Additional units} = \frac{10,000}{20.50} = 487.80 \text{ units}$$

$$\text{Total Units} = 10,000 + 487.80 = 10,487.80$$

31/3/2009

$$\begin{aligned} \text{Dividend amount} &= (10,487.80 \text{ units} \times 10 \times 20\%) \\ &= 20,975 \end{aligned}$$

$$\begin{aligned} \text{Additional units} &= (11,296.11 - 10,487.80) \\ &= 808.31 \text{ units} \end{aligned}$$

$$\text{NAV} = \frac{20,974}{808.31} = 25.95$$



31/3/2010

$$\text{Annualized Yield} = \text{HPR} \times \frac{12}{n}$$

$$73.52 = \text{HPR} \times \frac{12}{33}$$

$$\text{HPR} = 73.52 \times \frac{33}{12} = 202.18\%$$

$$202.18\% = \frac{x - 1,00,000}{1,00,000} = 3,02,180$$

$$\text{NAV} = \frac{3,02,180}{11,296.11} = ₹ 26.75$$

Question – 23

Mr. Alex, a practicing Chartered Accountant, can earn a return of 15 percent by investing in equity shares on his own. He is considering a recently announced equity based mutual fund scheme in which initial expenses are 6 percent and annual recurring expenses are 2 percent.

- (i) How much should the mutual fund earn to provide Mr. Alex a return of 15 percent per annum?
- (ii) Mr. Alex's current Annual Professional Income is ₹ 40 Lakhs. His portfolio value is ₹ 50 Lakhs and now he is spending 10% of his time to manage his portfolio. If he spends this time on profession, his professional income will go up in same proportion. He is thinking to invest his entire portfolio into a Multi-cap Fund, assuming the fund's NAV will grow at 13% per annum (including dividend).

You are requested to advise Mr. Alex, whether he can invest the portfolio into Multi-cap Funds? If so, what is the net financial benefit?

(Exam November – 2019) (8 Marks)

Solution:

(i) Earning of Mutual Fund

$$R_2 = \frac{R_1}{1 - \text{Initial Expenses}} + \text{Recurring Expenses}$$

$$= \frac{15}{1 - 0.06} + 2 = 17.96\%$$

(ii) Option 1: Managed by Himself ₹ in Lakh

Professional Income	40.00
Portfolio Income (50 × 15%)	7.50
	47.50

Option 2: Investment in Mutual Fund ₹ in Lakh

Professional Income (40 × 110%)	44
Portfolio Income (50 × 13%)	6.5
	50.50

It is better to invest the Portfolio in Multi Cap Fund due to higher return.



(3) HEDGE FUND

Question – 24

ANP Plan, a hedge fund currently has assets of ₹ 20 crore. CA. X, the manager of fund charges fee of 0.10% of portfolio asset. In addition to it he charges incentive fee of 2%. The incentive will be linked to gross return each year in excess of the portfolio maximum value since the inception of fund. The maximum value the fund achieved so far since inception of fund about one and half year ago was ₹ 21 crores.

You are required to compute the fee payable to CA. X, if return on the fund this year turns out to be

(a) 29%, (b) 4.5%, (c) -1.8%

(SM TYK – 23)

Solution:

(a) Return = 29%

Portfolio Value (20×1.29) = 25.80 Crore

Basic Fees ($20 \text{ Crore} \times 0.10\%$) = ₹ 2,00,000

Incentives ($(25.80 \text{ Crore} - 21 \text{ Crore}) \times 2\%$) = ₹ 9,60,000

Total = ₹ 11,60,000

(b) Return = 4.5%

Portfolio Value (20×1.045) = 20.90 Crore

Basic Fees = 2,00,000

Incentives = 0

= 2,00,000

(c) Return = -1.8%

Portfolio Value ($20 \text{ Crore} \times 0.982$) = 19.64 Crore

Basic Fees = 2,00,000

Incentives = 0

= 2,00,000

(4) PERFORMANCE EVALUATION OF MUTUAL FUND

Question – 25

Five portfolios experienced the following results during a 7- year period:

Portfolio	Average Annual Return (R_p) (%)	Standard Deviation (S_p)	Correlation with the market returns (r)
A	19.0	2.5	0.840
B	15.0	2.0	0.540
C	15.0	0.8	0.975
D	17.5	2.0	0.750
E	17.1	1.8	0.600
Market Risk (σ_m)		1.2	
Market Risk of Return (R_m)	14.0		
Risk-free rate (R_f)	9.0		

Rank the portfolios using (a) Sharpe's method, (b) Treynor's method and (c) Jensen's Alpha

(RTP November – 2021)

Solution:

Let portfolio standard deviation be σ_p



Market Standard Deviation = σ_m

Coefficient of correlation = r

$$\text{Portfolio beta } (\beta_p) = \frac{\sigma_p r}{\sigma_m}$$

$$\text{Required portfolio return } (R_p) = R_f + \beta_p (R_m - R_f)$$

Portfolio	Beta	Return from the portfolio (Rp) (%)
A	1.75	17.75
B	0.90	13.50
C	0.65	12.25
D	1.25	15.25
E	0.90	13.50

Portf olio	Sharpe Method		Treynor Method		Jensen's Alpha	
	Ratio	Rank	Ratio	Rank	Ratio	Rank
A	4.00	IV	5.71	V	1.25	V
B	3.00	V	6.67	IV	1.50	IV
C	7.50	I	9.23	I	2.75	II
D	4.25	III	6.80	III	2.25	III
E	4.50	II	9.00	II	3.60	I

Question – 26

There are two Mutual Funds viz. D Mutual Fund Ltd. and K Mutual Fund Ltd. Each having close ended equity schemes.

NAV as on 31-12-2014 of equity schemes of D Mutual Fund Ltd. is ₹ 70.71 (consisting 99% equity and remaining cash balance) and that of K Mutual Fund Ltd. is 62.50 (consisting 96% equity and balance in cash).

Following is the other information:

Particular	Equity Schemes	
	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Sharpe Ratio	2	3.3
Treynor Ratio	15	15
Standard deviation	11.25	5

There is no change in portfolios during the next month and annual average cost is ₹ 3 per unit for the schemes of both the Mutual Funds.

If Share Market goes down by 5% within a month, calculate expected NAV after a month for the schemes of both the Mutual Funds.

For calculation, consider 12 months in a year and ignore number of days for particular month.

(SM TYK – 22, RTP Nov – 2020 & May – 2019)

Solution:

(1) Calculation of Equity & Cash Component

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
NAV	₹ 70.71	₹ 62.50
Equity (%)	99%	96%
Equity per unit	70.00	60.00
Cash per unit	0.71	2.50

(2) Calculation of Beta

D Mutual Fund Ltd.



$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

$$2 = \frac{R_p - R_f}{11.25}$$

$$R_p - R_f = 22.5$$

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

$$15 = \frac{22.5}{\beta_p}$$

$$\beta_p = 1.5$$

K Mutual Fund Ltd.

$$3.3 = \frac{R_p - R_f}{5}$$

$$= 16.5$$

$$15 = \frac{16.5}{\beta} = 1.1$$

$$\beta = 1.1$$

(3) Equity component after 1 month

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Equity	70	60
Beta	1.50	1.10
Market Down	5 %	5%
Portfolio Down	7.5%	5.5%

Equity	64.75	56.70
--------	-------	-------

(4) Calculation of Cash

	D Mutual Fund	K Mutual Fund
Cash Opening	0.71	2.50
(-) Expenses $\left(\frac{3}{12}\right)$	0.25	0.25
	0.46	2.25

(5) Calculation of NAV after 1 month

Equity	64.75	56.70
(+) Cash	0.46	2.25
	65.21	58.95

Question – 27

Following is the information related to three mutual funds:

Year	MF-A	MF-B	MF-C
2020	10%	5%	14%
2021	8%	10%	10%
2022	12%	8%	18%

Correlation between market and mutual fund:

	MF-A	MF-B	MF-C
Correlation with market	0.45	0.25	0.65

Variance of the market is 9% and rate of return of government bond is 7%.

You are required to Rank the Mutual fund using Sharpe's ratio and Treynor's ratio.

(Exam Nov – 2022) (8 Marks)

Solution:

Working Note 1:

Mutual Fund A

$$R_p = \frac{10 + 8 + 12}{3} = 10\%$$

$$\sigma_p = \frac{\sqrt{(10-10)^2 + (8-10)^2 + (12-10)^2}}{3}$$

$$= 1.63\%$$

Mutual Fund B

$$R_p = \frac{5 + 10 + 18}{3} = 7.67\%$$

$$\sigma_p = \frac{\sqrt{(5-7.67)^2 + (10-7.67)^2 + (18-7.67)^2}}{3}$$

$$= 2.05\%$$

Mutual Fund C

$$R_p = \frac{14 + 10 + 18}{3} = 14\%$$

$$\sigma_p = \frac{\sqrt{(14-14)^2 + (10-14)^2 + (18-14)^2}}{3}$$

$$= 3.27\%$$

Working Note 2:

$$B_p = \frac{\sigma_p}{\sigma_m} \times r_m$$

$$B_A = \frac{1.63}{3} \times 0.45 = 0.244$$

$$B_B = \frac{2.05}{3} \times 0.25 = 0.171$$

$$B_C = \frac{3.27}{3} \times 0.65 = 0.709$$

Fund	Sharpe Ratio	Rank	Treynor Ratio	Rank
A	$\frac{10-7}{1.63} = 1.84$	II	$\frac{10-7}{0.244} = 12.30$	I
B	$\frac{7.67-7}{2.05} = 0.33$	III	$\frac{7.67-7}{0.171} = 3.92$	III
C	$\frac{14-7}{3.27} = 2.14$	I	$\frac{14-7}{0.709} = 9.87$	II

Question – 28

The following are the details of three mutual funds of MFL:

	Growth Fund	Balanced Fund	Regular Fund	Market
Average Return (%)	7	6	5	9
Variance	92.16	54.76	40.96	57.76
Coefficient of Determination	0.3025	0.6561	0.9604	

The yield on 182 days Treasury Bill is 9 per cent per annum.

You are required to:

- Rank the funds as per Sharpe's measure.
- Rank the funds as per Treynor's measure.
- Compare the performance with the market.

(MTP April – 2023)

Solution:

(i) Sharpe Ratio

$$SR = \frac{R_p - R_f}{\sigma_p}$$

$$\text{Growth Fund} = \frac{7 - 9}{\sqrt{92.16}} = -0.208 \quad \text{..... I}$$

$$\text{Balance Fund} = \frac{6 - 9}{\sqrt{54.76}} = -0.405 \quad \text{..... II}$$

$$\text{Regular Fund} = \frac{5 - 9}{\sqrt{40.96}} = -0.625 \quad \text{..... III}$$

(ii) Treynor's Ratio

$$\text{Coefficient of determination} = r^2$$

$$r_{AM} = \sqrt{0.3025} = 0.55$$

$$r_{BM} = \sqrt{0.6561} = 0.81$$

$$r_{CM} = \sqrt{0.9604} = 0.98$$

$$B_A = \frac{\sigma_A}{\sigma_M} \times r_{AM}$$

$$B_A = \frac{\sqrt{92.16}}{\sqrt{57.76}} \times 0.55 = 0.696$$

$$B_B = \frac{\sqrt{54.76}}{\sqrt{57.76}} \times 0.81 = 0.789$$

$$B_C = \frac{\sqrt{40.96}}{\sqrt{57.76}} \times 0.98 = 0.825$$

$$TR = \frac{R_p - R_f}{\beta}$$

$$\text{Growth Fund} = \frac{7 - 9}{0.696} = -2.87 \quad \text{..... I}$$

$$\text{Balance Fund} = \frac{6 - 9}{0.789} = -3.802 \quad \text{..... II}$$

$$\text{Regular Fund} = \frac{5 - 9}{0.825} = -4.849 \quad \text{..... III}$$

(iii) Sharpe Ratio & Treynor's Ratio of Market

$$\text{Sharpe Ratio} = \frac{9 - 9}{\sqrt{57.76}} = 0$$

$$\text{Treynor's Ratio} = \frac{9 - 9}{1} = 0$$

Market performance is better than Mutual Fund.

Question – 29

A Portfolio Manager (PM) has three mutual funds in his portfolio. Following the details of these three mutual funds:



Particulars	Growth Fund	Balanced Fund	Regular Fund	Market
Average Return (%)	7.5	6.3	5.4	
Variance				50.41
Sharpe Ratio	-0.15	-0.36	-0.48	
Treynor's Ratio	-2	-3	-4.80	

The yield on 182 days Treasury bill is 9 per cent per annum.

You are required to calculate

- Variance of the Funds
- Coefficient of Determination of the Funds

(Exam November – 2024) (8 Marks)

Solution:

- Variance of different Funds can be calculated by using Sharpe Ratio as follows:

(a) Growth Fund

Let σ_G be the Standard Deviation of Growth Fund. Accordingly

$$- 0.15 = \frac{7.50 - 9.00}{\sigma_G}$$

$$\sigma_G = 10$$

Hence the variance of Growth Fund $(\sigma_G)^2 = 100$

(b) Balanced Fund

Let σ_B be the Standard Deviation of Balanced Fund. Accordingly

$$- 0.36 = \frac{6.30 - 9.00}{\sigma_B}$$

$$\sigma_B = 7.50$$

Hence the variance of Growth Fund $(\sigma_G)^2 = 56.25$

(c) Regular Fund

Let σ_R be the Standard Deviation of Regular Fund. Accordingly

$$- 0.48 = \frac{5.40 - 9.00}{\sigma_R}$$

$$\sigma_R = 7.50$$

Hence the variance of Growth Fund $(\sigma_R)^2 = 56.25$

- To determine the Coefficient of Determination we need the Coefficient of Correlation which can be determined by computing β using the Treynor Ratio as follows:

(a) Growth Fund

Let β_G be the Beta of Growth Fund. Accordingly

$$- 2 = \frac{7.50 - 9.00}{\beta_G}$$

$$\beta_G = 0.75$$

$$\text{Since Beta} = \frac{\sigma_G \text{Corr}_{G,M}}{\sigma_M}$$

and $\sigma_G = 10$ and $\sigma_M = \sqrt{50.41} = 7.1$

$$\text{Hence } 0.75 = \frac{10 \text{Corr}_{G,M}}{7.10}$$

$\text{Corr}_{G,M} = 0.5325$ or 0.53 and

Coefficient of Determination of Growth Fund = $(0.5325)^2 = 0.2836$ or $(0.53)^2 = 0.28$

(b) Balanced Fund

Let β_B be the Beta of Balanced Fund. Accordingly

$$- 3 = \frac{6.30 - 9.00}{\beta_B}$$

$$\beta_B = 0.90$$

$$\text{Since Beta} = \frac{\sigma_B \text{Corr}_{B,M}}{\sigma_M}$$

and $\sigma_B = 7.50$ and $\sigma_M = \sqrt{50.41} = 7.1$

$$\text{Hence } 0.90 = \frac{7.50 \text{Corr}_{B,M}}{7.10}$$

$\text{Corr}_{B,M} = 0.852$ or 0.85 and

Coefficient of Determination of Balanced Fund = $(0.852)^2 = 0.7259$ or $(0.85)^2 = 0.73$

(c) Regular Fund

Let β_R be the Beta of Regular Fund. Accordingly

$$- 4.80 = \frac{5.40 - 9.00}{\beta_R}$$

$$\beta_R = 0.75$$

$$\text{Since Beta} = \frac{\sigma_R \text{Corr}_{R,M}}{\sigma_M}$$

and $\sigma_R = 7.50$ and $\sigma_M = \sqrt{50.41} = 7.1$

$$\text{Hence } 0.75 = \frac{7.50 \text{Corr}_{R,M}}{7.10}$$

$\text{Corr}_{R,M} = 0.71$ and

Coefficient of Determination of Regular Fund = $(0.71)^2 = 0.5041$ or 0.50

Question – 30

Mr. Potential has made investments in two mutual funds. The following information is available:

Mutual Fund	Smart	Growth
Jensen Alpha	1.10%	1.50%
Treynor's Ratio	0.0714	0.0775
Actual Return	8.50%	9.10%
Risk Premium	4%	

You are required to calculate:

- Beta (β) for both the funds
- Risk free Rate
- Security Market Line

(Exam May – 2023) (8 Marks)

Solution:

SMART

$$\text{Alpha} = R_p - R_e$$

$$1.10 = 8.50 - R_e$$

$$R_e = 7.40$$

(1) CAPM

$$R_e = R_f + \text{MRP } \beta$$

$$7.40 = R_f + 4 \beta$$

$$R_f = 7.40 - 4 \beta$$

$$\text{TR} = \frac{R_p - R_f}{\beta}$$

$$0.0714 = \frac{0.085 - [0.074 - 0.04 \beta]}{\beta}$$

$$0.0714 \beta = 0.085 - 0.074 + 0.04 \beta$$

$$0.031 \beta = 0.011$$

$$\beta = \frac{0.011}{0.031} = 0.35$$

(2) Put Beta Value in CAPM Equation

$$7.40 = R_f + (4 \times 0.35)$$

$$R_f = 6\%$$

(3) SML

$$6 + 4 \beta$$

GROWTH

$$\text{Alpha} = R_p - R_e$$

$$1.50 = 9.10 - R_e$$

$$R_e = 7.60$$

(1) CAPM

$$R_e = R_f + \text{MRP } \beta$$

$$7.60 = R_f + 4 \beta$$

$$R_f = 7.60 - 4 \beta$$

$$\text{TR} = \frac{R_p - R_f}{\beta}$$

$$0.0775 = \frac{0.0910 - [0.0760 - 0.04 \beta]}{\beta}$$

$$0.0755 \beta = 0.0910 - 0.0760 + 0.04 \beta$$

$$0.0375 \beta = 0.015$$



$$\beta = \frac{0.015}{0.0375} = 0.40$$

(2) Put Beta Value in CAPM Equation

$$7.60 = R_f + (4 \times 0.40)$$

$$R_f = 6\%$$

(3) SML

$$6 + 4 \beta$$

(5) RESIDUAL

Question – 31

On 1-4-2012 ABC Mutual Fund issued 20 lakh units at ₹ 10 per unit. Relevant initial expenses involved were ₹ 12 lakhs. It invested the fund so raised in capital market instruments to build a portfolio of ₹ 185 lakhs. During the month of April 2012 it disposed off some of the instruments costing ₹ 60 lakhs for ₹ 63 lakhs and used the proceeds in purchasing securities for ₹ 56 lakhs. Fund management expenses for the month of April 2012 was ₹ 8 lakhs of which 10% was in arrears. In April 2012 the fund earned dividends amounting to ₹ 2 lakhs and it distributed 80% of the realized earnings. On 30-4-2012 the market value of the portfolio was ₹ 198 lakhs.

Mr. Akash, an investor, subscribed to 100 units on 1-4-2012 and disposed off the same at closing NAV on 30-4-2012. What was his annual rate of earning?

(SM TYK – 16 & RTP May – 2020)

Solution:

W.N. 1: Calculation of Cash

Opening Cash (20 × 10 – 12 – 185)	3.00
(+) Investment Sold	63.00
(-) Purchase of Securities	56.00
(-) Expenses Paid (8 × 90%)	7.20
(+) Dividend Income	2.00
(-) Distribution of Income [2 + (63 – 60)] × 80%	4.00
Closing Cash	0.80 Lacs

Month end NAV

Closing Cash	0.80
Portfolio Value	198
Asset	198.80
(-) Outstanding Expenses	0.80
Net Asset	198.00
(÷) Units	20
NAV	9.90

Annual Rate of Earning

$$\text{Beginning NAV} = 10$$



$$\text{Income per unit} = \frac{4.00}{20} = 0.20$$

$$\text{Closing NAV} = 9.90$$

$$\text{HRP} = \frac{(9.90 - 10) + 0.20}{10} \times 100 = 1\%$$

$$\text{Annualized return} = 1 \times \frac{12}{1} = 12\% \text{ p.a.}$$

Question – 32

Stork Capital, a SEBI Registered Mutual Fund, launched its first New Fund Offer (NFO) on June 1, 2024, with a face value of ₹ 10 per unit. The fund received subscriptions for 180 lakh units.

An underwriting agreement was in place with Griffin Securities Ltd., which agreed to underwrite the entire issue of 200 lakh units for a commission of 2.0%.

The fund's financial activities are summarized below:

- Initial investments in various capital market instruments amounted to ₹ 1,780 lakhs.
- Marketing expenses for the NFO were ₹ 25 lakhs.
- During the financial year ended March 31, 2025, the fund sold securities with a cost of ₹ 250 lakhs for ₹ 280 lakhs.
- The fund subsequently purchased new securities for ₹ 265 lakhs.
- Management expenses are regulated by SEBI and cannot exceed 0.50% of the average funds invested during the year. The actual

management expenses incurred were ₹ 5.50 lakhs, of which ₹ 50,000 was outstanding at year-end.

- Dividends earned on investments amounted to ₹ 5.0 lakhs, of which ₹ 40,000 was yet to be collected.
- The fund's policy is to distribute 80% of all realized earnings (capital gains and dividends).
- The market value of the investment portfolio as of March 31, 2025, was ₹ 2,150.50 lakhs.

Required:

Determine the closing per unit Net Asset Value (NAV) of the fund as on March 31, 2025. Show all necessary workings.

(**Note:** Round off all intermediate and final calculations to two decimal places.)

(Exam Jan – 2026)

Solution:

Working Notes:

(1) Computation of Opening Cash Balance

Particulars	Amount (₹ in Lakhs)
Proceeds from NFO (200 Lakh units @ ₹ 10)	2,000.00
Less: Initial Purchase of Securities	(1780.00)
Less: Underwriting Commission (2% of 2,000)	(40.00)
Less: Meeting Expenses	(25.00)
Opening Cash Balance	155.00

(2) Management Expenses

	Amount (₹ in Lakhs)
A. Actual Expenses Incurred	5.50
B. SEBI Limit (0.50% of Average Funds)	1780.00



- Opening Investment	1795.00
- Closing Investment (1780 – 250 + 265)	1787.50
- Average Fund Invested	8.94
- SEBI Limit (0.50% of ₹ 1787.50)	5.50
Chargeable Expenses (Lower of A or B)	(0.50)
Less: Amount unpaid (Outstanding)	5.00
Management Expenses Paid	

(3) Computation of Closing Balance

Particulars	Amount (₹ in Lakhs)
Opening Cash Balance	155.00
Add: Cash Inflows	
Proceeds from Sale of Securities	280.00
Dividend Received (₹ 5.00 Lakh Earned – ₹ 0.40 Lakh uncollected)	4.60
Less: Cash Outflows	
Cost of New Securities Purchased	(265.00)
Management Expenses	(5.00)
Capital Gains Distributed [80% of (₹ 280 sale – ₹ 250 cost)]	(24.00)
Dividend Distributed (80% of ₹ 5.00 Lakh total earned)	(4.00)
Closing Cash Balance	141.60

Net Asset Value (NAV) as on March 31, 2025

Particulars	Amount (₹ in Lakhs)
Assets:	
Closing Cash Balance	141.60
Closing Market Value of Investment	2150.50
Accrued Dividends (Receivable)	0.40
Total Assets	2292.50
Less: Liabilities	
Outstanding Management Expenses	(0.50)

Closing Net Assets (A)	2292.00
Total Units Outstanding (in Lakhs) (B)	200.00
NAV per Unit (A/B)	₹ 11.46

Alternative Solution: In case dividend has been paid excluding the dividend receivable from the Dividend Income then solution will be as follows

Working Notes:

(1) Computation of Opening Cash Balance

Particulars	Amount (₹ in Lakhs)
Proceeds from NFO (200 Lakh units @ Rs. 10)	2,000.00
Less: Initial Purchase of Securities	(1,780.00)
Less: Underwriting Commission (2% of 2,000)	(40.00)
Less: Meeting Expenses	(25.00)
Opening Cash Balance	155.00

(2) Management Expenses

Particulars	Amount (₹ in Lakhs)
A. Actual Expenses Incurred	5.50
B. SEBI Limit (0.50% of Average Funds)	
- Opening Investment	1,780.00
- Closing Investment (1,780 – 250 + 265)	1,795.00
- Average Fund Invested	1,787.50
- SEBI Limit (0.50% of ₹ 1,787.50)	8.94
Chargeable Expenses (Lower of A or B)	5.50
Less: Amount unpaid (Outstanding)	(0.50)
Management Expenses Paid	5.00

(3) Computation of Closing Balance

Particulars	Amount (₹ in Lakhs)
-------------	------------------------

Opening Cash Balance	155.00
Add: Cash Inflows	
Proceeds from Sale of Securities	280.00
Dividend Received (₹ 5.00 Lakh Earned – ₹ 0.40 Lakh uncollected)	4.60
Less: Cash Outflows	
Cost of New Securities Purchased	(265.00)
Management Expenses	(5.00)
Capital Gains Distributed [80% of (₹ 280 sale – ₹ 250 cost)]	(24.00)
Dividend Distributed (80% of ₹ 4.60 Lakh total realised)	(3.68)
Closing Cash Balance	141.92

Net Asset Value (NAV) as on March 31, 2025

Particulars	Amount (₹ in Lakhs)
Assets:	
Closing Cash Balance	141.92
Closing Market Value of Investment	2150.50
Accrued Dividends (Receivable)	0.40
Total Assets	2292.82
Less: Liabilities	
Outstanding Management Expenses	(0.50)
Closing Net Assets (A)	2292.32
Total Units Outstanding (in Lakhs) (B)	200.00
NAV per Unit (A/B)	₹ 11.46

Question – 33

On 1st april, an open ended scheme of mutual fund had 300 lakh units outstanding with Net Assets Value (NAV) of ₹ 18.75. At the end of April, it issued 6 lakh units at opening NAV plus 2% load, adjusted for dividend equalization. At the end of May, 3 Lakh units were repurchased at opening NAV less 2% exit load adjusted for dividend equalization. At the end of June, 70% of its available income was distributed.

In respect of April-June quarter, the following additional information are available:

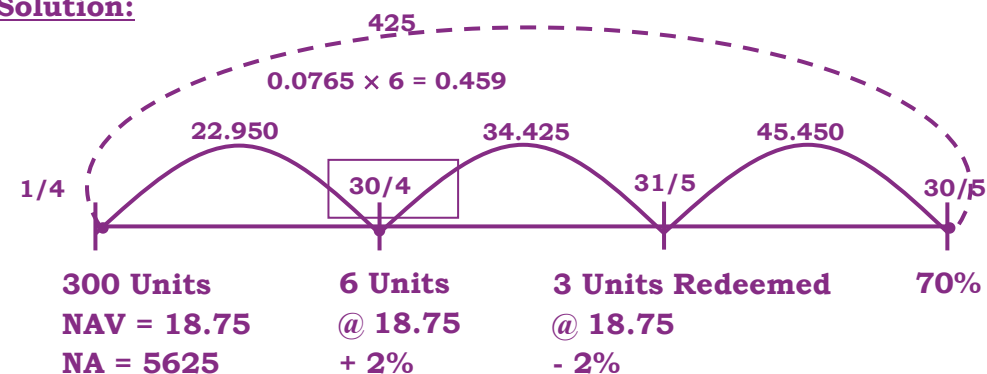
	₹ in lakh
Portfolio value appreciation	425.47
Income of April	22.950
Income for May	34.425
Income for June	45.450

You are required to calculate

- Income available for distribution;
- Issue price at the end of April;
- repurchase price at the end of May; and
- net asset value (NAV) as on 30th June.

(SM TYK & RTP November – 2018)

Solution:



1. Income available for distribution

Particulars	Units	Per unit	Total
Income from April	300	0.0765	22.950
(+) Dividend equalization of April	6	0.0765	0.459
Total	306	0.0765	23.409
Income from May	306	0.1125	34.425
	306	0.1890	57.834
(-) Dividend equalization for May	3	0.1890	(0.5670)
Total	303	0.1890	57.2670
Income from June	303	0.150	45.450
Total	303	0.3390	102.7170
Dividend (70%)	303	0.2373	(71.9019)
Income available for distribution	303	0.1017	30.8151

2. Issue price at the end of April

Opening NAV	= 18.75
(+) Entry load	= 2%
(+) Dividend equalization	= 0.0765
	<u>= 19.2015</u>

3. Repurchased price

Opening NAV	= 18.75
(-) Exit load	= 2%
(+) Dividend equalization	= 0.1890
	<u>= 18.5640</u>

4. NAV [300 lacs units]

Opening Net asset (300 × 18.75)	= 5,625
(+) Income [22.950 + 34.425 + 45.450]	= 102.825
(+) Appreciation in portfolio value	= 425.47

(+) Unit issued [6 × 19.2015]	= 115.209
(-) Units repurchased [3 × 18.5640]	= 55.692
(-) Dividend	= 71.9019
Net Asset	<u>= 6,140.9101</u>
÷ Units	= 303
NAV	<u>= 20.2670</u>

(6) FAMA NET SELECTIVITY MODEL

Question – 34

The returns of a portfolio A and market portfolio for the last 12 months are indicated as follows:

Month	Portfolio A	Market Portfolio
January	-0.52	0.82
February	2.20	0.04
March	2.17	2.80
April	4.17	1.72
May	2.04	0.27
June	3.00	0.39
July	1.99	1.95
August	4.00	0.64
September	-1.38	1.53
October	2.67	2.70
November	3.99	2.52
December	1.86	2.09
Standard Deviation (σ)	1.6223	0.9498

- (i) You are required to find out the monthly returns attributable to the sheer skill of the Portfolio Manager.

- (ii) What part of the monthly return is attributable to the higher risk assumed by the Portfolio Manager?

Assume that the risk-free rate of return is 12% per annum and the portfolio is fully diversified.

(Exam November – 2019) (8 Marks)

Solution:

- (i) The monthly risk free rate of return = $(12\%/12) = 1\%$

Month	R _A	R _M
January	-0.52	0.82
February	2.20	0.04
March	2.17	2.80
April	4.17	1.72
May	2.04	0.27
June	3.00	0.39
July	1.99	1.95
August	4.00	0.64
September	-1.38	1.53
October	2.67	2.70
November	3.99	2.52
December	1.86	2.09
Average Returns	26.19	17.47
	2.1825	1.4558

Average Portfolio Return (R_p) = 2.1825

Average Portfolio Return (R_m) = 1.4558

Portfolio Risk (σ_p) = 1.6223

Market Risk (σ_m) = 0.9498

Since portfolio A is fully diversified then it can be computed with a portfolio whose beta (β) can be found as follows:

$$\sigma_m^2 \times \beta^2 = \sigma_p^2$$

$\beta = \frac{\sigma_p}{\sigma_m} = \frac{1.6223}{0.9498} = 1.708$ Therefore, portfolio A is comparable to a portfolio whose Beta is 1.708.

Expected monthly returns on such portfolio can be calculated as follows:

$$R^1_p = R_f + \beta(R_m - R_f)$$

$$= 1\% + 1.708 (1.4558\% - 1.0000\%)$$

$$= 1.7785\%$$

Return due to the net selectivity = R_p – R¹_p

$$= 2.1825\% - 1.7785\%$$

$$= 0.404\% \text{ per month}$$

- (ii) The returns due to higher risk assumed by the portfolio manager

$$= 1.7785\% - 1.4558\% = 0.3227\% \text{ per month}$$

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

Mr. Y has invested in the three mutual funds (MF) as per the following details:

Particulars	MF 'X'	MF 'Y'	MF 'Z'
Amount of Investment (₹)	4,00,000	8,00,000	4,00,000
Net Assets Value (NAV) at the time of purchase (₹)	10.30	10.10	10
Dividend Received up to 31.03.2023 (₹)			
NAV as on 31.03.2023 (₹)	9,000	0	6,000
Effective Yield per annum as on 31.03.2023 (percent)	10.35	10	10.30
	9.66	-11.66	24.15

Assume 1 Year = 365 days

On the basis of above information, choose the most appropriate answer to the following questions:

- I. Total NAV of MF 'Y' as on 31.03.2023 would be approximately
- (a) ₹ 401941.73 (b) ₹ 412000.00
(c) ₹ 792079.20 (d) ₹ 82500.00
- II. Total Yield of MF 'X' in terms of ₹ would be approximately
- (a) ₹ 10941.73 (b) ₹ 7,920.80
(c) ₹ 18,000.00 (d) ₹ 12450.45

- III. Number of days for which MF 'X' is held would be approximately.....

- (a) 31 Days (b) 68 Days
(c) 103 Days (d) 85 Days

- IV. Number of days for which MF 'Y' is held would be.....

- (a) 31 Days (b) 68 Days
(c) 103 Days (d) 85 Days

(MTP March - 2024)

Answer: Case Scenario – 01

- I. (c) ₹ 792079.20
II. (a) ₹ 10941.73
III. (c) 103 Days
IV. (a) 31 Days

Case Scenario – 02

Mr. X on 1.7.2021, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31/03/2022, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31/12/2023, 20% dividend was given. On 31/03/2024, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%.

Based on the above information answer the following questions:

- I. NAV per unit of the Fund as on 31/03/2022 shall be approximately.....



- (a) ₹ 19.50 (b) ₹ 20.50
(c) ₹ 21.50 (d) ₹ 22.50

II. Total number of units as on 31/03/2022 shall be approximately.....

- (a) 10487.80 units (b) 12585.65 units
(c) 9465.35 units (d) 11575.40 units

III. Total Dividend received by Mr. X as on 31/03/2023 shall be

- (a) ₹ 20,625.50 (b) ₹ 20,870.45
(c) ₹ 20,975.60 (d) ₹ 21,565.75

IV. NAV per unit as on 31/03/2023 shall be approximately.....

- (a) ₹ 24.65 (b) ₹ 24.85
(c) ₹ 25.95 (d) ₹ 26.45

V. NAV as on 31/03/2024 shall be approximately.....

- (a) ₹ 20.50 (b) ₹ 25.95
(c) ₹ 26.75 (d) ₹ 27.20

(MTP September – 2024 & RTP November – 2024)

Answer: Case Scenario – 02

- I. (b) ₹ 20.50**
II. (a) 10,487.80 units

III. (c) ₹ 20,975.60

IV. (c) ₹ 25.95

V. (c) ₹ 26.75

Case Scenario – 03

The Asset Management Company of the mutual fund (MF) has declared a dividend of 9.98% on the units under the dividend reinvestment plan for the year ended 31st March 2021. The investors are issued additional units for the dividend at the rate of closing Net Asset Value (NAV) for the year as per the conditions of the scheme.

The closing NAV was ₹ 24.95 as on 31st March 2021. An investor Mr. X who is having 20,800 units at the year-end has made an investment in the units before the declaration of the dividend at the rate of opening NAV plus an entry load of ₹ 0.04. The NAV has appreciated by 25% during the year.

Assume the face value of the unit as ₹ 10.00.

Based on above Case Scenario, answer the following questions:

I. The Opening NAV of the Asset Management Company shall be

- (a) ₹ 20.24 (b) ₹ 19.96
(c) ₹ 18.75 (d) ₹ 17.65

II. The Number of the units purchased shall be

- (a) 18750 (b) 17500
(c) 20450 (d) 20000



III. Original amount of the investment shall be

- (a) ₹ 4,00,000 (b) ₹ 6,50,000
(c) ₹ 3,55,000 (d) ₹ 5,65,000

IV. Which of the following statement about Expense ratio is/ are incorrect:

- (i) It is the percentage of income that were spent to run a mutual fund.
(ii) It includes advisory fees, travel costs, registrar fees, custodian fees, etc.
(iii) It includes Brokerage costs for trading of Portfolio.
(iv) High Expense Ratio can seriously undermine the performance of a mutual fund scheme.

- (a) (i), (ii), (iii) (b) (i), (iii)
(c) only (iii) (d) only (i)

V.considers and uses downside deviation instead of total standard deviation in denominator.

- (a) Expense Ratio
(b) Sharpe Ratio
(c) Treynor Ratio
(d) Sortino Ratio

Answer: Case Scenario – 03

- I. (b) ₹ 19.96**
II. (d) 20000
III. (a) ₹ 4,00,000
IV. (d) only (i)
V. (d) Sortino Ratio

Case Scenario – 04

There are two Mutual Funds viz. D Mutual Fund Ltd. and K Mutual Fund Ltd. Each having close ended equity schemes.

NAV as on 31-12-2014 of equity schemes of D Mutual Fund Ltd. is ₹ 70.71 (consisting 99% equity and remaining cash balance) and that of K Mutual Fund Ltd. is 62.50 (consisting 96% equity and balance in cash).

Following is the other information:

Particular	Equity Schemes	
	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Sharpe Ratio	2	3.3
Treynor Ratio	15	15
Standard deviation	11.25	5

There is no change in portfolios during the next month and annual average cost is ₹ 3 per unit for the schemes of both the Mutual Funds.

If Share Market goes down by 5% within a month, calculate expected NAV after a month for the schemes of both the Mutual Funds.



For calculation, consider 12 months in a year and ignore number of days for particular month.

I. Beta of K Mutual Fund Ltd. is

- (a) 1.50 (b) 1.75
(c) 1.10 (d) 1.25

II. NAV after 1 month of K Mutual Fund is

- (a) 65.21 (b) 58.59
(c) 58.95 (d) 65.92

Answer: Case Scenario – 04

I. (c) 1.10

II. (c) 58.95

Case Scenario – 05

Steady Mutual Fund has the following assets in Scheme – Star Gold at the close of business as on 31st March, 2025:

Company	No. of Shares (Units)	Market Price per Share (₹)
A Ltd.	20,000	25
B Ltd.	30,000	350
C Ltd.	38,000	290
D Ltd.	50,000	400

The total numbers of units of Scheme – Star Gold are 20 lakhs. The Scheme – Star Gold has accrued expenses of ₹ 2,00,000 and other liabilities of ₹ 2,50,000.

From the information given above, choose the correct answer to the question no. **I** to **III**:

(EXAM MAY–2025)

I. Total gross value of the Scheme – Star Gold is –

- (A) ₹ 325.00 lakhs (B) ₹ 420.20 lakhs
(C) ₹ 480.40 lakhs (D) ₹ 520.30 lakhs

II. Total net value of the Scheme – Star Gold is –

- (A) ₹ 422.70 lakhs (B) ₹ 420.70 lakhs
(C) ₹ 415.70 lakhs (D) ₹ 424.70 lakhs

III. NAV per unit of the Scheme – Star Gold is –

- (A) ₹ 21.135 (B) ₹ 21.035
(C) ₹ 20.785 (D) ₹ 21.235

Answer: Case Scenario – 05

I. (B) ₹ 420.20 lakhs

II. (C) ₹ 415.70 lakhs

III. (C) ₹ 20.785

Case Scenario – 06

Fair return mutual fund made an issue of 5,00,000 units @ ₹ 10 each on 1st April 2024. No entry load was charged. The fund made the following investments:

Particulars	₹
25,000 Equity Shares of ₹ 100 each @ ₹ 160	40,00,000
	4,00,000



7% Government Securities (At Par)	2,50,000
9% Debentures (Unlisted) (At Par)	2,50,000
10% Debentures (Listed) (At Par)	49,00,000

During the year dividend of ₹ 6,00,000 were received on equity shares. Interest on all types of debt securities was received as and when due. Operating expenses paid during the year amounted to ₹ 2,50,000.

At the end of the year equity shares and 10% debentures are quoted at 175% and 90% respectively of the face value. Other investments are valued at par.

From the information given above, choose the correct answer to the following question No.:

I. What is net cash balance of the Mutual Fund at the end of the year?

- | | |
|----------------|----------------|
| (a) ₹ 1,00,000 | (b) ₹ 7,00,000 |
| (c) ₹ 7,75,500 | (d) ₹ 5,25,000 |

II. What is Total Net Assets Value (NAV) of the Mutual Fund at the end of the year?

- | | |
|-----------------|-----------------|
| (a) ₹ 52,50,000 | (b) ₹ 50,25,000 |
| (c) ₹ 57,75,500 | (d) ₹ 49,00,500 |

III. If the Mutual Fund distributes a dividend of ₹ 0.60 per unit during the year to the unit holders, the NAV per unit at the end of the year will be

(a) ₹ 11.551

(b) ₹ 10.951

(c) ₹ 10.051

(d) ₹ 12.751

(EXAM SEPTEMBER – 2025)

Answer: Case Scenario – 06

I. (d) ₹ 5,25,000

II. (c) ₹ 57,75,500

III. (b) ₹ 10.951



Student Notes



Student Notes



Student Notes



03

DERIVATIVES

PART – 01 : OPTION

I BASICS

Question – 01

The equity share of VCC Ltd. is quoted at ₹ 210. A 3-month call option is available at a premium of ₹ 6 per share and a 3-month put option is available at a premium of ₹ 5 per share. Ascertain the net payoffs to the option holder of a call option and a put option separately.

- The strike price in both cases is ₹ 220; and
- The share price on the exercise day is ₹ 200, 210, 220, 230, 240.

Also indicate the price range at which the call and the put options may be gainfully exercised.

(SM TYK – 20)

Solution:**Net payout for the holder of the call option**

Share Price	Action	Gross Payoff	Premium	Net Payoff
200	Lapsed	0	6	(6)
210	Lapsed	0	6	(6)
220	Lapsed	0	6	(6)
230	Exercised	10	6	4
240	Exercised	20	6	14

Net payoff for the holder of the put option

Share Price	Action	Gross Payoff	Premium	Net Payoff
200	Exercised	20	5	15
210	Exercised	10	5	5
220	Lapsed	0	5	(5)
230	Lapsed	0	5	(5)
240	Lapsed	0	5	(5)

The call option can be exercised gainfully for any price above ₹ 226 (₹ 220 + ₹ 6) and put option for any price below ₹ 215 (₹ 220 – ₹ 5).

Question – 02

Hari is holding 100 equity shares of VCC Ltd. which is being quoted at ₹ 210 per share. He is interested in hedging downside risk of his holding as he is going to sell them after 2 months. A 2-month Call option is available at a premium of ₹ 6 per share and a 2-month put option is available at a premium of ₹ 5 per share. The strike price in both cases is ₹ 220. You are required to:

- Suggest the position Hari should take in the option market to hedge his holding in the VCC Ltd.
- Calculate his final position after 2 months if after 2 months i.e. on the day of exercise the actual market price of per share of VCC Ltd. happens to be ₹ 200, ₹ 210, ₹ 220, ₹ 230 and ₹ 240.



(MTP October – 2023)

Solution:

- (i) Since Hari holds 100 equity shares, he should buy equal no. of Put option i.e. 100 put options in the same stock to hedge his position.

Total Premium amount to be paid = $5 \times 100 \text{ Put} = ₹ 500$

- (ii) Net Position after 2-months

Share Price	Action	Gross Payoff	Pre-mium	Net Pay off	Sell Share	Net Position	Total Position
200	Exercised	20	5	15	200	215	21,500
210	Exercised	10	5	5	210	215	21,500
220	Lapsed	0	5	(5)	220	215	21,500
230	Lapsed	0	5	(5)	230	225	22,500
240	Lapsed	0	5	(5)	240	235	23,500

Thus, from above table it can be observed in any case the value of holding of Hari in VCC Ltd. shall not go below ₹ 215 per share.

Question – 03

Mr. A is holding 1,000 shares of face value of ₹ 100 each of M/s. ABC Ltd. He wants to hold these shares for long term and have no intention to sell.

On 1st January 2020, M/s. XYZ Ltd. has made short sales of M/s. ABC Ltd.'s shares and approached Mr. A to lend his shares under Stock Lending Scheme with following terms:

- (i) Shares to be borrowed for 3 months from 1st January 2020 to 31st March 2020.

- (ii) Lending Charges/Fees of 1% to be paid every month on the closing price of the stock quoted in Stock Exchange and
- (iii) Bank Guarantee will be provided as collateral for the value as on 1st January 2020.

Other Information:

- (a) Cost of Bank Guarantee is 8% per annum.
- (b) On 29th February 2020 M/s. ABC Ltd. declared dividend of 25%.
- (c) Closing price of M/s. ABC Ltd.'s shares quoted in Stock Exchange on various dates are as follows:

Date	Share Price in Scenario – 1 Bullish	Share Price in Scenario – 2 Bearish
1 st January 2020	1,000	1,000
31 st January 2020	1,020	980
29 th February 2020	1,040	960
31 st March 2020	1,050	940

You are required to find out :

- (i) Earnings of Mr. A through Stock Lending Scheme in both the scenarios,
- (ii) Total earnings of Mr. A during 1st January 2020 to 31st March 2020 in both the scenarios,
- (iii) What is the profit or loss to M/s. XYZ by shorting the shares using through Stock Lending Scheme in both the scenarios ?

(Exam January – 2021) (8 Marks)

Solution:

1. Earnings of Mr. A

	I	II
Lending Charges		
31/01/2020	10.20	9.80
29/02/2020	10.40	9.60
31/03/2020	10.50	9.40
Lending Charges	31.10	28.80
(X) No. of Shares	1,000	1,000
Earnings	31,100	28,800

2. Total Earnings of Mr. A

	I	II
Lending Charges	31.10	28.80
(+) Dividend [100 × 25%]	25.00	25.00
	56.10	53.80
(X) No. of Shares	1,000	1,000
Total	56,100	53,800

3. Profit/Loss to M/s XYZ

	I	II
Lending Charges	(31.10)	(28.80)
Bank Guarantee [1,000 × 8% × 3/12]	(20.00)	(20.00)
	(50.00)	60.00
Gain/Loss on Short Selling	(101.10)	11.20
	1,000	1,000
(X) No. of Shares	(1,01,100)	11,200
Gain/Loss		

(II) OPTION STRATEGIES

Question – 04

Mr. X established the following strategy on the Delta Corporation's stock :

- (1) Purchased one 3-month call option with a premium of ₹ 30 and an exercise price of ₹ 550.
- (2) Purchased one 3-month put option with a premium of ₹ 5 and an exercise price of ₹ 450.

Delta Corporation's stock is currently selling at ₹ 500.

CALCULATE profit or loss, if the price of Delta Corporation's stock:

- remains at ₹ 500 after 3 months.
- falls at ₹ 350 after 3 months.
- rises to ₹ 600.

Assume the option size is 100 shares of Delta Corporation.

(MTP April – 2022, SM TYK – 19)

Solution:

Calculation of Profit & Loss

- Price of both shares on maturity ₹ 500:** In this situation, Mr. X will not exercise call option & put option, hence

Gross Payoff = 0

(-) Cost of Strategy = ₹ 35

Loss	<u>= ₹ 35</u>
------	---------------

Loss on 100 shares (₹ 35 × 100) = ₹ 3,500

- (ii) **Price of share ₹ 350:** In this situation, Mr. X will exercise his put option but call option will lapse

Gross Payoff (450 – 350) = ₹ 100

(-) Cost of Strategy = ₹ 35

Profit	<u>= ₹ 65</u>
--------	---------------

Profit on 100 shares (₹ 65 × 100) = ₹ 6,500

- (iii) **Price of share ₹ 600:** In this situation, Mr. X will exercise his call option but put option will lapse

Gross Payoff (600 – 550) = ₹ 50

(-) Cost of Strategy = ₹ 35

Profit	<u>= ₹ 15</u>
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Profit on 100 shares (₹ 15 × 100) = ₹ 1,500

Question – 05

The market received rumour about ABC corporation's tie-up with a multinational company. This has induced the market price to move up. If the rumour is false, the ABC corporation stock price will probably fall dramatically. To protect from this an investor has bought the call and put options.

He purchased one 3 months call with a striking price of ₹ 42 for ₹ 2 premium, and paid Re.1 per share premium for a 3 months put with a striking price of ₹ 40.

- (i) Determine the Investor's position if the tie up offer bids the price of ABC Corporation's stock up to ₹ 43 in 3 months.
(ii) Determine the Investor's ending position, if the tie up program me fails and the price of the stocks falls to ₹ 36 in 3 months.

(SM TYK – 16)

Solution:

Cost of Strategy

Assume = 100 shares

Cost = (₹ 2 + ₹ 1) × 100 shares
= ₹ 300

Price of Shares ₹ 43

In this situation call option will exercise but put option will lapse

Gross payoff (43 – 42) = ₹ 1

(-) Cost of Strategy = ₹ 3

Loss	<u>= ₹ 2</u>
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(X) No. of shares = 100

Loss	<u>= ₹ 200</u>
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Price of Shares ₹ 36

In this situation call option will Lapse but put option will Exercise

Gross payoff (40 – 36)	= ₹ 4
(–) Cost of Strategy	= ₹ 3
Gain	= ₹ 1
(X) No. of shares	= 100
Gain	= ₹ 100

Question – 06

Ms. Priya initiated the following option strategy on Omega Industries Limited's equity shares:

Transactions executed:

(1) Bought one European Call Option with the following terms:

- Premium paid : ₹ 42 per share
- Strike Price : ₹ 620
- Maturity : 3 months

(2) Bought one European Put Option with the following terms:

- Premium paid : ₹ 8 per share
- Strike Price : ₹ 480
- Maturity : 3 months

Additional Information:

- Current Market Price (CMP) of Omega Industries : ₹ 550 per share
- Lot size : 150 shares per contract
- Ms. Priya holds the positions until expiration

Required:

- (i) Calculate the net profit/loss in the following scenarios at expiration:
 - Share price remains unchanged at ₹ 550
 - Share price declines to ₹ 380
 - Share price appreciates to ₹ 680
- (ii) Determine the upper and lower breakeven points for this strategy.

(Note: Round off all intermediate and final calculations to four decimal places.)

(Exam Jan – 2026)

Solution:

Total premium paid $(₹ 42 + ₹ 8) \times 150 = ₹ 7,500$

(i) Net Profit/Loss under different scenarios:

- Share price remains unchanged at ₹ 550
 - * None of the options will be exercised and whatever premium paid i.e. ₹ 7,500 will be net loss.
- Share price declines to ₹ 380
 - * Call option is lapsed and put option is exercised. The net profit will be as follows:



Gain on per put option (₹ 480 - ₹ 380)	₹ 100
Lot size	150
Total gain	₹ 15,000
Less: Premium paid	₹ 7,500
Net Profit	₹ 7,500

- Share price appreciates to ₹ 680

* Put option is lapsed and call option is exercised. The net profit will be as follows:

Gain on per call option (₹ 680 - ₹ 620)	₹ 60
Lot size	150
Total gain	₹ 9,000
Less: Premium paid	₹ 7,500
Net Profit	₹ 1,500

(ii) Break-even Points

Upper: ₹ 620 + ₹ 50 = ₹ 670

Lower: ₹ 480 - ₹ 50 = ₹ 430

(III) OPTION PRICING & VALUATION

EXPECTED VALUE OF OPTION

Question – 07

Equity share of PQR Ltd. is presently quoted at ₹ 320. The Market Price of the share after 6 months has the following probability distribution:

Market Price ₹	180	260	280	320	400
Probability	0.1	0.2	0.5	0.1	0.1

A put option with a strike price of ₹ 300 can be written.

You are required to find out expected value of option at maturity (i.e. 6 months)

(SM TYK – 17)

Solution:

Expected Value of Option

Share Price	Action	Gross Pay Off	Probability	Gross Pay Off × Probability
180	Exercised	120	0.1	12
260	Exercised	40	0.2	8
280	Exercised	20	0.5	10
320	Lapsed	0	0.1	0
400	Lapsed	0	0.1	0
Expected Value of Option				₹ 30

Question – 08

You had purchased a 3 month call option on the Equity shares of Satya Ltd for a premium of ₹ 30 each, the current market price of the share is ₹ 560 and the exercise price is ₹ 590. You expect the price range between ₹ 540 to ₹ 640.

The expected share price of Satya Ltd and related probability is given below:

Expected price (₹)	540	560	580	600	620	640
Probability	0.10	0.15	0.05	0.35	0.20	0.15

Compute the followings:

(i) Expected share price at the end of 3 months,

- (ii) Value of call option at the end of 3 months, if the exercise price prevails,
- (iii) In case the option is held to its maturity, what will be the expected value of the call option?
- (iv) Find out the price of the shares quoted at the stock exchange to get the value of the call option as computed in (iii) above.

(Exam May – 2022) (8 Marks)

Solution:

(i) Expected Share Price

$$\begin{aligned}\text{Expected Price} &= (540 \times 0.10) + (560 \times 0.15) + (580 \times 0.05) + (600 \times 0.35) + (620 \times 0.20) + (640 \times 0.15) \\ &= ₹ 597\end{aligned}$$

(ii) Value of Call Option, If Exercise Price Prevails

[अगर Maturity को Exercise Price ही Market Price होगा]

$$\text{Value of Call} = 590 - 590 = 0$$

(iii) Expected Value of Option

Share Price	Action	Gross Pay Off	Probability	Gross Pay Off × Probability
540	Lapsed	0	0.10	0
560	Lapsed	0	0.15	0
580	Lapsed	0	0.05	0
600	Exercised	10	0.35	3.50

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620	Exercised	30	0.20	6.00
640	Exercised	50	0.15	7.50
				₹ 17

(iv) Price of Share so that Value of Call be ₹ 17

$$\text{Price of Share} = 590 + 17 = ₹ 60$$

BINOMINL MODULE

Question – 09

The current market price of an equity share of Penchant Ltd is ₹ 420. Within a period of 3 months, the maximum and minimum price of it is expected to be ₹ 500 and ₹ 400 respectively. If the risk free rate of interest be 8% p.a., what should be the value of a 3 months Call option under the “Risk Neutral” method at the strike rate of ₹ 450?

$$\text{Given } e^{0.02} = 1.0202$$

(SM TYK – 24)

Solution:

Step 1: Given

$$S = ₹ 420$$

$$uS = \frac{₹ 500}{₹ 420} = ₹ 1.1905$$

$$dS = \frac{₹ 400}{₹ 420} = ₹ 0.9524$$



$$E = ₹ 450$$

$$R = 8 \times \frac{3}{12} = 2\%$$

$$E^{0.02} = 1.0202$$

Step 2: Risk Neutral Probability

$$P = \frac{e^{rt} - d}{u - d} = \frac{1.0202 - 0.9524}{1.1905 - 0.9524} = 0.2844$$

Step 3: Binomial Tree



Step 4: Value of Call Option

$$\begin{aligned} C_0 &= \frac{C_u p + C_d (1 - p)}{e^{rt}} \\ &= \frac{(50 \times 0.2844) + (0 \times 0.7156)}{1.0202} \\ &= ₹ 13.94 \end{aligned}$$

Question – 10

Sumana wanted to buy shares of EIL which has a range of ₹ 411 to ₹ 592 a month later. The present price per share is ₹ 421. Her

broker informs her that the price of this share can sore up to ₹ 522 within a month or so, so that she should buy a one-month CALL of EIL. In order to be prudent in buying the call, the share price should be more than or at least ₹ 522 the assurance of which could not be given by her broker.

Though she understands the uncertainty of the market, she wants to know the probability of attaining the share price ₹ 592 so that buying of a one-month CALL of EIL at the execution price of ₹ 522 is justified. Advice her. Take the risk-free interest to be 3.60% and $e^{0.036} = 1.037$.

(SM TYK – 21)

Solution:

Risk Neutral Probability

$$P = \frac{e^{rt} - d}{u - d}$$

$$e^{rt} = e^{0.036} = 1.037$$

$$u = \frac{592}{421} = 1.4062$$

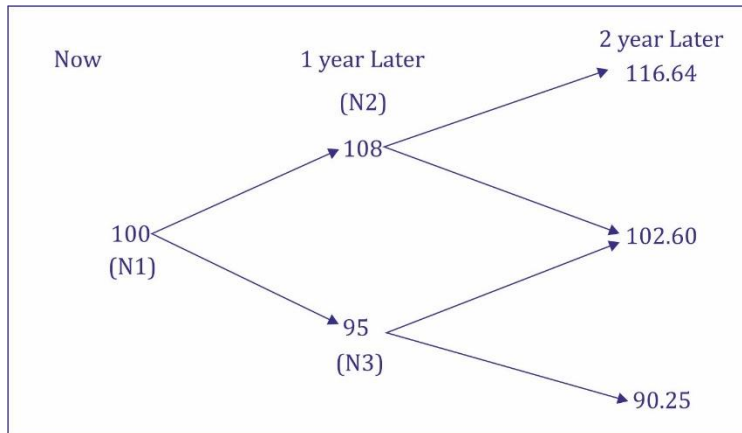
$$d = \frac{411}{421} = 0.9762$$

$$P = \frac{1.037 - 0.9762}{1.4062 - 0.9762} = 0.1414$$

TWO PERIOD BINOMIAL MODELS

Question – 11

A two year tree for a share of stock in ABC Ltd., is as follows:



Consider a two years American call option on the stock of ABC Ltd., with a strike price of ₹ 98. The current price of the stock is ₹ 100. Risk free return is 5 per cent per annum with a continuous compounding and $e^{0.05} = 1.05127$. Assume two time periods of one year each.

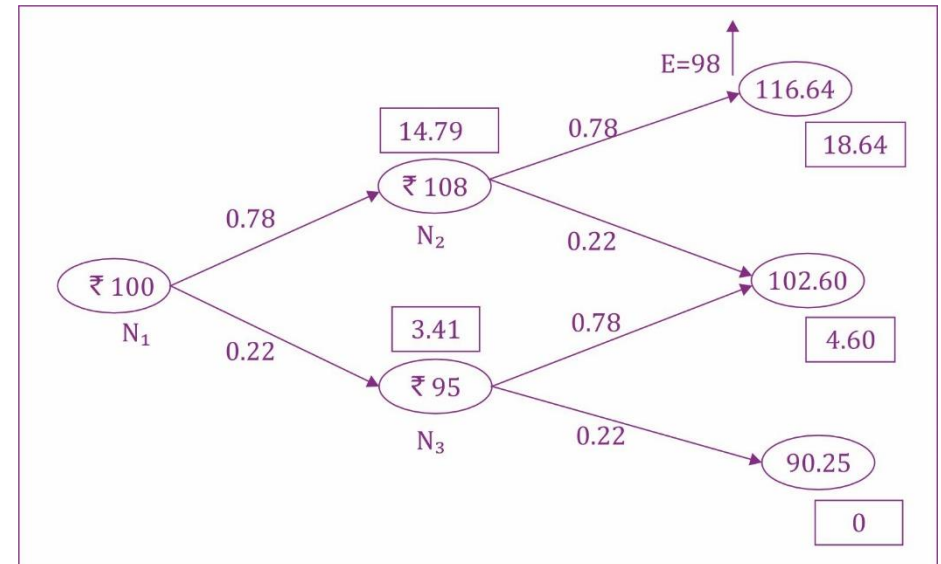
Using the Binomial Model, calculate:

- The probability of price moving up and down;
- Expected pay offs at each nodes i.e. N1, N2 and N3 (round off upto 2 decimal points).

(Exam Nov – 2020) (8 Marks)

Solution:

Step 1: Binomial Tree



Step 2: Risk Neutral Probability

$$P = \frac{e^{rt} - d}{u - d}$$

$$P = \frac{1.05127 - 0.95}{1.08 - 0.95} = 0.78$$

Step 3: Value of Option

N₂

$$C_0 = \frac{(18.64 \times 0.78) + (4.60 \times 0.22)}{1.05127} = 14.79$$

Intrinsic Value = $108 - 98 = ₹ 10$

Hence, value of option at node 2 is ₹ 14.79

N₃

$$C_0 = \frac{(4.60 \times 0.78) + (0 \times 0.22)}{1.05127} = 3.41$$

Intrinsic Value = 0

Hence, value of option at node 3 is ₹ 3.41

N₁

$$C_0 = \frac{(14.79 \times 0.78) + (3.41 \times 0.22)}{1.05127} = 11.69$$

Intrinsic Value = 2

Hence, value of call today is ₹ 11.69

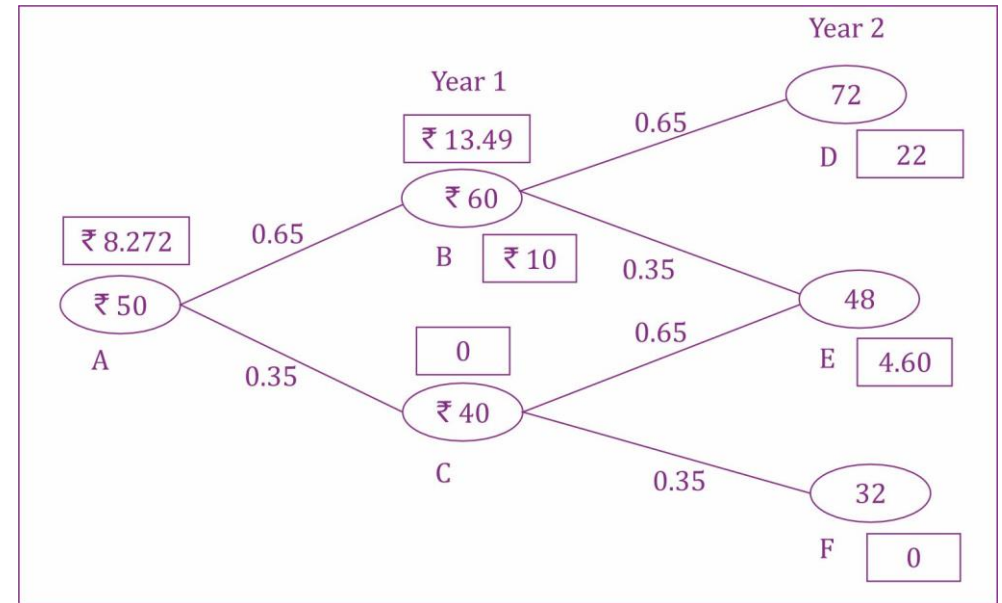
Question – 12

Consider a two-year call option with a strike price of ₹ 50 on a stock the current price of which is also ₹ 50. Assume that there are two-time periods of one year and in each year the stock price can move up or down by equal percentage of 20%. The risk-free interest rate is 6%. Using binominal option model, calculate the probability of price moving up and down. Also draw a two-step binomial tree showing prices and payoffs at each node.

(SM TYK – 23)

Solution:

(i) Binomial Tree



(ii) Risk Neutral Probability

$$R = 6\%, \quad u = 1.20, \quad d = 0.80$$

$$P = \frac{R-d}{u-d} = \frac{1.06 - 0.80}{1.20 - 0.80} = 0.65$$

(iii) Value of Option

$$\text{Node B} = \frac{(22 \times 0.65) + (0 \times 0.35)}{1.06} = ₹ 13.49$$

$$\text{P.V. of Expected Payoff} = ₹ 13.49$$

$$\text{Intrinsic Value} = ₹ 10$$



Value of option at Node B (Higher) = ₹ 13.49

Node C Value of Option = 0

Node A = $\frac{(\text{₹ } 13.49 \times 0.65) + (0 \times 0.35)}{1.06}$ = ₹ 8.272

Intrinsic Value = 50 – 50 = 0

Hence Value of Call Option today is ₹ 8.272

DELTA HEDGING BINOMIAL MODEL

Question – 13

Mr. Dayal is interested in purchasing equity shares of ABC Ltd. which are currently selling at ₹ 600 each. He expects that price of share may go upto ₹ 780 or may go down to ₹ 480 in three months. The chances of occurring such variations are 60% and 40% respectively. A call option on the shares of ABC Ltd. can be exercised at the end of three months with a strike price of ₹ 630.

- What combination of share and option should Mr. Dayal select if he wants a perfect hedge?
- What should be the value of option today (the risk free rate is 10% p.a.)?
- What is the expected rate of return on the option?

(SM TYK – 22)

Solution:

Given: S = ₹ 600

us = ₹ 780

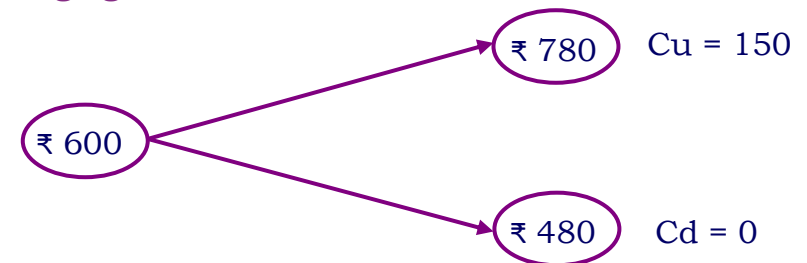
ds = ₹ 480

Period = 3 Months

E = ₹ 630

Option = Call

Delta Hedging



(i) Combination of Shares & Options

$$\text{Delta of Call} = \frac{C_u - C_d}{u_s - d_s}$$

$$= \frac{150 - 0}{\text{₹ } 780 - \text{₹ } 480} = 0.5$$

Write a Call Option & buy 0.5 share today for perfect hedge.

(ii) Cash Flow

(1) Price = ₹ 780

Call Option Exercised = (150)

Sell Shares (₹ 780 × 0.5) = 390

Cash Inflows = 240



(2) Price = ₹ 480

Call Option Lapsed	= 0
Sell Shares (₹ 480 × 0.5)	= 240
Cash Inflows	= 240
Present Value Cash Inflow	$= \frac{240}{1.025}$
	= 234.15

$$\text{Value of Call} = 0.5 \times 600 - 234.15 = ₹ 65.85$$

(3) Cost of option = 65.85

$$\begin{aligned} \text{Expected Profit} &= (150 \times 0.6) + (0 \times 0.5) \\ &= ₹ 90 \\ \text{Expected Rate of Return} &= \frac{90 - 65.85}{65.85} \times 100 \\ &= 36.67\% \end{aligned}$$

Question – 14

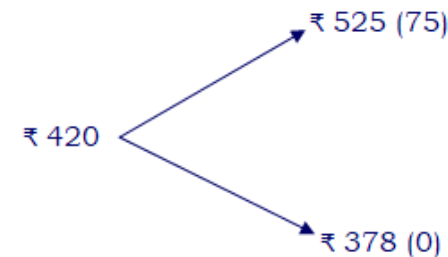
Following is the information available pertaining to shares of Omni Ltd.:

Current Market Price (₹)	₹ 420.00
Strike Price (₹)	₹ 450.00
Maximum Price (₹) expected in next 3 months' time	₹ 525.00
Minimum Price (₹) expected in next 3 months' time	₹ 378.00
Continuously Compounded Rate of Interest (p.a.) (%)	8.00%
e^{rt}	1.0202

From the above:

- Calculate the 3 months call option by using Binomial Method and Risk Neutral Method. Are the calculated values under both the models are same?
- State also clearly the basis of Valuation of options under these models.

(Exam Nov – 2023) (8 Marks)

Solution:**(i) (1) Call Option value using Binomial Model**

$$\Delta = \frac{₹ 75 - 0}{₹ 525 - ₹ 378} = 0.51$$

Delta of Call = 0.51 means write 1 call & buy 0.51 share.

$$\text{Initial Cash Outflows} = 420 \times 0.51 = 214.20$$

Cash Flows on Maturity**If price 525**

$$\text{Sell share } (525 \times 0.51) = 267.75$$

$$\text{Call Exercise} = (75)$$

$$\text{Cash Inflows} = 192.75$$

If price 378

$$\text{Sell Share } (378 \times 0.51) = 192.78$$

$$\text{Call Lapse} = 0$$

$$\text{Cash Inflows} = 192.78$$

$$\text{P.V. off payoff} = \frac{192.78}{1.0202} = 188.96$$

$$C_0 = 214.20 - 188.96 = ₹ 25.24$$

(2) Value of Call Option using Risk Neutral Method

Let 'P' be the probability of Price increase, then

$$p \times 525 + (1 - p) \times 378 = 420(1.0202)$$

$$147p = 50.48$$

$$p = 0.34$$

$$\text{Probability of Price increase} = 0.34$$

$$\text{Probability of Price decrease} = 0.66$$

$$\frac{0.34 \times 75 + 0.66 \times 0}{1.0202} = ₹ 25.24$$

Yes, the value of option under both Models is same.

(ii) Basis of valuation of options :

- Binomial model uses an approach called "Risk less Hedge Approach" to find the price of the option, by creating a portfolio which will have same value at expiration irrespective of any price. Hedge means to create an equal and opposite position for protecting the value of portfolio.
- In Risk Neutral Model, valuation of options is based on arbitrage and is therefore independent of risk preferences; one should be able to value options assuming any set of risk preferences and get the same answer.

Question – 15

- (i) On 1 April 2015, Sunidhi was holding a portfolio of 10 securities whose value was ₹ 9,94,450, the weighted average of beta of 9 securities of the portfolio was 1.10.

Since she was expecting a fall in the prices of the shares in near future to hedge her portfolio, she sold 5 contract of NIFTY Futures (Multiplier of 25) expiring in May 2015, which was trading at 8,767.07 on 1 April.

Required:

- (1) Calculate the beta of the 10th security.



(2) Reconcile the reasons in spite of 2% fall in the market as per Sunidhi's apprehension if she would have earned some profit on her cash position.

- (ii) A Futures contract is available on R Ltd. that pays an annual dividend of ₹ 4 and whose stock is currently priced at ₹ 125. Each Futures contract calls for delivery of 1,000 shares to stock in one year, daily marking to market. The treasury bill rate is 8%.

Required:

- (1) Given the above information, assess the price of one Futures contract.
- (2) If the company stock price decreases by 6%, then what will be the price of one Futures contract?
- (3) Suppose the company stock price decreases, then evaluate realizes gain or loss for an investor that has a long position in one Futures contract of R Ltd.

(Ignore margin and taxation, if any)

(RTP Jan – 2026)

Solution:

- (i) (1) To compute the beta of 10th security first we shall compute overall

weighted beta as follows:

Let weighted β of the Portfolio is w , then,

$$5 = \frac{9,94,450}{8,767.07 \times 25} \times w$$

$$w = 1.102 \text{ approximately}$$

Let beta of 10th security is β then,

$$1.102 = 0.90 \times 1.10 + 0.10 \times \beta$$

$$\beta = 1.12$$

- (2) The main reason for the profit in cash position might be due to reason that contrary to her expectation fall in the value of cash position there may be increase in value of cash position or decrease in the stock price may be lesser than 2%.

- (ii) (1) Future Price = Spot + Cost of Carry – Dividend

$$= ₹ 125 + (₹ 125 \times 0.08) - ₹ 4 = ₹ 131$$

$$\text{Price of one future contract} = 1,000 \text{ share} \times ₹ 131$$

$$= ₹ 1,31,000$$

- (2) Price decrease by 6%

$$\text{Market Price} = ₹ 125 \times 94\% = ₹ 117.50$$

Then, price of one future contract

$$= ₹ 117.50 + (₹ 117.50 \times 0.08) - ₹ 4 = ₹ 122.90$$

$$= ₹ 122.90 \times 1,000 = ₹ 1,22,900$$

- (3) If the investor has taken a long position, decrease in price will result in **loss** for the investor.



Amount of loss will be:

$$₹ 1,31,000 - ₹ 1,22,900 = ₹ 8,100$$

BLACK SCHOLES MODEL

Question – 16

From the following data for certain stock, find the value of a call option:

Price of stock now	=	₹ 80
Exercise price	=	₹ 75
Standard deviation of continuously compounded annual return	=	0.40
Maturity period months	=	6
Annual interest rate	=	12%

Given

Number of S.D. from Mean, (z)	Area of the left or right (one tail)
0.25	0.4013
0.30	0.3821
0.55	0.2912
0.60	0.2743

$$e^{0.12 \times 0.5} = 1.062$$

$$\ln 1.0667 = 0.0646$$

(SM TYK – 25)

Solution:

Working Note 1: d_1 & d_2

$$d_1 = \frac{\ln \frac{S_0}{E} + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln \frac{80}{75} + \left(0.12 + \frac{(0.40)^2}{2}\right)0.5}{0.40\sqrt{0.5}}$$

$$d_1 = \frac{\ln 1.0667 + 0.10}{0.2828}$$

$$d_1 = \frac{0.0646 + 0.10}{0.2828} = 0.5820$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

$$= 0.5820 - 0.2828 = 0.2992$$

Working Note 2: $n(d_1)$ & (d_2)

[BSM में हमेशा Cumulative Area देखना है। But d_1 & d_2 negative दिया है तो फिर हम Tail Area देखेंगे]

$$d_1 = 0.5820$$

$n(d_1)$	=	0.55	0.2912
		0.60	0.2743
		0.05	0.0169

$$n(d_1) = 0.2912 - \left(\frac{0.0169}{0.05} \times 0.032\right)$$



$$n(d_1) = 0.2804$$

$$n(d_1) = 1 - 0.2804 = 0.7196$$

$$d_2 = 0.2992$$

$$n(d_2) =$$

$$0.25 \quad 0.4013$$

$$0.30 \quad 0.3821$$

$$0.05 \quad 0.0192$$

$$n(d_2) = 0.4013 - \left(\frac{0.0192}{0.05} \times 0.0492 \right)$$

$$n(d_2) = 0.3824$$

$$n(d_2) = 1 - 0.3824 = 0.6176$$

$$C_0 = S_0 \times n(d_1) - \frac{E}{e^{rt}} \times n(d_2)$$

$$= 80 \times 0.7196 - \frac{75}{e^{0.12 \times 0.5}} \times 0.6176$$

$$= 80 \times 0.7196 - \frac{75}{1.062} \times 0.6176$$

$$= ₹ 13.952$$

Question – 17

The shares of TIC Ltd. are currently priced at ₹ 415 and call option exercisable in three months' time has an exercise rate of ₹ 400. Risk free interest rate is 5% p.a. and standard deviation (volatility) of the

share price is 22%. The TIC Ltd. is not going to declare any dividend over the next three months.

- (i) DECIDE whether the option worth buying for ₹ 25.
- (ii) CALCULATE the value of aforesaid call option if the current price of share is considered as ₹ 380.
- (iii) CALCULATE the value of aforesaid call option if present price of share is taken as ₹ 408 and a dividend of ₹ 10 is expected to be paid in the two months' time.

Given

$$\ln(1.0375) = 0.03681, \ln(0.95) = -0.05129 \text{ and } \ln(0.9952) = -0.00481$$

$$e^{0.0125} = 1.0126 \text{ and } e^{0.00833} = 1.0084$$

Cumulative Area of Number of S.D. from Mean (z)

Z	0.0150	0.1250	0.3933	0.5033	-0.2976	-0.4076
Area	0.5060	0.5497	0.6530	0.6926	0.3830	0.3418

(MTP Oct – 2022)

Solution:

(1) Current Price of share of TIC Ltd. = ₹ 415

Exercise rate = ₹ 400

Risk free interest rate is = 5% p.a.

SD (Volatility) = 22%



Based on the above bit is calculated value of an option based on Black Scholes Model:

$$d_1 = \frac{\ln\left(\frac{415}{400}\right) + \left[0.05 + \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}}$$

$$= \frac{0.03681 + 0.01855}{0.11} = 0.5033$$

$$d_2 = \frac{\ln\left(\frac{415}{400}\right) + \left[0.05 - \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}}$$

$$= \frac{0.03681 + 0.00645}{0.11} = 0.3933$$

$$V_o = V_s N(d_1) - \frac{E}{e^{rt}} N(d_2)$$

$$N(d_1) = N(0.5033) = 0.6926$$

$$N(d_2) = N(0.3933) = 0.6530$$

$$\begin{aligned}\text{Value of Option} &= 415(0.6926) - \frac{400}{e^{(0.05)(0.25)}} (0.6530) \\ &= 287.43 - \frac{400}{1.0126} (0.6530) \\ &= 287.43 - 257.95 = ₹ 29.48\end{aligned}$$

Since market price of ₹ 25 is less than ₹ 29.48 (Black Scholes Valuation Model) indicate that option is underpriced, hence worth buying.

Extra Work

Value of Put Option: अगर Value of Call दिया हुआ है तो Value of Put, Put Call Parity से Calculate किया जा सकता है।

Put Call Parity:

$$S_0 + P_0 = C_0 + \text{P.V. of EP}$$

$$415 + P_0 = 29.48 + \frac{400}{1.0126}$$

$$P_0 = 9.50$$

Black Scholes Model:

$$\frac{E}{e^{rt}} \times N(-d_2) - S_0 N(-d_1)$$

$$\frac{400}{1.0126} \times 0.347 - 415 \times 0.3074$$

$$P_0 = 9.50$$

(2) If the current price is taken as ₹ 380 the computations are as follows:

$$d_1 = \frac{\ln\left(\frac{380}{400}\right) + \left[0.05 + \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}}$$

$$= \frac{-0.05129 + 0.01855}{0.11} = -0.2976$$

$$d_2 = \frac{\ln\left(\frac{380}{400}\right) + \left[0.05 - \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}}$$

$$= \frac{-0.05129 + 0.00645}{0.11} = -0.4076$$



$$N(d_1) = N(-0.2976) = 0.3830$$

$$N(d_2) = N(-0.4076) = 0.3418$$

$$\begin{aligned}\text{Value of Option} &= 380(0.3830) - \frac{400}{e^{(0.05)(0.25)}} (0.3418) \\ &= 145.54 - \frac{400}{1.0126} (0.3418) \\ &= 145.54 - 135.02 = ₹ 10.52\end{aligned}$$

- (3) Since dividend is expected to be paid in two months time was we have to adjust the share price and then use Black Scholes model to value the option:

Present Value of Dividend (using continuous discounting)

$$= \text{Dividend} \div e^{rt}$$

$$= ₹ 10 / 1.0084 = ₹ 9.92$$

$$\text{Adjusted price of shares is } ₹ 408.00 - ₹ 9.92 = ₹ 398.08$$

This can be used in Black Scholes model

$$\begin{aligned}d_1 &= \frac{\ln\left(\frac{398.08}{400}\right) + \left[0.05 + \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}} \\ &= \frac{-0.00481 + 0.01855}{0.11} = 0.1250 \\ d_2 &= \frac{\ln\left(\frac{398.08}{400}\right) + \left[0.05 - \frac{1}{2}(0.22)^2\right]0.25}{0.22\sqrt{0.25}}\end{aligned}$$

$$= \frac{-0.00481 + 0.00645}{0.11} = 0.0150$$

$$N(d_1) = N(0.1250) = 0.5497$$

$$N(d_2) = N(0.0150) = 0.5060$$

$$\begin{aligned}\text{Value of Option} &= 398.08(0.5497) - \frac{400}{e^{(0.05)(0.25)}} (0.5060) \\ &= 218.82 - \frac{400}{1.0126} (0.5060) \\ &= 218.82 - 199.88 = ₹ 18.94\end{aligned}$$

REAL OPTION

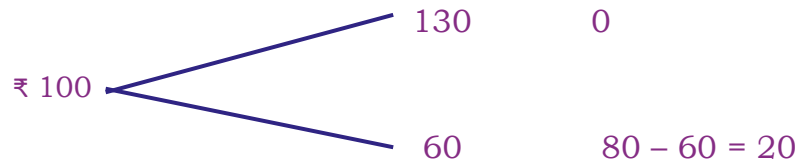
Question – 18

IPL already in production of Fertilizer is considering a proposal of building a new plant to produce pesticides. Suppose the PV of proposal is ₹ 100 crore without the abandonment option. However, if market conditions for pesticide turns out to be favorable the PV of proposal shall increase by 30%. On the other hand, market conditions remain sluggish the PV of the proposal shall be reduced by 40%. In case company is not interested in continuation of the project it can be disposed of for ₹ 80 crore.

If the risk-free rate of interest is 8% then what will be value of abandonment option.

Solution:

(i) Binomial Tree



(ii) Risk Neutral Probability

$$30P + (-40)(1 - P) = 8$$

$$30P - 40 + 40P = 8$$

$$70P = 48 \quad P = \frac{48}{70} = 0.686$$

(iii) Value of Abandonment Option

$$\text{Value} = \frac{(0 \times 0.686) + (20 \times 0.314)}{1.08}$$

$$= ₹ 5.81 \text{ Cr. (Put option)}$$

Question – 19

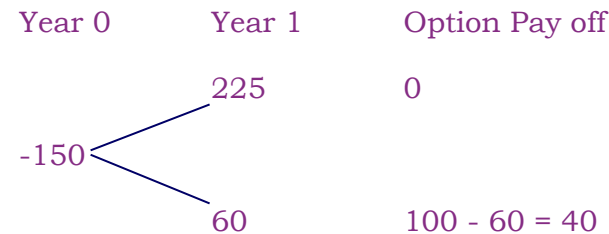
A firm is considering a proposal to set up a cement manufacturing plant with an initial investment of ₹ 150 crore. The firm has the option to abandon the project after one year by selling it to a competitor for ₹ 100 crore if the market conditions are unfavorable and the demand is low, the project's value will decline by 60%. However, if the market conditions turn out to be favorable and the demand for cement is high, the value of the project at the end of year 1 will increase by 50%.

Given that the risk free rate of interest as 8%, what will be the value of the abandonment option and the value of the project with abandonment option?

(Exam November – 2024) (4 Marks)

Solution:

Decision Tree showing pay off



First of all, we shall calculate probability of high demand (p) using risk neutral method as follows:

$$8\% = p \times 50\% + (1-p) \times (-60\%)$$

$$0.08 = 0.50p - 0.60 + 0.60p$$

$$p = 0.68 / 1.10 = 0.618 \text{ or } 0.62$$

The value of abandonment option will be as follows:

Expected Payoff at Year 1

$$= p \times 0 + [(1-p) \times 40]$$

$$= 0.618 \times 0 + [0.382 \times 40]$$

$$= ₹ 15.28 \text{ crore}$$

Since expected pay off at year 1 is ₹ 15.28 crore. Present value of expected pay off will be:

$$\frac{15.28}{1.08} = ₹ 14.15 \text{ crore}$$

Thus, the value of abandonment option is ₹ 14.15 crore.

Value of Project with abandonment option -

(i) If PVF is not considered

Expected Value of the Project without abandonment option
 $= (0.618 \times ₹ 225 \text{ crore} + 0.382 \times ₹ 60 \text{ crore}) - ₹ 150 \text{ crore}$
 $= ₹ 11.97 \text{ crore}$

Value of Project with abandonment Option

$= ₹ 11.97 \text{ crore} + ₹ 14.15 \text{ crore}$
 $= ₹ 26.12 \text{ crore}$

Or

Expected Value of the Project without abandonment option
 $= (0.62 \times ₹ 225 \text{ crore} + 0.38 \times ₹ 60 \text{ crore}) - ₹ 150 \text{ crore}$
 $= ₹ 12.30 \text{ crore}$

Value of Project with abandonment Option

$= ₹ 12.30 \text{ crore} + ₹ 14.15 \text{ crore}$
 $= ₹ 26.45 \text{ crore}$

(ii) If PVF is considered

Expected Value of the Project without abandonment option
 $= (0.618 \times ₹ 225 \text{ crore} + 0.382 \times ₹ 60 \text{ crore})/1.08 - ₹ 150 \text{ crore}$

$= - ₹ 0.028 \text{ crore OR } - ₹ 0.03$

Value of Project with abandonment Option

$= - ₹ 0.028 \text{ crore} + ₹ 14.15 \text{ crore}$
 $= ₹ 14.122 \text{ crore OR } ₹ 14.12 \text{ crore}$

Or

Expected Value of the Project without abandonment option
 $= (0.62 \times ₹ 225 \text{ crore} + 0.38 \times ₹ 60 \text{ crore})/1.08 - ₹ 150 \text{ crore}$

$= ₹ 00.28 \text{ crore}$

Value of Project with abandonment Option

$= ₹ 0.28 \text{ crore} + ₹ 14.15 \text{ crore}$
 $= ₹ 14.43 \text{ crore}$

Question – 20

Suppose MIS Ltd. is considering installation of solar electricity generating plant for light the staff quarters. The plant shall cost ₹ 2.50 crore and shall lead to saving in electricity expenses at the current tariff by ₹ 21 lakh per year forever.

However, with change in Government in state, the rate of electricity is subject to change. Accordingly, the saving in electricity can be of ₹ 12 lakh or ₹ 35 lakh per year and forever.

Assuming WACC of MIS Ltd. is 10% and risk-free rate of rate of return is 8%.

Decide whether MIS Ltd. should accept the project or wait and see.

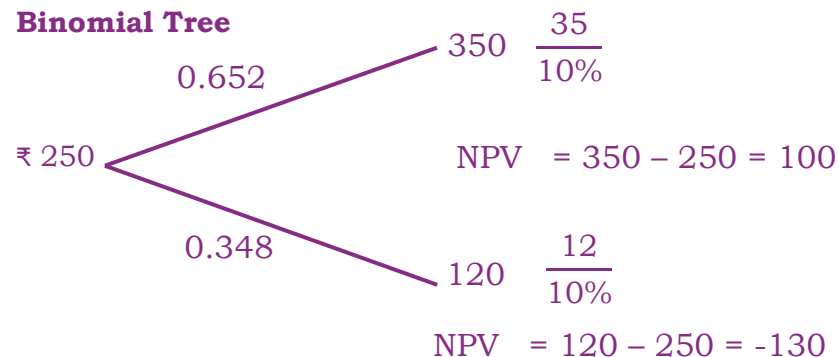
Solution:

If Project Install New

$$\begin{aligned} \text{NPV} &= \text{PVC I} - \text{PVCO} \\ &= \frac{21}{10\%} - 250 \text{ lacs} \\ &= -40 \text{ lacs (0.4 Crore)} \end{aligned}$$

Wait for 1 year

(i) Binomial Tree



(ii) Risk Neutral Probability

$$\begin{aligned} - \quad \text{Return if PVC I} &= 350 \\ &= \frac{350 - 250}{250} \times 100 = 40\% \end{aligned}$$

$$\begin{aligned} - \quad \text{Return if PVC I} &= 120 \\ &= \frac{120 - 250}{250} \times 100 = -52\% \\ 40 P + (-52) (1-P) &= 8 \\ 40 P - 52 + 52 P &= 8 \\ 92 P &= 60 \\ P &= \frac{60}{92} \\ &= 0.652 \end{aligned}$$

(iii) Value of Abandonment Option

$$\begin{aligned} \text{Value} &= \frac{(100 \times 0.652) + (-130 \times 0.348)}{1.08} \\ &= ₹ 18.48 \text{ lacs} \end{aligned}$$

It is better to wait for 1 year due to higher NPV.

Question – 21

Indian manufacturer TEJ Ltd. has just completed new project. Based on capital budgeting evaluation of the project Present Value is ₹ 500 Lakhs without the abandonment option.

However, due to geopolitical issues and international trade war the project may be at risk and company may have to discontinue the project. It is estimated that if favourable conditions remain, Present Value of the project shall increase by 20%. Whereas, if international trade war continues due to geopolitical issues the Present Value of proposal shall reduce by 30%. In case company is not interested to

continue the project it can be disposed off for ₹ 450 Lakhs to avoid future huge losses.

The risk-free rate of interest is 8%.

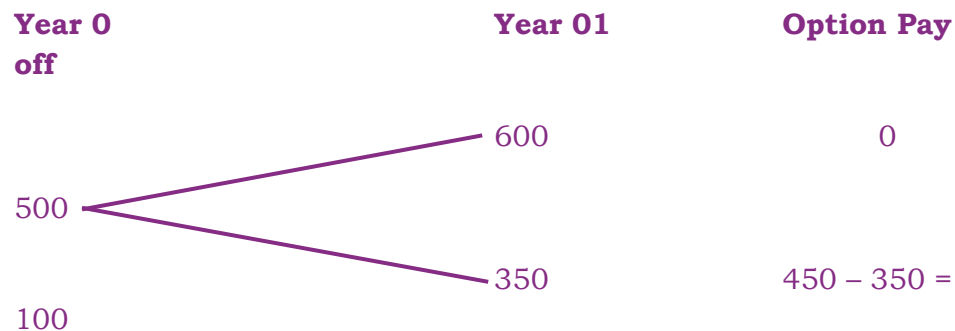
Required:

Calculate the value of abandonment option using risk neutral method.

(Exam Jan – 2026)

Solution:

Decision Tree showing pay off



First of all we shall calculate probability of high demand (p) using risk neutral method as follows:

$$8\% = p \times 20\% + (1-p) \times (-30\%)$$

$$0.08 = 0.20p - 0.30 + 0.30p$$

$$p = \frac{0.38}{0.50} = 0.76$$

The value of abandonment option will be as follows:

Expected Payoff at Year 1

$$= p \times 0 + [(1 - p) \times 100]$$

$$= 0.76 \times 0 + [0.24 \times 100] = ₹ 24 \text{ Lakhs}$$

Since expected pay off at year 1 is ₹ 24 Lakhs. Present value of expected pay off will be:

$$\frac{24.00}{1.08} = ₹ 22.22 \text{ Lakhs}$$

This is the value of abandonment option (Put Option).

Question – 22

ABC Ltd. is a pharmaceutical company possessing a patent of a drug called 'Aidrex', a medicine for aids patient. Being an approach drug ABC Ltd. holds the right of production of drugs and its marketing. The period of patent is 15 years after which any other pharmaceutical company produce the drug with same formula. It is estimated that company shall require to incur \$ 12.5 million for development and market of the drug. As per a survey conducted the expected present value of cash flows from the sale of drug during the period of 15 years shall be \$ 16.7 million. Cash flow from the previous similar type of drug have exhibited a variance of 26.8% of the present value of cash flows. The current yield on Treasury Bonds of similar duration (15 years) is 7.8%.

Determine the value of the patent.

$$\text{Given } \ln(1.336) = 0.2897$$

$$e^{-1.0005} = 0.3677 \text{ and } e^{-1.17} = 0.3104$$

Solution:

Given

$$E = \$ 12.5$$

$$S_0 = \$ 16.7$$

$$t = 15 \text{ years}$$

$$\text{Variance} = 0.268$$

$$\sigma = \sqrt{0.268} = 0.5177$$

$$r = 0.078$$

$$y = 1/15 = 0.0667$$

Step 1 d_1 & d_2

$$\begin{aligned} d_1 &= \frac{\ln \frac{S_0}{E} + \left[(r - y) + \frac{\sigma^2}{2} \right] t}{\sigma \sqrt{t}} \\ &= \frac{\ln \frac{16.7}{12.50} + \left[(0.078 - 0.0667) + \frac{0.268}{2} \right] 15}{\sqrt{0.268} \times \sqrt{15}} \\ &= \frac{\ln 1.336 + 2.1795}{2.005} \\ &= \frac{0.2897 + 2.1795}{2.005} = 1.2315 \end{aligned}$$

$$\begin{aligned} d_2 &= d_1 - \sigma \sqrt{t} \\ &= 1.2315 - 2.005 \\ &= -0.7735 \end{aligned}$$

$N(d_1)$

$$N(1.2315)$$

$$1.20 \text{ ————— } 0.1151$$

$$1.25 \text{ ————— } 0.1056$$

$$\underline{0.05} \qquad \qquad \underline{0.0095}$$

$$0.1151 - \left(\frac{0.0095}{0.05} \times 0.0315 \right) = 0.1091$$

$$N(d_1) = 1 - 0.1091 = 0.8909$$

$N(d_2)$

$$N(-0.7735)$$

$$N(d_2) = 0.2196$$

Value of Patent

$$\begin{aligned} \text{Value of patent} &= S_0 e^{-yt} \times N(d_1) - E e^{-rt} \times N(d_2) \\ &= 16.7 \times e^{-0.0667 \times 15} \times 0.8909 - 12.5 \times \\ &\quad e^{-0.078 \times 15} \times 0.2196 \\ &= 16.7 \times e^{-1.0005} \times 0.8909 - 12.5 \times e^{-1.17} \times \\ &\quad 0.2196 \\ &= 16.7 \times 0.3677 \times 0.8909 - 12.5 \times 0.3104 \times \\ &\quad 0.2196 \\ &= \$ 5.4707 - \$ 0.8520 \\ &= \$ 4.619 \text{ Millions.} \end{aligned}$$



PART – 02: FUTURE

(V) MARGIN A/C

Question – 23

On 31/08/2021 Mr. R has taken a Long position of Two lots of Nifty Futures at 17300.

One lot of Nifty future is 50 units.

Initial Margin required is 10% of Contract Value.

Maintenance Margin required is 80% of Initial Margin.

The closing price of 5 days are given below –

Date	Closing Price of Nifty Future
01/09/2021	17340
02/09/2021	17180
03/09/2021	16990
06/09/2021	16900
07/09/2021	17120

You are required to-

- Prepare a statement showing the daily balances in the margin account & payment on margin calls, if any.
- Compute the Gain or Loss of Mr. R, if contract squared off on 07/09/2021.
- What would be the Gain or Loss if Mr. R, had taken the short position? **(Exam December – 2021) (8 Marks)**

Solution:

Initial Margin = $[17,300 \times 50 \times 2] \times 10\%$

= ₹ 1,73,000

Maintenance Margin = ₹ 1,73,000 $\times$ 80%

= ₹ 1,38,400

(i) Margin Account

Date	Closing Price	Profit/Loss	Balance	Margin Call
31/08/2021	17,300	—	1,73,000	—
01/09/2021	17,340	40 $\times$ 100 = +4,000	1,77,000	—
02/09/2021	17,180	-160 $\times$ 100 = -16,000	1,61,000	—
03/09/2021	16,990	-190 $\times$ 100 = -19,000	1,42,000	—
06/09/2021	16,900	-90 $\times$ 100 = -9,000	1,33,000	—
07/09/2021	17,120	+220 $\times$ 100 = 22,000	1,55,000	40,000

(ii) Gain or Loss

Closing Balance = ₹ 1,55,000

(-) Initial Margin = ₹ 1,73,000

(-) Variation Margin = ₹ 40,000

Loss = ₹ 18,000

OR

$(17,300 - 17,120) \times 100 = 18,000$ (Loss)

DERIVATIVES

(iii) Gain or Loss (Short Position)

Margin Account

Date	Closing Price	Profit/Loss	Balance	Margin Call
31/08/2021	17,300	—	1,73,000	—
01/09/2021	17,340		1,69,000	—
02/09/2021	17,180	$-40 \times 100 = -4,000$	1,85,000	—
03/09/2021	16,990	$160 \times 100 = 16,000$	2,04,000	—
06/09/2021	16,900	$190 \times 100 = 19,000$	2,13,000	—
07/09/2021	17,120	$90 \times 100 = 9,000$	1,91,000	—
		$-220 \times 100 = -22,000$		—

Gain or Loss

Closing Balance = ₹ 1,91,000

(-) Initial Margin = ₹ 1,73,000

Gain = ₹ 18,000

Question – 24

Sensex futures are traded at a multiple of 50. Consider the following quotations of Sensex futures in the 10 trading days during February, 2009:

Day	High	Low	Closing
4-2-09	3306.40	3290.00	3296.50
5-2-09	3298.00	3262.50	3294.40
6-2-09	3256.20	3227.00	3230.40

7-2-09	3233.00	3201.50	3212.30
10-2-09	3281.50	3256.00	3267.50
11-2-09	3283.50	3260.00	3263.80
12-2-09	3315.00	3286.30	3292.00
14-2-09	3315.00	3257.10	3309.30
17-2-09	3278.00	3249.50	3257.80
18-2-09	3118.00	3091.40	3102.60

Abhishek bought one sensex futures contract on February, 04. The average daily absolute change in the value of contract is ₹ 10,000 and standard deviation of these changes is ₹ 2,000. The maintenance margin is 75% of initial margin.

You are required to determine the daily balances in the margin account and payment on margin calls, if any.

(SM TYK – 14)

Solution:

$$\begin{aligned}\text{Initial Margin} &= \mu + 3 \sigma \\ &= 10,000 + (3 \times 2000) \\ &= ₹ 16,000 \\ \text{Maintenance Margin} &= 16,000 \times 75\% \\ &= ₹ 12,000\end{aligned}$$



Margin A/c (Long)

Day	Closing Price	Profit Loss	Marg in A/c (₹)	Balance
04/02/09	3,296.50	---	-	16,000
05/02/09	3,294.40	$(3,294.40 - 3,296.50) \times 50 = -105$	-	15,895
06/02/09	3,230.40	$(3,230.40 - 3,294.40) \times 50 = -3,200$	-	12,695
07/02/09	3,212.30	$(3,212.30 - 3,230.40) \times 50 = -905$	4210	16,000
10/02/09	3,267.50	$(3,267.50 - 3,212.30) \times 50 = 2,760$	-	18,760
11/02/09	3,263.80	$(3,263.80 - 3,267.50) \times 50 = -185$	-	18,575
12/02/09	3,292.00	$(3,292.00 - 3,263.8) \times 50 = 1,410$	-	19,985
14/02/09	3,309.30	$(3,309.30 - 3,292.00) \times 50 = 865$	-	20,850
17/02/09	3,257.80	$(3,257.80 - 3,309.30) \times 50 = -2,575$	-	18,275
18/02/09	3,102.60	$(3,102.60 - 3,257.80) \times 50 = -7,760$	5485	16,000

(VI) VALUATION OF FUTURE**Question – 25**

The following data relate to Anand Ltd.'s share price:

Current price per share	₹ 1,800
6 months future's price/share	₹ 1,950

Assuming it is possible to borrow money in the market for transactions in securities at 12% per annum, you are required:

- to calculate the theoretical minimum price of a 6-months forward purchase; and
- to explain arbitrage opportunity.

(SM TYK – 02)**Solution:****(1) Theoretical Future Price**

$$\begin{aligned}
 F &= S(1 + r) - D \\
 &= 1,800(1.06) \\
 &= ₹ 1,908
 \end{aligned}$$

(2) Arbitrage

(i) Action: Since future is overpriced, hence sell future & buy spot

(ii) Process**Today**

- Borrow ₹ 1,800 & buy stock
- Contract to sell such stock at future price ₹ 1,950

After 6 Months

Cash Inflows



Sell share at future price = ₹ 1,950

Cash outflows

Repayment of Borrowing ₹ 1,800 (1.06) = ₹ 1,908

Arbitrage Gain = ₹ 42

Question – 26

Calculate the price of 3 months PQR futures, if PQR (FV ₹ 10) quotes ₹ 220 on NSE and the three months future price quotes at ₹ 230 and the one month borrowing rate is given as 15 percent per annum and the expected annual dividend is 25 percent, payable before expiry. Also examine arbitrage opportunities.

(SM TYK – 04)**Solution:****(1) Theoretical Future Price**

$$\begin{aligned} F &= S(1 + r) - D \\ &= ₹ 220(1.0375) - 2.5 \\ &= ₹ 225.75 \end{aligned}$$

(2) Arbitrage

(i) **Action:** Since future is overpriced. Hence, buy spot & sell future.

(ii) **Process:**

Today:



- Borrow ₹ 220 @ 15% p.a. for 3 months & buy spot
- Sell future @ 230

After 3 Months

Cash Flows =	Sell Stock	= +230
	Dividend	= +2.50
	Repay 220(1.0375)	= -228.25
	Arbitrage Gain	= 4.25

Question – 27

The share of X Ltd. is currently selling for ₹ 300. Risk free interest rate is 0.8% per month. A three month futures contract is selling for ₹ 312. Develop an arbitrage strategy and show what your riskless profit will be 3 months hence assuming that X Ltd. will not pay any dividend in the next three months.

(SM TYK – 06)**Solution:****(I) Theoretical Future Price**

$$\begin{aligned} F &= S(1 + r) \\ &= 300 \times (1.008)^3 \\ &= ₹ 307.26 \end{aligned}$$

(II) Arbitrage

(i) **Action:** Since future is overpriced, hence buy spot & sell future.



(ii) Process

Today

- Borrow ₹ 300 @ 0.8% p.m. & buy share
- Sell future at ₹ 312

After 3 Months

Sell Stock = + 312

Repay $300(1.008)^3$ = 307.27

Arbitrage Gain = 4.74

Note: यदि rate per month दिया है तो monthly compounding लेना है।

Question – 28

The 6-months forward price of a security is ₹ 208.18. The borrowing rate is 8% per annum payable with monthly rests. What should be the spot price?

(SM TYK – 01)

Solution:

Forward Price = $S(1 + r)^n$

208.18 = $S(1.0067)^6$

S = ₹ 200

Question – 29

On 31-8-2011, the value of stock index was ₹ 2,200. The risk free rate of return has been 8% per annum. The dividend yield on this Stock Index is as under:

Month	Dividend Paid p.a.
January	3%
February	4%
March	3%
April	3%
May	4%
June	3%
July	3%
August	4%
September	3%
October	3%
November	4%
December	3%

Assuming that interest is continuously compounded daily, find out the future price of contract deliverable on 31-12-2011. Given: $e^{0.01583} = 1.01593$

(SM TYK – 03)

Solution:

Average Dividend Yield

$$= \frac{3 + 3 + 4 + 3}{4}$$

= 3.25% p.a.

$$F = S \times e^{(r - d)t}$$

$$= 2,200 \times e^{(0.08 - 0.0325)4/12}$$



$$= 2,200 \times e^{0.01583}$$

$$= 2,200 \times 1.01593$$

$$= 2,235.046$$

Question – 30

The NSE-50 Index futures are traded with rupee value being ₹100 per index point. On 15th September, the index closed at 1195, and December futures (last trading day December 15) were trading at 1225. The historical dividend yield on the index has been 3% per annum and the borrowing rate was 9.5% per annum.

- Determine whether on September 15, the December futures were under-priced or overpriced?
- What arbitrage transaction is possible to gain out this mispricing?
- Calculate the gains and losses if the index on 15th December closes at (a) 1260 (b) 1175.

Assume 365 days in a year for your calculations

(Exam November – 2019) (8 Marks)

Solution:**(I) Theoretical Future Price**

$$\text{Future Price} = S \times [1 + (r - d)t]$$

$$= 1,195 \left[1 + (0.095 - 0.03) \times \frac{91}{365} \right]$$

$$= 1,214.37$$

$$\text{Future Price} = 1,214.37 \times 100 = 1,21,437$$

$$\text{Actual Future Price} = 1,225 \times 100 = 1,22,500$$

(II) Action

- Since actual future price is more than Theoretical Future Price hence future is overpriced.
- Since future is overpriced, hence buy spot & sell future [short position]

(III) Arbitrage Process

- Borrow ₹ (1,195 × 100) @ 9.5% p.a. for 91 days & buy index

- Sell future (short position) at 1,225 × 100

After 91 Days

	1260	1175
Sell Index	(1,260 × 100)	(1,175 × 100)
Dividend Received	+ 1,26,000	+ 1,17,500
$\left[1,195 \times 3\% \times \frac{91}{365} \right] \times 100$	+ 893.79	+ 893.79
Repayment	- 1,22,330.35	- 1,22,330.35
$1,195 \left[1 + \left(0.095 \times \frac{91}{365} \right) \right] \times 100$	(-35 × 100)	(+50 × 100)
Gain/Loss on short position	= -3,500	= +5,000
Arbitrage	1,063.44	1,063.44



Question – 31

Suppose current price of an index is ₹ 13,800 and yield on index is 4.8% (p.a.). A 6 months future contract on index is trading at ₹ 14,340.

Assuming that risk free rate of interest is 12%. Show Mr. X (an arbitrageur) can earn an abnormal rate of return irrespective of outcome after 6 months. You can assume that after 6 months index closes at ₹ 10,200 and ₹ 15,600 and 50% of stock included in index shall pay dividend in next 6 months. Also Calculate implied risk free rate.

Solution:**Theoretical Future Price**

$$\begin{aligned}
 F &= 13,800(1.06) \\
 &= 13,800 \times 50\% \times 4.8\% \\
 &= 14,628 - 331.20 \\
 &= 14,296.80
 \end{aligned}$$

Since future is overpriced hence buy spot & sell future (short position)

After 6 Months **14340**

	10,200	15,600
Sell Index	+ 10,200	+ 15,600
Dividend	+ 331.20	+ 331.20
Buy Index (Spot)	- 13,800	- 13,800
Gain/Loss on Short Position	+ 4,140	- 1,260
Arbitrage Gain	871.20	871.20

$$\begin{aligned}
 \text{Implied Risk Free Rate} &= \frac{871.20}{13,800} \times 100 \times \frac{12}{6} \\
 &= 12.63\% \text{ p.a.}
 \end{aligned}$$

Note: जब भी Implied Risk Free Rate मांगेगा Borrowing नहीं लेना है।

Question – 32

A future contract is available on R Ltd. that pays an annual dividend of ₹4 and whose stock is currently priced at ₹125. Each future contract calls for delivery of 1,000 shares to stock in one year, daily marking to market. The corporate treasury bill rate is 8%.

Required:

- Given the above information, what should the price of one future contract be?
- If the company stock price decreases by 6%, what will be the price of one futures contract?
- As a result of the company stock price decrease, will an investor that has a long position in one futures contract of R Ltd. realizes a gain or loss ? What will be the amount of his gain or loss?

(Ignore margin and taxation, if any)

(Exam Nov – 2019) (6 Marks)

Solution:**(i) Price of one future contract**

$$\begin{aligned}
 F &= S (1 + r) - D \\
 &= ₹ 125 (1.08) - 4
 \end{aligned}$$

$$= ₹ 131$$

$$= 131 \times 1,000 \text{ shares}$$

$$= ₹ 1,31,000$$

(ii) If stock price decrease by 6%

$$S = 125 \times 0.94 = 117.50$$

$$F = 117.50 (1.08) - 4$$

$$= ₹ 122.90$$

$$= ₹ 122.90 \times 1,000 \text{ shares}$$

$$= ₹ 1,22,900$$

(iii) Gain or Loss on Long Position

If stock price decrease then loss on long position

$$(1,31,000 - 1,22,900) = 8,100 \text{ (Loss)}$$

Question – 33

The price of ACC stock on 31 December 2010 was ₹ 220 and the futures price on the same stock on the same date, i.e., 31 December 2010 for March 2011 was ₹ 230. Other features of the contract and related information are as follows:

Time to expiration	- 3 months (0.25 year)
Borrowing rate	- 15% p.a.
Annual dividend on the stock	- 25% payable before 31.03.2011
Face Value of the Stock	- ₹ 10

- Based on the above information, what should be the futures price?
- Show the process of arbitrage

Solution:

- Futures price = $220 + (220 \times 0.15 \times 0.25) - (0.25 \times 10) = 225.75$
- He will buy the ACC stock at ₹ 220 by borrowing the amount @ 15 % for a period of 3 months and at the same time sell the March 2011 futures on ACC stock. By 31st March 2011, he will receive the dividend of ₹ 2.50 per share. On the expiry date of 31st March, he will deliver the ACC stock against the March futures contract sales.

The arbitrager's inflows/outflows are as follows:

Sale proceeds of March 2011 futures	₹ 230.00
Dividend	₹ 2.50
Total (A)	₹ 232.50
Pays back the Bank	₹ 220.00
Cost of borrowing	₹ 8.25
Total (B)	₹ 228.25
Balance (A) – (B)	₹ 4.25

Thus, the arbitrage earns ₹ 4.25 per share without involving any risk.



(VII) BETA MANAGEMENT**Question – 34**

On April 1, 2015, an investor has a portfolio consisting of eight securities as shown below:

Security	Market Price	No. of Shares	Value
A	29.40	400	0.59
B	318.70	800	1.32
C	660.20	150	0.87
D	5.20	300	0.35
E	281.90	400	1.16
F	275.40	750	1.24
G	514.60	300	1.05
H	170.50	900	0.76

The cost of capital for the investor is 20% p.a. continuously compounded. The investor fears a fall in the prices of the shares in the near future. Accordingly, he approaches you for the advice to protect the interest of his portfolio.

You can make use of the following information:

- (1) The current NIFTY value is 8500.
- (2) NIFTY futures can be traded in units of 25 only.
- (3) Futures for May are currently quoted at 8700 and Futures for June are being quoted at 8850.

You are required to calculate:

- (i) The beta of his portfolio.
- (ii) The theoretical value of the futures contract for contracts expiring in May and June. Given ($e^{0.03} = 1.03045$, $e^{0.04} = 1.04081$, $e^{0.05} = 1.05127$)

- (iii) The number of NIFTY contracts that he would have to sell if he desires to hedge until June in each of the following cases:

- (A) His total portfolio
- (B) 50% of his portfolio
- (C) 120% of his portfolio

(SM TYK – 13)

Solution:**(i) Calculation of Beta of Portfolio**

Stoc ks	Market Price	No. of Sha res	Value	Weight	Beta	W × B
A	29.40	400	11,760	0.01182	0.59	0.0070
B	318.70	800	2,54,960	0.2564	1.32	0.3384
C	660.20	150	99,030	0.0996	0.87	0.0866
D	5.20	300	1,560	0.00157	0.35	0.0005
E	281.90	400	1,12,760	0.1134	1.16	0.1315
F	275.40	750	2,06,550	0.2077	1.24	0.2575
G	514.60	300	1,54,380	0.15524	1.05	0.1630
H	170.50	900	1,53,450	0.1543	0.76	0.1173
			9,94,450		Bp = 1.102	

(ii) Calculation of theoretical future price**May future (2 months)**

$$\begin{aligned}
 F &= S e^{rt} \\
 &= 8,500 \times e^{0.20 \times 2/12} \\
 &= 8,500 \times e^{0.0333}
 \end{aligned}$$



$$= 8,500 \times 1.03387$$

$$= 8,787.89$$

Interpolation

$e^{0.03}$	1.03045
$e^{0.0333}$	
$e^{0.04}$	1.04081
$e^{0.01}$	0.01036
$1.03045 + \frac{0.01036}{0.01} \times 0.0033$	
$= 1.03387$	

June Future

$$F = 8,500 \times e^{(0.20 \times 3/12)}$$

$$= 8,500 \times 1.05127$$

$$= ₹ 8,935.79$$

(iii) Calculation of no. of contacts

$$\text{No. of Contract} = \frac{V_P \times (B_T - B_P)}{F \times M}$$

(A) Total portfolio

$$\text{No of contracts} = \frac{9,94,450 \times (0 - 1.102)}{8,850 \times 25}$$

$$= 4.95 \text{ Contracts}$$

5 contracts sold

(B) 50% of Portfolio

$$\text{No of contracts} = \frac{9,94,450 \times 50\% \times (0 - 1.102)}{8850 \times 25}$$

$$= 2.48 \text{ Contracts}$$

2 contracts sold or 3 contracts

(C) 120% of Portfolio

$$\text{No of contracts} = \frac{(9,94,450 \times 120\%) \times (0 - 1.102)}{8850 \times 25}$$

$$= 5.94 \text{ Contracts}$$

6 contracts sold.

Question – 35

Details about portfolio of shares of an investor is as below:

Shares	No. of shares (lakh)	Price per share	Beta
A.Ltd.	3.00	₹ 500	1.40
B Ltd.	4.00	₹ 750	1.20
C Ltd.	2.00	₹ 250	1.60

The investor thinks that the risk of portfolio is very high and wants to reduce the portfolio beta to 0.91. He is considering two below mentioned alternative strategies:

- (i) Dispose off a part of his existing portfolio to acquire risk free securities, or

- (ii) Take appropriate position on Nifty Futures which are currently traded at 8125 and each Nifty points is worth ₹ 200.

You are required to determine:

- (1) Portfolio beta,
- (2) The value of risk free securities to be acquired,
- (3) The number of shares of each company to be disposed off,
- (4) The number of Nifty contracts to be bought/sold; and
- (5) The value of portfolio beta for 2% rise in Nifty.

(SM TYK – 12)

Solution:

1. Beta of Portfolio

Shares	MPS	No.	Amount	Weights	Beta	$\beta \times w$
A	3.00	500	1,500	0.3	1.40	0.42
B	4.00	750	3,000	0.6	1.20	0.72
C	2.00	250	500	0.1	1.60	0.16
		V _p =	5,000		B _p =	1.30

2. Value of Risk Free Securities

$$0.91 = \frac{(5,000 \times 1.30) - (x \times 1.30) + (x \times 0)}{5,000}$$

$$x = 1,500$$

ICAI

$$W_p = \frac{B_T}{B_P} = \frac{0.91}{1.30} = 0.70$$

$$R_f = 5,000 \times 0.3 = 1,500$$

$$\text{Investment in } R_f = ₹ 1,500$$

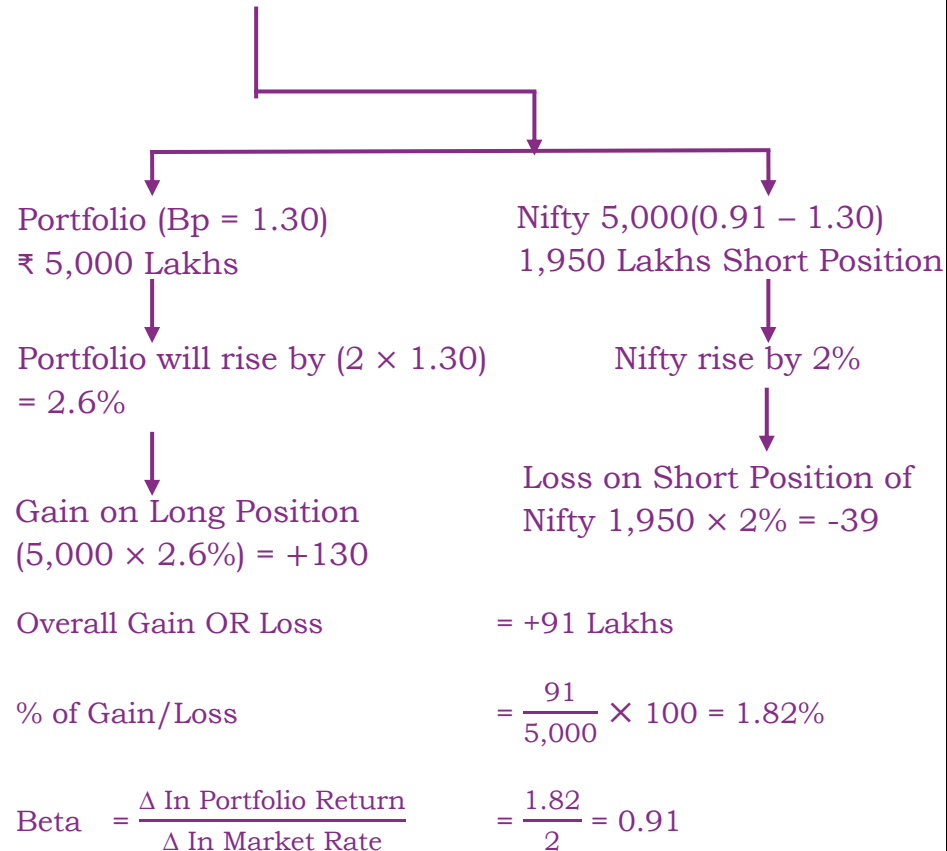
3. No. of Shares

Shares	Weights	Amount (Lakh)	MPS	No. of Shares in (Lakh)
A	0.3	450	500	0.9
B	0.6	900	750	1.20
C	0.1	150	250	0.6
		1,500		

4. No. of Nifty Contracts

$$\begin{aligned} \text{No.} &= \frac{V_P \times (B_T - B_P)}{F \times M} \\ &= \frac{5,000 \times (0.91 - 1.30)}{8,125 \times 200} \\ &= 120 \text{ Contracts Short} \end{aligned}$$

5. Value of Portfolio Beta



Question – 36

On January 1, 2013 an investor has a portfolio of 5 shares as given below:

Security	Price	No. of Shares	Beta
A	349.30	5,000	1.15
B	480.50	7,000	0.40
C	593.52	8,000	0.90
D	734.70	10,000	0.95
E	824.85	2,000	0.85

The cost of capital to the investor is 10.5% per annum.

You are required to calculate:

- The beta of his portfolio.
- The theoretical value of the NIFTY futures for February 2013.
- The number of contracts of NIFTY the investor needs to sell to get a full hedge until February for his portfolio if the current value of NIFTY is 5900 and NIFTY futures have a minimum trade lot requirement of 200 units. Assume that the futures are trading at their fair value.
- The number of future contracts the investor should trade if he desires to reduce the beta of his portfolios to 0.6.

No. of days in a year be treated as 365.

Given: $\ln(1.105) = 0.0998$ and $e^{(0.015858)} = 1.01598$

(SM TYK – 11)

Solution:

1. Beta of Portfolio

Shares	MPS	No.	Amount	Weights	Beta	$\beta \times w$
A	349.30	5,000	17,46,500	0.093	1.15	0.107
B	480.50	7,000	33,63,500	0.178	0.40	0.071
C	593.52	8,000	47,48,160	0.252	0.90	0.227
D	734.70	10,000	73,47,000	0.390	0.95	0.370
E	824.85	2,000	16,49,700	0.087	0.85	0.074
		Vp =	1,88,54,860		Bp =	0.849



2. Theoretical Future

$$\begin{aligned}
 F &= S \times e^{rt} \\
 &= 5,900 \times e^{0.105 \times 58/365} \\
 &= 5,900 \times e^{0.01668} \\
 &= 5,900 \times 1.01682 \\
 &= 5,999.24
 \end{aligned}$$

3. No. of Contracts ($B_T = 0$)

$$\begin{aligned}
 \text{No.} &= \frac{V_P \times (B_T - B_P)}{F \times M} \\
 &= \frac{1,88,54,860 \times (0 - 0.849)}{5,999.24 \times 200} \\
 &= 13.35 \text{ or 13 Contracts Short}
 \end{aligned}$$

4. $B_T = 0.6$

$$\begin{aligned}
 \text{No.} &= \frac{1,88,54,860 \times (0.6 - 0.849)}{5,999.24 \times 200} \\
 &= 3.91 \text{ or 4 Contracts Short}
 \end{aligned}$$

Question – 37

Following information is available for consideration:

BSE Index	25,000
Value of portfolio	₹ 50,50,000

Risk free interest rate 9% p.a.

Dividend yield on Index 6% p.a.

Beta of portfolio 1.5

We assume that a future contract on the BSE index with 4 months maturity is used to hedge the value of portfolio over next 3 months. One future contract is for delivery of 50 times the index.

Based on the above information calculate:

- Price of future contract.
- Gain on short futures position if index turns out to be 22,500 in 3 months.

Note: Daily compounding (exponential) formula is not required to be used.

(RTP May – 2022, Exam July – 2021) (8 Marks)

Solution:**(i) Price of Future Contracts**

$$\begin{aligned}
 F &= S [1 + (r - d)t] \\
 &= 25,000 [1 + (0.09 - 0.06)4/12] \\
 &= ₹ 25,250
 \end{aligned}$$

Price of 1 Future Contract = ₹ 25,250 × 50

$$= ₹ 12,62,500$$

$$\begin{aligned}
 \text{No. of Contracts} &= \frac{V_p(B_T - B_p)}{F \times M} \\
 &= \frac{50,50,000 \times (0 - 1.5)}{25,250 \times 50} \\
 &= 6 \text{ Contracts Short}
 \end{aligned}$$

(ii) Gain or Loss on Short Position

$$3 \text{ months spot} = 22,500$$

[हमने Contract 4 month future पर किया है तो, 3 month के End पर काटना है Spot पर नहीं कटेगा, ये Contract 1 month future पर सौदा कटेगा]

$$\begin{aligned}
 F &= 22,500 \left[1 + (0.09 - 0.06) \frac{1}{12} \right] \\
 &= 22,556.25
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain on Short Position} &= (25,250 - 22,556.25) \times 50 \times 6 \\
 &= ₹ 8,08,125
 \end{aligned}$$

Question – 38

Mr. X, is a Senior Portfolio Manager at ABC Asset Management Company. He expects to purchase a portfolio of shares in 90 days. However, he is worried about the expected price increase in shares in coming day and to hedge against this potential price increase he decides to take a position on a 90-day forward contract on the Index. The index is currently trading at 2290. Assuming that the continuously compounded dividend yield on the same index is

1.75% and risk-free rate of interest is 4.16%, you are required to determine:

- The justified forward price on this contract.
- The position Mr. X should take in forward contract on the Index.
- Gain/ loss on the position taken if after 28 days of the purchase of the contract the Index value stands at 2450.
- Gain/ loss on the position taken if at expiration of 90 days the Index Value is 2470.

Note: Take 365 days in a year and value of $e^{0.005942} = 1.005960$, $e^{0.001849} = 1.001851$.

(RTP May – 2025)**Solution:**

- The Forward Price shall be $= S_0 e^{n(r - y)}$

Where S_0 = Spot price
 n = period
 r = risk free rate of interest
 y = dividend yield

Accordingly,

$$\begin{aligned}
 \text{Forward Price} &= 2290 e^{90/365(0.0416 - 0.0175)} \\
 &= 2290 e^{0.005942} \\
 &= 2290(1.005960)
 \end{aligned}$$



$$= 2303.65$$

(b) Mr. X shall take long position in the Forward Contract on Index.

(c) Gain/loss on Long Position after 28 days

$$= 2450 - 2290 e^{(0.0416 - 0.0175)28/365}$$

$$= 2450 - 2290 e^{0.001849}$$

$$= 2450 - 2290(1.001851)$$

$$= 2450 - 2294.24$$

$$= 155.76$$

(d) Gain/loss on Long Position at maturity

$$= S_n - S_0 e^{n(r - y)}$$

$$= 2470.00 - 2303.65$$

$$= 166.35$$

Question – 39

A Future contract on BSE Index with 4 months maturity is used to hedge the value of the portfolio over the next 3 months. One future contract for delivery is 50 times of the index.

The following information is available:

Value of the portfolio ₹ 1,16,00,000

BSE Sensex on 1st January 2022 58,580

(Anticipated on 1st September 2021)

BSE Sensex on 1st January 2022 56641.25

(Anticipated on 1st December 2021)

Dividend Yield of Index 6% p.a

181 day's treasury bills offers a rate of interest 9% p.a.

Beta of the portfolio 1.5

You are required to calculate

(i) The present value of the Sensex as on 1st September 2021

(ii) Turned out value of the Sensex on 1st December 2021

(iii) The number of contracts to hedge the portfolio.

(Exam December – 2021) (8 Marks)

Solution:

(I) Spot (01/09/2021)

$$F = S [1 + (r - d)t]$$

$$₹ 58,580 = S [1 + (0.09 - 0.06)4/12]$$

$$S = ₹ 58,000$$

(II) Spot Rate (01/12/2021)

$$56,641.25 = S [1 + (0.09 - 0.06) \times 1/12]$$

$$S = ₹ 56,500$$



(III) No. of Contracts

$$\begin{aligned}\text{No. of Contracts} &= \frac{V_P(B_T - B_P)}{F \times M} \\ &= \frac{1,16,00,000 (0 - 1.50)}{58,580 \times 50} \\ &= 6 \text{ Contracts Short}\end{aligned}$$

Question – 40

Mr. X is having a portfolio of shares worth ₹ 170 lakhs at current price and cash ₹ 30 lakhs. The beta of share portfolio is 1.6. After 3 months the price of shares dropped by 3.2%.

Determine:

- Current portfolio beta.
- Portfolio beta after 3 months if Mr. X on current date goes for long position on ₹ 200 lakhs Nifty futures.

(Exam July – 2021) (8 Marks)

Solution:

(I) Beta of Portfolio

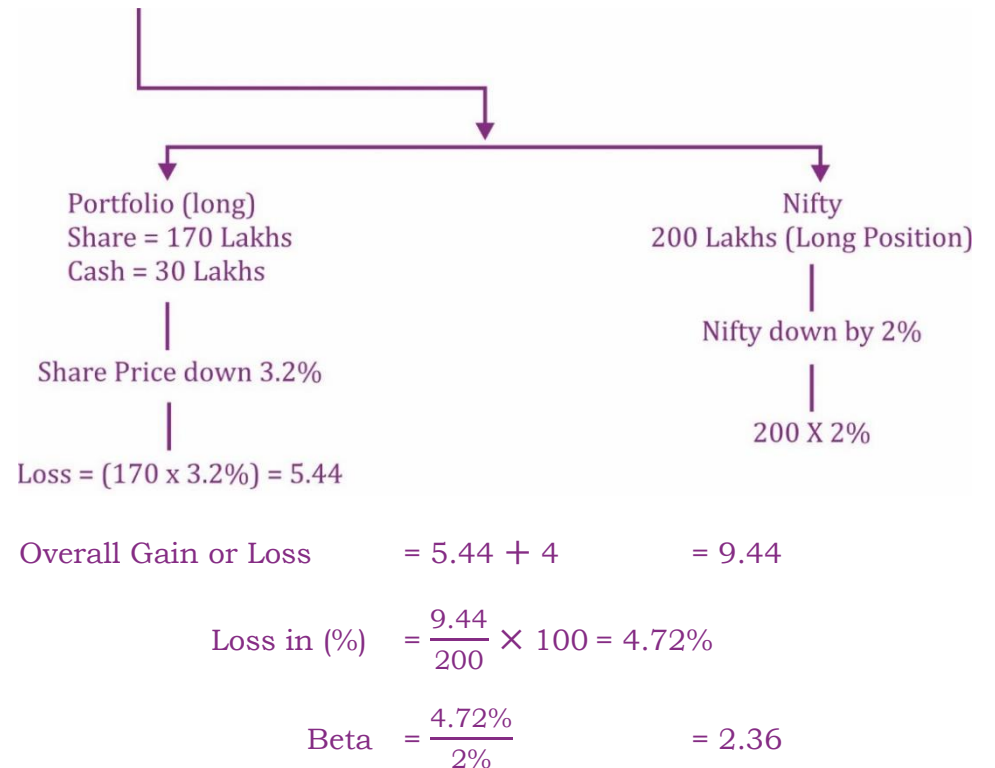
$$\begin{aligned}B_P &= \frac{(170 \times 1.60) + (30 \times 0)}{200} \\ &= 1.36\end{aligned}$$

(II) Beta After 3 Months

Price of shares dropped by 3.2%, It means Nifty dropped by

$$\beta = \frac{\Delta \text{ Stock}}{\Delta \text{ Market}} = 1.6 = \frac{3.2}{\Delta \text{ Market}} = 2\%$$

Gain or Loss



Question – 41

Mr. SG sold five 4-Month Nifty Futures on 1st February 2020 for ₹ 9,00,000. At the time of closing of trading on the last Thursday of May 2020 (expiry), Index turned out to be 2100. The contract multiplier is 75.

Based on the above information calculate:

- (i) The price of one Future Contract on 1st February 2020.
- (ii) Approximate Nifty Sensex on 1st February 2020 if the Price of Future Contract on same date was theoretically correct. On the same day Risk Free Rate of Interest and Dividend Yield on Index was 9% and 6% p.a. respectively.
- (iii) The maximum Contango/Backwardation.
- (iv) The pay-off of the transaction.

Note: Carry out calculation on month basis.

(RTP November – 2020)

Solution:

(i) Price of One Future Contract

$$\text{Price} = \frac{\text{₹ } 9,00,000}{5} = \text{₹ } 1,80,000$$

$$F = \frac{\text{₹ } 1,80,000}{75} = 2,400$$

(ii) Spot Price

$$F = S [1 + (r - d)t]$$

$$2,400 = S [1 + (0.09 - 0.06)4/12]$$

$$S = 2,376$$

(iii) Contango/Backwardation

Generally, Future is more than Spot, if Future is more than Spot it is called “सीधा बदला” [Contango]. If $S > F$, उल्टा बदला [Backwardation]

$$\begin{aligned} \text{Basis} &= S - F \\ \text{Negative} &\rightarrow \text{Contango} \\ \text{Positive} &\rightarrow \text{Backwardation} \\ \text{Maximum Contango} &= (2,376 - 2,400) \\ &= \text{₹ } 24 \end{aligned}$$

(iv) Gain or Loss

$$\begin{aligned} \text{Gain on Short Position} &= (2,400 - 2,100) \times 75 \times 5 \\ &= \text{₹ } 1,12,500 \end{aligned}$$

Question – 42

A Mutual Fund is holding the following assets in ₹ Crores :

Investments in diversified equity shares	90.00
Cash and Bank Balances	<u>10.00</u>
	100.00

The Beta of the equity shares portfolio is 1.1. The index future is selling at 4300 level. The Fund Manager apprehends that the index will fall at the most by 10%. How many index futures he should short for perfect hedging? One index future consists of 50 units.

Substantiate your answer assuming the Fund Manager's apprehension will materialize.



(SM TYK – 07)**Solution:**

Note: In this question, ICAI assumed that cash is not part of the portfolio.

No. of Contracts

$$= \frac{V_p(B_T - B_P)}{F \times M \times B_F}$$

$$= \frac{₹ 90 \text{ Cr. } (0 - 1.10)}{4,300 \times 50 \times 1} = 4,605 \text{ Contracts Short}$$

Substantiate

If Nifty down by 10%

Loss on Equity	(90 × 10% × 1.1)	9.90 Cr.
Gain on Nifty	(4,300 × 50 × 4,605) × 10%	9.90 Cr.
		<u>0</u>

Question – 43

Shyam buys 10,000 shares of X Ltd., @ ₹ 25 per share and obtains a complete hedge of shorting 400 Nifty at ₹ 1,100 each. He closes out his position at the closing price of the next day when the share of X Ltd., has fallen by 4% and Nifty Future has dropped by 2.5%.

What is the overall profit or loss from this set of transaction?

(Exam January – 2021)**Solution:****Today Investment**

Long Position in X Ltd. (10,000 × 25)	= 2,50,000
Short Position in Nifty (400 × 1,100)	= 4,40,000

Calculation of Profit/Loss

Loss on Long Position of X Ltd (2,50,000 × 4%)	= 10,000
Gain on Short Position of Nifty (4,40,000 × 2.5%)	= 11,000
Overall Gain	<u>= 1,000</u>

Alternative [ICAI]**Cash Flow Today (Initial Cashflows)**

Buy X Ltd share	(2,50,000)
Sell Nifty	<u>+ 4,40,000</u>
Net Cash Inflow	<u>1,90,000</u>

Cash Flow Next Day (Cashflow at Closes Out)

Sell X Ltd. (10,000 × 25 × 96%)	+ 2,40,000
Buy Nifty (4,40,000 × 97.5%)	<u>(4,29,000)</u>
Net Cash Outflow	<u>1,89,000</u>
Gain or Loss = 1,90,000 – 1,89,000	= 1,000 Profit.



Question – 44

Which position on the index future gives a speculator, a complete hedge against the following transactions:

- The share of Right Limited is going to rise. He has a long position on the cash market of ₹ 50 lakhs on the Right Limited. The beta of the Right Limited is 1.25.
- The share of Wrong Limited is going to depreciate. He has a short position on the cash market of ₹ 25 lakhs on the Wrong Limited. The beta of the Wrong Limited is 0.90.
- The share of Fair Limited is going to stagnant. He has a short position on the cash market of ₹ 20 lakhs of the Fair Limited. The beta of the Fair Limited is 0.75.

(SM TYK – 09)**Solution:****Statement Showing Position in Nifty**

Company	Position in Cash Market	Amount	Beta	Position in Future
Right Ltd.	Long	50,00,000	1.25	62,50,000 Short
Wrong Ltd.	Short	25,00,000	0.9	22,50,000 Long
Fair Ltd.	Short	20,00,000	0.75	15,00,000 Long
Net Position in Future				25,00,000 Short

Question – 45

Mr. SK buys 5,000 shares of P Ltd. @ ₹ 50 per share whose beta value is 1.5 and sells 2,500 shares of Q Ltd. @ ₹ 80 per share having a beta value of 2. He obtains a complete hedge by Nifty Futures at ₹ 1,000 each.

He closes out his position at the closing price of the next day when the share of P Ltd. has fallen by 2%, share of Q Ltd. appreciated by 5% and Nifty Futures has dropped by 2.50%.

You are required to compute the overall profit or loss to Mr. SK from these set of transactions.

(Exam September – 2025) (4 Marks)**Solution:**

No. of the Future Contract to be obtained to get a complete hedge

$$= \frac{5,000 \times 50 \times 1.50 - 2,500 \times 80 \times 2}{1,000}$$

$$= \frac{3,75,000 - 4,00,000}{1,000} = - 25 \text{ Contracts or } 25 \text{ contracts}$$

Thus, by purchasing 25 Nifty Futures contracts Mr. SK can obtain a complete hedge.

Cash Outlay

$$= 5,000 \times ₹ 50 - 2,500 \times ₹ 80 + 25 \times ₹ 1,000$$

$$= ₹ 2,50,000 - ₹ 2,00,000 + ₹ 25,000$$

$$= ₹ 75,000$$

Cash Inflow at Close Out

$$\begin{aligned}
 &= 5,000 \times ₹ 50 \times 0.98 - 2500 \times ₹ 80 \times 1.05 + 25 \times ₹ 1,000 \\
 &\quad \times 0.975 \\
 &= ₹ 2,45,000 - ₹ 2,10,000 + ₹ 24,375 \\
 &= ₹ 59,375 \text{ Gain/ Loss} = ₹ 59,375 - ₹ 75,000 \\
 &= - ₹ 15,625 \text{ (Loss)}
 \end{aligned}$$

Question – 46

Ram buys 10,000 shares of X Ltd. at a price of ₹ 22 per share whose beta value is 1.5 and sells 5,000 shares of A Ltd. at a price of ₹ 40 per share having a beta value of 2. He obtains a complete hedge by Nifty futures at ₹ 1,000 each. He closes out his position at the closing price of the next day when the share of X Ltd. dropped by 2%, share of A Ltd. appreciated by 3% and Nifty futures dropped by 1.5%.

What is the overall profit/loss to Ram?

(SM TYK – 10)**Solution:****Statement Showing Net Position in Nifty**

Company	Position in Cash Market	Amount	Beta	Position in Future
X Ltd.	Long	2,20,000	1.5	3,30,000 Short
A Ltd.	Short	2,00,000	2	4,00,000 Long
Net Position in Nifty				70,000 Long

$$\text{No. of Contract} = \frac{70,000}{1,000} = 70 \text{ Contracts Long}$$

Overall Profit/Loss

Loss in Long Position of X Ltd. (2,20,000 × 2%)	= 4,400
Loss on Short Position of A Ltd. (2,00,000 × 3%)	= 6,000
Loss on Long Position of Nifty (70,000 × 1.5%)	= 1,050
Overall Loss	= 11,450

Question – 47

Mohan buys 10,000 shares of X Ltd. @ ₹ 25 per share whose beta value is 1.5 and sells 5,000 shares of A Ltd. @ ₹ 40 per share having a beta value of 2. He obtains a complete hedge by buying 25 Nifty Futures. He closes out his position at the closing price of the next day when the share of X Ltd. has fallen by 4% and Nifty Futures has dropped by 2.50%. In the process he suffered a loss of ₹ 16,625.

You are required to determine

- The value of the Nifty future
- Initial cash outlay
- Cash inflow at the close out
- Percentage Gain/loss to Shares of A Ltd. at the time of closure

(Exam November – 2024) (4 Marks)

Solution:

- (i) Let N be the value of Nifty Future Contract then:

$$\frac{10,000 \times 25 \times 1.50 - 5,000 \times 40 \times 2}{N} = 25$$

$$N = -1,000$$

Since Mohan has bought Nifty Futures the above value shall be considered as positive i.e. value of per one Nifty Future is ₹ 1,000.

Alternatively, it can also be computed as follows:

$$\frac{5,000 \times 40 \times 2 - 10,000 \times 25 \times 1.50}{N} = 25$$

$$N = 1,000$$

Accordingly, the value of Nifty Future is ₹ 1,000.

- (ii) Initial Cash Outlay

$$= 10,000 \times ₹ 25 + 1,000 \times ₹ 25 - 5,000 \times ₹ 40$$

$$= ₹ 2,50,000 + ₹ 25,000 - ₹ 2,00,000$$

$$= ₹ 75,000$$

- (iii) Cash inflow at the closeout

$$= ₹ 75,000 - ₹ 16,625 = ₹ 58,375$$

- (iv) Percentage Gain/loss to Shares of A Ltd. at the time of closure

Let the amount realized from the sale of share of A Ltd. is A.

Accordingly, next day at the time of closing out the position will be as follows:

$$10,000 \times ₹ 25 (1 - 0.04) + 25 \times ₹ 1,000 \times (1 - 0.025) - A = ₹ 58,375$$

$$A = ₹ 2,06,000$$

Thus, percentage of loss to shares of A Ltd.

$$= \frac{2,06,000 - 2,00,000}{2,00,000}$$

$$= 0.03 \text{ i.e. } 3\%$$

Alternative Presentation-

Percentage Gain / Loss to shares of A Ltd. at the time of closure:

Loss suffered by Mohan when he closes out his position at the closing price of the next day		₹ 16,625
Less:		
a. Loss suffered in purchase of shares of X Ltd.	10,000	₹ 10,625
(10,000 × 25 × 0.04)		
b. Loss suffered in Nifty Futures	625	
(1,000 × 25 × 0.025)		
Loss suffered in sale of shares of A Ltd.		₹ 6,000

Thus, percentage of loss to shares of A Ltd. = $(6,000 / 2,00,000) \times 100 = 0.03 \text{ i.e. } 3\%$.



Question – 48

On 1 April 2015, Sunidhi was holding a portfolio of 10 securities whose value was ₹ 9,94,450, the weighted average of beta of 9 securities was 1.10.

Since she was expecting a fall in the prices of the shares in near future to hedge her portfolio she sold 5 contract of NIFTY Futures (Multiplier of 25) expiring in May 2015, which was trading at 8767.07 on 1 April.

- (a) Calculate the beta of the 10th security.
- (b) Reconcile the reasons in spite of 2% fall in the market as per Sunidhi's apprehension if she would have earned some profit on her cash position.

(MTP March – 2018)

Solution:

- (i) To compute the beta of 10th security first we shall compute overall weighted beta as follows:

Let weighted β be w, then

$$5 = \frac{9,94,450}{8,767.07 \times 25} \times w$$

$w = 1.102$ approximately

Let beta of 10th security is β then,

$$1.102 = 0.90 \times 1.10 + 0.10 \times \beta$$

$$\beta = 1.12$$

- (ii) The main reason for the profit in cash position might due to reason that contrary to her expectation fall in the value of cash position there may be increase in value of cash position.

Question – 49

A speculator purchases BFL Ltd. May Futures (lot of 125 shares) at 7,750 and chooses to Write BFL 7,790 May call option with a premium of ₹ 30 (lot of 125 shares). As on May 18, spot prices rise and so the futures price and call premium. Futures price rise to 7,780. Call premium also rises to ₹ 36. Brokerage for the transaction is 0.02% for the transaction value of futures and strike price net of call premium for options.

You are required to calculate:

- (i) Profit/Loss on Futures net of transaction costs.
- (ii) Profit/Loss on options net of transaction costs.
- (iii) Overall profit from both the positions net of costs.
- (iv) Total Brokerage cost.

(Exam Jan – 2026)

Solution:**Working Notes:****(1) Brokerage Payable on Initial Position**

On Futures (7,750 × 125 × 0.02%)	₹ 193.75
On Options (₹ 7,790 – ₹ 30) × 0.02% × 125	₹ 194.00
Total	₹ 387.75

(2) Brokerage Payable on Closing Position

On Futures (7,780 × 125 × 0.02%)	₹ 194.50
On Options (₹ 7,790 – ₹ 36) × 0.02% × 125	₹ 193.85
Total	₹ 388.35

(i) Profit/Loss on Futures (₹)

Selling Futures	7,780
Buying Futures	7,750
Profit per share	30
Total Profit 125 × ₹ 30	3,750.00
Less: Brokerage on Initial Position	193.75
Brokerage on Closing Position	194.50
Net Profit	3,361.75

(ii) Profit/Loss on Options (₹)

Loss on writing Call Option (₹ 36 – ₹ 30)	6
Loss on Call Option (125 × ₹ 6)	750.00
Add: Brokerage Paid on Initial Position	194.00
Brokerage on Closing Position	193.85
Net Loss	1,137.85

(iii) Overall Profit (₹)

Profit/ Loss on Futures (Net Profit)	3,361.75
Less: Profit/Loss on Options (Net Loss)	1,137.85
Overall Profit	2,223.90

(iv) Total Broking Cost (₹)

On Initial Position	387.75
On Options	388.35
Total	776.10

(VIII) COMDITY FUTURE

Question – 50

The following information is available about standard gold.

Spot Price (SP) ₹ 15,600 per 10 gms.

Future Price (FP) ₹ 17,100 for one year future contract

Risk free interest Rate (R)^f 8.5%

Present Value of Storage Cost ₹ 900 per year

From the above information you are requested to calculate the Present Value of Convenience yield (PVC) of the standard gold.

Solution:

Future Price = (Spot Price + Present Value of Storage Cost – Present Value of Convenience Yield) (1 + r)

17,100 = (15,600 + 900 – x) (1.085)

PVCY = 740

Question – 51

A company is long on 10 MT of copper @ ₹ 534 per kg (spot) and intends to remain so for the ensuing quarter. The variance of change in its spot and future prices are 16% and 36% respectively, having correlation coefficient of 0.75. The contract size of one contract is 1,000 kgs.

Required:

- (i) Calculate the Optimal Hedge Ratio for perfect hedging in Future Market.
- (ii) Advice the position to be taken in Future Market for perfect hedging.
- (iii) Determine the number and the amount of the copper futures to achieve a perfect hedge.

(RTP November – 2021)

Solution:

- (i) The optional hedge ratio to minimize the variance of Hedger's position is given by:

$$H = \rho \frac{\sigma_S}{\sigma_F}$$

Where,

σ_S = Standard deviation of ΔS (Change in Spot Prices)

σ_F = Standard deviation of ΔF (Change in Future Prices)

ρ = Coefficient of correlation between ΔS and ΔF

H = Hedge Ratio

ΔS = Change in spot price.

ΔF = Change in Future price.

Accordingly

Standard deviation of $\Delta S = \sqrt{16\%} = 4\%$ and

Standard deviation of $\Delta F = \sqrt{36\%} = 6\%$ and

$$H = 0.75 \times \frac{0.04}{0.06} = 0.5$$

- (ii) Since the company is long position in Spot (Cash) Market it shall take Short Position in Future Market.
- (iii) Since contract size of one contract is 1,000 Kg,

$$\text{No. of contract to be short} = \frac{10,000 \text{ Kgs}}{1,000 \text{ Kgs}} \times 0.50 = 5 \text{ Contracts}$$

$$\text{Amount} = ₹ 5,000 \times 534 = ₹ 26,70,000$$

Question – 52

A Rice Trader has planned to sell 22,000 kg of Rice after 3 months from now. The spot price of the Rice is ₹ 60 per kg and 3 months Future on the same is trading at ₹ 59 per kg. Size of the contract is 1000 kg. The price is expected to fall as low as ₹ 56 per kg, 3 months hence.

Required:

- (i) To interpret the position of trader in the Cash Market.
- (ii) To advise the trader the trader should take in Future Market to mitigate its risk of reduced profit.
- (iii) To demonstrate effective realized price for its sale if he decides to make use of future market and after 3 months, spot price is ₹ 57 per kg and future contract price for closing the contract is ₹ 58 per kg.

(RTP Nov – 2020 & MTP May – 2019)

Solution:

- (1) Rice Trader hold the stock for 3 months & expects that price rise. Hence, he has Long Position in cash market.
- (2) He should take Short Position in future market to hedge risk of price fall.
- (3) Effective Realized Price per kg.

$$\text{Sell (22,000 kg} \times \text{₹ 57)} = \text{₹ 12,54,000}$$

(+) Gain on short position

$$(59 - 58) \times 22,000 = \text{₹ 22,000}$$

$$\text{Total} = \text{₹ 12,76,000}$$

$$(\div) \text{ Quantity} = \text{₹ 22,000}$$

$$\text{Effective Price} = \text{₹ 58/kg}$$

Question – 53

A call option on gold with exercise price ₹ 26,000 per ten gram and three months to expire is being traded at a premium of ₹ 1,010 per ten gram. It is expected that in three months time the spot price might change to ₹ 27,300 or 24,700 per ten gram. At present this option is at-the-money and the rate of interest with simple compounding is 12% per annum. Is the current premium for the option justified?

Evaluate the option and comments.

Solution:**Risk Neutral****Step 1: Risk Neutral Probability**

$$R = 1.03$$

$$u = \frac{27,300}{26,000} = 1.05$$

$$d = \frac{24,700}{26,000} = 0.95$$

$$P = \frac{R - d}{u - d} = \frac{1.03 - 0.95}{1.05 - 0.95} = 0.8$$

Step 2: Value of Call

$$C_0 = \frac{(1,300 \times 0.8) + (0 \times 0.2)}{1.03} = 1,010$$



MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

X and Y are two friends. since Y has earned a profit from trading in financial derivative market, X is also considering speculating on Gamma corporation's share which is currently trading at ₹ 700 per share through taking positions in options on Gama corporation's stock:

- (1) Purchasing one contract of 2-month call option with a premium of ₹ 35 and an exercise price of ₹ 750
- (2) Purchasing one contract of 2-month put option with a premium of ₹ 25 and an exercise price of ₹ 600

After some time, trading in Option market and understanding the nitty-gritties of same, X being CEO in an organization advised his team to implement the concept of financial option in the capital budgeting decision called 'Real option'.

Based on the above information answer the following questions:

- I.** Assuming that the contract size of each option contract is 100 and the price of Gama corporation's share after two month falls to ₹550, the net pay-off of X will be_____

Answer-1 : ₹ 1,000 loss

Answer-2 : ₹ 1,000 profit

Answer-3 : ₹ 3,000 profit

Answer-4: ₹ 3,000 loss

- II.** The per share price of Gama corporation's stock after 2 month at which X shall be at Break even is_____

Answer-1 : ₹ 540

Answer-2 : ₹ 600

Answer-3 : ₹ 625

Answer-4: ₹ 785

(MTP November – 2025)

Answer: Case Scenario – 01

I. Answer-1 : ₹ 1,000 loss

II. Answer-1 : ₹ 540

Case Scenario – 02

You as an investor had purchased a 4-month European Call Option on the equity shares of X Ltd. for ₹ 10, of which the current market price is ₹ 132 per share and the exercise price ₹ 150. You expect the price to range between ₹ 120 to ₹ 190. The expected share price of X Ltd. and related probability is given below:

Expected Price (₹)	120	140	160	180	190
Probability	0.05	0.20	0.50	0.10	0.15

Based on above case scenario answer the following questions:

- I.** Expected price of share of X Ltd. at the end of 4 months shall be.....
- | | |
|--------------|--------------|
| (a) ₹ 160.00 | (b) ₹ 160.50 |
| (c) ₹ 158.00 | (d) ₹ 140.00 |



- II.** Suppose if the exercise price prevails at the end of 4 months the Value of Call Option shall be.....
- (a) ₹ 0 (b) ₹ 18
(c) ₹ 10 (d) ₹ 14
- III.** In case the option is held to its maturity, the expected value of the call option shall be.....
- (a) ₹ 0 (b) ₹ 18
(c) ₹ 10 (d) ₹ 14
- IV.** In the given different scenarios of expected prices of share of X Ltd. at the time of maturity the option shall be in-the-money in scenarios.
- (a) two (b) three
(c) five (d) In none of the scenario
- V.** In the given different scenarios of expected prices of share of X Ltd. at the time of maturity the option shall be at-the-money in scenarios.
- (a) two (b) three
(c) five (d) In none of the scenario

(MTP September – 2024)**Answer: Case Scenario – 02**

- I. (b) ₹ 160.50**
II. (a) ₹ 0
III. (d) ₹ 14
IV. (b) three
V. (d) In none of the scenario

Case Scenario – 03

Suppose you are a risk manager at a financial institution, and your company has loaned a significant amount of ₹ 500 crore to a company X Ltd. for a period of 3 years at 6-month at MCLR plus 200 bps. You are concerned about X Ltd.'s ability to repay the debt due to recent market volatility. To protect your institution from potential default, you decide to purchase a Credit Default Swap (CDS) from ABC Bank Ltd. for same notional amount at a premium quoted at 1% per year through cash settlement

On the respective reset dates for the same period actual MCLR interest rate comes out as follows:

Reset	MCLR
1	9.75%
2	10.00%
3	10.25%
4	10.35%
5	10.50%
6	10.60%

Based on above case scenario answer the following questions:

- I.** The primary purpose of a Credit Default Swap (CDS) is.....
- (a) to increase the value of bonds.
 (b) to protect against default risk of a debt obligation.
 (c) to provide guaranteed profit to the buyer.
 (d) to create a new form of loan.
- II.** Which of the following statements is true about CDS contracts?

(MTP October – 2024 & April & Aug – 2025)**Answer: Case Scenario – 03****I. (b) to protect against default risk of a debt obligation.****II. (c) CDS contracts are private agreements between two parties.****III. (b) International Swap and Derivative Association (ISDA)****IV. (c) ₹ 15 crore****V. (c) ₹ 125 crore****Case Scenario – 04**

Based on the following information, choose the correct answer from the following questions:

Situation	Action	Exercise Price	Premium	Spot Price
I	Exercised	140	20	160
II	Exercised	200	15	175
III	Lapsed	300	25	400

From the information given above, choose the correct answer to the Question no. 10 to 12:

I.In situation 1, the investor's position and amount of profit or loss is:

- (a) Put option and ₹ 20
- (b) Call option and ₹ 0
- (c) Put option and ₹ 0

- (a) CDS contracts cannot be used for speculation.
- (b) CDS contracts are governed by government regulations.
- (c) CDS contracts are private agreements between two parties.
- (d) CDS contracts eliminate all risks for the buyer.

III. Which organization publishes the guidelines and rules for conducting Credit Default Swap transactions?

- (a) Federal Reserve
- (b) International Swap and Derivative Association (ISDA)
- (c) Securities and Exchange Commission (SEC)
- (d) World Trade Organization (WTO)

IV. Assuming no default occurs the total premium your company will pay during the designated loan period shall be.....

- (a) ₹ 5 crore
- (b) ₹ 10 crore
- (c) ₹ 15 crore
- (d) ₹ 30 crore

V. Suppose if the lender defaults somewhere in the beginning of third year of loan (after payment of interest upto 2 years) and the market value of a reference loans falls to 75% of its par value, then ABC Bank will pay your companyin a cash settlement.

- (a) ₹ 15 crore
- (b) ₹ 30 crore
- (c) ₹ 125 crore
- (d) ₹ 500 crore



- (d) Call option and ₹ 20

II. In Situation III, the investor's position and the amount of profit/loss is:

- (a) Put option, ₹ (25)
 (b) Call option, ₹ 75
 (c) Short position, ₹ 100
 (d) Long position, ₹ (100)

III. In situation II, the investor's position and the amount of profit/loss is :

- (a) Put option and ₹ 10
 (b) Call option and ₹ 10
 (c) Put option and ₹ 25
 (d) Call option and ₹ 25

(EXAM NOVEMBER 2024)

Answer: Case Scenario – 04

I. (b) Call option and ₹ 0

II.(a) Put option, ₹ (25)

III.(a) Put option and ₹ 10

Case Scenario – 05

Mr. X an investor buys the stocks of WBL Limited worth ₹ 21,60,000 due to very strong fundamentals. Since last 3 months, the market sentiment is weak and witnessed a significant volatility and considered to remain weak for about the next three months. Keeping in the mind volatility in the market, Mr. X is planning to hedge his portfolio in the future market. The Beta of WBL stock is 1.3 and the current value of NIFTY is 2250 and 3 months future is selling at 2310. The current market price of the WBL stock is ₹ 240. Each Nifty future can be trade in units of 240 only.

Assume there is no transaction cost and M to M Margin.

From the information given above, choose the correct answer to the question no. **12** to **14**:

(EXAM MAY - 2025)

12. Number of future contract to be buy/sell to hedge WBL stock against expected fall in the market (rounded of contracts) –

- (A) Buy 6 future contracts
 (B) Sell 5 future contracts
 (C) Buy 5 future contracts
 (D) Sell 6 future contracts

13. If Nifty index fall by 10% from 2250 to 2025 and WBL stock falls to ₹ 212, what will be net Gain/Loss if portfolio was hedged on Nifty future?

- (A) Net Gain ₹ 3,42,000



- (B) Net Gain ₹ 3,02,000
- (C) Net Gain ₹ 50,100
- (D) Net Gain ₹ 90,000

14. If NIFTY index rises by 6% from 2250 to 2385 and WBL stock rise to ₹ 225, what will be Net Gain/Loss when portfolio was in hedged?

- (A) Net Loss ₹ 90,000
- (B) Net Gain ₹ 45,000
- (C) Net Gain 1,35,000
- (D) Net Gain ₹ 90,000

Answer: Case Scenario – 05

- 12. (B) Sell 5 future contracts**
- 13. (D) Net Gain ₹ 90,000**
- 14. (B) Net Gain ₹ 45,000**

Case Scenario — 06

HHM & Co., is a famous consultant. It has established the following strategy on the Stock of PLS Ltd

- (a) Purchased one 3 – month call option with a premium of ₹ 25 and an exercise price of ₹ 750.

- (b) Purchased one 3 – month put option with a premium of ₹ 8 and an exercise price of ₹ 600.

The stock of PLS Ltd ; is currently selling at ₹680 The option size is 100 shares of PLS Ltd.

From the information given above, choose the correct answer to the following question No. I to III:

(EXAM SEPTEMBER - 2025)

I. What is the Net gain or loss, if the price of the stock rises to ₹ 820?

- (a) Net loss of ₹ 3,700
- (b) Net gain of ₹ 3,700
- (c) Net gain of ₹ 10,300
- (d) Net loss of ₹ 10,300

II. What is the Net gain or loss, if the price of the stock remains at ₹ 680 after 3 months ?

- (a) Net loss of ₹ 3,300
- (b) Net gain of ₹ 3,300
- (c) Net gain of ₹ 6,700
- (d) Net loss of ₹ 6,700

III. What is the Net gain or loss, if the price of the stock falls to ₹ 500 after 3 months ?

- (a) Net loss of ₹ 3,300
- (b) Net gain of ₹ 3,300
- (c) Net gain of ₹ 6,700
- (d) Net loss of ₹ 6,700

Answer: Case Scenario – 06

- I. (b) Net gain of ₹ 3,700**
- II. (a) Net loss of ₹ 3,300**
- III. (c) Net gain of ₹ 6,700**

Case Scenario – 07

Following information is related to actions taken in different situation of financial options:

Situation	Action	Exercise Price	Premium	Spot Price
I	Exercised	140	20	160
II	Exercised	200	15	175
III	Lapsed	300	25	400

From the information given above, choose the correct answer to the following questions:

- I.** In situation III, the investor's position and the amount of profit/loss is.....

- (a) Put option, ₹ (25)
- (b) Call option, ₹ 75
- (c) Short position, ₹ 100
- (d) Long position, ₹ (100)

- II.** In situation I, the investor's position and the amount of profit or loss is.....

- (a) Put option and ₹ 20
- (b) Call option, and ₹ 0
- (c) Put option and ₹ 0
- (d) Call option and ₹ 20

- III.** In situation II, the investor's position and the amount of profit/loss is.....

- (a) Put option and ₹ 10
- (b) Call option, and ₹ 10
- (c) Put option and ₹ 25
- (d) Call option and ₹ 25

(MTP December – 2025)

Answer Case Scenario – 07

- I. (a) Put option, ₹ (25)**



II. (b) Call option, and ₹ 0**III. (a) Put option and ₹ 10****Case Scenario – 08**

A and B are two friends. Since B has earned a lot of profit from trading in financial derivative market, A is also considering speculating on G Ltd.'s shares which is currently trading at ₹ 1,400 per share through taking positions in options in stocks of same company. Accordingly, A took following contract positions in the options on G Ltd.'s stock:

- (i) Selling one contract of 2-month call option with a premium of ₹ 70 and an exercise price of ₹ 1,500.
- (ii) Purchasing one contract of 2-month put option with a premium of ₹ 50 and an exercise price of ₹ 1,200.

From the information given above, choose the correct answer to the following questions:

- I.** Assuming that the contract size of each option contract is 200 and the price of G Ltd.'s share after two months falls to ₹ 1,100, the net pay-off of A will be.....
- (a) ₹ 24,000 loss (b) ₹ 24,000 profit
- (c) ₹ 40,000 profit (d) ₹ 40,000 loss
- II.** The per share price of G Ltd.'s stock after 2 months at which A shall be at Break Even is.....
- (a) ₹ 1,180 (b) ₹ 1,200

(c) ₹ 1,250

(d) ₹ 1,570

- III.** Assuming that the contract size of each option contract is 300 and the price of G Ltd.'s share after two months raises to ₹ 1,600, the net pay-off of A will be.....

- (a) ₹ 24,000 loss (b) ₹ 66,000 loss
- (c) ₹ 24,000 profit (d) ₹ 66,000 profit

Answer Case Scenario – 08

- I. (b) ₹ 24,000 profit**
- II. (a) ₹ 1,180**
- III. (a) ₹ 24,000 loss**

Case Scenario – 09

Following information is related to actions taken in different situation of financial options:

Situation	Action	Exercise Price	Premium	Spot Price
I	Exercised	2,800	400	3,200
II	Exercised	4,000	300	3,500
III	Lapsed	6,000	1,000	8,000

From the information given above, choose the correct answer to the following questions:

- I.** The investor incurs a loss equal to the premium paid in.....
- (a) Situation I



- (b) Situation II
- (c) Situation III
- (d) Situation I and II

II. In which situation, the option is classified as in-the-money.

- (a) Situation I
- (b) Situation II
- (c) Situation III
- (d) Situations I and II

III. In Situation II, the intrinsic value of the option is.....

- (a) ₹ 0
- (b) ₹ 200
- (c) ₹ 300
- (d) ₹ 500

IV. represents a breakeven outcome for the investor.

- (a) Situation I
- (b) Situation II
- (c) Situation III
- (d) Situations I and II

V. The relationship between spot price and exercise price in Situation II indicates.....

- (a) Call option in-the-money

- (b) Call option out-of-the-money
- (c) Put option in-the-money
- (d) Put option out-of-the-money

Answer Case Scenario – 09

- I. (c) Situation III**
- II. (d) Situations I and II**
- III. (d) ₹ 500**
- IV. (a) Situation I**
- V. (c) Put option in-the-money**



Student Notes





Student Notes



Student Notes



04

INTEREST RATE RISK MANAGEMENT

PART 1: FORWARD RATE AGREEMENT (FRA)

Question – 01

M/s. Parker & Co. is contemplating to borrow an amount of ₹ 60 crores for a Period of 3 months in the coming 6 months' time from now. The current rate of interest is 9% p.a., but it may go up in 6 months' time. The company wants to hedge itself against the likely increase in interest rate.

The Company's Bankers quoted an FRA (Forward Rate Agreement) at 9.30% p.a.

What will be the Final settlement amount, if the actual rate of interest after 6 months happens to be

- (i) 9.60% p.a. and
- (ii) 8.80% p.a.?

(Exam May – 2013, SM TYK – 01) (8 Marks)

Solution:

M/s Parker & Co. Buy FRA 6 × 9 at 9.30% because it afraid of interest rate rising

- (i) **Rate of Interest = 9.60%**

FRA bank will pay to Parker & Co.

$$\begin{aligned}
 \text{Net settlement amount} &= \frac{N (RR - FR) Dtm/DY}{1 + (RR \times Dtm/DY)} \\
 &= \frac{₹ 60,00,00,000 \times (9.60\% - 9.30\%) \times 3/12}{1 + (9.60\% \times 3/12)} \\
 &= \frac{₹ 60,00,00,000 \times 0.075\%}{1.024} \\
 &= ₹ 4,39,453
 \end{aligned}$$

- (ii) **If Rate of Interest = 8.80%**

In this situation Parker & Co. will pay to FRA bank

Net settlement amount

$$\begin{aligned}
 &= \frac{₹ 60,00,00,000 \times (9.3\% - 8.80\%) \times 3/12}{1 + (8.80\% \times 3/12)} \\
 &= \frac{₹ 60,00,00,000 \times 0.125\%}{1.022} \\
 &= ₹ 7,33,855
 \end{aligned}$$

Question – 02

P Ltd. is contemplating to borrow an amount of ₹ 50 crores for a period of 3 months in the coming 6 months' time for now. The current rate of interest is 8% per annum but it may go up in 6 months' time.



The company wants to hedge itself against the likely increase in interest rate.

The Company's Bankers quoted an FRA (Forward Rate Agreement) at 8.30% per annum.

Compute the effect of FRA and actual rate of interest cost to the company, if the actual rate of interest during consideration period happens to be

- (i) 8.60% p.a., or
- (ii) 7.80% p.a.

(Show your working on the basis of months)

(Exam November – 2019) (8 Marks)

Solution:

(1) Actual Rate of Interest 8.60%

In this situation FRA bank will pay to P Ltd.

Effect of FRA

$$\begin{aligned}\text{Net settlement amount} &= \frac{(N)(RR - FR) \times (Dtm/Dy)}{1 + (RR \times Dtm/Dy)} \\ &= \frac{50 \text{ Cr.} (8.60\% - 8.30\%) \times 3/12}{1 + (8.60\% \times 3/12)} \\ &= \frac{\text{₹ } 3,75,000}{1.0215} \\ &= \text{₹ } 3,67,107\end{aligned}$$

Effective Rate of Interest

$$\begin{aligned}\text{Interest paid on loan } (50 \text{ Cr} \times 8.6\% \times \frac{3}{12}) \\ = \text{₹ } 1,07,50,000\end{aligned}$$

INTEREST RATE RISK MANAGEMENT

$$\text{Net settlement } (\text{₹ } 3,67,107 \times 1.0215) = \text{₹ } 3,75,000$$

$$\text{Effective Cost} = \text{₹ } 1,03,75,000$$

$$\begin{aligned}\text{Effective rate} &= \frac{1,03,75,000}{50 \text{ Cr.}} \times 100 \times \frac{12}{3} \\ &= 8.30\% \text{ p.a.}\end{aligned}$$

(2) If Rate 7.8%

In this situation P Ltd. will pay to FRA bank

Effect of FRA

$$\begin{aligned}\text{Final settlement} &= \frac{50 \text{ Cr.} (7.8\% - 8.30\%) \times 3/12}{1 + (7.8\% \times 3/12)} \\ &= \frac{\text{₹ } 6,25,000}{1.0195} \\ &= \text{₹ } 6,13,046\end{aligned}$$

Effective Rate of Interest

$$\begin{aligned}\text{Interest paid on Loan } (50 \text{ Cr.} \times 7.8\% \times 3/12) \\ = \text{₹ } 97,50,000\end{aligned}$$

$$\begin{aligned}(+) \text{ Net Settlement } (\text{₹ } 6,13,046 \times 1.0195) &= \text{₹ } 6,25,000 \\ &= \text{₹ } 1,03,75,000\end{aligned}$$

$$\begin{aligned}\text{Effective Rate} &= \frac{1,03,75,000}{50 \text{ Cr.}} \times 100 \times \frac{12}{3} \\ &= 8.30\% \text{ p.a.}\end{aligned}$$



INTEREST RATE RISK MANAGEMENT

Question – 03

XY Ltd. is planning to expand its operations in view of growing demand for its products. For this purpose, it is considering to borrow an amount of ₹ 100 crores for a period of 3 months in the coming 6 months' time from now. The current rate of interest is 8% per annum but due to inflation it may go up in 6 months' time. The company wants to hedge itself against the likely increase in interest rate.

The company's Bankers quoted an FRA (Forward Rate Agreement) at 8.20% per annum.

You are required to calculate due to FRA:

- (i) The actual interest rate if the Banker pays to XY Ltd. an amount of ₹ 9,78,952.52
- (ii) The actual interest rate if XY Ltd. will pay to the Banker a sum of ₹ 9,80,872.98

(Exam November – 2024) (6 Marks)

Solution:

Final settlement amount shall be computed by using formula:

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]}$$

Where,

N = the notional principal amount of the agreement;

RR = Actual Reference Rate for the maturity specified by the contract prevailing on the contract settlement date;

FR = Agreed-upon Forward Rate; and

dtm = maturity period in days or months of the forward rate agreement

DY = Total number of days or month in a year as per convention

Accordingly,

- i. If Banker pays to XY Ltd. an amount of ₹ 9,78,952.52 then actual interest shall be computed as follows:

$$₹ 9,78,952.52 = \frac{(\text{₹ } 100 \text{ crore})(RR - 0.082)(3/12)}{[1 + RR(3/12)]}$$

$$RR = 0.086$$

Thus, the actual interest rate happens to be 8.60% on the settlement date.

- ii. If XY Ltd. pays to Banker an amount of ₹ 9,80,872.98 then actual interest shall be computed as follows:

$$₹ 9,80,872.98 = \frac{(\text{₹ } 100 \text{ crore})(RR - 0.082)(3/12)}{[1 + RR(3/12)]}$$

$$RR = 0.078$$

Thus, the actual interest rate happens to be 7.80% on the settlement date.

Question – 04

TM Fincorp has bought a 6 × 9 ₹ 100 crore Forward Rate Agreement (FRA) at 5.25%. On fixing date reference rate i.e. MIBOR turns out be as follows:

Period	Rate (%)
3 Months	5.50



6 Months	5.70
9 Months	5.85

You are required to determine:

- Profit/Loss to TM Fincorp. in terms of basis points.
- The settlement amount.

(Assume 360 days in a year)

(SM TYK – 02)

Solution:

(i) Calculation of profit/Loss to TM Fincorp.

TM bought FRA 6 × 9 at 5.25%. On maturity 3 months MIBOR is 5.50% hence TM will Make profit (5.50% – 5.25%) i.e., 0.25% on 25 basic.

(ii) Settlement Amount

$$\begin{aligned}
 \text{Settlement Amount} &= \frac{N (RR - FR) DT_m / 360}{1 + (RR \times DT_m / 360)} \\
 &= \frac{1,00,00,00,000 (5.50\% - 5.25\%) 90 / 360}{1 + (5.50\% \times 90 / 360)} \\
 &= \frac{1,00,00,00,000 \times 0.0625\%}{1.01375} \\
 &= ₹ 6,16,523
 \end{aligned}$$

Question – 05

The following market data is available:

Spot USD/JPY 116.00

Deposit Rates p.a.	USD	JPY
3 Months	4.50%	0.25%
6 Months	5.00%	0.25%

Forward Rate Agreement (FRA) for Yen is Nil.

- What should be 3 months FRA rate at 3 months forward?
- The 6 & 12 months LIBORS are 5% & 6.5% respectively. A bank is quoting 6/12 USD FRA at 6.50 – 6.75%. Is any arbitrage opportunity available?

Calculate profit in such case.

Solution:

(i) Calculation of FRA 3 × 6

$$\begin{aligned}
 1 + (5\% \times 6/12) &= [1 + (4.5\% \times 3/12)] \times (1 + r) \\
 1.025 &= 1.01125 \times (1 + r) \\
 r &= \left(\frac{1.025}{1.01125} \right) - 1 \times 100 \\
 &= 1.3595\% \\
 \text{FRA } 3 \times 6 &= 1.3597 \times \frac{12}{3} \\
 &= 5.44\% \text{ P.a.}
 \end{aligned}$$

(ii) Calculation of Theoretical 6 × 12 FRA

PART 2: INTEREST RATE GUARANTEE

Question – 06

Two companies ABC Ltd. and XYZ Ltd. approach the DEF Bank for FRA (Forward Rate Agreement). They want to borrow a sum of ₹ 100 crores after 2 years for a period of 1 year. Bank has calculated Yield Curve of both companies as follows:

Year	XYZ Ltd.	ABC Ltd.
1	3.86	4.12
2	4.20	5.48
3	4.48	5.78

*The difference in yield curve is due to the lower credit rating of ABC Ltd. compared to XYZ Ltd.

- You are required to calculate the rate of interest DEF Bank would quote under 2V3 FRA, using the company's yield information as quoted above.
- Suppose bank offers Interest Rate Guarantee for a premium of 0.1% of the amount of loan, you are required to calculate the interest payable by XYZ Ltd. if interest rate in 2 years turns out to be

(a) 4.50% (b) 5.50%

(RTP November – 2020)

Solution:

- Calculation of Theoretical FRA 2 × 3 XYZ**

$$(1.0448)^3 = (1.0420)^2 (1 + r)$$

$$1 + \left(6.5\% \times \frac{12}{12}\right) = \left[1 + \left(5\% \times \frac{6}{12}\right)\right] \times (1 + r)$$

$$1.065 = (1.025) (1 + r)$$

$$r = \left(\frac{1.065}{1.025} - 1\right) \times 100 \times \frac{12}{6}$$

$$= 7.80\% \text{ p.a.}$$

Arbitrage:

Action: Since Actual FRA is less than theoretical FRA, it means FRA is underpriced, hence buy FRA 6.75% p.a. i.e. contract to borrow.

Process:

Today

- Borrow \$ 1 @ 5% p.a. for 6 months
- Invest \$ 1 @ 6.5% p.a. for 12 Months
- Contract to borrow @ 6.75% p.a. after 6 months for 6 months

After 12 Months

Cash Inflows

$$\text{\$ } 1 (1.065) = \text{\$ } 1.065$$

Cash outflows

$$\text{\$ } 1 (1.025) (1.03375) = \underline{\text{\$ } 1.060}$$

$$\text{Arbitrage} = \underline{\text{\$ } 0.005}$$



$$1.14057 = 1.08576 (1 + r)$$

$$r = \left(\frac{1.14057}{1.08576} \right) - 1 \times 100$$

$$= 5.04\% \text{ p.a.}$$

ABC

$$(1.0578)^3 = (1.0548)^2 (1 + r)$$

$$1.1836 = 1.1126 (1 + r)$$

$$r = \left(\frac{1.1836}{1.1126} - 1 \right) \times 100$$

$$= 6.38\% \text{ p.a.}$$

(ii) Calculation of Interest payable by XYZ: XYZ buys interest rate guarantee for 1 year after 2 years @ 5.04% & paid premium ₹ 100 Cr. $\times 0.1\% = ₹ 0.1 \text{ Cr.}$ (Call Option)

(a) 4.5% p.a.

Interest paid on loan (₹ 100 Cr. $\times 4.5\%$) = ₹ 4.50 Cr.

Interest rate guarantee payoff = Lapsed

Premium = ₹ 0.10 Cr.

Interest payable = ₹ 4.60 Cr.

(b) 5.50% p.a.

Interest paid on loan (₹ 100 Cr. $\times 5.50\%$) = ₹ 5.50 Cr.

Interest rate guarantee (5.50% – 5.04%)

$\times 100 \text{ Cr.} = (0.46 \text{ Cr.})$

Premium = ₹ 0.10 Cr.

Interest payable = ₹ 5.14 Cr.

PART 3: INTEREST RATE FUTURE (EURO DOLLAR FUTURE)

Question – 07

Electra space is consumer electronics wholesaler. The business of the firm is highly seasonal in nature. In 6 months of a year, firm has a huge cash deposits and especially near Christmas time and other 6 months firm cash crunch, leading to borrowing of money to cover up its exposures for running the business.

It is expected that firm shall borrow a sum of €50 million for the entire period of slack season in about 3 months.

A Bank has given the following quotations:

Spot 5.50% - 5.75%

3 $\times$ 6 FRA 5.59% - 5.82%

3 $\times$ 9 FRA 5.64% - 5.94%

3 month €50,000 future contract maturing in a period of 3 months is quoted at 94.15 (5.85%).

You are required to determine:

- (a) How a FRA, shall be useful if the actual interest rate after 3 months turnout to be:



INTEREST RATE RISK MANAGEMENT

(i) 4.5% (ii) 6.5%

- (b) How 3 months Future contract shall be useful for company if interest rate turns out as mentioned in part (a) above.

Solution:

(I) FRA Hedging:

Electra space wants to borrow after 3 months for 6 months & afraid from interest rate rising. In order to hedge risk, it should take long position (Buy FRA) 3×9 @ 5.94% p.a.

Calculation of Interest payable

(i) Rate 4.5%

Interest paid on loan

$$(\text{€ } 5,00,00,000 \times 4.5\% \times 6/12) = \text{€ } 11,25,000$$

Electra space paid to FRA

$$[5.94\% - 4.5\%] \times \text{€ } 5,00,00,000 \times 6/12 = \text{€ } 3,60,000$$

$$\text{Interest payable} = \text{€ } 14,85,000$$

(ii) Rate 6.5%

Interest paid on loan

$$(\text{€ } 5,00,00,000 \times 6.5\% \times 6/12) = \text{€ } 16,25,000$$

Electra space will receive

$$[0.065 - 0.0594] \times \text{€ } 5,00,00,000 \times 6/12 = \text{€ } 1,40,000$$

$$\text{Interest payable} = \text{€ } 14,85,000$$

(II) Interest Rate Future

$$\text{Effective Cost} = \frac{\text{€ } 14,85,000}{\text{€ } 5,00,00,000} \times 100 \times \frac{12}{6} = 5.94\% \text{ p.a.}$$

*Electra space should take short position at 94.15 on IRF

$$\begin{aligned} \text{*No. of contracts} &= \frac{\text{Borrowing Amount}}{\text{Contract Size}} \times \frac{\text{Duration of Loan}}{3 \text{ Months}} \\ &= \frac{\text{€ } 5,00,00,000}{\text{€ } 50,000} \times \frac{6 \text{ Months}}{3 \text{ Months}} \\ &= 2,000 \text{ Contract short} \end{aligned}$$

Interest Payable

(i) Rate 4.5% (100 – 4.5 = 95.50)

$$\text{Interest paid on loan } (\text{€ } 5,00,00,000 \times 4.5\% \times 6/12) = \text{€ } 11,25,000$$

$$\text{Loss on IRF } (94.15 - 95.50) = 1.35\%$$

$$\begin{aligned} (1.35\% \times 2,000 \times \text{€ } 50,000 \times 3/12) &= \text{€ } 3,37,500 \\ &= \text{€ } 14,62,500 \end{aligned}$$

(ii) Rate 6.5% (100 – 6.5 = 93.50)

Interest paid on loan

$$(\text{€ } 5,00,00,000 \times 6.5\% \times 6/12) = \text{€ } 16,25,000$$

$$\text{Gain on IRF } (94.15 - 93.50) = 0.65\%$$

$$(0.65\% \times 2,000 \times \text{€ } 50,000 \times 3/12) = (\text{€ } 1,62,500)$$



$$\text{Effective Cost} = \frac{\text{€ } 14,62,500}{\text{€ } 5,00,00,000} \times 100 \times \frac{12}{6} = 5.85\% \text{ p.a.}$$

$$= \text{€ } 14,62,500$$

PART 4: FINANCIAL SWAP

(I) EQUITY SWAP

Question – 08

CMT Pension Fund has a portfolio of shares of diversified companies valued at ₹ 800 crore enters into a swap arrangement with Boom Bank on the terms that it will get 1.15% quarterly on notional principal of ₹ 800 crore in exchange of return on portfolio which is exactly tracking the Sensex which is presently 43,200.

You are required to determine the net payment to be received/ paid if Sensex turns out to be 43,720, 43,560, 44,160 and 43,920 at the end of each quarter.

Note: Make calculation in ₹ crore and round off calculations upto 4 decimal points.

(MTP October – 2021)

Solution:

(i) Calculation of net settlement in each quarter

Quarter	Sensex Return (%)	Sensex Return (₹)	Fixed return (₹) (1.15%)	Net payable/receivable
I	$\frac{43,720 - 43,200}{43,200} \times 100 = 1.2037\%$	9.6296	9.20 Cr.	- 0.4296 Paid

INTEREST RATE RISK MANAGEMENT

II	$\frac{43,560 - 43,720}{43,700} \times 100 = -0.3660\%$	-2.9280	9.20 Cr.	12.128 Received
III	$\frac{44,160 - 43,560}{43,560} \times 100 = 1.3774\%$	11.0192	9.20 Cr.	-1.8192 Paid
IV	$\frac{43,920 - 44,160}{44,160} \times 100 = -0.5435\%$	-4.3480	9.20 Cr.	13.5480 Received

(II) PLAIN VANILLA SWAP

Question – 09

Suppose a dealer quotes 'All-in-cost' for a generic swap at 8% against six month SOFR flat. If the notional principal amount of swap is ₹ 5,00,000.

- Calculate semi-annual fixed payment.
- Find the first floating rate payment for (i) above if the six month period from the effective date of swap to the settlement date comprises 181 days and that the corresponding SOFR was 6% on the effective date of swap.

In (ii) above, if the settlement is on 'Net' basis, how much the fixed rate payer would pay to the floating rate payer?

Generic swap is based on 30/360 days basis.

(SM TYK – 04, Exam November – 2018) (8 Marks)

Solution:

- Calculation of semi Amount fixed payment
 $= ₹ 5,00,000 \times 8\% \times 6/12$



$$= ₹ 20,000$$

(ii) Calculation of floating rate payment

$$= ₹ 5,00,000 \times 6\% \times 181/360$$

$$= ₹ 15,083$$

(iii) Calculation of Net settlement

$$\begin{aligned} \text{Net Amount paid by fixed rate payer} &= 20,000 - 15,083 \\ &= ₹ 4,917 \end{aligned}$$

Question – 10

Suppose a dealer bank quotes for a generic swap "AIC 8%/8.20% vs. 6M LIBOR Flat". Notional principal amount of swap is ₹ 1 Million, and the same is for a period of three years, reset after every six months.

In this context, answer the following questions:

- (1) Interpret the dealer bank quote.
- (2) If a firm is buying a swap, what is the nature of cash flows?
- (3) If a firm is selling a swap, what is the nature of cash flows?
- (4) Calculate semi-annual fixed payment for the buyer of swap at the end of every six months.
- (5) If the six month period from the effective date of swap to the settlement date comprises of 181 days and that the corresponding LIBOR was 5% on the effective date of swap, then what will be the first floating rate payment for the buyer?

- (6) If the settlement is on "Net Basis", how much the buyer of swap has to pay or receive at the end of first six months? [Assume 30/360 days basis]

(Exam November – 2023) (8 Marks)

Solution:

(1) Interpretation of dealer bank quote:

- AIC in the dealer bank quote refers to 'All in cost' i.e. cost of swap all inclusive.
- First part of the quote i.e. '8%/8.20%' refers to the fixed leg part and the second part of the quote i. e. '6m LIBOR Flat' refers to the floating leg part.
- The difference in the fixed rates i.e. 20 bps refers to the margin charged by the Bank on the fixed leg of transactions.
- The term 'flat' on the floating leg quote, indicates that the Bank does not charge any commission on the floating leg. Therefore, bank charges 20 bps for transacting swap as a whole.

(2) A buyer of swap pays 'Fixed' cash flows and receives 'Floating'. As per the quote, the buyer would pay 8.2% (higher of 8%, 8.2%) to the Bank and would receive '6M LIBOR' against it.

(3) A seller of swap pays 'floating' cash flows and receives 'fixed'. As per the quote, the seller would pay '6M LIBOR' to the bank and would receive 8% (lower of 8%, 8.2%) against it.

(4) Semi-annual Payment every six-month for buyer of Swap:

$$₹ 10,00,000 \times 8.20\% \times \frac{1}{2} = ₹ 41,000$$



(5) Floating Rate Payment

$$= N (\text{LIBOR}) \left(\frac{dt}{360} \right)$$

$$= 10,00,000 \times 0.05 \times \frac{181}{360}$$

$$= ₹ 25,138.89$$

(6) Net Settlement

$$\text{Or, } ₹ 41,000 - ₹ 25,138.89 = ₹ 15,861.11$$

(III) OVERNIGHT INDEX SWAP**Question – 11**

Derivative Bank entered into a swap arrangement on a principal of ₹ 10 crores and agreed to receive MIBOR overnight floating rate for a fixed payment on the principal. The swap was entered into on Monday, 19th August, 2019 and was to commence on 20th August, 2019 and run a period of 7 days.

Respective MIBOR rates for Tuesday to Monday were:

8.15%, 7.98%, 7.95%, 8.12%, 8.15%, 7.75%.

If Fixed Rate of Interest is 8%, then evaluate

- The nature of this swap arrangement.
- The Net Settlement amount.

Note:

- Sunday is Holiday.

- Work in rounded rupees and avoid decimal working.

- Consider 365 days in a year.

(RTP November – 2021)

Solution:**(I) Nature of Swap**

This swap is Plain Vanilla Overnight Index swap.

(II) The Net Settlement Amount

Fixed Rate Payment

$$= 10,00,00,000 \times 8\% \times 7/365$$

$$= ₹ 1,53,425$$

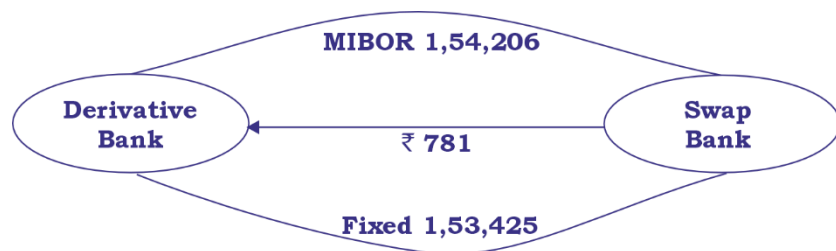
Floating Rate (MIBOR) Payment

Day	MIBOR	Calculation	Interest
Tuesday	8.15 %	10,00,00,000 × 8.15% ×	22,329
Wednesday	7.98 %	1/365	21,868
Thursday	7.95 %	10,00,22,329 × 7.98% ×	21,790
Friday	8.12 %	1/365	22,261
Sat. & Sun.	8.15 %	10,00,44,197 × 7.95% ×	44,697
Monday	7.75 %	1/365	21,261
		10,00,65,987 × 8.12% ×	
		1/365	
		10,00,88,248 × 8.15% ×	
		2/365	
		10,01,32,945 × 7.75% ×	
		1/365	
			1,54,206



INTEREST RATE RISK MANAGEMENT

(Exam November – 2010, SM TYK – 05) (8 Marks)



In Settlement, ₹ 781 will be paid by swap bank to Derivative Bank $(1,54,206 - 1,53,425) = 781$.

Note: Saturday को (SATURDAY & SUNDAY) दो दिनों का INTEREST लेना है।

Question – 12

Derivative Bank entered into a plain vanilla swap through on OIS (Overnight Index Swap) on a principal of ₹ 10 crores and agreed to receive MIBOR overnight floating rate for a fixed payment on the principal. The swap was entered into on Monday, 2nd August, 2010 and was to commence on 3rd August, 2010 and run for a period of 7 days.

Respective MIBOR rates for Tuesday to Monday were:

7.75%, 8.15%, 8.12%, 7.95%, 7.98%, 8.15%.

If Derivative Bank received ₹ 317 net on settlement, calculate Fixed rate and interest under both legs.

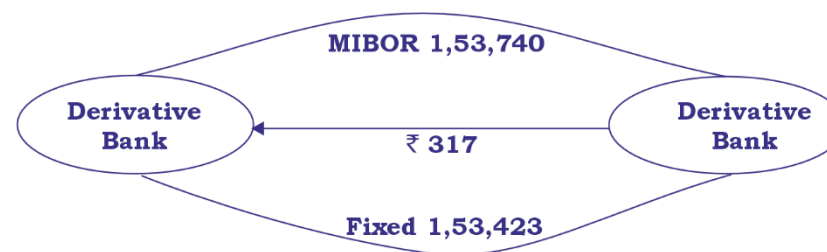
Notes:

- (i) Sunday is Holiday.
- (ii) Work in rounded rupees and avoid decimal working.

Solution:

(i) Calculation of floating Rate payment (MIBOR)

Days	MIBOR	Calculation	Interest
Tuesday	7.75 %	$10,00,00,000 \times 7.75 \% \times \frac{1}{365}$	21,233
Wednesday	8.15 %		22,333
Thursday	8.12 %	$10,00,21,233 \times 8.15 \% \times \frac{1}{365}$	22,256
Friday	7.95 %		21,795
Sat. & Sun.	7.98 %	$10,00,43,567 \times 8.12 \% \times \frac{1}{365}$	43,764
Monday	8.15 %		22,358
		$10,00,65,823 \times 7.95 \% \times \frac{1}{365}$	
		$10,00,87,618 \times 7.98 \% \times \frac{2}{365}$	
		$10,01,31,382 \times 8.15 \% \times \frac{1}{365}$	
			1,53,740



$$\begin{aligned} \text{Fixed Rate Payment} &= 1,53,740 - 317 \\ &= 1,53,423 \end{aligned}$$

$$\text{Rate of Interest} = \frac{1,53,423}{10,00,00,000} \times 100 \times \frac{365}{7}$$



= 8% p.a.

(IV) TWO PARTY SWAP

Question – 13

IB an Indian firm has its subsidiary in Japan and Zaki a Japanese firm has its subsidiary in India and face the following interest rates:

Company	IB	Zaki
INR floating rate	BPLR + 0.50%	BPLR + 2.50%
JPY (Fixed rate)	2%	2.25%

Zaki wishes to borrow Rupee Loan at a floating rate and IB wishes to borrow JPY at a fixed rate. The amount of loan required by both the firms is same at the current exchange rate. A financial institution may arrange a swap and requires 25 basis points as its commission. Gain, if any, is to be shared by the firms equally.

You are required to find out:

- Whether a swap can be arranged which may be beneficial to both the firms?
- What rate of interest will the firms end up paying?

(Exam November – 2020) (8 Marks)

Solution:

- IB has absolute advantage in floating rate as well as fixed rate but it has comparative advantage in floating rate.

Hence,

* IB should borrow at floating rate.

INTEREST RATE RISK MANAGEMENT

* ZAKI should borrow at fixed rate.

Total potential gain

= Difference in floating rate – Difference in fixed rate

$$= 2\% - 0.25\% = 1.75\%$$

(-) Commission = 0.25%

Net gain = 1.50%

Effective Cost

$$\text{IB} = 2 - (1.5 \times 1/2) = 1.25\%$$

$$\text{ZAKI} = \text{BPLR} + 2.5 - (1.5 \times 1/2) = \text{BPLR} + 1.75$$

(ii) Interest Rate Payable [Process]

IB

Interest paid on loan BPLR + 0.5%

Commission $(0.25 \times 1/2)$ 0.125%

Received from Zaki – BPLR + 0.5%

Paid to ZAKI $(1.25 - 0.125\%)$ 1.125%

Effective Cost 1.25%

ZAKI

Interest paid on loan 2.25%

Commission 0.125%



INTEREST RATE RISK MANAGEMENT

Paid to IB	BPLR + 0.50%
Received from IB	– 1.125%
Effective Cost	<u>BPLR + 1.75</u>

Question – 14

IF an Indian firm has its subsidiary in Singapore and SF a Singapore firm has its subsidiary India and face the following interest rates:

Company	IF	SF
INR Floating Rate	BPLR + 0.5%	BPLR + 1.5%
SGD (Fixed Rate)	3%	3.50%

SF wishes to borrow Rupee loan at a floating rate and IF wishes to borrow SGD at a fixed rate. The amount of loan required by both the companies is same at the current exchange rate. A Bank arranges a swap and requires 50 basis points as its commission, which is to be shared equally. IF requires a minimum gain of 20 basis points and SF requires a minimum gain of 10 basis points for structuring the deal. The Bank is very keen to structure the deal, even if, it has to forego a part of its commission.

You are required to find out

- Whether there are any advantages available to IF and SF?
- Whether a swap can be arranged which may be beneficial to both the firms?
- What rate of interest will they end up paying? Show detailed working.

(Exam May – 2023) (8 Marks)

Solution:

- (i) IF has absolute advantage in floating rate as well as fixed rate but it has comparative advantage in floating rate.

Hence,

* IF should borrow at floating rate.

* SF should borrow at fixed rate.

Total potential gain

= Difference in floating rate – Difference in fixed rate

$$= 1\% - 0.5\% = 0.5\%$$

(–) Commission = 0.2%

Net gain = 0.3%

Effective Cost

IF = 3% – 0.20%

$$= 2.8\%$$

SB = BPLR + 1.5% – 0.10%

$$= \text{BPLR} + 1.4\%$$

- (ii) **Design**

IF

Interest paid on loan BPLR + 0.5%

Commission 0.10%



INTEREST RATE RISK MANAGEMENT

Received from SF – BPLR + 0.5%

Paid to SF (2.8% – 0.10%) 2.7%

Effective Cost 2.8%

SF

Interest paid on loan 3.5%

Commission 0.10%

Paid to IF BPLR + 0.5%

Received from IF – 2.7%

BPLR + 1.4%

Question – 15

A Inc. and B Inc. intend to borrow \$ 200,000 and \$200,000 in ¥ respectively for a time horizon of one year. The prevalent interest rates are as follows:

Company	¥ Loan	\$ Loan
A Inc	5%	9%
B Inc	8%	10%

The prevalent exchange rate is \$1 = ¥120.

They entered in a currency swap under which it is agreed that B Inc will pay A Inc @ 1% over the ¥ Loan interest rate which the later will have to pay as a result of the agreed currency swap whereas A Inc will reimburse interest to B Inc only to the extent of 9%. Keeping the exchange rate invariant, quantify the opportunity gain or loss

component of the ultimate outcome, resulting from the designed currency swap.

(Exam May – 2011, SM TYK – 06) (8 Marks)

Solution:

A Incorporation

Today

Borrow yen from bank + ¥ 2,40,00,000

Yen transfer to B incorporation – ¥ 2,40,00,000

\$ Transfer from B Incorporation + \$ 2,00,000

\$ 2,00,000

After 1 year

Interest paid to bank (¥ 2,40,00,000 × 5%) – ¥12,00,000

Interest transferred from B incorporation
(¥ 2,40,00,000 × 6%) + 14,40,000

Interest transferred to B Incorporation
(\$2,00,000 × 9%) – \$ 18,000

– \$ 18,000 + ¥ 2,40,000

Net Interest paid in \$ = \$18,000 – $\left(\frac{¥ 2,40,000}{120}\right)$

= \$16,000



$$\text{Effective Cost} = \frac{\$ 1,600}{\$ 2,00,000} \times 100 = 8\%$$

$$\text{Without Swap} = 9\%$$

$$\text{Gain} = 1\%$$

B Incorporation

Today

Borrow \$ from Bank + \$ 2,00,000

Transfer \$ to A Inc. – \$ 2,00,000

Transfer ¥ from A Inc. ¥ 2,40,00,000

¥ 2,40,00,000

After 1 Year

Interest paid to bank (\$ 2,00,000 × 10%) – \$ 20,000

Interest transfer from A Inc. (\$ 200000 × 9%) + \$ 18,000

Interest transfer to A Inc. (¥ 2,40,00,000 × 6%) – ¥ 14,40,000

– ¥ 14,40,000 – \$2,000

Interest paid in ¥ = ¥ 14,40,000 + (\$ 2,000 × 120)

= ¥ 16, 80,000

Effective Cost = $\frac{¥ 16,80,000}{¥ 2,40,00,000} \times 100$

= 7%

Without Swap = 8%

Gain = 1%

PART 5 : CAP, COLOUR & FLOOR

Question – 16

MPD Ltd. issues a ₹ 50 Million Floating Rate Loan on July 1, 2018 with resetting of coupon rate every 6 Months equal to LIBOR + 50 bps.

MPD is interested in an Interest rate Collar Strategy of selling a Floor and buying a cap. MPD buys the 3 years cap and sell 3 years Floor as per the following details on July 1, 2018:

Principal Amount	₹ 50 Million
Strike Rate	5% for Floor & 8% for Cap
Reference Rate	6 months LIBOR
Premium	NIL, since premium paid for cap = premium received for Floor

The Reset dates & Interest rates p.a., on that dates are:

Reset Date	LIBOR(%)
31/12/2018	7.00
30/06/2019	8.00
31/12/2019	6.00
30/06/2020	4.75
31/12/2020	4.25
30/06/2021	5.25

Using the above data, you are required to determine:

- Effective Interest paid out at each six reset dates, (Round off to the nearest rupee)



INTEREST RATE RISK MANAGEMENT

- (ii) Average overall effective rate of interest p.a. (round off to 2 decimals)

(Exam May – 2022) (8 Marks)

Solution:

(i) Calculation of Effective Cost

Reset Date	LIBOR	Date of payment	No. of days	Intt paid (L+0.50 %)	Cap Receipt	Floor payment	Net
31/12/2018	7.00%	30/06/2019	181	18,59,589	0	0	18,59,589
30/06/2019		31/12/2019		21,42,466			21,42,466
31/12/2019	8.00%	30/06/2020	184	16,16,120	0	0	16,16,120
30/06/2020		31/12/2020		13,19,672			13,82,514
31/12/2020	6.00%	30/06/2021	182	11,77,740	0	0	13,63,699
30/06/2021		31/12/2021		14,49,315			14,49,315
	4.75%		184		0	62,842	
	4.25%		181		0	1,85,959	
	5.25%		184		0	0	
			1,096				98,13,703

(ii) Average Annual Rate = $\frac{98,13,703}{5,00,00,000} \times 100 \times \frac{365}{1,096} = 6.54\% \text{ p.a.}$

Question – 17

A textile manufacturer has taken floating interest rate loan of ₹ 40,00,000 on 1st April, 2012. The rate of interest at the inception of loan is 8.5% p.a. interest is to be paid every year on 31st March, and the duration of loan is four years. In the month of October 2012, the

Central bank of the country release following projections about the interest rates likely to prevail in future.

Dates	Interest Rate
31 st March, 2013	8.75%
31 st March, 2014	10.00%
31 st March, 2015	10.50%
31 st March, 2016	7.75%

- (i) Advise how borrower can hedge the risk arising out of expected rise in the rate of interest when he is interested in pegging his interest cost at 8.50% p.a. and if option on Interest Rate is available at 0.75% p.a.
- (ii) Assume that the premium negotiated by both the parties at the above-mentioned rate which is to be paid on upfront basis and the actual rate of interest on the respective due dates happens to as follows:

Dates	Interest Rate
31 st March, 2013	10.20%
31 st March, 2014	11.50%
31 st March, 2015	9.25%
31 st March, 2016	8.25%

EVALUATE how the settlement will be executed on the respective interest due dates.

(Exam Nov – 2017, SM TYK – 07, RTP May – 2022 & MTP March – 22) (8 Marks)

Solution:

As borrower does not want to pay more than 8.5% p.a., on this loan where the rate of interest is likely to rise beyond this, hence, he has



hedge the risk by entering into an agreement to buy interest rate caps with the following parameters:

- Notional Principal: ₹ 40,00,000/-
- Strike rate: 8.5% p.a.
- Reference rate: the rate of interest applicable to this loan
- Calculation and settlement date: 31st March every year
- Duration of the caps: till 31st March 2016
- Premium for caps: negotiable between both the parties

To purchase the caps this borrower is required to pay the premium upfront at the time of buying caps. The payment of such premium will entitle him with right to receive the compensation from the seller of the caps as soon as the rate of interest on this loan rises above 8.5%. The compensation will be at the rate of the difference between the rate of none of the cases the cost of this loan will rise above 8.5% calculated on ₹ 40,00,000/-. This implies that in none of the cases the cost of this loan will rise above 8.5%. This hedging benefit is received at the respective interest due dates at the cost of premium to be paid only once.

The premium to be paid on 1st October 2012 is 30,000/- ($₹ 40,00,000 \times 0.75/100$). The payment of this premium will entitle the buyer of the caps to receive the compensation from the seller of the caps whereas the buyer will not have obligation. The compensation received by the buyer of caps will be as follows:

On 31st March 2013

The buyer of the caps will receive the compensation at the rate of 1.70% ($10.20 - 8.50$) to be calculated on ₹ 40,00,000, the amount of compensation will be ₹ 68,000/- ($40,00,000 \times 1.70/100$).

On 31st March 2014

The buyer of the caps will receive the compensation at the rate of 3.00% ($11.50 - 8.50$) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 1,20,000/- ($40,00,000 \times 3.00/100$).

On 31st March 2015

The buyer of the caps will receive the compensation at the rate of 0.75% ($9.25 - 8.50$) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 30,000 ($40,00,000 \times 0.75/100$).

On 31st March 2016

The buyer of the caps will not receive the compensation as the actual rate of interest is 8.25% whereas strike rate of caps is 8.5%. Hence, his interest liability shall not exceed 8.50%.

Thus, by paying once the premium upfront buyer of the caps gets the compensation on the respective interest due dates without any obligations.

Question – 18

DEF Ltd. has implemented a strategy to manage its exposure to fluctuating interest rates by engaging in both interest rate caps and floors.

The company has purchased \$ 50,00,000 (i.e. call options on interest rates) cap of 8% at a premium of 0.75% of the face value to protect against rising interest rates. \$ 50,00,000 (i.e. put options on interest

INTEREST RATE RISK MANAGEMENT

rates) floor of 5% is also available at a premium of 0.85% of face value.

You are required to analyze the following situation:

- (i) If interest rate rise to 10 percent, what is the amount received by DEF Ltd.? What are the net saving from the cap?
- (ii) If DEF Ltd. also purchase a floor, what are net savings if interest rate rise to 10%?
- (iii) Calculate net savings if interest rates fall to 4 percent considering cap & floor both purchase.
- (iv) If DEF Limited has purchase the cap and sell the floor and there is price rise is 11%, what will be net saving to the company?

(Exam May – 2025) (7 Marks)

Solution:

- (i) Premium for purchasing the Cap

$$= 0.0075 \times \$ 50,00,000 = \$ 37,500$$

If interest rate rises to 10%, Cap purchase receive (10% - 8% = 2%) i.e.,

$$= \$ 50,00,000 \times 2/100 = \$ 1,00,000,$$

$$= (\$ 1,00,000 - \$ 37,500) = \$ 62,500$$

$$\text{Net saving} = \$ 62,500$$

- (ii) Purchase Floor, premium for Floor

$$= 0.0085 \times \$ 50,00,000 = \$ 42,500 \text{ and}$$

Total premium will be $\$ 37,500 + \$ 42,500 = \$ 80,000$ and

$$= (\$ 50,00,000 \times 0.02) - \$ 80,000 = \$ 20,000$$

$$\text{Net saving} = \$ 20,000$$

- (iii) If interest rate falls to 4 percent, there will be no payoff from the Cap, but there will be payoff from Floor.

However, the total premium $(\$ 37,500 + \$ 42,500) = \$ 80,000$ is the cost.

$$\text{Payoff from floor} = \$ 50,00,000 \times 0.01 (5\% - 4\% = 1\%)$$

$$= \$ 50,000$$

$$= \$ 50,000 - \$ 80,000 = (\$ 30,000)$$

$$\text{Net loss} = (\$ 30,000)$$

- (iv) Sell the Floor, company will receive

$$= 0.0085 \times \$ 50,00,000 = \$ 42,500 \text{ and}$$

Pay premium for Cap

$$= 0.0075 \times \$ 50,00,000 = \$ 37,500,$$

Net premium receiving will be

$$= \$ 42,500 - \$ 37,500 = \$ 5,000$$



INTEREST RATE RISK MANAGEMENT

If interest rate rises to 11%, Cap purchase receive

$$= 0.03 \times \$ 50,00,000 = \$ 1,50,000$$

$$= \$ 1,50,000 + \$ 5,000 = \$ 1,55,000$$

$$\text{Net saving} = \$ 1,55,000$$

Question – 19

XYZ Limited borrows £ 15 Million of six months LIBOR + 10.00% for a period of 24 months. The company anticipates a rise in LIBOR, hence it proposes to buy a Cap Option from its Bankers at the strike rate of 8.00%. The lump sum premium is 1.00% for the entire reset periods and the fixed rate of interest is 7.00% per annum. The actual position of LIBOR during the forthcoming reset period is as under:

Reset Period	LIBOR
1	9.00%
2	9.50%
3	10.00%

You are required to show how far interest rate risk is hedged through Cap Option.

For calculation, work out figures at each stage up to three decimal points and amount nearest to £. It should be part of working notes.

(SM TYK – 03)

Solution:

(i) Calculation of Amortization of Premium

$$\text{Premium Amount} = €1,50,00,000 \times 1\% = €1,50,000$$

$$\text{Rate of discount} = 7\% \text{ p.a.}$$

$$6 \text{ months} = 3.5\%$$

$$\text{Cumulative factor} = 3.5\%, 4 \text{ Period}$$

$$= 3.671$$

$$\text{Amortization Amount} = \frac{£ 1,50,000}{3.671(\text{PVAF}, 3.5\%)}$$

$$= £ 40,861$$

Statement showing effective interest Rate 8% ↑

Period	Rate (LIBOR)	Interest paid to Bank (L + 10%)	Cap Receipt	Premium	Effective Interest
I	9.00	£14,25,000	(£75,000)	£40,861	£13,90,861
II	9.50	£14,62,500	(£1,12,500)	£40,861	£13,90,861
III	10.00	£1,50,000	(£1,50,000)	£40,861	£13,90,861
		£43,87,500			£41,72,583

$$\text{Interest saved due to cap } (43,87,500 - 41,72,583) = £ 2,14,917$$

Question – 20

Name	Status	Principal Amount (₹ in millions)	Duration of loan / deposit (time)	Interest rates of Borrowing / Lending	Strike Rate (PLR) (K)	Premium (%) (lump sum) (P)	If PLR rate at the end of first 6 months (Reset Period)
AB Ltd.	Borrower	₹ 5.00	5 Years	PLR + 0.5	8% p.a.	0.4%	10% p.a.
XY Ltd.	Depositor	₹ 2.00	3 Years	PLR – 0.5	8% p.a.	0.5%	6% p.a.



INTEREST RATE RISK MANAGEMENT

You are required to:

- (i) Elaborate the strategy to be adopted by both the companies to hedge against the risk of interest rate fluctuations.
- (ii) Premium paid/received based on the strategy to be adopted in (i) using 8% p.a. as the reference rate.
- (iii) Net Gain/loss due to hedging to both the companies.

(Exam Jan – 2026)

Solution:

- (i) Strategy to be adopted by both the companies:
 - AB Ltd. – Buying Cap Option
 - XY Ltd. – Buying Floor Option
- (ii) Premium paid by both companies First of all we shall calculate premium payable to bank as follows:

$$P = \frac{rp}{\left[(1 + i) - \frac{1}{i \times (1 + i)^t} \right]} \times A \text{ or } \frac{rp}{PVA(4.00\%, 10)} \times A$$

Where,

P = Premium

A = Principal Amount

rp = Rate of Premium

i = Fixed Rate of Interest

t = Time

For AB Ltd.

$$= \frac{0.004}{\left[(1 + 0.04) - \frac{1}{0.04 \times 1.04^{10}} \right]} \times 50,00,000 \text{ or } \frac{1}{8.1108} \times 50,00,000 \times 0.4\%$$

$$= ₹ 2,465.85 \text{ or } ₹ 2,466$$

For XY Ltd.

$$= \frac{0.005}{\left[(1 + 0.04) - \frac{1}{0.04 \times 1.04^6} \right]} \times 20,00,000 \text{ or } \frac{1}{5.2421} \times 20,00,000 \times 0.5\%$$

$$= ₹ 1,907.63 \text{ or } ₹ 1,908$$

- (iii) **Net Gain/Loss due to hedging to both the companies**

AB Ltd.

Gain on Interest for Cap Option	2%
Amount of Gain ₹ 5 millions × 2% × (1/2)	₹ 50,000
Less: Premium Paid	₹ 2,466
Net Gain	₹ 47,534

XY Ltd.

Gain on Interest for Cap Option	2%
	₹ 20,000
	₹ 1,908



Amount of Gain ₹ 2 millions × 2% × (1/2)	₹ 18,092
Less: Premium Paid	
Net Gain	

Alternative Solution if students have considered reset period (n-1)

(i) Strategy to be adopted by both the companies

- AB Ltd. – Buying Cap Option
- XY Ltd. – Buying Floor Option

(ii) Premium paid by both companies

First of all we shall calculate premium payable to bank as follows:

$$P = \frac{rp}{\left[(1 \div i) - \frac{1}{i \times (1+i)^t} \right]} \times A \text{ or } \frac{rp}{PVA(4.00\%, 10)} \times A$$

Where,

P = Premium

A = Principal Amount

rp = Rate of Premium

i = Fixed Rate of Interest

t = Time

For AB Ltd.

$$= \frac{0.004}{\left[(1 \div 0.04) - \frac{1}{0.04 \times 1.04^9} \right]} \times 50,00,000 \text{ or } \frac{1}{7.4353} \times 50,00,000 \times 0.4\%$$

$$= ₹ 2,689.87 \text{ or } ₹ 2,690$$

For XY Ltd.

$$= \frac{0.005}{\left[(1 \div 0.04) - \frac{1}{0.04 \times 1.04^5} \right]} \times 20,00,000 \text{ or } \frac{1}{4.4518} \times 20,00,000 \times 0.5\%$$

$$= ₹ 2,246.28 \text{ or } ₹ 2,246$$

(iii) Net Gain/Loss due to hedging to both the companies

AB Ltd.

Gain on Interest for Cap Option	2%
Amount of Gain ₹ 5 millions × 2% × (1/2)	₹ 50,000
Less: Premium Paid	₹ 2,690
Net Gain	₹ 47,310

XY Ltd.

Gain on Interest for Cap Option	2%
Amount of Gain ₹ 2 millions × 2% × (1/2)	₹ 20,000
Less: Premium Paid	₹ 2,246
Net Gain	₹ 17,754

(6) T. BOND FUTURE

Question – 21

In March 2020, XYZ bank sold some 7% interest rate futures underlying notional 7.50% coupon bonds. The exchange provides following details of eligible securities that can be delivered:

Security	Quoted Price of Bond	Conversion Factor
7.96 GOI 2023	1037.40	1.0370
6.55 GOI 2025	926.40	0.9060
6.80 GOI 2029	877.50	0.9195
6.85 GOI 2026	972.30	0.9643
8.44 GOI 2027	1146.30	1.1734
8.85 GOI 2028	1201.70	1.2428

Recommend the Security that should be delivered by the XYZ bank if future settlement price is 1,000.

(MTP April – 2021)

Solution:

Bond	Future Value	Conversion Factor	Future Value × Conversion Factor	Price	Gain/Loss
7.96%	1,000	1.037	1037	1037.40	(0.40)
6.55%	1,000	0.9060	906	926.40	(20.40)
6.80%	1,000	0.9195	919.50	877.50	42.00
6.85%	1,000	0.9643	964.30	972.30	(8)
8.44%	1,000	1.1734	1173.40	1146.30	27.10
8.85%	1,000	1.2428	1242.80	1201.70	4.11

Bond 6.80% GOI 2029 Bond should be delivered due to maximum profit.

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

P Ltd. is planning to borrow an amount of ₹ 60 crores for a period of 3 months in the coming 6 month's time from now. The current rate of interest is 9% p.a., but it is likely to go up in 6 month's time. The company wants to hedge itself against the likely increase in interest rate.

You as CFO has been asked to suggest both traditional as well as modern methods to hedge interest rate risk.

Suppose the banker of P Ltd. has quoted the following Forward Rate Agreement (FRA) rates:

3 × 6	9.10%	9.15%
6 × 9	9.20%	9.30%
9 × 12	9.35%	9.45%

P Ltd. has already issued redeemable bonds or debentures.

From the information given above, choose the correct answer to the following questions:

MULTIPLE CHOICE QUESTIONS

I. Which of the following rate shall be applicable for FRA to hedge the interest rate risk if P Ltd. agrees to adopt this method to hedge interest rate risk?

- (a) 9.10% p.a.
- (b) 9.30% p.a.



INTEREST RATE RISK MANAGEMENT

Answer: Case Scenario – 01

- I. (b) 9.30% p.a.
II. (c) ₹ 4,39,453 shall be paid by Banker to P Ltd.
III. (a) ₹ 7,33,855 shall be paid by P Ltd. to its Banker.
IV. (d) Forward Rate Agreement

Case Scenario – 02

A Ltd. is planning to borrow an amount of ₹ 300 crores for a period of 3 months in the coming 6 months' time from now. The current rate of interest is 9% p.a., but it is likely to go up in 6 months' time. The company wants to hedge itself against the likely increase in interest rate.

You as CFO has been asked to suggest both traditional as well as modern methods to hedge interest rate risk.

Supposed the banker of A Ltd. has quoted the following Forward Rate Agreement (FRA) rates:

3 × 6	8.10%	8.15%
6 × 9	8.20%	8.30%
9 × 12	8.35%	8.45%

A Ltd. has already issued redeemable bonds or debentures.

- (c) 9.35% p.a.
(d) 9.45% p.a.
- II. If the actual rate of interest after 6 months happens to be 9.60%, approximately the settlement amount of.....
(a) ₹ 7,33,855 shall be paid by P Ltd. to its Banker
(b) ₹ 4,39,453 shall be paid by P Ltd. to its Banker.
(c) ₹ 4,39,453 shall be paid by Banker to P Ltd.
(d) ₹ 7,33,855 shall be paid by Banker to P Ltd.
- III. If the actual rate of interest after 6 months happens to be 8.80%, approximately the settlement amount of.....
(a) ₹ 7,33,855 shall be paid by P Ltd. to its Banker.
(b) ₹ 4,39,453 shall be paid by P Ltd. to its Banker
(c) ₹ 4,39,453 shall be paid by Banker to P Ltd.
(d) ₹ 7,33,855 shall be paid by Banker to P Ltd.
- IV. Which of the following technique is not the modern technique to hedge the interest rate risk.
(a) Interest Rate Futures
(b) Interest Rate Options
(c) Interest Rate Swaps
(d) Forward Rate Agreement

(RTP May – 2025)



INTEREST RATE RISK MANAGEMENT

From the information given above, choose the correct answer to the following questions:

- I.** In an FRA contract, the company intending to borrow funds in future should.....
- (a) Buy FRA at bid rate.
 - (b) Sell FRA at bid rate.
 - (c) Buy FRA at offer rate.
 - (d) Sell FRA at offer rate.
- II.** If interest rates rise above the FRA rate, the A Ltd. under FRA will.....
- (a) Incur loss and pay banker.
 - (b) Neither gain nor lose.
 - (c) Receive compensation from banker.
 - (d) Cancel the contract.
- III.** The settlement under an FRA is made.....
- (a) at the end of the loan period.
 - (b) at the beginning of the loan period.
 - (c) at the maturity of FRA.
 - (d) along with repayment of principal.
- IV.** The FRA settlement amount is discounted because.....
- (a) interest rates are uncertain

- (b) FRA is settled in advance
- (c) principals are not exchanged
- (d) only interest differential is paid

- V.** Interest Rate Futures differ from FRAs mainly because Futures are.....
- (a) OTC instruments
 - (b) tailor-made contracts
 - (c) standardized and exchange traded
 - (d) settled only at maturity

Answer Case Scenario – 02

- I. (c) Buy FRA at offer rate.**
- II. (c) Receive compensation from banker.**
- III. (b) at the beginning of the loan period.**
- IV. (b) FRA is settled in advance**
- V. (c) standardized and exchange traded**





Student Notes



Student Notes



05

FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

(1) BASICS

Question – 01

Given:

$$\text{US\$ } 1 = \text{¥ } 107.31$$

$$\text{£ } 1 = \text{US\$ } 1.26$$

$$\text{A\$ } 1 = \text{US\$ } 0.70$$

- (i) Calculate the cross rate for Pound in Yen terms
- (ii) Calculate the cross rate for Australian Dollar in Yen terms
- (iii) Calculate the cross rate for Pounds in Australian Dollar terms

(RTP November – 2020)**Solution:****(i) Calculation the cross rate for Pounds in ¥ terms**

$$\text{¥}/\$ = 107.31$$

$$\$/\text{£} = 1.26$$

$$\text{¥}/\text{£} = ?$$

$$\text{¥}/\text{£} = 107.31 \times 1.26$$

(ii) Calculation of cross rate for A\$ in ¥ terms

$$\text{¥}/\$ = 107.31$$

$$\$/\text{A\$} = 0.70$$

$$\begin{aligned}\text{¥}/\text{A\$} &= 107.31 \times 0.70 \\ &= \text{¥ } 75.12\end{aligned}$$

(iii) Calculation of cross rate for £ in A\$

$$\$/\text{£} = 1.26$$

$$\$/\text{A\$} = 0.7$$

$$\begin{aligned}\text{A\$}/\text{£} &= \frac{1}{0.7} \times 1.26 \\ &= \text{A\$ } 1.80.\end{aligned}$$

Question – 02

On January 28, 2013 an importer customer requested a Bank to remit Singapore Dollar (SGD) 2,500,000 under an irrevocable Letter of Credit (LC). However, due to unavoidable factors, the Bank could effect the remittances only on February 4, 2013. The inter-bank market rates were as follows:



	January 28, 2013	February 4, 2013
US\$ 1 =	₹ 45.85/45.90	₹ 45.91/45.97
GBP £ 1 =	US\$ 1.7840/1.7850	US\$ 1.7765/1.7775
GBP £ 1 =	SGD 3.1575/3.1590	SGD 3.1380/3.1390

The Bank wishes to retain an exchange margin of 0.125%

Required:

How much does the customer stand to gain or lose due to the delay?

(Note: Calculate the rate in multiples of 0.0001)

(SM TYK – 09 & Exam November – 2011) (4 Marks)

Solution:

Calculation of Exchange Rate ₹/SGD

28/01/2013

$$₹/\$ = 45.90$$

$$\$/\pounds = 1.7850$$

$$\text{SGD}/\pounds = 3.1575$$

$$₹/\text{SGD} = ₹/\$ \times \$/\pounds \times \text{SGD}/\pounds$$

$$= 45.90 \times 1.7850 \times \frac{1}{3.1575}$$

$$= 25.9482 + 0.125\%$$

$$= ₹ 25.9806.$$

04/02/2013

$$₹/\$ = 45.97$$

$$\$/\pounds = 1.7775$$

$$\text{SGD}/\pounds = 3.1380$$

$$₹/\text{SGD} = 45.97 \times 1.7775 \times \frac{1}{3.1380}$$

$$= 26.0341 + 0.125\%$$

$$= ₹ 26.0719$$

Loss due to delay

$$= (25.9806 - 26.0719) \times \text{SGD } 25,00,000$$

$$= ₹ 2,28,250.$$

Question – 03

The price of a bond just before a year of maturity is \$ 5,000. Its redemption value is \$ 5,250 at the end of the said period. Interest is \$ 350 p.a. The Dollar appreciates by 2% during the said period. Calculate the rate of return.

(SM TYK – 01 & Exam May – 2012) (4 Marks)

Solution:

Rate of return

US investor



$$\text{Return} = \frac{\$ 5,250 + \$ 350 - \$ 5,000}{\$ 5,000} \times 100 = 12\%$$

Other than US investor

$$\text{Return} = [(1.12 \times 1.02) - 1] \times 100 = 14.24\%$$

Question – 04

Mr. Mammen, an Indian investor invests in a listed bond in USA. If the price of the bond at the beginning of the year is USD 100 and it is USD 103 at the end of the year. The coupon rate is 3% payable annually.

Find the return on investment in terms of home country currency if:

- (i) USD is Flat.
- (ii) USD appreciates during the year by 3%.
- (iii) USD depreciates during the year by 3%.
- (iv) Indian Rupee appreciates during the year by 5%.

(RTP May – 2022 & Exam July – 2021) (4 Marks)

Solution:**Calculation of the return on investment**

(i) If USD is flat

$$\text{Return on Investment} = \frac{\$ 106 - \$ 100}{\$ 100} \times 100 = 6\% \text{ p.a.}$$

(ii) If \$ appreciator by 3%

$$\text{Return on Investment} = [(1.06 \times 1.03) - 1] \times 100 = 9.18\%$$

(iii) If \$ deprecator by 3%

$$\text{Return on Investment} [(1.06 \times 0.97) - 1] \times 100 = 2.82\%$$

(iv) If ₹ appreciator by 5% (If \$ depreciates by 5%)

$$\text{Return on Investment} = [(1.06 \times 0.95) - 1] \times 100 = 0.7\%$$

(2) SPOT MARKET**Question – 05**

Followings are the spot exchange rates quoted at three different forex markets:

USD/INR	48.30 in Mumbai
GBP/INR	77.52 in London
GBP/USD	1.6231 in New York

The arbitrageur has USD 1,00,00,000. Assuming that there are no transaction costs, explain whether there is any arbitrage gain possible from the quoted spot exchange rates.

(SM TYK – 13)

Solution:**Arbitrage process**

Sell \$ in Mumbai

$$= \$1,00,00,000 \times 48.30 = ₹ 48,30,00,000$$



Buy £ in London

$$= \frac{\text{₹ } 48,30,00,000}{77.52} = \text{£ } 62,30,650.15$$

Sell £ in New York

$$= \text{£ } 62,30,650.15 \times 1.6231 = \$ 1,01,12,968.27$$

Arbitrage

$$= \$ 1,01,12,968.27 - \$1,00,00,000 = \$ 1,12,968.27$$

Question – 06

USD 10,000 is lying idle in your Bank Account. You are able to get the following quotes from the dealers:

Dealer	Quote
A	EUR/USD 1.1539
B	EUR/GBP 0.9094
C	GBP/USD 1.2752

Is there an opportunity of gain from these quotes?

(Exam November – 2020) (4 Marks)

Solution:

The arbitrageur can proceed as started below to realize arbitrage gains.

(i) Buy € from US\$ 10,000 from Dealer A

$$(10,000/1.1539) = \text{€ } 8,666.26$$

(ii) Convert these € to £ by selling to Dealer B

$$(\text{€ } 8,666.26 \times 0.9094) = \text{£ } 7,881.09$$

(iii) Convert £ to US\$ by selling to Dealer C

$$(\text{£ } 7,881.09 \times 1.2752) = \text{US\$ } 10,049.97$$

There is net gain of US\$ 10,049.97 less US\$ 10,000 i.e. US\$ 49.97 or US\$ 50.00.

Question – 07

Citi Bank quotes JPY/ USD 105.00 - 106.50 and Honk Kong Bank quotes USD/JPY 0.0090- 0.0093.

- Are these quotes identical if not then how they are different?
- Is there a possibility of arbitrage?
- If there is an arbitrage opportunity, then show how would you make profit from the given quotation in both cases if you are having JPY 1,00,000 or US\$ 1,000.

(RTP November – 2020)

Solution:

City bank	Honk Kong Bank
¥/\$ 105.00/106.50	\$/¥ 0.0090/0.0093
\$/¥ 0.0094/0.0095	¥/\$107.53/111.11



- (i) No. these quote are not identical, City bank's quote direct quote for yen & Honk Kong bank's quote direct quote for \$.
- (ii) Since City bank rates of \$/¥ 0.0094/0.0095, both rate are higher than Honk Kong bank's rate, hence there is a possibility of arbitrage.

Arbitrage Gain

If we have ¥ 1,00,000

Buy \$ from City Bank

$$= \frac{¥ 1,00,000}{106.50} = \$ 938.967$$

Sell \$ to Honk Kong Bank

$$= \$ 938.967 \times 107.53 = ¥ 1,00,967$$

$$\begin{aligned} \text{Gain} &= ¥ 1,00,967 - ¥ 1,00,000 \\ &= ¥ 967 \end{aligned}$$

It we have \$ 1,000

Buy ¥ from Honk Kong Bank

$$= \$ 1,000 \times 107.53 = ¥ 1,07,530$$

Sell ¥ to City bank

$$\frac{¥ 1,07,530}{106.50} = \$ 1,009.67$$

$$\text{Gain} = \$ 1,009.67 - \$ 1,000$$

$$= \$ 9.67$$

Question – 08

Following are the direct quotes available in the international market:

$$\text{GBP 1} = \text{EURO} \quad 1.2950/65 \text{ (Direct rate)}$$

$$\text{GBP 1} = \text{USD } 1.6025/6,000$$

$$\text{EURO 1} = \text{USD } 1.2375/9,000$$

You are required to:

- (i) Calculate Bid & Ask Cross Rates for Euro per Pound (Euro/Pound)
- (ii) Prove that arbitrage gains are not possible if –
- (a) You buy Pounds against Euro under direct route and sell through cross rate route.
- (b) You sell Pounds against Euro under direct route and buy through cross rate route.

(Exam Jan – 2026)

Solution:**(i) Ask Cross Rate for Euro Per Pound**

$$= \frac{\text{USD/GBP}_{\text{Ask}}}{\text{USD/Euro}_{\text{Bid}}} = \frac{1.6025}{1.2375}$$

$$= \text{Euro } 1.2950/\text{GBP}$$



Bid Cross Rate for Euro per Pound

$$= \frac{\text{USD/GBP}_{\text{Bid}}}{\text{USD/Euro}_{\text{Ask}}} = \frac{1.6000}{1.9000}$$

= Euro 0.8421/ GBP

Final Quote: GBP 1 = EUR 0.8421 / 1.2950

(ii) Calculation of Arbitrage Gain/ Loss

(1)

Buying Pound against Euro under direct route	Euro 1.2965
Selling Pound against Euro through Cross Route	Euro 0.8421
Loss per Pound	Euro 0.4544

(2)

Buying Pound against Euro under Cross Route	Euro 1.2950
Selling Pound against Euro through Direct Route	Euro 1.2950
Loss per Pound	Nil

Arbitrage is not possible – Proof

- (1) If you want to buy Pounds against Euro, under the direct route, you can acquire Pound at the quoting bank's selling rate of Euro 1.2965/Pound. If you sell Pounds under cross rate route, you can sell Pound at the Cross-rate Bid price calculated in (i) above i.e. 0.8421/ Pound resulting in a loss of 0.4544 Euros. Hence arbitrage is not possible.
- (2) If you want to sell Pounds against Euro, under the direct route, you can sell Pounds at the quoting bank's buying rate of Euro 1.2950/ Pound. If you buy Pounds under cross rate route, you can get Pounds at the Cross rate Ask price

calculated in (i) above i.e. 1.2950/ Pound resulting in no profit no loss hence arbitrage is not possible.

(3) FORWARD CONTRACT

Question – 09

The following 2-way quotes appear in the foreign exchange market:

	Spot	2-months forward
RS/US \$	₹ 46.00/₹ 46.25	₹ 47.00/₹ 47.50

Required:

- (i) How many US dollars should a firm sell to get ₹ 25 lakhs after 2 months?
- (ii) How many Rupees is the firm required to pay to obtain US \$ 2,00,000 in the spot market?
- (iii) Assume the firm has US \$ 69,000 in current account earning no interest. ROI on Rupee investment is 10% p.a. Should the firm encase the US \$ now or 2 months later?

(SM TYK – 11)

Solution:

- (i) \$ Sell by firm to get ₹ 25,00,000

$$\frac{\text{₹ } 25,00,000}{47} = \$ 53,191.49$$

- (ii) Rupees required to buy \$ 2,00,000



$$\text{\$ } 2,00,000 \times \text{₹ } 46.25 = \text{₹ } 92,50,000$$

(iii) Encase Now

$$\text{Sell \$ } 69,000 \times 46 = \text{₹ } 31,74,000$$

Invest ₹ 31,74,000 @ 10% p.a for 2 Months

$$\begin{aligned} \text{Cash Inflow} &= 31,74,000 + (31,74,000 \times 10\% \times \frac{2}{12}) \\ &= \text{₹ } 32,26,900 \end{aligned}$$

Encase after 2 months

Sell \$ 69,000 at 2 months FR

$$\text{\$ } 69,000 \times 47 = \text{₹ } 32,43,000$$

Encase after 2 months is better due to higher cash inflows.

Question – 10

In March, 2009, the Multinational Industries make the following assessment of dollar rates per British pound to prevail as on 01/09/2009:

\$/Pound	Probability
1.60	0.15
1.70	0.20
1.80	0.25
1.90	0.20
2.00	0.20

(i) What is the expected spot rate for 01/09/2009?

(ii) If, as of March, 2009, the 6-month forward rate is \$ 1.80, should the firm sell forward its pound receivables due in September, 2009?

(SM TYK – 15)

Solution:

(i) Expected Spot Rate

$$\begin{aligned} &= (\text{\$ } 1.60 \times 0.15) + (1.70 \times 0.20) + (1.80 \times 0.25) + (1.90 \times 0.20) + (2.00 \times 0.20) \\ &= \text{\$ } 1.81. \end{aligned}$$

(ii) Since Expected spot rate is more than forward rate, hence firm should not sell its pound receivable.

Question – 11

JKL Ltd., an Indian company has an export exposure of JPY 10,000,000 receivable August 31, 2014. Japanese Yen (JPY) is not directly quoted against Indian Rupee.

The current spot rates are:

$$\text{INR/US \$} = \text{₹ } 62.22$$

$$\text{JPY/US\$} = \text{JPY } 102.34$$

It is estimated that Japanese Yen will depreciate to 124 level and Indian Rupee to depreciate against US \$ to ₹ 65.

Forward rates for August 2014 are

$$\text{INR/US \$} = \text{₹ } 66.50$$



JPY/US\$ = JPY 110.35

Required:

- (i) Calculate the expected loss, if the hedging is not done. How the position will change, if the firm takes forward cover?
- (ii) If the spot rates on August 31, 2014 are:

INR/US \$ = ₹ 66.25

JPY/US\$ = JPY 110.85

Is the decision to take forward cover justified?

(SM TYK – 06 & Exam May – 2014) (8 Marks)

Solution:

- (i) Calculation of SR, Expected SR & FR of ₹/¥**

Spot Rate:

₹/\$ 62.22

¥/\$ 102.34

₹/¥ $62.22 \times \frac{1}{102.34} = 0.6080$

Expected SR:

₹/\$ 65

¥/\$ 124

₹/¥ $65 \times \frac{1}{124} = 0.5242$

Forward Rate:

₹/\$ 66.50

¥/\$ 110.35

₹/¥ $66.50 \times \frac{1}{110.35} = 0.6026$

Calculation of Expected Loss if hedging is not done

If sell ¥ at SR (¥ 1,00,00,000 × 0.6080) = ₹ 60,80,000

If Sell ¥ at Expected SR (1,00,00,000 × 0.5242) = ₹ 52,42,000

Expected Loss	= ₹ 8,38,000
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Calculation of loss if forward cover is done

If Sell ¥ at SR = ₹ 60,80,000

If Sell ¥ at FR (¥ 1,00,00,000 × 0.6026) = ₹ 60,26,000

Loss	= ₹ 54,000
------	------------

Due to the forward contract loss is reduced to ₹ 54,000.

- (ii) Spot Rate as on August 31st, 2014**

₹/\$ = 66.25

¥/\$ = 110.85

₹/¥ $66.25 \times \frac{1}{110.85} = 0.5977$

Calculation of loss on the basis actual SR



If Sell ¥ at SR = ₹ 60,80,000

If Sell ¥ at SR on 31/08/2014
(¥ 1,00,00,000 × 0.5977) = ₹ 59,77,000

Loss = ₹ 1,03,000

Least loss in forward cover, hence forward cover is justified.

Question – 12

A company operating in Japan has today effected sales to an Indian company, the payment being due 3 months from the date of invoice. The invoice amount is 108 lakhs yen. At today's spot rate, it is equivalent to ₹ 30 lakhs. It is anticipated that the exchange rate will decline by 10% over the 3 months period and in order to protect the yen payments, the importer proposes to take appropriate action in the foreign exchange market. The 3 months forward rate is presently quoted as 3.3 yen per rupee. You are required to calculate the expected loss and to show how it can be hedged by a forward contract.

(SM TYK – 17)

Solution:

Calculation of Spot Rate & Estimated SR

Spot Rate: ¥ 108 Lakhs = ₹ 30 Lakhs

¥/₹ = $\frac{¥ 108}{₹ 30} = 3.60$

Expected SR: ¥/₹ = $3.60 \times 0.90 = 3.24$

Forward Rate ¥/₹ = 3.30

Calculation of Expected Loss, if hedging is not done

If Buy ¥ at SR = $\frac{¥ 108}{3.60}$ = ₹ 30 Lakhs

If Buy ¥ at Estimated SR = $\frac{¥ 108 \text{ Lakhs}}{3.24}$ = ₹ 33.33 Lakhs

Expected Loss = ₹ 3.33 Lakhs

Calculation of loss, if covered through forward contract

If Buy ¥ at SR = ₹ 30 Lakhs

If Buy ¥ at FR = $\frac{¥ 108 \text{ Lakhs}}{3.30}$ = ₹ 32.73 Lakhs

Loss = ₹ 2.73 Lakhs

Loss can be reduced to ₹ 2.73 Lakhs due to the forward contract.

Question – 13

KK Ltd. operating in Japan has today affected sales to an Indian company 'LM Limited, the payment being due in 3 months from the date of invoice. The invoice amount is 855 lakh yen which at today's spot rate is equivalent to ₹ 500 lakh. Looking to the present market conditions, it is anticipated that the exchange rate will decline by 10% over the 3 months period. In order to protect the yen payments, the importer proposes to take appropriate action in the foreign exchange market. The 3 months forward rate is presently quoted as 1.68 yen per rupee.

You are required to



- (i) Calculate the Expected Gain/(Loss) under present situation.
- (ii) Calculate the Expected Gain/(Loss) if hedged by forward contract.
- (iii) Recommend which option from above (i) and (ii) will best for the company.

Calculation up to 3 decimal points.

(Exam September – 2025) (4 Marks)

Solution:

Spot rate of ₹ 1 against yen = 855 lakhs yen/₹ 500 lakhs = 1.71 yen

3 months forward rate of ₹ 1 against yen = 1.68 yen

Anticipated decline in Exchange rate = 10%.

Expected spot rate after 3 months = 1.71 yen – 10% of 1.71

= 1.71 yen – 0.171 yen

= 1.539 yen per rupee

	₹ (in lakhs)
Present cost of 855 lakhs yen	500.000
Cost after 3 months: 855 lakhs yen/1.539 yen	<u>555.556</u>
Expected exchange loss	<u>55.556</u>

If the expected exchange rate risk is hedged by a Forward contract:

Present cost	500.000
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Cost after 3 months if forward contract is taken

855 lakhs yen/ 1.68 yen

508.929

Expected loss

8.929

Suggestion: If the exchange rate risk is not covered with forward contract, the expected exchange loss is ₹ 55.556 lakhs. This could be reduced to ₹ 8.929 lakhs if it is covered with Forward contract. Hence, taking forward contract is suggested.

Question – 14

A company is considering hedging its foreign exchange risk. It has made a purchase on 1st July, 2016 for which it has to make a payment of US\$ 60,000 on December 31, 2016. The present exchange rate is 1 US \$ = ₹ 65. It can purchase forward 1 \$ at ₹ 64. The company will have to make an upfront premium @ 2% of the forward amount purchased. The cost of funds to the company is 12% per annum.

In the following situations, compute the profit/loss the company will make if it hedges its foreign exchange risk with the exchange rate on 31st December, 2016 as:

- (i) ₹ 68 per US \$.
- (ii) ₹ 62 per US \$.
- (iii) ₹ 70 per US \$.
- (iv) ₹ 65 per US \$.

(SM TYK – 23 & Exam November – 2016) (8 Marks)



Solution:

Calculation of Cost of Hedging

Premium Amount = \$ 60,000 × 64 × 2% = ₹ 76,800

(+) Opportunity Cost = 76,800 × 12% × $\frac{6}{12}$ = ₹ 4,608

Cost of Hedging ₹ 81,408

Calculation of Profit /Loss to the Company

(i) Exchange Rate ₹ 68 per \$

Profit (68 – 64) × \$ 60,000 = ₹ 2,40,000

(–) Hedging Cost = ₹ 81,408

Net Profit = ₹ 1,58,592

(ii) If Exchange Rate ₹ 62 per \$

Loss = (₹ 62 – ₹ 64) × \$60,000 = ₹ 1,20,000

(+) Hedging Cost = ₹ 81,408

Net Loss = ₹ 2,01,408

(iii) If Exchange Rate ₹ 70 per \$

Profit = (₹ 70 – ₹ 64) × \$ 60,000 = ₹ 3,60,000

(–) Hedging Cost = ₹ 81,408

Net Profit = ₹ 2,78,592

(iv) If Exchange Rate is ₹ 65 per \$

Profit = (₹ 65 – ₹ 64) × \$ 60,000 = ₹ 60,000

(–) Hedging Cost = ₹ 81,408

Net Loss = ₹ 21,408

Question – 15

ZX Ltd. has made purchases worth USD 80,000 on 1st May 2020 for which it has to make a payment on 1st November 2020. The present exchange rate is INR/USD 75. The company can purchase forward dollars at INR/USD 74. The company will have to make an upfront premium @ 1 per cent of the forward amount purchased. The cost of funds to ZX Ltd. is 10 per cent per annum.

The company can hedge its position with the following expected rate of USD in foreign exchange market on 1st May 2020:

	Exchange Rate	Probability
(i)	INR/USD 77	0.15
(ii)	INR/USD 71	0.25
(iii)	INR/USD 79	0.20
(iv)	INR/USD 74	0.40

You are required to advise the company for a suitable cover for risk.

(Exam November – 2020) (8 Marks)



Solution:

- (i) ZX Ltd. not takes Forward Position

$$\text{Expected rate} = (77 \times 0.15) + (71 \times 0.25) + (79 \times 0.20) + (74 \times 0.40)$$

$$= 74.70$$

$$\text{Expected amount payable} = \$ 20,000 \times 74.70$$

$$= ₹ 59,76,000$$

- (ii) Hedge Position by Forward Cover

Company purchase \$ 80,000 forward premium (80,000 × 74 × 1%)	59,200
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Interest on forward premium (59,200 × 10% × 6/12)	2,960
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Total hedging cost	62,160
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Amount to be paid for \$ 80,000 (\$ 80,000 × 74)	₹ 59,20,000
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Total	₹ 59,82,160
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No hedging is better due to lower cash outflow.

Question – 16

Excel Exporters are holding an Export bill in United States Dollar (USD) 1,00,000 due 60 days hence. They are worried about the falling USD value which is currently at ₹ 45.60 per USD. The concerned

Export Consignment has been priced on an Exchange rate of ₹ 45.50 per USD. The Firm's Bankers have quoted a 60-day forward rate of ₹ 45.20.

Calculate:

- (i) Rate of discount quoted by the Bank
- (ii) The probable loss of operating profit if the forward sale is agreed to.

(SM TYK – 19)

Solution:

- (i) Rate of discount (1 year = 365 days)

$$\begin{aligned} \text{Premium/(Discount)} &= \frac{F - S}{S} \times 100 \times \frac{365}{60} \\ &= \frac{₹ 45.20 - 45.60}{45.60} \times 100 \times \frac{365}{60} \\ &= 5.34\% \end{aligned}$$

- (ii) Probable loss of operating profit if hedging is done

$$\begin{aligned} \text{Probable loss} &= (45.20 - 45.50) \times \$ 1,00,000 \\ &= ₹ 30,000 \end{aligned}$$

Question – 17

ABC Co. have taken a 6 month loan from their foreign collaborators for US Dollars 2 millions. Interest payable on maturity is at LIBOR plus 1.0%. Current 6-month LIBOR is 2%.



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

v) Variable Cost (iii × iv)	₹ 27,00,000	₹ 39,50,000	₹ 40,80,000
vi) Forward Rate	2.427	0.0216	0.0178
vii) Sale if hedging (i/vi)	₹ 32,13,844	₹ 47,36,111	53,88,764
viii) Expected SR	2.459	0.02156	0.0179
ix) Sales if hedging not done	31,72,021	47,44,898	53,58,659

Total VC (27,00,000 + 39,50,000 + 40,80,000) = 1,07,30,000

Sales FC (32,13,844 + 47,36,111 + 53,88,764) = 1,33,38,719

Sales ESR (31,72,021 + 47,44,898 + 53,58,659) = 1,32,75,578

Average Contribution to Sales Ratio

If hedging is done (Forward Contract)

$$\frac{1,33,38,719 - 1,07,30,000}{1,33,38,719} \times 100 = 19.56\%$$

If Hedging is not done $\frac{1,32,75,578 - 1,07,30,000}{1,32,75,578} \times 100 = 19.17\%$

Forward cover is better due to higher contribution to sales ratio.

Question – 19

You have following quotes from Bank A and Bank B:

	Bank A	Bank B
SPOT	USD/CHF 1.4650/55	USD/CHF 1.4653/60
3 Months	5/10	
6 Months	10/15	
SPOT	GBP/USD 1.7645/60	GBP/USD 1.7640/50

3 Months	25/20	
6 Months	35/25	

Calculate:

- How much minimum CHF amount you have to pay for 1 Million GBP spot?
- Considering the quotes from Bank A only, for GBP/CHF what are the Implied Swap points for Spot over 3 months?

(SM TYK – 47)

Solution:

- Calculation of CHF Amount to pay £ 1 Million

Buy \$ from Bank A = CHF/\$ 1.4655

Sell \$ & Buy £ from Bank B = \$/£ 1.7650

$$\begin{aligned} \text{Calculation} &= \text{CHF}/\text{£} = 1.4655 \times 1.7650 \\ &= 2.5866 \end{aligned}$$

Minimum CHF required to buy £ 10,00,000

$$£ 10,00,000 \times 2.5866 = \text{CHF } 25,86,600$$

- Calculation of 3 months Swap

Spot Rate

CHF/\$ 1.4650/1.4655

\$/£ 1.7645/1.7660



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Bid Rate

$$\text{CHF/£} \quad 1.4650 \times 1.7645 = 2.5850$$

$$1.4655 \times 1.7660 = 2.5881$$

3 Months FR

$$\text{CHF/\$} \quad 1.4655/1.4665$$

$$\text{\$/£} \quad 1.7620/1.7640$$

Bid Rate

$$\text{CHF/£} \quad 1.4655 \times 1.7620 = 2.5822$$

$$1.4665 \times 1.7640 = 2.5869$$

Swap Points

$$3 \text{ Month FR} \quad 2.5822/2.5869$$

$$\begin{array}{r} (-)\text{Spot} \quad \quad \quad 2.5850/2.5881 \\ \hline \quad \quad \quad -0.0028/-0.0012 \end{array}$$

$$3 \text{ Month Swap Points} \quad = 28/12.$$

Question – 20

An importer customer of your bank wishes to book a forward contract with your bank on 3rd September for sale to him of SGD 5,00,000 to be delivered on 30th October.

The spot rates on 3rd September are USD/INR 49.3700/3800 and USD/SGD 1.7058/68. The swap points are:

USD/INR		USD/SGD	
Spot/September	0300/0400	1 st Month Forward	48/49
Spot/October	1100/1300	2 nd Month Forward	96/97
Spot/November	1900/2200	3 rd Month Forward	138/140
Spot/December	2700/3100		
Spot/January	3500/4000		

Calculate the rate to be quoted to the importer by assuming an exchange margin of 5 paisa.

(SM TYK – 16 & Exam May – 2018) (8 Marks)

Solution:

Calculation of 2 Month FR

₹/\$

$$\text{SR} \quad 49.3700/49.3800$$

$$(+)\text{ Swap} \quad 0.1100/0.1300$$

$$\hline 49.4800/49.5100$$

SGD/\$

$$\text{SR} \quad 1.7058/1.7068$$

$$(+)\text{ Swap} \quad 0.0096/0.0097$$

$$\hline 1.7154/1.7165$$

$$\text{₹/SGD} \quad = 49.5100 \times \frac{1}{1.7154}$$



$$= 28.8621$$

$$\text{Customer Rate} = 28.8621 + 0.05$$

$$= 28.9121$$

Question – 21

In International Monetary Market an international forward bid for December, 15 on pound sterling is \$ 1.2816 at the same time that the price of IMM sterling future for delivery on December, 15 is \$ 1.2806. The contract size of pound sterling is £ 62,500. How could the dealer use arbitrage in profit from this situation and how much profit is earned?

(SM TYK – 20)**Solution:**

In this situation, we should buy £ 62500 from future & sell in forward market.

Calculation of arbitrage gain

Buy £ 62,500 in forward market (£ 62,500 × 1.2806)	\$ 80,037.50
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Sell £ 62,500 in future market (£ 62,500 × 1.2816)	\$ 80,100.00
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Arbitrage Gain	\$ 62.50
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Question – 22

The current spot exchange rate is \$1.35/£ and the three-month forward rate is \$1.30/£. According to your analysis of the exchange

rate, you are quite confident that the spot exchange rate will be \$1.32/£ after 3 months.

- Suppose you want to speculate in the forward market then what course of action would be required and what is the expected dollar Profit (Loss) from this speculation?
- What would be your Profit (Loss) in Dollar terms on the position taken as per your speculation if the spot exchange rate turns out to be \$1.26/£.

Assume that you would like to buy or sell £1,000,000.

(RTP November – 2020)**Solution:**

- Since expected spot rate is more than forward rate hence we should buy £ at forward rate i.e., long position on £.

Expected Profit	= (\$1.32 – \$1.30) × £ 10,00,000
	= \$ 20,000

- If actual spot rate is \$/£ in 1.26 then loss on long position of pound

Loss	= (\$1.26 – \$1.30) × £ 10,00,000
	= \$ 40,000

Question – 23

On April 3, 2016, a Bank quotes the following:

Spot exchange Rate (US \$ 1)	INR 66.2525	INR 67.5945
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FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

2 months' swap points	70	90
3 months' swap points	160	186

In a spot transaction, delivery is made after two days.

Assume spot date as April 5, 2016.

Assume 1 swap point = 0.0001,

You are required to:

- Ascertain swap points for 2 months and 15 days. (For June 20, 2016),
- Determine foreign exchange rate for June 20, 2016, and
- Compute the annual rate of premium/discount of US\$ on INR, on an average rate.

(SM TYK – 05 & Exam November – 2016) (8 Marks)

Solution:

- Calculation of swap points for 2 month & 15 days.

	Bid Rate	Ask Rate
2 Month Swap points	70	90
(+) 15 days swap point	$\left(\frac{160 - 70}{2}\right)$	$\left(\frac{186 - 90}{2}\right)$
	45	48
2 months & 15 days swap	115	138

- Calculation of 2 months 15 days FR

Spot Rate	66.2525
67.5945	

(+) Swap	0.0115	0.0138
Forward rate	66.2640	67.6083

Calculation of Premium/Discount of \$

$$\text{Premium/Discount} = \frac{F - S}{\text{Average}} \times 100 \times \frac{12}{2.5}$$

$$\begin{aligned} \text{Bid rate} &= \frac{66.2640 - 66.2525}{\frac{66.2640 + 66.2525}{2}} \times 100 \times \frac{12}{2.5} \\ &= 0.0833\% \text{ Premium} \end{aligned}$$

$$\begin{aligned} \text{Ask rate} &= \frac{67.6083 - 67.5945}{\frac{67.6083 + 67.5945}{2}} \times 100 \times \frac{12}{2.5} \\ &= 0.0980\% \text{ Premium.} \end{aligned}$$

(4) COVER DEAL

Question – 24

You sold Hong Kong Dollar 1,00,00,000 value spot to your customer at ₹ 5.70 & covered yourself in London market on the same day, when the exchange rates were

US\$ 1 = H.K. \$ 7.5880 7.5920

Local inter bank market rates for US\$ were

Spot US\$ 1 = ₹ 42.70 42.85

Calculate cover rate and ascertain the profit or loss in the transaction. Ignore brokerage.



(SM TYK – 07)**Solution:****Calculation of Cover Rate**

Buy \$ from local Inter Bank Market @ ₹/\$ 42.85

Buy HK \$ from London Market @ HK\$/\$ 7.5880

$$\text{₹/HK\$ Cover Rate} = 42.85 \times \frac{1}{7.5880} = 5.6471$$

Calculation Profit/Loss

Amount received from customer (HK\$1,00,00,000 × 5.70)

$$= ₹ 5,70,00,000$$

Amount paid on cover deal (HK\$1,00,00,000 × 5.6471)

$$= ₹ 5,64,70,000$$

$$\text{Profit} = ₹ 5,29,000$$

Question – 25

Mr. H as Treasurer for your bank working under you sold HK\$ 10 million value Spot to your customer at ₹ 10.55/ HK\$ and covered yourself in the London market on the same day when the exchange rates were:

$$\text{US\$ 1} = \text{HK\$ } 7.8880 / 7.8920$$

Local interbank market rates for US\$ were:

$$\text{Spot US\$ 1} = ₹ 82.70 / 82.85$$

Required:

- (i) Calculate Cover Rate
- (ii) Calculate Profit or loss in the transaction
- (iii) Do you agree with the views of the Internal Auditor that Mr. H has a speculative nature?

Note: Ignore brokerage.**(MTP September – 2024)****Solution:**

- (i) Rupee – Dollar Selling Rate: = ₹ 82.85
 Dollar – Hong Kong Dollar Buying Rate: = H.K.\$ 7.8880
 Hong Kong Dollar (Selling) Cross Rate: = ₹ 82.85 / 7.8880

$$= ₹ 10.5033$$

- (ii) Profit / Loss to the Bank
 Amount received from customer
 (HK\$ 10 million × 10.55) ₹ 10,55,00,000
 Amount paid on cover deal
 (HK\$ 10 million × ₹ 10.5033) ₹ 10,50,33,000
 Profit to Bank ₹ 4,67,000

- (iii) To some extent, we agree with views of Internal Auditor as the gain on the same transaction is bit lesser keeping in view the amount involved.



Question – 26

You, a foreign exchange dealer of your bank, are informed that your bank has sold a T.T. on Copenhagen for Danish Kroner 10,00,000 at the rate of Danish Kroner 1 = ₹ 6.5150. You are required to cover the transaction either in London or New York market. The rates on that date are as under:

Mumbai – London	₹ 74.3000	₹ 74.3200
Mumbai – New York	₹ 49.2500	₹ 49.2625
London – Copenhagen	DKK 11.4200	DKK 11.4350
New York – Copenhagen	DKK 07.5670	DKK 07.5840

In which market will you cover the transaction, London or New York, and what will be the exchange profit or loss on the transaction? Ignore brokerages.

(SM TYK – 08 & Exam November – 2013) (8 Marks)

Solution:**Calculation of Cover Rate****London Market**

- Buy £ from Bank @ ₹/£ 74.3200
- Buy DKK from Bank @ DKK/£ 11.4200

$$\begin{aligned}\text{₹/DKK} &= ₹ 74.3200 \times \frac{1}{11.4200} \\ &= 6.5079\end{aligned}$$

New York Market

- Buy \$ from Bank @ ₹/\$ 49.2625

- Buy DKK from Bank @ DKK/\$ 7.5670

$$\begin{aligned}\text{₹/DKK} &= ₹ 49.2625 \times \frac{1}{7.5670} \\ &= 6.5102\end{aligned}$$

Covered from London Market is better & Cover Rate ₹/DKK is 6.5079.

Calculation of Profit/Loss

$$\begin{aligned}\text{Amount received from customer (DKK 10,00,000} \times 6.5150) \\ &= ₹ 65,15,000\end{aligned}$$

$$\text{Amount paid on cover deal (DKK 10,00,000} \times 6.5079) = ₹ 65,07,900$$

$$\text{Profit to Bank} = ₹ 7,100$$

(5) EXCHANGE RATE DETERMINATION**(I) INTEREST RATE PARITY****Question – 27**

On April 1, 3 months interest rate in the UK £ and US \$ are 7.5% and 3.5% per annum respectively. The UK £/US \$ spot rate is 0.7570. What would be the forward rate for US \$ for delivery on 30th June?

(SM TYK – 31)



Solution:

$$£/\$ = £ 0.7570$$

Calculation of 3 Months FR using IRP**IRP Equation**

$$\frac{F}{S} = \frac{1 + r_A}{1 + r_B}$$

$$\frac{F}{£ 0.7570} = \frac{1 + (7.5\% \times 3/12)}{1 + (3.5\% \times 3/12)}$$

$$\frac{F}{£ 0.7570} = \frac{1.01875}{1.00875}$$

$$F = \frac{£ 0.7570 \times 1.01875}{1.00875}$$

$$F = £ 0.7645.$$

Question – 28

The US dollar is selling in India at ₹ 55.50. If the interest rate for 6 months borrowing in India is 10% per annum and the corresponding rate in USA is 4%.

- (i) Do you expect that US dollar will be at a premium or at discount in the Indian Forex Market?
- (ii) What will be the expected 6-months forward rate for US dollar in India? and
- (iii) What will be the rate of forward premium or discount?

(SM TYK – 14)**Solution:**

- (i) As per IRP, Currency of a country having lower rate of interest will be at premium in comparison to currency of country having higher rate of Interest. In this question US \$ will be at premium due to lower rate of Interest.

- (ii) Calculation of 6 months FR using IRP

$$\frac{F}{S} = \frac{1 + r_a}{1 + r_b}$$

$$\frac{F}{₹ 55.50} = \frac{1 + (10\% \times 6/12)}{1 + (4\% \times 6/12)}$$

$$\frac{F}{₹ 55.50} = \frac{1.05}{1.02}$$

$$F = \frac{55.50 \times 1.05}{1.02}$$

$$= ₹ 57.13$$

- (iii) Calculation of premium a discount in \$

$$= \frac{F - S}{S} \times 100 \times \frac{12}{6}$$

$$= \frac{57.13 - 55.50}{55.50} \times 100 \times \frac{12}{6}$$

$$= 5.87\% \text{ premium in \$}.$$



(II) COVERED INTEREST ARBITRAGE**Question – 29**

Spot rate 1 US \$ = ₹ 48.0123

180 days Forward rate for 1 US \$ = ₹ 48.8190

Annualized interest rate for 6 months – Rupee = 12%

Annualized interest rate for 6 months – US \$ = 8%

Is there any arbitrage possibility? If yes how an arbitrageur can take advantage of the situation, if he is willing to borrow ₹ 40,00,000 or US \$83,312.

(SM TYK – 26)

Solution:**Calculation of Premium is \$ & Interest Rate Difference**

$$\begin{aligned}\text{Premium is \$} &= \left(\frac{48.8190 - 48.0123}{48.0123} \right) \times 100 \times \frac{12}{6} \\ &= 3.36\% \text{ p.a.}\end{aligned}$$

$$\text{Interest Rate difference} = 12\% - 8\% = 4\%$$

Yes, there is a possibility of arbitrage because premium in \$ is not equal to interest rate difference.

Since premium in \$ is less than interest rate difference, hence borrow from USA and invest in India.

Arbitrage Process**Today**

- Borrow \$ 83,312 from USA @ 8% p.a. for 6 months.
- Sell \$ at SR (\$ 83,312 × ₹ 48.0123) = ₹ 40,00,000
- Invest ₹ 40,00,000 @ 12% p.a. for 6 Months.
- Contract to buy \$ at 6 months FR.

After 6 Months**Cash Inflows****Investment Amount (₹)**

$$(\text{₹ } 40,00,000 \times 1.06) = \text{₹ } 42,40,000$$

$$\text{Buy \$ at FR } \left(\frac{\text{₹ } 42,40,000}{48.8190} \right) = \$ 86,851.43$$

Cash Outflows**Repayment of loan**

$$\$ 83,312 (1.04) = \$ 86,644.48$$

$$\begin{aligned}\text{Arbitrage Gain} &= (\$ 86,851.43 - \$ 86,644.48) \\ &= \$ 206.95\end{aligned}$$

$$\text{Or } \$ 206.95 \times 48.8190 = \text{₹ } 10,103$$

Question – 30

Given the following information:

Exchange rate – Canadian dollar 0.665 per DM (spot)

Canadian dollar 0.670 per DM (3 months)

Interest rates – DM 7% p.a.



Canadian Dollar – 9% p.a.

What operations would be carried out to take the possible arbitrage gains?

(SM TYK – 27, Exam May – 2011) (4 Marks)

Solution:

Calculation of Premium in DM & Interest Rate Difference

$$\begin{aligned}\text{Premium in DM} &= \left(\frac{\$ 0.670 - \$ 0.665}{\$ 0.665} \right) \times 100 \times \frac{12}{3} \\ &= 3.007\% \text{ p.a.}\end{aligned}$$

$$\text{Interest Rate Difference} = 9\% - 7\% = 2\%$$

Since premium in DM is more than interest rate difference, hence borrow in Can \$ & Invest in DM.

Arbitrage Process

It is assumed that we borrow Can \$1,000.

Today

Borrow Can \$ 1,000 @ 9% p.a. for 3 months.

$$\text{Sell Can \$ at SR } \frac{\text{Can \$ 1000}}{0.665} = \text{DM 1,503.7594}$$

Invest DM 1,503.7594 @ 7% p.a. for 3 Months.

After 3 Months

Cash Inflows

Investment Amount

$$\begin{aligned}\text{DM} &= ₹ 1,503.7594 (1.0175) \\ &= 1,530.0752\end{aligned}$$

$$\begin{aligned}\text{Sell DM at FR} &= 1,530.0752 \times 0.670 \\ &= \text{Can \$ 1,025.15}\end{aligned}$$

Cash Outflows

$$\begin{aligned}\text{Repayment} &= \text{Can \$ 1,000 (1.0225)} \\ &= \text{Can \$ 1,022.50}\end{aligned}$$

$$\begin{aligned}\text{Arbitrage gain} &= (\text{Can \$ 1,025.15} - \text{Can \$ 1,022.50}) \\ &= \$ 2.65.\end{aligned}$$

Question – 31

$$\text{Spot rate 1 US\$} = ₹ 68.50$$

USD premium on a six month forward in 3%. The annualized interest in US is 4% and 9% in India.

Is there any arbitrage possibility? If yes, how a trader can take advantage of the situation if he is willing to borrow USD 3 million.

(Exam November – 2018) (8 Marks)

Solution:

(i) Calculation of Forward Rate

$$\text{Premium in \$} = \frac{F - S}{S} \times 100 \times \frac{12}{6}$$



$$\begin{aligned}
 6 &= \frac{F - 68.50}{68.50} \times 100 \times \frac{12}{6} \\
 6 \times \frac{6}{12} &= \frac{F - 68.50}{68.50} \times 100 \\
 0.03 &= \frac{F - 68.50}{68.50} \\
 2.055 &= F - 68.50 \\
 F &= ₹ 70.56
 \end{aligned}$$

(ii) Premium in \$ = 6% P.a.

Interest Rate Different = 9 – 4 = 5%

Yes, there is an arbitrage possibility because premium in \$ is not equal to interest rate difference.

Since premium in \$ is more than interest rate difference hence borrow from India & invest in USA.

Arbitrage Process.

Today

- Borrow equivalent ₹ (\$ 30,00,000)
- Buy \$ at SR $\left(\frac{₹ 20,55,00,000}{68.50}\right) = \$ 30,00,000$
- Invest \$ in USA & 4% p.a. for 6 months.
- Contract to sell \$ at 6 months FR

After 6 Months

Cash Inflows

Investment Amount in (\$) [\$ 30,00,000 (1.02)] = \$ 30,60,000

Sell \$ at 6 Months FR [\$ 30,60,000 × 70.56] = ₹ 21,59,13,600

Cash outflows

Repayment [₹ 20,55,00,000 (1.045)] = ₹ 21,47,47,500

Arbitrage Gain = ₹ 11,66,100

or $\frac{11,66,100}{70.56} = \$ 16,526$

(III) PURCHASING POWER PARITY

Question – 32

The rate of inflation in India is 8% per annum and in the U.S.A. it is 4%. The current spot rate for USD in India is ₹ 46. What will be the expected rate after 1 year and after 4 years applying the Purchasing Power Parity Theory.

(SM TYK – 30, Exam May – 2010) (4 Marks)

Solution:

Calculation of expected rate applying PPP

1 year

$$\frac{ES_1}{₹ 46} = \frac{1.08}{1.04}$$

$$ES_1 = \frac{46 \times 1.08}{1.04} = ₹ 47.77$$



2 year

$$\frac{ES_2}{47.77} = \frac{1.08}{1.04}$$

$$ES_2 = \frac{₹ 47.77 \times 1.08}{1.04} = ₹ 49.61$$

3 year

$$\frac{ES_3}{49.61} = \frac{1.08}{1.04}$$

$$ES_3 = \frac{₹ 49.61 \times 1.08}{1.04} = ₹ 51.52$$

4 year

$$\frac{ES_4}{51.52} = \frac{1.08}{1.04}$$

$$ES_4 = \frac{₹ 51.52 \times 1.08}{1.04} = ₹ 53.50$$

(IV) INTERNATIONAL FISHER EFFECT

Question – 33

A US investor chose to invest in Sensex for a period of one year. The relevant information is given below.

Size of investment (\$)	20,00,000
Spot rate 1 year ago (₹/\$)	42.50/60
Spot rate now (₹/\$)	43.85/90

Sensex 1 year ago	3,256
Sensex now	3,765
Inflation in US	5%
Inflation in India	9%

- Compute the nominal rate of return to the US investor.
- Compute the real depreciation / appreciation of Rupee.
- What should be the exchange rate if relevant purchasing power parity holds good?
- What will be the real return to an Indian investor in Sensex?

(RTP May – 2022 & January – 2021)

Solution:

(i) Calculation of nominal Rate of Return to US Investor

Size of investment = \$ 20,00,000

Sell \$ 1 year ago = \$ 20,00,000 × ₹ 42.50
= ₹ 8,50,00,000

Invest ₹ 8,50,00,000 in sensex

After 1 year Investment value = $\frac{₹ 8,50,00,000}{3,256} \times 3,765$
= ₹ 9,82,87,776

Buy \$ at SR = $\frac{₹ 9,82,87,776}{43.90}$
= \$ 22,38,901



Nominal rate of return to US investor

$$= \frac{\$ 22,38,901 - \$ 20,00,000}{\$ 20,00,000} \times 100$$

$$= ₹ 11.94\% \text{ P.a.}$$

(ii) Calculation of real appreciation or deprecation in ₹

Real rate of \$ (Without Inflation)

$$\text{Bid Rate} = \frac{₹ 43.85}{s} = \frac{1.09}{1.05}$$

$$S = \frac{43.85 \times 1.05}{1.09} = 42.24$$

$$= \frac{S - F}{F} \times 100$$

$$= \frac{42.50 - 42.24}{42.24} \times 100 = 0.61\%$$

$$\text{Ask Rate} = \frac{₹ 43.90}{S} = \frac{1.09}{1.05}$$

$$S = \frac{₹ 43.90 \times 1.05}{1.09} = 42.29$$

$$= \frac{42.60 - 42.29}{42.29} \times 100 = 0.73\% \text{ Premium in ₹}$$

(iii) Calculation of Estimate SR as per PPP

$$\text{Bid Rate} = \frac{E(S)}{42.50} = \frac{1.09}{1.05} = ₹ 44.12$$

$$\text{Ask Rate} = \frac{E(S)}{42.60} = \frac{1.09}{1.05} = ₹ 44.22$$

(iv) Real rate of return on sensex for Indian Investor

$$\text{Sensex Return} = \frac{3,765 - 3,256}{3,256} \times 100$$

$$= 15.63\%$$

$$\text{Real Return} = \left[\frac{1.1563}{1.09} - 1 \right] \times 100$$

$$= 6.08\%$$

Question – 34

Shoe Company sells to a wholesaler in Germany. The purchase price of a shipment is 50,000 deutsche marks with term of 90 days. Upon payment, Shoe Company will convert the DM to dollars. The present spot rate for DM per dollar is 1.71, whereas the 90-day forward rate is 1.70.

You are required to calculate and explain:

- (i) If Shoe Company were to hedge its foreign-exchange risk, what would it do? What transactions are necessary?
- (ii) Is the deutsche mark at a forward premium or at a forward discount?
- (iii) What is the implied differential in interest rates between the two countries? (Use interest-rate parity assumption).



Solution:

- (i) It Shoe company want to hedge the risk then it should enter into forward contract & Sell its DK receivable at forward rate i.e. DM/\$ 1.70

$$\begin{aligned}\text{Amount Receivable in \$} &= \frac{\text{DM } 50,000}{1.70} \\ &= \$ 29,411.76\end{aligned}$$

- (ii) DM is at premium because DM is appreciating in forward market

$$\begin{aligned}\text{Premium in DM} &= \frac{S - F}{F} \times 100 \\ &= \frac{1.71 - 1.70}{1.70} \times 100 \times \frac{365}{90} \\ &= 2.386\% \text{ P.a.}\end{aligned}$$

- (iii) Since IRP hold good, hence premium in DM is equal to interest rate difference. Therefore Interest Rate Difference is 2.386% p.a. & rate of interest in DM is less than rate of interest in \$ by 2.386% p.a.

Question – 35

XYZ has taken a six-month loan from its foreign collaborator for USD 2 millions. Interest is payable on maturity @ LIBOR plus 1%. The following information is available:

Spot Rate	INR/USD	68.5275
6 months Forward rate	INR/USD	68.4575

6 months LIBOR for USD 2%

6 months LIBOR for INR 6%

You are required to :

- Calculate Rupee requirements if forward cover is taken.
- Advise the company on the forward cover.

What will be your opinion if spot rate of INR/USD is 68.4275 ?

(Exam Jan – 2021) (8 Marks)

Solution:

- (i) Rupee requirement if forward cover is taken:

6 Month Forward rate
68.4575

Interest amount $(20,00,000 \times 3\% \times \frac{6}{12})$ US\$ 30,000

Principal amount	US\$ 20,00,000
	US\$ 20,30,000

Rupee Requirement = INR 68.4575 × US\$ 20,30,000

= INR 13,89,68,725

- (ii) Forward Rate as per Interest Rate Parity after 6 months is expected to be:

$$= 68.5275 \times \frac{(1.03)}{(1.01)} = 69.8845/\text{US\$}$$



The company should take forward cover because as per Interest Rate Parity, the rate after 6 months is expected to be higher than forward rate.

However, if spot rate is 68.4275, the expected rate as per Interest Rate Parity shall be:

$$= 68.4275 \times \frac{(1.03)}{(1.01)} = 69.7825/\text{US\$}$$

Thus, still the company should take forward cover.

(6) FOREIGN CURRENCY EXPOSURES

(1) LEADING & LAGGINGS

Question – 36

An Indian importer has to settle an import bill for \$ 1,30,000. The exporter has given the Indian exporter two options:

- (i) Pay immediately without any interest charges.
- (ii) Pay after three months with interest at 5 percent per annum.

The importer's bank charges 15 percent per annum on overdrafts. The exchange rates in the market are as follows:

Spot rate (₹ / \$) : 48.35 / 48.36

3-Months forward rate (₹ / \$) : 48.81 / 48.83

The importer seeks your advice. Give your advice.

(SM TYK – 21 & Exam November – 2011) (4 Marks)

Solution:

Calculation of Cash Outflows (₹)

(i) Pay Immediately

Rupees required to buy \$ 1,30,000 at SR

$$(\$ 1,30,000 \times 48.36) = ₹ 62,86,800$$

Borrow ₹ 62,86,800 @ 15% p.a. for 3 Month

$$\begin{aligned} \text{Cash Outflows} &= 62,86,800 (1.0375) \\ &= ₹ 65,22,555 \end{aligned}$$

(ii) Pay after 3 months

$$\begin{aligned} \$ \text{ Payable with interest} &= \$ 1,30,000 (1.0375) \\ &= \$ 1,31,625 \end{aligned}$$

Buy \$ 1,31,625 at 3 months FR

$$\text{Cash outflow} = \$1,31,625 \times 48.83 = ₹ 64,27,249$$

Pay after 3 months is better due to the lower cash outflows.

Question – 37

Z Ltd. importing goods worth USD 2 million, requires 90 days to make the payment. The overseas supplier has offered a 60 days interest free credit period and for additional credit for 30 days an interest of 8% per annum.



The bankers of Z Ltd offer a 30 days loan at 10% per annum and their quote for foreign exchange is as follows:

	₹
Spot 1 USD	56.50
60 days forward for 1 USD	57.10
90 days forward for 1 USD	57.50

You are required to evaluate the following options:

- Pay the supplier in 60 days, or
- Avail the supplier's offer of 90 days credit.

(SM TYK – 12)

Solution:

Calculation of Cash Outflows (₹)

(i) Pay the supplier in 60 days

Rupees required to buy \$ 20,00,000 at 60 days FR

$$(\$ 20,00,000 \times 57.10) = ₹ 11,42,00,000$$

Borrow ₹ 11,42,00,000 @ 10% P.a. for 30 day cash outflows

$$(₹ 11,42,00,000 \times 1.00833) = 11,51,51,667$$

(ii) Pay the supplier in 90 days

$$\begin{aligned} \$ \text{ Payable with interest} &= \$ 20,00,000 \times 1.00667 \\ &= \$ 20,13,333 \end{aligned}$$

Buy \$ 20,13,333 at 90 days FR

$$\begin{aligned} \text{Cash outflow} &= \$ 20,13,333 \times 57.50 \\ &= ₹ 11,57,66,667 \end{aligned}$$

Pay the supplier in 60 days is better due to the lower cash outflows.

Question – 38

XYZ Ltd. has imported goods to the extent of US\$ 8 Million. The payment terms are as under:

- 1% discount if full amount is paid immediately; or
- 60 days interest free credit. However, in case of a further delay up to 30 days, interest at the rate of 8% p.a. will be charged for additional days after 60 days. M/s XYZ Ltd. has ₹ 25 Lakh available and for remaining it has an offer from bank for a loan up to 90 days @ 9.0% p.a.

The quotes for foreign exchange are as follows:

Spot Rate INR/ US\$ (buying)	₹ 66.98
60 days Forward Rate INR/ US\$ (buying)	₹ 67.16
90 days Forward Rate INR/ US\$ (buying)	₹ 68.03

Advise which one of the following options would be better for XYZ Ltd.

- Pay immediately after utilizing cash available and for balance amount take 90 days loan from bank.
- Pay the supplier on 60th day and avail bank's loan (after utilizing cash) for 30 days.

FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

(iii) Avail supplier offer of 90 days credit and utilize cash available.

Further presume that the cash available with XYZ Ltd. will fetch a return of 4% p.a. in India till it is utilized.

Assume year has 360 days. Ignore Taxation.

Compute your working upto four decimals and cash flows in Crore.

(RTP November – 2021)

Solution:

Option 1: Pay Immediately

Particulars	Amount
\$ Payable After Discount (\$ 8 Million × 0.99)	7.92 Million
Rupees required to buy (7.92 Million × 66.98) ₹	₹ 53.0482 Cr.
530.486 Million	
(-) Cash Available	₹ 0.25 Cr.
Fund Required	₹ 52.7982 Cr.
Borrow ₹ 52.7982 Cr. @ 9% p.a. for 90 days	
Cash Outflow ₹ 52.7982 × $\left[1 + \left(0.09 \times \frac{90}{360}\right)\right]$	53.9862 Cr.

Option 2: Pay in 60 days

Particulars	Amount
\$ Payable	\$ 8 Million
₹ Required to buy \$ 8 Million at 60 days FR	
\$ 8 Millions × 67.16	₹ 53.7280 Cr.

(-) Available Amount $0.25 \times \left[1 + 0.04 \times \frac{60}{360}\right]$	₹ 0.2517 Cr.
Fund required	₹ 53.4763
Borrow ₹ 53.4763 Cr. @ 9% p.a. for 30 days	
Cash Outflow ₹ 53.4763 Cr. × $\left[1 + \left(0.09 \times \frac{30}{360}\right)\right]$	₹ 53.8774 Cr.

Option 3: Pay in 90 Days

Particulars	Amount
\$ Payable with interest \$ 8 × $\left[1 + \left(0.08 \times \frac{30}{360}\right)\right]$	8.0533 Million
Rupees required to buy \$ 8.0533 at 90 days FR	
\$ 8.0533 × 68.03	₹ 54.7866 Cr.
(-) Available Cash 0.25 Cr × $\left[1 + 0.04 \times \frac{90}{360}\right]$	0.2525 Cr
Cash Outflow	54.5341

Pay the supplier on 60th days is the best option due to lower cash outflows.

Question – 39

XYZ Ltd. has imported goods to the extent of US \$ 8 Million. The payment terms are as under:

(1) 1% discount if full amount is paid immediately, or



- (2) 60 days interest free credit. However, in case of a further delay up to 30 days, interest at the rate of 8% p.a. will be charged for additional days after 60 days. XYZ Ltd. has ₹ 25 Lakh available and for remaining it has an offer from bank for a loan upto 90 days @ 9.0% p.a.

The quotes for foreign exchange are as follows:

Spot Rate INR/US\$ (buying) ₹ 66.98

60 days Forward Rate INR/US\$ (buying) ₹ 67.16

90 days Forward Rate INR/US\$ (buying) ₹ 68.03

Advise: which one of the following options would be better for XYZ Ltd.:

- Pay immediately after utilizing cash available and for balance amount take 90 days loan from bank.
- Pay the supplier on 60th day and avail bank's loan (after utilizing cash) for 30 days.
- Avail supplier's offer of 90 days credit and utilize cash available.

Further presume that the cash available with XYZ Ltd. will fetch a return of 4% p.a. in India till it is utilized.

Note:

- Assume year has 360 days.
- Ignore Taxation.
- Cash flows in ₹ in Crore.

- Round off all intermediate and final calculations to four decimal places.

(Exam Jan – 2026)

Solution:

(i) Pay immediately to avail the discount

Spot Rate	₹ 66.98
Payment Due	USD 79,20,000
Outflow in Rupees (USD 79,20,000 × ₹ 66.98)	(₹ Crore) 53.0482
Less: Cash Available (a)	0.2500
Balance amount to be borrowed from bank	52.7982
Add: Interest on loan for 90 days @ 9% p.a.	1.1880
Add: Opportunity cost of surplus cash (b)	0.0025
Total outflow in ₹ crore (a + b)	53.9887
	54.2387

OR

If Opportunity Cost of Surplus Cash is not considered then, the Total Outflow will be $54.2387 - 0.0025 = ₹ 54.2362$ Crore

(ii) Pay the supplier in 60 days

If the payment is made to supplier in 60 days the applicable forward rate for 1 USD	₹ 67.16
Payment Due	USD 80,00,000
Outflow in Rupees (USD 80,00,000 × ₹ 67.16)	(₹ Crore) 53.7280
	0.2500



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Less: Cash Available (a)	0.0017
Interest on the same for 60 days @ 4% p.a.	53.4763
Balance amount to be borrowed from bank	0.4011
Add: Interest on loan for 30 days @9% p.a.	53.8774
Loan Outflow in ₹ Crore (b)	54.1274
Total Outflow in ₹ Crore (a) + (b)	

(iii) **Availing supplier's offer of 90 days credit**

Amount payable	USD
Add: Interest on credit period for 30 days @ 8% p.a.	80,00,000 USD 53,333
Total outflow in USD	USD 80,53,33
Applicable forward rate for 1 USD	₹ 68.03
Outflow in ₹ crore (USD 80,53,333 × ₹ 68.03)	54.7868 0.0025
Less: Interest Earned on cash for 90 days @ 4% in ₹ crore	
Total outflow in ₹ crore	54.7843

Alternative (ii) is better as it entails lower cash outflow.

(2) INVOICING

Question – 40

XP Pharma Ltd., has acquired an export order for ₹ 10 million for formulations to a European company. The Company has also planned to import bulk drugs worth ₹ 5 million from a company in UK. The proceeds of exports will be realized in 3 months from now and the payments for imports will be due after 6 months from now. The invoicing of these exports and imports can be done in any currency i.e. Dollar, Euro or Pounds sterling at company's choice. The following market quotes are available.

	Spot Rate	Annualized Premium
₹/\$	67.10/67.20	\$ - 7%
₹ /Euro	63.15/63.20	Euro - 6%
₹ /Pound	88.65/88.75	Pound - 5%

Advice XP Pharma Ltd. about invoicing in which currency.

(Calculation should be upto three decimal places).

(Exam July – 2021) (8 Marks)

Solution:

(i) Proceeds of Exports in INR = ₹ 10 Million

Position of Inflow under three currencies will be as follows:

Currency	Invoice at Spot Rate	Expected Rate after 3-months	Conversion in INR after 3-months
\$	₹ 100,00,000 / ₹ 67.10 = \$ 149031.297	₹ 67.10 (1 + 0.07/4) = ₹ 68.27	₹ 68.27 × \$ 149031.297 = ₹ 1,01,74,367
€	₹ 100,00,000 / ₹ 63.15 = € 1,58,353.127	₹ 63.15 (1 + 0.06/4) = ₹ 64.10	₹ 64.10 × € 1,58,353.127 = ₹ 1,01,50,435
£	₹ 100,00,000 / ₹ 88.65 = £ 1,12,803.158	₹ 88.65 (1 + 0.05/4) = ₹ 89.76	₹ 89.76 × £ 1,12,803.158 = ₹ 1,01,25,211

(ii) Payment of Import in INR = ₹ 5 Million

Position of outflow under three currencies will be as follows:



Currency	Invoice at Spot Rate	Expected Rate after 6-months	Conversion in INR after 6-months
\$	₹ 50,00,000 / ₹ 67.20 = \$ 74404.762	₹ 67.20 (1 + 0.07/2) = ₹ 69.55	₹ 69.55 × \$ 74404.762 = ₹ 51,74,851
€	₹ 50,00,000 / ₹ 63.20 = € 79,113.924	₹ 63.20 (1 + 0.06/2) = ₹ 65.10	₹ 65.10 × € 79,113.924 = ₹ 51,50,316
£	₹ 50,00,000 / ₹ 88.75 = £ 56,338.028	₹ 88.75 (1 + 0.05/2) = ₹ 90.97	₹ 90.97 × £ 56,338.028 = ₹ 51,25,070

Advice: Since cash inflow is highest (1,01,74,367) in case of \$ hence invoicing for Export should be in \$. However, cash outflow is least (51,25,070) in case of £ the invoicing for import should be in £.

(3) MONEY MARKET COVER

Question – 41

An exporter is a UK based company. Invoice amount is \$3,50,000. Credit period is three months. Exchange rates in London are :

Spot Rate (\$/£) 1.5865 – 1.5905

3-month Forward Rate (\$/£) 1.6100 – 1.6140

Rates of interest in Money Market:

	Deposit	Loan
\$	7%	9%
£	5%	8%

Compute and show how a money market hedge can be put in place. Compare and contrast the outcome with a forward contract.

Solution:

Option 1: Forward Cover

Sell \$ 3,50,000 at 3 months FR

$$\text{Cash Inflows} = \frac{\$ 3,50,000}{1.6140} = \text{£ } 2,16,852.54$$

Option 2: Money Market Cover

- Amount to be borrowed from US money market @ 9% p.a. for 3 months

$$\frac{\$ 3,50,000}{1 + (0.09 \times 3/12)} = \$ 3,42,298.29$$

- Sell \$ 3,42,298.29 at SR = $\frac{\$ 3,42,298.29}{1.5905} = \text{£ } 2,15,214.27$

- Invest £ 2,15,214.27 in UK Money Market @ 5% for 3 months

$$\begin{aligned} \text{Cash Inflows} &= \text{£ } 2,15,214.27 \times [1 + (0.05 \times 3/12)] \\ &= \text{£ } 2,17,904.44. \end{aligned}$$

Money Market Cover is better due to higher cash Inflows.

Question – 42

Columbus Surgicals Inc. is based in US, has recently imported surgical raw materials from the UK and has been invoiced for £ 480,000, payable in 3 months. It has also exported surgical goods to India and France.

The Indian customer has been invoiced for £ 138,000, payable in 3 months, and the French customer has been invoiced for € 590,000, payable in 4 months.

Current spot and forward rates are as follows:

£ / US\$

Spot: 0.9830 – 0.9850

Three months forward: 0.9520 – 0.9545

US\$ / €

Spot: 1.8890 – 1.8920

Four months forward: 1.9510 – 1.9540

Current money market rates are as follows:

UK: 10.0% – 12.0% p.a.

France: 14.0% – 16.0% p.a.

USA: 11.5% – 13.0% p.a.

You as Treasury Manager are required to show how the company can hedge its foreign exchange exposure using Forward markets and Money markets hedge and suggest which the best hedging technique is.

(SM TYK – 38)

Solution:

- Import from UK & £ 4,80,000 payable in 3 months.
- Export to India & £ 1,38,000 receivable in 3 months hence netting is possible.

Net £ Payable to UK party = (£ 4,80,000 – £ 1,38,000)

= £ 3,42,000

We have to hedge for £ 3,42,000

Payable £ 3,42,000

(i) Forward Cover

Buy £ 3,42,000 at 3 months FR

$$\frac{£ 3,42,000}{0.9520} = \$ 3,59,243.70$$

(ii) Money Market Cover

- Amount to be invested in UK money market @ 10% p.a. for 3 months

$$\frac{£ 3,42,000}{1 + (0.10 \times 3/12)} = £ 3,33,658.54$$

- \$ required to buy £ 3,33,658.54 at SR

$$\frac{£ 3,33,658.54}{0.9830} = \$ 3,39,428.83$$

- Borrow \$ 3,39,428.83 from US money market @ 13% p.a. for 3 months.

Cash Outflow

$$\$ 3,39,428.83 \times \left[1 + 0.13 \times \frac{3}{12} \right] = \$ 3,50,460.27$$

Money Market Cover is better due to lower cash outflow.

Hedging of receivable € 5,90,000



(i) Forward Cover

Sell € 5,90,000 at 3 months FR

$$\begin{aligned}\text{Cash Inflows} &= \text{€ } 5,90,000 \times 1.9510 \\ &= \$ 11,51,090\end{aligned}$$

(ii) Money Market Cover

- Amount to be borrowed from France money market @ 16% p.a. for 4 months

$$\frac{\text{€ } 5,90,000}{1 + (0.16 \times 4/12)} = \text{€ } 5,60,126.58$$

- Sell € 5,60,126.58 at SR

$$\text{€ } 5,60,126.58 \times 1.8890 = \$ 10,58,079.11$$

- Invest \$ 10,58,079.11 in US money market @ 11.5% for 4 months

$$\begin{aligned}\text{Cash Inflows} &= \$ 10,58,079.11 \times [1 + (0.115 \times 4/12)] \\ &= \$ 10,98,638.81\end{aligned}$$

Forward cover is better due to higher cash inflows.

Question – 43

An Indian exporting firm, Rohit and Bros., would be covering itself against a likely depreciation of pound sterling. The following data is given:

Receivables of Rohit and Bros : £ 500,000

Spot rate : ₹ 56.00/£

Payment date : 3-months

3 months interest rate : India : 12 per cent per annum
: UK : 5 per cent per annum

What should the exporter do?

(SM TYK – 28)

Solution:

Exporter should hedge with money market cover

- Amount to be borrowed from UK money market @ 5% p.a. for 3 months

$$\frac{\text{£ } 500,000}{1 + (0.05 \times 3/12)} = \text{£ } 4,93,827.72$$

- Sell £ 4,93,827.72 at SR = £ 4,93,827.72 × 56 = ₹ 2,76,54,352

- Invest ₹ 2,76,54,352 in Indian money market @ 12% p.a for 3 months

$$\text{₹ } 2,76,54,352 \times [1 + (0.12 \times 3/12)] = \text{₹ } 2,84,83,982.$$

Question – 44

H Ltd. is an Indian firm exporting handicrafts to North America. All the exports are invoiced in US\$. The firm is considering the use of money market or forward market to cover the receivable of \$ 50,000 expected to be realized in 3 months time and has the following information from its banker:



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

	Exchange Rates
Spot	₹/\$ 72.65/73
3 – months forward	₹/\$ 72.95/73.40

The borrowing rates in US and India are 6% and 12% p.a. and the deposit rates are 4% and 9% p.a. respectively.

- (i) Which option is better for H Ltd?
- (ii) Assume the H Ltd. anticipates the spot exchange rate in 3 – months time to be equal to the current 3 – months forward rate. After 3 – months the spot exchange rate turned out to be ₹/\$: 73/73.42.

What is the foreign exchange exposure and risk of H Ltd.?

(Exam November – 2019) (8 Marks)

Solution:

(i) Option 1: Forward Cover

Sell \$ 50,000 at 3 months FR

$$\begin{aligned}\text{Cash Inflows} &= \$ 50,000 \times 72.95 \\ &= ₹ 36,47,500\end{aligned}$$

Option 2: Money Market Cover

- Amount to borrow from US Money market @ 6% for 3 months

$$\frac{\$ 50,000}{1 + (0.06 \times 3/12)} = \$ 49,261.08$$

$$\begin{aligned}- \text{ Sell } \$ 49,261.08 \text{ at SR} &= \$ 49,261.08 \times 72.65 = ₹ \\ &35,78,817\end{aligned}$$

- Invest ₹ 35,78,817 in Indian money market @ 9% p.a. for 3 months

$$\begin{aligned}\text{Cash Inflows} &= ₹ 35,78,817 \times [1 + (0.09 \times 3/12)] \\ &= ₹ 36,59,340\end{aligned}$$

Money market cover is better due to higher cash Inflows.

(ii) Foreign Exchange Exposure = \$ 50,000

Risk (Gain) due to anticipated exchange rate

$$\begin{aligned}&= (73 - 72.95) \times \$ 50,000 \\ &= ₹ 2,500\end{aligned}$$

Question – 45

Quick & Smart Inc. is a leading software development company in the UK. It has a substantial portfolio of its trade in various countries including the USA. It has recently invoiced a USA customer the sum of USD (\$) 75,00,000 receivable in one year's time. Quick & Smart Inc.'s Chief Finance Officer (CFO) is considering two alternatives for hedging the exchange risk:

Alternative I : Borrowing present value of USD (\$) 75,00,000 now for one year, converting the amount into GBP (£), and repaying the loan out of eventual receipts.

Alternative II : Entering into a 12 month forward exchange contract with the company's bank to sell the USD (\$) 75,00,000.

The spot exchange rate is GBP (£) 1 = USD (\$) 1.3288



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

The 12 month forward exchange rate is GBP (£) 1 = USD (\$) 13128

Interest rates for 12 months are = USA 4.50%; and UK 5%,

You are required to –

Calculate net proceeds in GBP (£) under both the alternatives and advise the company.

Note : Ignore bank commission and decimals,

(Exam September – 2025) (7 Marks)

Solution:

Alternative I

Borrowing PV of US\$ 75,00,000 (US\$ 75,00,000/1.045)	\$ 71,77,033
Converting it into GBP (£) at Spot Rate	\$ 1.3288
Converted Amount	£ 54,01,139
Add: Interest on the same @ 5%	£ 2,70,057
	£ 56,71,196

Alternative II

Applicable Forward Rate of 1 \$	\$ 1.3128
Amount Receivable in \$ after one year	\$ 75,00,000
Amount Receivable in \$ after	\$ 57,12,980

Advise: Since amount receivable is higher under alternative II the company should opt for it.

(4) CURRENCY FUTURE

Question – 46

EFD Ltd. is an export business house. The company prepares invoice in customers' currency. Its debtors of US\$. 10,000,000 is due on April 1, 2015.

Market information as at January 1, 2015 is:

Exchange rates US\$/INR		Currency Futures US\$/INR	
Spot	0.016667	Contract size: ₹ 24,816,975	
1-month forward	0.016529	1-month	0.016519
3-months forward	0.016129	3-month	0.016118
Initial Margin		Interest rates in India	
1-Month	₹ 17,500	6.5%	
3-Months	₹ 22,500	7%	

On April 1, 2015 the spot rate US\$/INR is 0.016136 and currency future rate is 0.016134.

Which of the following methods would be most advantageous to EFD Ltd?

- (i) Using forward contract
- (ii) Using currency futures
- (iii) Not hedging the currency risk

(SM TYK – 25, MTP April – 2022 & Exam May – 2015) (8 Marks)



Solution:

(i) Forward Contract

$$= \frac{\$ 1,00,00,000}{0.016129} = ₹ 62,00,01,240$$

(ii) Currency Future

Step 1: EFD Ltd. should long position on ₹ at \$/₹ 0.016118

Step 2: No. of contracts

$$\text{Exposure amount} = \frac{\$ 1,00,00,000}{0.016118} = ₹ 62,04,24,370$$

$$\text{No. of contracts} = \frac{\$ 62,04,24,370}{2,48,16,975} = 25 \text{ Contracts long}$$

Step 3: Cash Flows

Variation margin

Gain on long position

$$(\$ 0.016134 - 0.016118) \times ₹ 2,48,16,975 \times 25 = \$ 9,926.79$$

$$\text{Total receipt } (\$ 1,00,00,000 + \$ 9,926.79) = \$ 1,00,09,926.79$$

$$\text{Cash inflows (₹)} = \frac{\$ 1,00,09,926.79}{0.016136} = ₹ 6,20,34,7471$$

$$\begin{aligned} (-) \text{ Interest on margin } (₹ 22,500 \times 25 \times 7\% \times 3/12) &= ₹ 9,844 \\ \hline &= ₹ 62,03,37,627 \end{aligned}$$

(iii) No Hedging

$$\frac{\$ 1,00,00,000}{0.016136} = ₹ 61,97,32,276$$

Decision: Currency future is the best option due to highest cash inflows.

Question – 47

Nitrogen Ltd, a UK company is in the process of negotiating an order amounting to €4 million with a large German retailer on 6 months credit. If successful, this will be the first time that Nitrogen Ltd has exported goods into the highly competitive German market. The following three alternatives are being considered for managing the transaction risk before the order is finalized.

- (i) Invoice the German firm in Sterling using the current exchange rate to calculate the invoice amount.
- (ii) Alternative of invoicing the German firm in € and using a forward foreign exchange contract to hedge the transaction risk.
- (iii) Invoice the German firm in € and use sufficient 6 months sterling future contracts (to the nearly whole number) to hedge the transaction risk.

Following data is available:

Spot Rate € 1.1750 - €1.1770/£

6 months forward premium 0.55-0.60 Euro Cents

6 months future contract is currently trading at €1.1760/£

6 months future contract size is £62500



Spot rate and 6 months future rate €1.1785/£

Required:

- (a) Calculate to the nearest £ the receipt for Nitrogen Ltd, under each of the three proposals.
- (b) In your opinion, which alternative would you consider to be the most appropriate and the reason thereof.

(SM TYK – 41 & Exam November – 2011) (8 Marks)

Solution:

(i) Invoicing

Invoicing the German firm in £ at current Exchange Rate

€/£ = 1.1770

Invoice amount in £ = $\frac{€ 40,00,000}{1.1770} = £ 33,98,471$

(ii) Forward Cover

Sell € 40,00,000 at 6 months FR

FR = $(1.1770 + 0.0060) = 1.1830$

Cash Inflows = $\frac{€ 40,00,000}{1.1830} = £ 33,81,234$

(iii) Currency Future

Step 1: € receivables & we afraid from € falling but rate is given of £ means afraid from £ rising hence we take long position of £ future at €/£ 1.1760

Step 2: No. of Contracts

£ Equivalent € 40,00,000 = $\frac{€ 40,00,000}{1.1760}$
= € 34,01,360

No. of contracts = $\frac{€ 34,01,360}{€ 62,500} = 54$ contract long

Step 3: Variation Margin

Currency future rate on settlement rate is € 1.1760/£, hence gain on long position $(€ 1.1785 - € 1.1760) \times 54 \times € 62,500$
= € 8,438

Cash inflows

Amount received = € 40,00,000

(-) Gain on future = € 8,438

€ 40,08,438

Cash inflows in £ = $\frac{€ 40,08,438}{1.1785}$

= £ 34,01,305

Currency future is the best option due to higher cash inflows.

Question – 48

ABC Technologic is expecting to receive a sum of US\$ 4,00,000 after 3 months. The company decided to go for future contract to hedge against the risk. The standard size of future contract available in the market is \$1000. As on date spot and futures \$ contract are quoting

at ₹ 44.00 & ₹ 45.00 respectively. Suppose after 3 months the company closes out its position futures are quoting at ₹ 44.50 and spot rate is also quoting at ₹ 44.50. You are required to calculate effective realization for the company while selling the receivable. Also calculate how company has been benefitted by using the future option.

(SM TYK – 34)

Solution:

Currency Future

Step 1: Since \$ is receivable, & we afraid from \$ falling hence we should take short position of \$ future at ₹/\$ 45.

Step 2: No. of Contracts

$$\text{No. of Contracts} = \frac{\$ 4,00,000}{\$ 1,000} = ₹ 400 \text{ Contract short}$$

Step 3: Variation Margin

\$ future rate on settlement date is 44.50, hence gain on short position

$$\begin{aligned} \text{Variation Margin} &= (\text{₹ } 45 - 44.50) \times 400 \times \$ 1000 \\ &= ₹ 2,00,000 \end{aligned}$$

Step 4: Cash Inflows

Sell \$ 4,00,000 at 3 months SR

$$(\$ 4,00,000 \times 44.50) = ₹ 1,78,00,000$$

$$(+)\text{ Variation Margin} = ₹ 2,00,000$$

$$\text{Cash Inflows} = ₹ 1,80,00,000$$

$$\text{Effective Realization per \$} = \frac{₹ 1,80,00,000}{\$ 4,00,000}$$

$$= ₹ 45$$

(5) CURRENCY OPTION

Question – 49

XYZ Ltd. a US firm will need £ 3,00,000 in 180 days. In this connection, the following information is available:

Spot rate 1 £ = \$ 2.00

180 days forward rate of £ as of today = \$1.96

Interest rates are as follows:

	U.K.	US
180 days deposit rate	4.5%	5%
180 days borrowing rate	5%	5.5%

A call option on £ that expires in 180 days has an exercise price of \$ 1.97 and a premium of \$ 0.04.

XYZ Ltd. has forecasted the spot rates 180 days hence as below:

Future rate	Probability
\$ 1.91	25%
\$ 1.95	60%



\$ 2.05 15%

Which of the following strategies would be most preferable to XYZ Ltd.?

- (a) A forward contract;
- (b) A money market hedge;
- (c) An option contract;
- (d) No hedging.

Show calculations in each case

(SM TYK – 39 & Exam November – 2015) (8 Marks)

Solution:

(a) Forward Cover

Buy £ 3,00,000 at 180 day FR

$$£ 3,00,000 \times 1.96 = \$ 5,88,000$$

(b) Money Market Cover

- Calculation of Amount to be invested in UK money market @ 4.5% for 180 days

$$= \frac{£ 3,00,000}{1.045} = £ 2,87,081.34$$

- \$ required to buy £ 2,87,081.34 at SR

$$£ 2,87,081.34 \times 2.00 = \$ 5,74,162.68$$

- Borrow \$ 5,74,162.68 @ 5.5% for 180 days from US Money market

Cash outflow

$$= \$ 5,74,162.68 (1.055)$$

$$= \$ 6,05,741.63$$

(c) Option Contract

Price	Exercised or not	Cost of \$	Premium	Effective Cost per \$	Cash outflows £ 3,00,000 × EC	Probability	Expected Cash outflows
1.91	No	\$ 1.91	\$ 0.04	\$ 1.95	\$ 5,85,000	25%	1,46,250
1.95	No	\$ 1.95	\$ 0.04	\$ 1.99	\$ 5,97,000	60%	3,58,200
2.01	Yes	\$ 1.97	\$ 0.04	\$ 2.01	\$ 6,03,000	15%	90,450
							\$ 5,94,900

(+) Opportunity Cost on premium

$$(£ 3,00,000 \times 0.04) \times 5.5\% = \$ 660$$

$$\text{Cash Outflow } (5,94,900 + 660) = \$ 5,95,560$$

(d) No Hedging

$$\begin{aligned} \text{Expected SR} &= (1.91 \times 0.25) + (1.95 \times 0.60) + (2.05 \times 0.15) \\ &= 1.955 \end{aligned}$$



$$\begin{aligned}\text{Cash outflows} &= \text{£ } 3,00,000 \times 1.955 \\ &= \$ 5,86,500\end{aligned}$$

No hedging is the best due to lower cash outflows.

Question – 50

A Ltd. of U.K. has imported some chemical worth of USD 3,64,897 from one of the U.S. suppliers. The amount is payable in six months time. The relevant spot and forward rates are:

Spot rate	USD 1.5617–1.5673
6 months' forward rate	USD 1.5455 –1.5609

The borrowing rates in U.K. and U.S. are 7% and 6% respectively and the deposit rates are 5.5% and 4.5% respectively.

Currency options are available under which one option contract is for GBP 12,500. The option premium for GBP at a strike price of USD 1.70/GBP is USD 0.037 (call option) and USD 0.096 (put option) for 6 months period.

The company has 3 choices:

- (i) Forward cover
- (ii) Money market cover, and
- (iii) Currency option

Which of the alternatives is preferable by the company?

(SM TYK – 40)

Solution:**Option 1: Forward Cover**

Buy \$ 3,64,897 at 6 months FR

$$\text{Cash outflows} = \frac{\$ 3,64,897}{1.5455} = 2,36,103$$

Option 2: Money Market Cover

- Amount to be invested in US money market @ 4.5% p.a. for 6 month

$$\begin{aligned}\frac{\$ 3,64,897}{1 + (0.045 \times 6/12)} &= \$ 3,56,867 \\ &= \text{£ } 2,28,512\end{aligned}$$

- Borrow £ 2,28,512 from UK money market @ 7% p.a. for 6 months

$$\begin{aligned}\text{Cash outflows} &= \text{£ } 2,28,512 \times [1 + (0.07 \times 6/12)] \\ &= \text{£ } 2,36,510\end{aligned}$$

Option 3: Currency Option

Step 1: Since \$ payables & we afraid from \$ rising but rate is given of £. It means £ falling hence we should we should buy put option £ at EP \$/£ 1.70.

Step 2: No. of Contract

$$\begin{aligned}\text{Contract size in \$} &= \text{£ } 12,500 \times 1.70 \\ &= \$ 21,250\end{aligned}$$



$$\begin{aligned}\text{No. of contracts} &= \frac{\$ 3,64,897}{\$ 21,250} \\ &= 17.17 \text{ "Means 17 Contracts"}\end{aligned}$$

Step 3: Cash Outflows**Covered through option hedging**

$$\text{Put option EP} = \$/\text{£} = \$ 1.70$$

$$\text{Contract size} = \$ 21,250 \quad \text{No.} = 17 \text{ contracts}$$

$$\text{Covered through option hedging } (\$ 21,250 \times 17) = \$ 3,61,250$$

$$\text{It means total £ required to buy } \frac{\$ 3,61,250}{\$ 1.70} = \text{£ } 2,12,500.$$

Covered through forward contract

$$\begin{aligned}\text{Uncovered portion} &= \$ 3,64,897 - \$ 3,61,250 \\ &= \$ 3,647\end{aligned}$$

$$\text{Buy \$ 3,647 at 6 months FR} = \frac{\$ 3,647}{1.5455} = \text{£ } 2,360.$$

Premium Amount

$$\text{Premium of put option} = \$ 0.0096 \text{ per £}$$

$$\text{Contract size} = \text{£ } 12,500,$$

$$\text{No. of contracts} = 17$$

$$\begin{aligned}\text{Total premium in \$} &= \$ 0.096 \times 12,500 \times 17 \\ &= \$ 20,400\end{aligned}$$

$$\begin{aligned}\text{Buy \$ 20,400 at SR} &= \frac{\$ 20,400}{1.5617} \\ &= \text{£ } 13,063\end{aligned}$$

$$\begin{aligned}\text{Total Cash Outflows} &= \text{£ } 2,12,500 + \text{£ } 2,360 + \text{£ } 13,063 \\ &= \text{£ } 2,27,923.\end{aligned}$$

Currency option is best due to lower cash outflows.

Question – 51

XYZ, an Indian firm, will need to pay JAPANESE YEN (JY) 5,00,000 on 30th June. In order to hedge the risk involved in foreign currency transaction, the firm is considering two alternative methods i.e. forward market cover and currency option contract.

On 1st April, following quotations (JY/INR) are made available:

Spot	3 months forward
1.9516/1.9711.	1.9726./1.9923

The prices for forex currency option on purchase are as follows:

Strike Price	JY 2.125
Call option (June)	JY 0.047
Put option (June)	JY 0.098

For excess or balance of JY covered, the firm would use forward rate as future spot rate. You are required to recommend cheaper hedging alternative for XYZ.



(SM TYK – 33 & Exam November – 2015) (8 Marks)

Solution:

Option 1: Forward Cover

Buy ¥ 5,00,000 at 3 Month FR

$$\text{Cash Outflow} = \frac{\text{¥ 5,00,000}}{1.9726} = \text{₹ 2,53,472}$$

Option 2: Currency Option

Step 1: Since ¥ Payable & We afraid from ¥ rising but rate is given of ₹ means ₹ falling hence we buy put option at

$$\text{EP } \text{¥/ ₹} = \text{¥ 2.125}$$

Step 2: Cash Outflow

Option Hedging

$$\text{Exposure} = \text{¥ 5,00,000}$$

$$\text{EP } \text{¥/ ₹} = \text{¥ 2.125}$$

$$\frac{\text{¥ 5,00,000}}{2.125} = 2,35,294$$

Premium

$$\text{Premium of put option} = \text{JY 0.098 per ₹}$$

$$\text{Total premium (¥)} = 0.098 \times 2,35,294$$

$$= \text{¥ 23,059}$$

$$\text{Buy ¥ 23,059 at SR} = \frac{\text{¥ 23,059}}{1.9516}$$

$$= \text{₹ 11,815}$$

$$\text{Cash Outflows } (2,35,294 + 11,815) = \text{₹ 2,47,109}$$

Currency option is better due to lower cash outflow.

Question – 52

An American firm is under obligation to pay interests of Can\$ 10,10,000 and Can\$ 7,05,000 on 31st July and 30th September respectively. The Firm is risk averse and its policy is to hedge the risks involved in all foreign currency transactions. The Finance Manager of the firm is thinking of hedging the risk considering two methods i.e. fixed forward or option contracts.

It is now June 30. Following quotations regarding rates of exchange, US\$ per Can\$, from the firm's bank were obtained:

Spot	1 Month Forward	3 Months Forward
0.9284-0.9288	0.9301	0.9356

Price for a Can\$ /US\$ option on a U.S. stock exchange (cents per Can\$, payable on purchase of the option, contract size Can\$ 50,000) are as follows:

Strike Price (USD/Can\$)	Calls		Puts	
	July	Sept.	July	Sept.
0.93	1.56	2.56	0.88	1.75
0.94	1.02	NA	NA	NA
0.95	0.65	1.64	1.92	2.34

According to the suggestion of finance manager if options are to be used, one month option should be bought at a strike price of 94 cents and three month option at a strike price of 95 cents and for the remainder uncovered by the options the firm would bear the risk itself. For this, it would use forward rate as the best estimate of spot. Transaction costs are ignored.



Recommend, which of the above two methods would be appropriate for the American firm to hedge its foreign exchange risk on the two interest payments.

(SM TYK – 36, MTP March – 2022 & Exam Nov – 2013) (8 Marks)

Solution:

Option 1: Forward Cover

1 Month Exposure

Cash Outflow = Can \$ 10,10,000 × 0.9301 = \$ 9,39,401

3 Month Exposure

Cash Outflow = Can \$ 7,05,000 × 0.9356 = \$ 6,59,598

Option 2: Currency Option

Step 1: Since Can \$ Payable & we Afraid from Can \$ rising
Hence we should buy call option

For 1 Month EP \$ 0.94 & Premium = \$ 0.0102

For 3 Month EP \$ 0.95 & Premium = \$ 0.0164

Step 2: No. of Contracts

$$1 \text{ Month} = \frac{\text{Can \$ } 10,10,000}{\text{Can \$ } 50,000} = 20 \text{ Contracts}$$

$$3 \text{ Month} = \frac{\text{Can \$ } 7,05,000}{\text{Can \$ } 50,000} = 14 \text{ Contracts}$$

	1 Month	3 Month
Option Hedging	(20 Contract × Can \$ 50,000) Can \$ 10,00,000 × 0.94 US \$ 9,40,000	(14 Contract × Can \$ 50,000) Can \$ 7,00,000 × 0.95 US \$ 6,65,000
Uncovered position using forward cover	(Can \$ 10,10,000 – Can \$ 10,00,000) Can \$ 10,000 × 0.9301 US \$ 9,301	(Can \$ 7,05,000 – 7,00,000) Can \$ 5,000 × 0.9356 US \$ 4,678
Premium	\$ 0.0102 per Can \$ (Can \$ 10,00,000 × 0.0102) = US \$ 10,200	\$ 0.0164 per Can \$ (Can \$ 7,00,000 × 0.0164) = US \$ 11,480
USD	9,59,501	6,81,158

Forward Contract is better due to lower cash outflows.

Question – 53

A Japanese company imports hi-tech printer cartridges from US worth \$1 million. The chief financial officer of the company wishes to know the best strategy for protection against uncertainty, for the payment that has to be made at the end of 3 months. Financial team of the company has collected the following options for evaluation:

Table-1: Exchange rates quoted in FOREX Market:

¥/\$ Quotations	Bid Price	Offer/Ask Price
Spot Rates	146.03	146.63
3M – Forward Rates	144.03	145.00
6M – Forward Rates	146.35	146.70

FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Table-2 : Options Market rates for European options with 3 months expiry :

Type of Option	Strike Price (X) (¥/\$)	Premium (%) for Call & Put Options
Call & Put	145.20	1.6766% (Call) & 1.7414% (Put)
Call & Put	146.00	1.3505% (Call) & 2.1006% (Put)

The expected spot price at expiry is ¥/\$: 144.90/145.05

Suggest the best strategy for CFO of the Japanese Company to protect against uncertainty, with respect to the following alternatives:

- (i) Forward Hedge
- (ii) Buy 3 months call, X = 145.20
- (iii) Sell 3 months put, X = 145.20
- (iv) Buy call & sell put both having X = 146.00

(Exam November – 2023) (8 Marks)

Solution:

Option 1: Forward Hedge

Buy \$ 10,00,000 at 3 months FR

$$\text{\$ } 10,00,000 \times 145 = \text{¥ } 14,50,00,000$$

Option 2: Buy 3 months call, X = 145.20

Call = Right to buy

If expected spot price on maturity is ¥ 145.05, then company would not exercise its option

Buy \$ 10,00,000 at ¥ 145.05

$$(\$ 10,00,000 \times 145.05) = \text{¥ } 14,50,50,000$$

$$\begin{aligned} (+) \text{ Premium } (\text{¥ } 145.20 \times 1.6766\%) \times \$ 10,00,000 &= \text{¥ } 24,34,423 \\ \hline &= \text{¥ } 14,74,84,423 \end{aligned}$$

Option 3: Sell 3 months put, X = 145.20

Put = Right to sell

If expected price after 3 months is ¥ 144.90, then put option will be exercised

Buy \$ 10,00,000 at strike price

$$(\$ 10,00,000 \times 145.20) = \text{¥ } 14,52,00,000$$

(-) Received premium

$$\begin{aligned} (\text{¥ } 145.20 \times 1.7414\%) \times \$ 10,00,000 &= \text{¥ } 25,28,513 \\ \hline &= \text{¥ } 14,26,71,487 \end{aligned}$$

Option 4: Buy call & sell put at, X = 146

Premium on call

$$(\text{¥ } 146 \times 1.3505\%) \times \$ 10,00,000 = \text{¥ } 19,71,730$$

Premium received on put



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

$$(\text{¥ } 146 \times 2.1006\%) \times \$ 10,00,000 = \text{¥ } 30,66,876$$

$$\text{Net premium received} = \text{¥ } 10,95,146$$

If expected price to be ¥ 144.90/145.05, then call option will lapsed & put will exercise

$$\text{Buy \$ at EP } (\$ 10,00,000 \times 146) = \text{¥ } 14,60,00,000$$

$$(-) \text{ Premium} = \text{¥ } 10,95,146$$

$$= \text{¥ } 14,49,04,854$$

Decision: Option 3 is the best due to lowest cash outflows.

(7) CANCELLATION OF FORWARD CONTRACT

(I) CANCELLATION & EXTENTION OF FORWARD CONTRACT

Question – 54

An importer requests his bank to extend the forward contract for US\$ 20,000 which is due for maturity on 30th October, 2010, for a further period of 3 months. He agrees to pay the required margin money for such extension of the contract.

$$\text{Contracted Rate} - \text{US\$ } 1 = \text{₹ } 42.32$$

The US Dollar quoted on 30-10-2010:- Spot – 41.5000/41.5200

3 months' Premium -0.87% /0.93%

Margin money for buying and selling rate is 0.075% and 0.20% respectively.

Compute:

- (i) The cost to the importer in respect of the extension of the forward contract, and
- (ii) The rate of new forward contract.

(SM TYK – 32 & Exam May – 2017) (4 Marks)

Solution:

(i) Extension Cost

$$\text{Selling Rate of Bank} = \text{₹ } 42.32$$

$$(-) \text{ Buying Rate of Bank } (41.5000 - 0.075\%) = \text{₹ } 41.47$$

$$\text{Gain to Bank} = \text{₹ } 0.85$$

$$(\times) \text{ Contract Size} = \$ 20,000$$

$$\text{Extension Cost} = \text{₹ } 17,000$$

(ii) Rate of New Forward Contract

$$\text{Selling SR} = 41.5200$$

$$(+) 3 \text{ Month Swap} = 0.93\%$$

$$= 41.91$$

$$(+) \text{ Margin} = 0.20\%$$

$$\text{New FR} = 41.99$$



Question – 55

On 15th January 2015 you as a banker booked a forward contract for US\$ 2,50,000 for your import customer deliverable on 15th March 2015 at ₹ 65.3450. On due date customer request you to cancel the contract. On this date quotation for US\$ in the inter-bank market is as follows:

Spot	₹ 65.2900/2975 per US\$
Spot/April	3,000/3,100
Spot/May	6,000/6,100

Assuming that the flat charges for the cancellation is ₹ 100 and exchange margin is 0.10%, then determine the cancellation charges payable by the customer.

Solution:**Calculation of Cancellation Charges**

Selling Rate of Bank	= ₹ 65.3450
Buying Rate of Bank (65.2900 – 0.10%)	= ₹ 65.2250
Gain to Bank	= ₹ 0.1200
(X) Contract Size	= \$ 2,50,000
Gain to Bank	= ₹ 30,000
(+) Flat Charges	= ₹ 100
	= ₹ 30,100

Question – 56

A customer with whom the bank had entered into 3 months forward purchase contract for Swiss Francs 1,00,000 at the rate of ₹ 36.25 comes to the bank after two months and requests cancellation of the contract. On this date, the rates are:

Spot	CHF 1 = ₹ 36.30	36.35
One month forward	36.45	36.52

Determine the amount of profit or loss to the customer due to cancellation of the contract.

(SM TYK – 44)**Solution:****Profit/Loss to Customer**

Buying rate	= 36.25
Selling rate	= 36.52
Gain to bank	= ₹ 0.27
(X) Contract Size	= SF 1,00,000
Loss to the customer	= ₹ 27,000

Question – 57

Suppose you are a banker and one of your export customer has booked a US\$ 1,00,000 forward sale contract for 2 months with you at the rate of ₹ 62.5200 and simultaneously you covered yourself in the interbank market at 62.5900. However on due date, after 2 months you customer comes to you and requests for cancellation of

the contract and also requests for extension of the contract by one month. On this date quotation for US\$ in the market was as follows:

Spot	₹ 62.7200/62.6800
1 month forward	₹ 62.6400/62.7400

Determine the extension charges payable by the customer assuming exchange margin of 0.10% on buying as well as selling.

Solution: Calculation of Extension Charge

Particular	Amount
Buying Rate of Bank	₹ 62.5200
Selling Rate of Bank (62.7200 + 0.10%)	₹ 62.7825
Gain to Bank	₹ 0.2625
(X) Contract Size	\$ 1,00,000
Extension Charge	₹ 26,250

New Forward Rate

Particular	Amount
Buying Rate (1 Month)	₹ 62.6400
(-) Margin	0.10%
	62.5774
New FR	62.5775

Question – 58

A bank enters into a forward purchase TT covering an export bill for Swiss Francs 1,00,000 at ₹ 32.4000 due 25th April and covered itself for same delivery in the local inter bank market at ₹ 32.4200. However, on 25th March, exporter sought for cancellation of the contract as the tenor of the bill is changed.

In Singapore market, Swiss Francs were quoted against dollars as under:

Spot	USD 1 = Sw. Fcs.	1.5076/1.5120
One month forward		1.5150/ 1.5160
Two months forward		1.5250 / 1.5270
Three months forward		1.5415/ 1.5445

and in the interbank market US dollars were quoted as under:

Spot	USD 1 = ₹	49.4302/.4455
Spot / April		4100/.4200
Spot/May		.4300/.4400
Spot/June		.4500/.4600

Calculate the cancellation charges, payable by the customer if exchange margin required by the bank is 0.10% on buying and selling.

(SM TYK – 45 & Exam November – 2015) (8 Marks)

Solution: Cancellation Rate

First customer buy \$ from ₹

$$\text{SR} = 49.4455$$

$$(+)\text{ 1 Month Swap} = 0.4200$$

$$= 49.8655$$

$$(+)\text{ Margin} = 0.10\%$$

$$\text{₹/ \$} = 49.9154$$

After that Customer Buy SF from \$

$$\text{SF/ \$} = 1.5150$$

$$\text{Cancellation Rate } \text{₹/SF} = \frac{49.9154}{1.5150}$$

$$= ₹ 32.9475$$

Calculation of Cancellation Charge

$$\text{Buying rate of Bank } \left(\frac{\text{₹}}{\text{SF}} \right) = ₹ 32.4000$$

$$\text{Selling rate of Bank } \left(\frac{\text{₹}}{\text{SF}} \right) = ₹ 32.9475$$

$$\text{Gain to Bank} = ₹ 0.5475$$

$$(\times)\text{ Contract Size} = \text{SF } 1,00,000$$

$$\text{Cancellation Charges} = ₹ 54,750$$

Question – 59

NP and Co. has imported goods for US \$ 7,00,000. The amount is payable after three months. The company has also exported goods for US \$ 4,50,000 and this amount is receivable in two months. For receivable amount a forward contract is already taken at ₹ 48.90.

The market rates for Rupee and Dollar are as under:

$$\text{Spot} = ₹ 48.50/70$$

$$\text{Two months} = 25/30 \text{ points}$$

$$\text{Three months} = 40/45 \text{ points}$$

The company wants to cover the risk and it has two options as under:

- (A) To cover payables in the forward market and
- (B) To lag the receivables by one month and cover the risk only for the net amount. No interest for delaying the receivables is earned. Evaluate both the options if the cost of Rupee Funds is 12%. Which option is preferable?

(SM TYK – 43 & Exam May – 2012) (8 Marks)

Solution:

(A) Cover payable in forward market

Payable

$$\text{Buy \$ 7,00,000 at 3 Month FR } (48.70 + 0.45) = 49.15$$

$$\text{Cash Outflows } (7,00,000 \times 49.15) \quad (\text{A}) = ₹ 34,40,500$$

Receivable

$$\text{Sell \$ 4,50,000 at 48.90 } (4,50,000 \times 48.90) = ₹ 2,20,05,000$$



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Interest $(2,20,05,000 \times 12 \times 1/12)$	= ₹ 2,20,050
Cash Inflows (B)	= ₹ 2,22,25,050
Net Cash Outflow (A – B)	= ₹ 1,21,79,950

(B) Lag the receivable by one month

Forward contract cancelled	
Buying Rate of Bank	= ₹ 48 .90
(–) Selling Rate of Bank $(48.70 + 0.30)$	= ₹ 49.00
Gain to Bank	= ₹ 0.10
(X) Contract Size	= \$ 4,50,000
Cancellation Charges	= ₹ 45,000
Net \$ payable after 3 month	
\$ Payable	= \$ 7,00,000
(–) \$ Receivable	= \$ 4,50,000
Net \$ Payable	= \$ 2,50,000
Cash Outflows	
Buy \$ 2,50,000 at 3 month FR	
$(\$ 2,50,000 \times 49.15)$	= ₹ 1,22,87,500
(+) Cancellation Charges	= ₹ 45,000
Cash Outflow	= ₹ 1,23,32,500

Option A is better due to lower cash outflow.

(II) EARLY DELIVERY

Question – 60

On 19th January, Bank A entered into forward contract with a customer for a forward sale of US \$ 7,000, delivery 20th March at ₹ 46.67. On the same day, it covered its position by buying forward from the market due 19th March, at the rate of ₹ 46.655. On 19th February, the customer approaches the bank and requests for early delivery of US \$. Rates prevailing in the interbank markets on that date are as under:

Spot (₹/\$)	46.5725/5800
March	46.3550/3650

Interest on outflow of funds is 16 % and on inflow of funds is 12 %.

Flat charges for early delivery are ₹ 100.

What is the amount that would be recovered from the customer on the transaction?

Note: Calculation should be made on months basis than on days basis.

(Exam November – 2018) (8 Marks)

Solution:

Amount receivable from customer

Sell \$ 7,000 at contracted rate $(7,000 \times 46.67) = ₹ 3,26,690$



Swap Loss/Gain

Bank enter into buy-sell swap where bank buy spot from other bank & sell one months forward to other Bank.

Buy spot	= ₹ 46.5800
Sell forward	= ₹ 46.3550
Swap loss	= ₹ 0.225
(X) Contract size	= ₹ 7,000
Swap loss	= ₹ 1,575

Note: अगर Swap Gain होता तो Customer को Transfer करना पड़ता।

Interest on outlay of fund/inflow of fund (19/2)

Cash outflow on buy spot (\$ 7,000 × 46.5800)	=	₹ 3,26,060
Cash Inflows on sell to customer (\$ 70,000×46.67)	= ₹ 3,26,690	
Net cash inflows	= ₹ 630	
Interest on Inflows of fund = ₹ 630 × 12% × 1/12	= ₹ 6.30	
Total amount receivable from customer		
(₹ 3,26,690 + 1,575 – 6.30)	=	₹ 3,28,258.70

Question – 61

On 1st October, 2020 Mr. Guru, and exporter, enters into a forward contract with the Bank to sell USD 1,00,000 on 31st December 2020

at INR/USD 75.40. However, at the request of the importer, Mr. Guru received the amount on 30th November, 2020. Mr. Guru requested the bank take delivery of the remittance on 30th November, 2020 i.e. before due date.

The inter-bank rate on 30th November, 2020 was as follows:

Spot INR/USD 75.22 - 75.27

One Month Premium 10/15

Assume 365 days in a year.

- If bank agrees to take early delivery then what will be net inflow to Mr. Guru assuming that the prevailing prime lending rate is 18% per annum.
- If Mr. Guru can deploy these funds in USD, he gets return at the rate of 3% per annum. Which is better? Why?

(Exam July – 2021) (8 Marks)

Solution:

Option 1: Early Delivery

- Swap Loss or Gain

Swap difference on 30/11 Bank sell spot & buy one month forward

Selling rate	= 75.22
Buying rate	= 75.42
Swap loss	= 0.20
(X) Contract size	= \$ 1,00,000



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Swap loss = ₹ 20,000

- Interest on inflows or outflows

Cash outflows (\$ 1,00,000 × 75.40) = ₹ 75,40,000

Cash inflows (\$ 1,00,000 × 75.22) = ₹ 75,22,000

Net outflows = 18,000

Interest = $18,000 \times 18\% \times \frac{31}{365}$ = 275.18

Net inflow to Guru (\$ 1,00,000 × 75.40) = ₹ 75,40,000

Swap loss = - 20,000

Interest on outflow = - 275.18

= ₹ 75,19,724.62

Option 2: Invest in \$ Fund at 3% p.a.

Cash inflow in \$ on 31/12

Principal = \$ 1,00,000

Interest (\$ 1,00,000 × 3% × 31/365) = \$ 254.79

Cash inflows (₹)

\$ 1,00,000 × 75.40 = 75,40,000

\$ 254.79 × 75.32 (75.22 + 0.10) = 19,190.78

= 75,59,190.78

Option 2 is better due to higher cash inflows

(III) OVERDUE FORWARD CONTRACT

Question – 62

An importer booked a forward contract with his bank on 10th April for USD 2,00,000 due on 10th June @ ₹ 64.4000. The bank covered its position in the market at ₹ 64.2800.

The exchange rates for dollar in the interbank market on 10th June and 13th June were:

	10 th June	13 th June
Spot	USD 1= ₹ 63.8000/8200	₹ 63.6800/7200
Spot/June	₹ 63.9200/9500	₹ 63.8000/8500
July	₹ 64.0500/0900	₹ 63.9300/9900
August	₹ 64.3000/3500	₹ 64.1800/2500
September	₹ 64.6000/6600	₹ 64.4800/5600

Exchange Margin 0.10% and interest on outlay of funds @ 12%. The importer requested on 14th June for extension of contract with due date on 10th August.

Rates rounded to 4 decimal in multiples of 0.0025.

On 10th June, Bank Swaps by selling spot and buying one month forward.

Calculate:



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

- (i) Cancellation rate
- (ii) Amount payable on \$ 2,00,000
- (iii) Swap loss
- (iv) Interest on outlay of funds, if any
- (v) New contract rate
- (vi) Total Cost

Solution:

(i) Cancellation Rate

Particular	Amount
Buying Rate of Bank (13/6)	₹ 63.6800
(-) Margin	0.10%
Customer Rate	₹ 63.6163
Round Off	₹ 63.6175

(ii) Cancellation Charge

Particular	Amount
Selling Rate of Bank	₹ 64.4000
(-) Buying rate of Bank	₹ 63.6175
Gain to Bank	0.7825
(X) Contract Size	\$ 2,00,000
Amount Payable by Customer	₹ 1,56,500

If gain to customer then amounts can't be paid to customer

(iii) Swap Loss

On 10th June bank sell spot to other Bank & buy forward from other Bank

Particular	Amount
Sell Spot	₹ 63.8000
(-) Buy Forward	₹ 63.9500
Swap Loss	₹ 0.15
(-) Contract Size	₹ 2,00,000
Swap Loss	₹ 30,000

Note: अगर Swap Gain होता तो Ram को Transfer नहीं करते।

(iv) Interest on Outlay of Fund

On 10th June bank buy (\$ 2,00,000 at Cover Rate hence Cash Outflows & sell at \$ 2,00,000 at SR)

Particular	Amount
Cash Outflow (\$ 2,00,000 × 64.2800)	₹ 1,28,56,000
Cash Inflow (\$ 2,00,000 × 63.8000)	₹ 1,27,60,000
Net Cash Outflow	₹ 96,000
Interest on Net Cash Outlay (₹ 96,000 12% × 3/360)	₹ 96

Note: अगर Net Cash Inflow होता तो उस पर Interest Ram को Transfer नहीं किया जाता।

(v) New Contract Rate (Available on 13/6)

Particular	Amount
Selling Rate of Bank	₹ 64.2500
(+) Margin	0.10%
Customer Rate	₹ 64.3142
Round off	₹ 64.3150



(vi) Total Cost

Particular	Amount
Cancellation Charges	₹ 1,56,500
(+) Swap Loss	₹ 30,000
(+) interest on Outlay of Fund	96
	₹ 1,86,596

Question – 63

On 10th July, an importer entered into a forward contract with bank for US \$ 50,000 due on 10th September at an exchange rate of ₹ 66.8400. The bank covered its position in the interbank market at ₹ 66.6800.

How the bank would react if the customer requests on 13th September:

- (i) To cancel the contract?
- (ii) To execute the contract?
- (iii) To extend the contract with due date to fall on 10th November?

The exchange rates for US\$ in the interbank market were as below:

	10 th September	13 th September
Spot	US\$1=66.1500/1700	65.9600/9900
Spot/September	66.2800/3200	66.1200/1800
Spot/October	66.4100/4300	66.2500/3300
Spot/November	66.5600/6100	66.4000/4900

Exchange margin was 0.1% on buying and selling.

Interest on outlay of funds was 12% p.a.

You are required to show the calculations to:

- (i) Cancel the Contract,
- (ii) Execute the Contract, and
- (iii) Extend the Contract as above.

Solution:

(i) Cancellation Rate

Particular	Amount
Buying Rate of Bank	₹ 65.9600
(-) Margin	₹ 0.10%
Rounded Off	₹ 65.8940

(ii) Cancellation Charge

Particular	Amount
Selling Rate of Bank	₹ 66.8400
(-) Buying rate of Bank	₹ 65.8950
Gain to Bank	0.945
(x) Contract Size	\$ 50,000
Cancellation Charges	₹ 47,250

(iii) Swap Loss

On 10th Sep Bank Sold \$ 50,000 at SR & Buy \$ 50,000 at Forward Rate means Sell- Buy

Particular	Amount
Selling Rate (SR)	₹ 66.1500
Buying Rate	₹ 66.3200
Swap Loss	₹ 0.17



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

(X) Contract Size	₹ 50,000
Swap Loss	₹ 8,500

(iv) Interest on Outlay of Fund

On 10th Sep

Particular	Amount
Cash Outflow (Buy\$ at Cover Rate) (\$ 50,000 × 66.68)	₹ 33,34,000
Cash Inflow (Sell \$ at SR) (\$ 50,000 × ₹ 66.1500)	₹ 33,07,500
Net Cash Outlay	₹ 26,500
Interest on Outlay of fund (₹ 26,500 × 12% × 3/360)	₹ 26

(v) New FR

Particular	Amount
Selling Rate (Nov)	₹ 66.4900
(+) Margin	0.1 %
	₹ 66.5565
Rounded Off	₹ 66.5575

(I) Cancel the Contract

Particular	Amount
Total amount payable by customer	
Cancellation charges	₹ 47,250
(+) Swap Loss	₹ 8,500
(+) Interest on Outlay of Fund	₹ 26

(II) Execute the Contract

Particular	Amount
Total amount of cancellation	₹ 55,776
(+) Buy \$ 50,000 at SR on 13 th Sep (65.9900 + 0.1%) 66.0560 (\$ 50,000 × 66.0550)	₹ 33,02,750
	₹ 33,58,526

(III) Extension

Particular	Amount
Total Amount of Cancellation	₹ 55,776
Book New Contract for 13 th Nov	₹ 66.5575

Question – 64

An import customer booked a forward contract with the bank on 10th April for USD 20,000 due on 10th June at ₹ 49.4000. The bank covered its position in the market at ₹ 49.2800.

The exchange rate for dollar in the interbank market on 10th June and 20th June were:

	10 th June	20 th June
Spot	USD 1 = ₹ 48.8000/8200	48.6800/7200
Spot/ June	48.9200/9500	48.8000/8500
Spot/ July	49.0500/0900	48.9300/9900
Spot/ August	49.3000/3500	49.1800/2500
Spot/ September	49.6000/6600	49.4800/5600
Exchange margin is 0.10%		
Interest on outlay of funds 12%		



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Calculate how the bank will react, if the customer requires on 20th June:

- (i) To cancel the contract.
 - (a) Exchange difference,
 - (b) Swap loss,
 - (c) Interest on outlay of funds and
 - (d) Cancellation charges
- (ii) To Execute the contract.
- (iii) To Extend the contract with due date to fall on 10th August.

(Exam November – 2023) (8 Marks)

Solution:

(i) Cancellation of Contract

(a) Exchange Difference

The forward sale contract shall be cancelled at Spot TT Purchase for \$ prevailing on the date of cancellation as follows:

\$/ ₹ Market Buying Rate	₹ 48.6800
Less: Exchange Margin @ 0.10%	₹ 0.0487
	₹ 48.6313

Rounded off to ₹ 48.6325

Exchange Difference Payable

Bank sells \$ 20,000 @ ₹ 49.4000	₹ 9,88,000
Bank buys \$ 20,000 @ ₹ 48.6325	₹ 9,72,650
Amount payable by customer	₹ 15,350

(b) Swap Loss

On 10th June the bank does a swap sale of \$ at market buying rate of ₹ 48.8000 and forward purchase for June at market selling rate of ₹ 48.9500.

Bank buys at	₹ 48.9500
Bank sells at	₹ 48.8000
Amount payable by customer	₹ 0.1500

Swap Loss for \$ 20,000 is = ₹ 3,000

(c) Interest on Outlay of Funds On 10th June, the bank receives delivery under cover contract at ₹ 49.2800 and sell spot at ₹ 48.8000.

Bank buys at	₹ 49.2800
Bank sells at	₹ 48.8000
Amount payable by customer	₹ 0.4800

Outlay for \$ 20,000 is ₹ 9,600

Interest on ₹ 9,600 @ 12% for 10 days ₹ 31.56 or ₹ 32.00

(d) Cancellation Charges

Particulars	Amount (₹)
Exchange Difference	15,350
Swap Loss	3,000
Interest on Outlay of Funds	32.00
Cancellation Charges payable by Customer	18,382



OR,

Particulars	Amount (₹)
Exchange Difference	15,350
Swap Loss	3,000
Interest on Outlay of Funds	31.56
Cancellation Charges payable by Customer	18,381.56

(ii) Execution of Contract

Cancellation charges of ₹ 18,382 or ₹ 18,381.56 as computed above will be recovered. The contract will be executed at the spot TT selling rate calculated as follows:

Dollar/₹ interbank spot selling rate	₹ 48.7200
Add: exchange margin at 0.10%	+ 0.0487
	₹ 48.7687

(iii) Extension of Contract

Cancellation charges of ₹ 18,382 or ₹ 18,381.56 as computed above will be recovered.

The contract will be extended at the current rate.

Dollar/₹ market forward selling rate for August	₹ 49.2500
Add: exchange margin at 0.10%	+ 0.0492
	₹ 49.2992

The exchange rate applied for the extended contract is ₹ 49.3000 or ₹ 49.2992.

(8) FOREIGN CURRENCY A/C

Question – 65

You as a dealer in foreign exchange have the following position in Swiss Francs on 31st October, 2009:

	Swiss Francs
Balance in the Nostro A/c Credit	1,00,000
Opening Position Overbought	50,000
Purchased a bill on Zurich	80,000
Sold forward TT	60,000
Forward purchase contract cancelled	30,000
Remitted by TT	75,000
Draft on Zurich cancelled	30,000

What steps would you take, if you are required to maintain a credit Balance of Swiss Francs 30,000 in the Nostro A/c and keep as overbought position on Swiss Francs 10,000?

(SM TYK – 51 & MTP March – 2022)

Solution:

Cash Position

	Cr.	Dr.
Opening balance credit	1,00,000	---
TT sales	---	75,000
	1,00,000	75,000
Closing balance (credit)	---	25,000

Exchange Position

	Long	Short
Opening balance over brought	50,000	---
Purchased bill	80,000	---
Forward sales	---	60,000
Forward purchase contract cancelled	---	30,000
TT sales	---	75,000
Draft cancelled	30,000	---
	1,60,000	1,65,000
Oversold position	5,000	---

The bank has to buy spot Swiss Francs 5,000 to increase the balance in NOSTRO A/c Swiss Francs 30,000 this would bring down the oversold position as nil.

Since bank requires overbought position of Swiss Francs 10,000 it has to buy forward Swiss Francs 10,000.

Question – 66

ABN-Amro Bank, Amsterdam, wants to purchase ₹ 15 million against US\$ for funding their NOSTRO account with Canara Bank, New Delhi. Assuming the inter-bank, rates of US\$ is ₹ 51.3625/3700, what would be the rate Canara Bank would quote to ABN-Amro Bank? Further, if the deal is struck, what would be the equivalent US\$ amount.

(SM TYK – 02)**Solution:**

\$ Required to buy ₹ 15,000,000

$$= \frac{₹ 15,000,000}{51.3625} = \$ 2,92,042$$

Question – 67

XYZ Bank, Amsterdam, wants to purchase ₹ 25 million against £ for funding their NOSTRO account and they have credited LORO account with Bank of London, London.

Calculate the amount of £'s credited. Ongoing inter-bank rates are per \$, ₹ 61.3625/3700 & per £, \$ 1.5260/70.

(SM TYK – 03)**Solution:**

$$₹/\$ = 61.3625$$

$$\$/\pounds = 1.5260$$

$$\begin{aligned} ₹/\pounds &= 61.3625 \times 1.5260 \\ &= 93.3692 \end{aligned}$$

Amount of £ to be credited

$$= \frac{₹ 2,50,00,000}{93.3692} = \pounds 2,67,754.25$$

(9) CURRENCY OF BORROWING**Question – 68**

Sun Ltd. is planning to import equipment from Japan at a cost of 3,400 lakh yen. The company may avail loans at 18 percent per annum with quarterly rests with which it can import the equipment. The company has also an offer from Osaka branch of an India based bank extending credit of 180 days at 2 percent per annum against opening of an irrecoverable letter of credit.



Additional information:

Present exchange rate ₹ 100 = 340 yen

180 day's forward rate ₹ 100 = 345 yen

Commission charges for letter of credit at 2 percent per 12 months.

Advice the company whether the offer from the foreign branch should be accepted.

(SM TYK – 42 & Exam January – 2021) (8 Marks)

Solution:

Option 1: Loan from Indian Bank

Payable = ₹ 3,400 lacs

Buy ₹ 3,400 lacs at SR ₹/₹ = 3.40

$$\frac{₹ 3,400}{3.40} = ₹ 1,000 \text{ lacs}$$

Borrow ₹ 1,000 lacs at 18%. p.a. for 6 months

Cash outflows = $1,000 (1.045)^2 = ₹ 1,092.03 \text{ lacs}$

Option 2: Loan from Japan

Loan amount = ₹ 3400 lacs

Borrow ₹ 3,400 lacs at 2 % p.a. for 180 days

Cash outflows (₹ 3,400 lacs × 1.01) = ₹ 3,434 lacs

Buy ₹ 3,434 lacs at 180 day FR ₹/₹ = 3.45

$$\frac{₹ 3,434}{3.45} = ₹ 995.36$$

Commission [Payment आज होता है]

$$\text{Loan amount} = \frac{₹ 3,400}{3.40} = ₹ 1,000 \text{ lacs}$$

Commission = 1,000 lacs × 1% = ₹ 10 lacs

$$\begin{aligned} \text{Commission after 6 month} &= 10 (1.045)^2 \\ &= ₹ 10.92 \text{ lacs} \end{aligned}$$

$$\begin{aligned} \text{Total cash outflows} &= ₹ 995.36 + 10.92 \\ &= ₹ 1,006.28 \text{ lacs} \end{aligned}$$

Loan from Japan is better due to lower cash outflows.

Question – 69

An Indian company obtains the following quotes (₹/\$)

Spot: 35.90/36.10

3 Months forward rate: 36.00/36.25

6 Months forward rate: 36.10/36.40

The company needs \$ funds for six months, Determine whether the company should borrow in \$ or ₹. Interest rates are:

3 Months interest rate: ₹: 12%, \$: 6%

6 Months interest rate: ₹: 11.50%, \$: 5.5%



Also determine what should be the rate of interest after 3 – months to make the company indifferent between 3- months borrowings and 6- months borrowings in the case of:

- (i) Rupee borrowing
- (ii) Dollar borrowing

Note: For the purpose of calculation you can take the units of dollar and rupee as 100 each.

(Exam November – 2018) (8 Marks)

Solution:

Suppose we need \$ 100 for 6 months

(i) Borrow from US

Principal	= \$ 100
Interest (\$ 100 × 5.5% × 6/12)	= \$ 2.75
	<u>= \$ 102.75</u>

Buy \$ 102.75 at 6 month FR

Cash outflows	= \$ 102.75 × 36.40
	= ₹ 3,740.10

(ii) Borrow from India

₹ Required to buy \$100 at SR (\$100 × 36.10) = ₹ 3,610

Interest (3,610 × 11.5% × 6/12) = ₹ 207.575

Cash outflows = ₹ 3,817.575

Borrow in \$ is better due to lower cash outflows.

Indifference Rate

₹ Borrowing

$$3 \text{ months rate} = \left[\frac{1.0575}{1.03} - 1 \right] \times 100 \times \frac{12}{3} = 10.68\%$$

\$ Borrowing

$$3 \text{ months FRA} = \left[\frac{1.0275}{1.015} - 1 \right] \times 100 \times \frac{12}{3} = 4.93\% \text{ p.a.}$$

Question – 70

A German subsidiary of an US based MNC has to mobilize 1,00,000 Euro's working capital for the next 12 months. It has the following options:

Loan from German Bank : @ 5% p.a.

Loan from US Parent Bank : @ 4% p.a.

Loan from Swiss Bank : @ 3% p.a.

Banks in Germany charge an additional 0.25% p.a. towards loan servicing. Loans from outside Germany attract withholding tax of 8% on interest payments. If the interest rates given above are market determined, examine which loan is the most attractive using interest rate differential.

(Exam November – 2019) (5 Marks)



Solution:

Calculation of cost of loan

(i) Borrow from German Bank

$$(5\% + 0.25\%) = 5.25\%$$

(ii) Borrow from US Bank Interest

$$\frac{4\%}{1 - 0.08} = 4.35\%$$

$$\text{Premium in \$} = \left[\frac{1.05}{1.04} - 1 \right] = 0.96\%$$

$$= 5.31\%$$

(iii) Borrow from Swiss Bank

$$\text{Interest} = \left[\frac{3\%}{1 - 0.08} \right] = 3.26\%$$

$$\begin{aligned} \text{Premium in SF} &= \left[\frac{1.05}{1.03} - 1 \right] \times 100 = \\ 1.94\% & \\ &= 5.20\% \end{aligned}$$

Borrow from Swiss Bank is better due to lower cash outflows.

$$\begin{aligned} \text{Cash outflows} &= \text{€ } 1,00,000 \times 1.052 \\ &= \text{€ } 1,05,200 \end{aligned}$$

(10) CURRENCY OF INVESTMENT

Question – 71

Your bank's London office has surplus funds to the extent of USD 5,00,000/- for a period of 3 months. The cost of the funds to the bank is 4% p.a. It proposes to invest these funds in London, New York or Frankfurt and obtain the best yield, without any exchange risk to the bank. The following rates of interest are available at the three centers for investment of domestic funds there at for a period of 3 months.

London 5 % p.a.

New York 8% p.a.

Frankfurt 3% p.a.

The market rates in London for US dollars and Euro are as under:

London on New York

Spot 1.5350/90

1 month 15/18

2 months 30/35

3 months 80/85

London on Frankfurt

Spot 1.8260/90

1 month 60/55

2 months 95/90

3 months 145/140



At which centre, will be investment be made & what will be the net gain (to the nearest pound) to the bank on the invested funds?

(SM TYK – 49, RTP Nov – 2021 & Exam Nov – 2013) (8 Marks)

Solution:

Option 1: Investment in London

Fund available = \$ 5,00,000

Convert \$ 5,00,000 at SR \$/£ 1.5390

$$\frac{\$ 5,00,000}{\text{£ } 1.5390} = \text{£ } 3,24,886.29$$

Invest @ 5% p.a. for 3 months

$$(\text{£ } 3,24,886.29 + (3,24,886.29 \times 5\% \times 3/12)) = \text{£ } 3,28,947.37$$

Repayment in \$ after 3 months [\$ 5,00,000 (1.01)] = \$ 5,05,000

Buy \$ 5,05,000 at 3 months FR [1.5350 + 0.0080] = \$/£ 1.5430

$$\text{Cash outflow in £} = \frac{\$ 5,05,000}{\text{£ } 1.5430} = \text{£ } 3,27,284.51$$

$$\text{Gain } (\text{£ } 3,28,947.37 - \text{£ } 3,27,284.51) = \text{£ } 1,663$$

Option 2: Invest in New York

Invest \$ 5,00,000 @ 8% p.a. for 3 months

Cash inflows (\$) [\$ 5,00,000 (1.02)] = \$ 5,10,000

Cash outflow \$ = \$ 5,05,000

Net Gain (\$) = \$ 5,000

Convert \$ 5,000 at 3 months FR

$$(1.5390 + 0.0085) = 1.5475$$

$$\text{Net Gain in £} = \frac{\$ 5,000}{1.5475} = \text{£ } 3,231$$

Option 3: Invest in Frankfurt

\$ Available = \$ 5,00,000

$$\text{Convert } \$ 5,00,000 \text{ at SR} = \frac{\$ 5,00,000}{1.5390} = \text{£ } 3,24,886.2$$

Convent £ 3,24,886 at in € at SR €/£ - 1.8260

$$(\text{£ } 3,24,886 \times 1.8260) = \text{€ } 5,93,242$$

Invest € 5,93,242 @ 8% p.a. for 3 months

$$(\text{€ } 5,93,242 \times 1.0075) = \text{€ } 5,97,691$$

Convert € 5,97,691 in £ at 3 months FR

$$(1.8290 - 0.0140) = \text{€/£ } 1.8150$$

$$= \frac{\$ 5,97,691}{1.8150} = \text{£ } 3,29,306$$

Buy \$ 5,05,000 at FR

$$\text{Gain} = \text{£ } 3,29,306 - \text{£ } 3,27,285$$

$$= \text{£ } 2,021$$

Invest in New York is the best option due to highest gain.



Question – 72

ICL an Indian MNC is executing a plant in Sri Lanka. It has raised ₹ 400 billion. Half of the amount will be required after six months' time. ICL is looking an opportunity to invest this amount on 1st April, 2020 for a period of six months. It is considering two underlying proposals:

Market	Japan	US
Nature of Investment	Index Fund (JPY)	Treasury Bills (USD)
Dividend (in billions)	25	-
Income from stock lending (in billions)	11.9276	-
Discount on initial investment at the end	2%	-
Interest	-	5 per cent per annum
Exchange Rate (1 st April, 2020)	JPY/INR 1.58	USD/INR 0.014
Exchange Rate (30 th September, 2020)	JPY/INR 1.57	USD/INR 0.013

You, as an Investment Manager, is required to suggest the best course of option.

(Exam November – 2020) (12 Marks)

Solution:

Option 1: Investment in japan (index fund)

Fund available (₹ 400 billions × 50%) = ₹ 200 billions

Convert ₹ 200 billions in ¥ at SR

₹ 200 billions × 1.58 = ¥ 316 billions

Investment value after 6 months

Index fund (¥ 316 × 98%) = ¥ 309.68 billions

(+) Dividend = ¥ 25 billions

(+) Stock lending income = ¥ 11.9276 billions

¥ 346.6076 billions

Convert ¥ into ₹ at FR

$\frac{¥ 346.6076}{1.57} = ₹ 220.7692 \text{ billions}$

Income = ₹ 220.7692 – ₹ 200 = 20.7692 billions

Option 2: Investment in US [T. Bill]

Amount available = ₹ 200 billions

Convert ₹ 200 billions in \$ at SR

₹ 200 × 0.014 = \$ 2.80 billions

Investment value after 6 months

\$ 2.80 × (1.025) = \$ 2.87 billions

Convert \$ 2.87 billions in ₹ at FR

$\frac{\$ 2.87}{0.013} = ₹ 220.7692$

Income = ₹ 220.7692 – 200 = ₹ 20.7692 billions



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

- The equivalent amount is same in both the options so ICL is indifferent.
- However, USD is more stable, and Treasury Bills are risk free, so investment in Treasury Bills (USD) is suggested.

Question – 73

(i) Interest rates for 3 months in USA and Canada are as follows:

Currency	Borrow	Invest
US \$	4%	2.5%
Canadian \$	4.5%	3.5 %

(ii) Can \$/ US \$ spot 1.235 ---- 1.240

3 months forward 1.255 ---- 1.260

Advice, the currency in which borrowing and lending for 3 months needs to be done for the US company. Take 3 months = 90/360 days.

(Exam Jan – 2021) (8 Marks)

Solution:

Assume \$ 1,000

Lending

Option 1: Lending in US

Principal		\$ 1,000.00
(+) Interest (\$ 1,000 × 2.5% × 90/360)	\$	6.25
		\$ 1,006.25

Option 2: Lending in Canada

Convert \$ 1,000 in Can \$ at SR

\$ 1,000 × 1.235	C\$ 1,235
Interest (1,235 × 3.5% × 90/360)	C\$ 10.80
	C\$ 1,245.80

Convert C\$ 1,245.80 in US \$ at FR

$$\frac{\text{Can \$ 1,245.80}}{1.260} = \$ 988.74$$

Lending in US is better due to higher cash inflows.

Borrowing

Option -1 Borrow from US

Borrowing amount	=	\$ 1,000
Interest (\$ 1,000 × 4% × 90/360)		\$ 10
		\$ 1,010

Option -2 Borrowing from Canada

Can \$ required to buy US \$ 1,000

\$ 1,000 × 1.240	C\$ 1,240
Interest (C\$ 1,240 × 4.5% × 90/360)	C\$ 13.95
	C\$ 1,253.95



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Buy C\$ 1,253.95 at FR

$$\frac{\text{Can \$ } 1,253.95}{1.255} = \$ 999.16$$

Borrow from Canada is better due to lower cash outflows.

(11) INTERNATIONAL CASH MANAGEMENT

Question – 74

Suppose you are a treasurer of XYZ plc in the UK. XYZ have two overseas subsidiaries, one based in Amsterdam and one in Switzerland. The Dutch subsidiary has surplus Euros in the amount of 725,000 which it does not need for the next three months but which will be needed at the end of that period (91 days). The Swiss subsidiary has a surplus of Swiss Francs in the amount of 998,077 that, again, it will need on day 91. The XYZ plc in UK has a net balance of £75,000 that is not needed for the foreseeable future.

Given the rates below, what is the advantage of swapping Euros and Swiss Francs into Sterling?

Spot Rate (€) £0.6858— 0.6869

91 day Pts 0.0037 0.0040

Spot Rate(£) CHF 2.3295— 2.3326

91 day Pts 0.0242 0.0228

Interest rates for the Deposits

Amount of Currency	91 Days Interest Rate % p.a.		
	£	€	CHF
0 – 1,00,000	1	¼	0

1,00,001 – 5,00,000	2	1 ½	¼
5,00,001 – 10,00,000	4	2	½
Over 10,00,000	5.375	3	1

Solution:

Option 1: [Individual basis] assume 1 year = 360

Without swapping		Individuals basis			
Currency	Principal	Interest	Total	Calculation of interest in £	6 months
£	£ 75,000	£ 75,000 × 1% × 91/360	₹ 75,189.58	---	£ 75,189.58
€	€ 7,25,000	£ 7,25,000 × 2% × 91/360	€ 7,28,665.28	€ 7,28,665.28 × 0.6895	£ 5,02,414.71
CHF	CHF 9,98,077	9,98,077 × 2% × 91/360	CHF 9,99,338.46	$\frac{9,99,338.46}{2.3098}$	£ 4,32,651.51
					£ 10,10,255.80

Option 2: Swapping

Available £ in UK = £ 75,000

Convert € in £ at SR (€ 7,25,000 × 0.6858) = £ 4,97,205

Convert CHF in £ at SR $\left(\frac{9,98,077}{2.3326} \right)$ = £ 4,27,881.76

= £ 10,00,086.76

Invest £ at 5.375% for 91 days



$$\text{Interest} = \text{£ } 10,00,086.76 \times 5.375\% \times 91/360$$

$$= \text{£ } 13,587.98$$

$$\text{Total} = \text{£ } 10,00,086.76 + \text{£ } 13,587.98$$

$$= \text{£ } 10,13,674.74$$

Swapping is better due to higher cash inflows.

(12) FX SWAP

Question – 75

Drilldip Inc. a US based company has won a contract in India for drilling oil field. The project will require an initial investment of ₹ 500 crore. The oil field along with equipments will be sold to Indian Government for ₹ 740 crore in one year time. Since the Indian Government will pay for the amount in Indian Rupee (₹) the company is worried about exposure due exchange rate volatility.

You are required to:

- Construct a swap that will help the Drilldip to reduce the exchange rate risk.
- Assuming that Indian Government offers a swap at spot rate which is 1US\$ = ₹ 50 in one year, then should the company should opt for this option or should it just do nothing. The spot rate after one year is expected to be 1US\$ = ₹ 54. Further you may also assume that the Drilldip can also take a US\$ loan at 8% p.a.

(SM TYK – 50)

Solution:

- Drilldip will enter into buy sell swap in which Drilldip will buy ₹ 500 cr now & sell ₹ 500 cr after 1 year.

Option 1 FX Swap

\$

$$\text{Buy ₹ 500 Cr. today } \left(\frac{\text{₹ } 500 \text{ cr}}{\text{₹ } 50} \right) = (10 \text{ Cr.})$$

$$\text{Sell ₹ 500 Cr. after 1 year } \left(\frac{\text{₹ } 500 \text{ cr}}{\text{₹ } 50} \right) = 10 \text{ Cr.}$$

$$\text{Sell ₹ 240 Cr. at 1 year expected rate } \left(\frac{\text{₹ } 240}{54} \right) = 4.444 \text{ Cr.}$$

$$\text{Interest } (\$ 10 \text{ Cr.} \times 8\%) = (0.8 \text{ Cr.})$$

$$\text{Profit} = \$ 3.644 \text{ Cr.}$$

Option 2 Do Nothing

$$\text{Buy 500 Cr. today at SR } \left(\frac{\text{₹ } 500 \text{ cr}}{\text{₹ } 50} \right) = (10 \text{ Cr.})$$

$$\text{Sell 740 Cr. at expected SR } \left(\frac{\text{₹ } 740 \text{ cr}}{\text{₹ } 54} \right) = 13.704$$

$$\text{Interest} = (0.8 \text{ Cr.})$$

$$\text{Profit} = \$ 2.904 \text{ Cr.}$$

FX Swap is better due to higher profit.



(13) ECONOMIC EXPOSURE**Question – 76**

M/s Omega Electronics Ltd. exports air conditioners to Germany by importing all the components from Singapore. The company is exporting 2,400 units at a price of Euro 500 per unit. The cost of imported components is S\$ 800 per unit. The fixed cost and other variables cost per unit are ₹ 1,000 and ₹ 1,500 respectively. The cash flows in Foreign currencies are due in six months. The current exchange rates are as follows:

₹/Euro 51.50/55

₹/S\$ 27.20/25

After six months the exchange rates turn out as follows:

₹/Euro 52.00/05

₹/S\$ 27.70/75

(A) You are required to calculate loss/gain due to transaction exposure.

(B) Based on the following additional information calculate the loss/gain due to transaction and operating exposure if the contracted price of air conditioners is ₹ 25,000 :

(i) the current exchange rate changes to

₹/Euro 51.75/80

₹/S\$ 27.10/15

(ii) Price elasticity of demand is estimated to be 1.5

(iii) Payments and receipts are to be settled at the end of six months.

(SM TYK – 48)

Solution:**(A) Calculation of Transaction Exposure****Current Exchange rate**

$$(2,400 \text{ units} \times \text{€ } 500 \times 51.50) - (2,400 \times \text{S\$ } 800 \times 27.25) - (2,400 \times 2,500)$$

$$= \text{₹ } 34,80,000$$

6 Months Exchange rate

$$(2,400 \text{ units} \times \text{€ } 500 \times 52) - (2,400 \times \text{S\$ } 800 \times 27.75) - (2,400 \times 2,500)$$

$$= \text{₹ } 31,20,000$$

Loss due to Exposure

$$\text{₹ } 34,80,000 - \text{₹ } 31,20,000 = \text{₹ } 3,60,000$$

(B) If current exchange rate change then calculation of loss due to Transaction Exposure**Current exchange rate**

$$(2,400 \text{ units} \times 25,000) - (2,400 \times \text{S\$ } 800 \times 27.15) - (2,400 \times 2,500)$$

$$= \text{₹ } 18,72,000$$

6 Months Exchange rate

$$(2,400 \text{ units} \times 25,000) - (2,400 \times \text{S\$ } 800 \times 27.75) - (2,400 \times 2,500)$$



$$= ₹ 7,20,000$$

$$\text{Loss due to Transaction Exposure} = ₹ 11,52,000$$

Calculation of new demand of units

Price of unit of German customer at old rate

$$\frac{₹ 25,000}{51.50} = € 485.44$$

Price of unit at new exchange rate

$$\frac{₹ 25,000}{51.75} = € 483.09$$

$$\% \text{ decrease in price} = \frac{€ 485.44 - € 483.09}{€ 485.44} \times 100 = 0.48\%$$

$$\% \text{ Increase in demand} = 0.48 \times 1.5 = 0.72\%$$

$$\text{New units} = 2,400 \times 0.72\% = 17 \text{ Units}$$

$$\text{Total Unit} = 2,400 + 17 = 2,417 \text{ unit}$$

Profit due to increase in new unit 6 month

$$(2,417 \text{ units} \times 25,000) - (2,417 \times \$ 800 \times 27.75) - (2,400 \times 1,000) - (2,417 \times 1,500)$$

$$= 7,42,100$$

$$\begin{aligned} \text{Loss due to economics exposure} &= 18,72,000 - 7,42,100 \\ &= 11,29,900 \end{aligned}$$

(14) NDF CONTRACT**Question – 77**

On 1st February 2020, XYZ Ltd. a laptop manufacturer imported a particular type of Memory Chips from SKH Semiconductor of South Korea. The payment is due in one month from the date of Invoice, amounting to 1190 Million South Korean Won (SKW). Following Spot Exchange Rates (1st February) are quoted in two different markets:

USD/ INR 75.00/ 75.50 in Mumbai

USD/ SKW 1190.00/ 1190.75 in New York

Since hedging of Foreign Exchange Risk was part of company's strategic policy and no contract for hedging in SKW was available at any in-shore market, it approached an off-shore Non Deliverable Forward (NDF) Market for hedging the same risk.

In NDF Market a dealer quoted one-month USD/ SKW at 1190.00/1190.50 for notional amount of USD 100,000 to be settled at reference rate declared by Bank of Korea.

After 1 month (1st March 2020) the dealer agreed for SKW 1185/ USD as rate for settlement and on the same day the Spot Rates in the above markets were as follows:

USD/ INR 75.50/ 75.75 in Mumbai

USD/ SKW 1188.00/ 1188.50 in New York

Analyze the position of company under each of the following cases, comparing with Spot Position of 1st February:

(i) Do Nothing.



- (ii) Opting for NDF Contract. Note: Both Rs./ SKW Rate and final payment (to be computed in Rs. Lakh) to be rounded off upto 4 decimal points. (10 Marks)

(MTP April – 2021)**Solution:****At Spot Rate**

If SKS 1,190 million today

$$\text{₹}/\$ = 75.50$$

$$\text{SKW}/\$ = 1,190$$

$$\begin{aligned}\text{₹}/\text{SKW} &= 75.50 \times \frac{1}{1190} \\ &= 0.0634\end{aligned}$$

$$\begin{aligned}\text{SKW } 1,190 \times 0.0634 \\ &= ₹ 75.446 \text{ m} \\ &= ₹ 754.46 \text{ lacs}\end{aligned}$$

Option 1: Do Nothing

Buy SKW 1,190 millions at 1 months SR

$$\text{₹}/\$ = 75.50$$

$$\text{SKW}/\$ = 1188$$

$$\text{₹}/\text{SKW} = 75.50 \times \frac{1}{1,188}$$

$$= 0.0638$$

$$\text{SKW } 1,190 \text{ millions} \times 0.0638 = 75.922 \text{ millions}$$

$$= 75.922 \text{ lacs}$$

$$\text{Loss} = 759.22 - 754.46 = 4.76 \text{ lacs}$$

Option 2: NDF Contract

Since SKW payable & we afraid from SKW rising but we have to take position on \$, hence we should take short position (sell) on \$ at SKW/\$ 1,190

$$\begin{aligned}\text{Amount of short position (\$)} &= \frac{\text{SKW } 1,190 \text{ m}}{1,190} = \$ 1 \text{ million} \\ &= \text{or } \$ 10,00,000\end{aligned}$$

Gain or Loss

On maturity, SKW/\$ is 1,185 hence gain on short position

$$(\text{SKW } 1,190 - \text{SKW } 1,185) \times \$ 10,00,000 = \$ \text{SKW } 50,00,000$$

$$\text{Gain in \$} = \frac{\text{SKW } 50,00,000}{1,185} = \$ 4,219.4093$$

$$\text{Buy SKW } 1190 \text{ millions at 1 month SR} = ₹ 759.22 \text{ lacs}$$

$$\begin{aligned}(-) \text{ Sell } \$ 4,219.4093 \text{ at 1 month SR} \\ (\$ 4,219.4093 \times 75.50) &= ₹ 3.1856 \text{ lacs}\end{aligned}$$

$$= ₹ 756.0343 \text{ lacs}$$

$$\text{Loss} = ₹ 756.0343 - 754.46 = 1.5743 \text{ lacs}$$



NDF contract is better due to less loss.

(15) RESIDUAL

Question – 78

M/s. Sky products Ltd., of Mumbai, an exporter of sea foods has submitted a 60 days bill for EUR 5,00,000 drawn under an irrevocable Letter of Credit for negotiation. The company has desired to keep 50% of the bill amount under the Exchange Earners Foreign Currency Account (EEFC). The rates for ₹/USD and USD/EUR in inter-bank market are quoted as follows:

	₹/ USD	USD/EUR
Spot	67.8000 – 67.8100	1.0775 – 1.8000
1 month forward	10/11 Paise	0.20/0.25 Cents
2 months forward	21/22 Paise	0.40/0.45 Cents
3 months forward	32/33 Paise	0.70/0.75 Cents

Transit Period is 20 days. Interest on post shipment credit is 8% p.a.

Exchange Margin is 0.1%. Assume 365 days in a year.

You are required to calculate:

- Exchange rate quoted to the company
- Cash inflow to the company
- Interest amount to be paid to bank by the company.

(Exam January – 2021) (8 Marks)

Solution:

(i) Exchange rate quoted to the company

60 pay FR

\$/€ 1.0775

(+) 2 months swap 0.0040

\$/€ 1.0815

₹/\$ 67.8000

(+) 2 months swap 0.2100

₹ 68.0100

(-) margin - 0.1%

₹ 67.9420

₹/€ = ₹ 67.9420 × 1.0815

= ₹ 73.4793

(ii) Cash inflow = € 2,50,000 × 73.4793

= ₹ 1,83,69,825

(iii) Interest on post shipment credit

₹ 1,83,69,825 × 8% × 80/365 = ₹ 3,22,101

Question – 79

ABC Ltd. of UK has exported goods worth Can \$ 5,00,000 receivable in 6 months. The exporter wants to hedge the receipt in the forward market. The following information is available:

Spot Exchange Rate	Can \$ 2.5/£
Interest Rate in UK	12%
Interest Rate In Canada	15%

The forward rates truly reflect the interest rates differential. Find out the gain/loss to UK exporter if Can \$ spot rates (i) declines 2%, (ii) gains 4% or (iii) remains unchanged over next 6 months.

(SM TYK – 04)

Solution:

Calculation of FR

$$\begin{aligned} \text{FR} &= \text{SR}_A \times \frac{1 + r_A}{1 + r} \\ &= \text{C\$ } 2.5 \times \frac{1.075}{1.06} \end{aligned}$$

$$\text{C\$/£} = \text{C\$ } 2.535$$

Gain/Loss

(i) C\$ decline by 2 % (2.5×1.02) = 2.55

Receivable under FC	=	$\frac{\text{C\$ } 5,00,000}{2.535}$	=	£
1,97,238.66				

Receivable under	=	$\frac{\text{C\$ } 5,00,000}{2.55}$	=	£
1,96,078.43				
Gain			=	£ 1,160.23
(ii) C\$ gain by 4% [$2.5 \times 0.96 = 2.40$]				
Forward cover				\$ 1,97,238.66
Expected SR $\left(\frac{\text{C\$ } 5,00,000}{2.40}\right)$				\$
2,08,333.33				
Loss				\$ 11,094.67
(iii) Remains unchanged				
Forward cover				\$ 1,97,238.66
Expected SR $\left(\frac{\text{C\$ } 5,00,000}{2.5}\right)$				\$
2,00,000				
Loss				\$ 2,761.34

Question – 80

Your forex dealer had entered into a cross currency deal and had sold US \$ 10,00,000 against EURO at US \$ 1 = EURO 1.4400 for spot delivery.

However, later during the day, the market became volatile and the dealer in compliance with his management's guidelines had to square – up the position when the quotations were:

Spot US \$ 1 INR 31.4300/4500



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

1 month margin 25/20

2 months margin 45/35

Spot US \$ 1 EURO 1.4400/4450

1 month forward 1.4425/4490

2 months forward 1.4460/4530

What will be the gain or loss in the transaction?

(SM TYK – 46)

Solution:

Gain/loss in €

Selling rate = € 1.4400

Buying rate = € 1.4450

Loss = € 0.0050 per \$

(X) Contract size = \$ 10,00,000

Loss = € 5,000

Loss in ₹ (Ask Rate)

₹/€ = $31.4500 \times \frac{1}{1.4400}$

= 21.8403

= € 5,000 × 21.8403

= ₹ 1,09,201

Question – 81

Following are the details of cash inflows and outflows in foreign currency denominations of MNP Co. an Indian export firm, which have no foreign subsidiaries:

Currency	Inflow	Outflow	Spot Rate	Forward Rate
US \$	4,00,00,000	2,00,00,000	48.01	48.82
French Franc (FFr)	2,00,00,000	80,00,000	7.45	8.12
U.K. £	3,00,00,000	2,00,00,000	75.57	75.98
Japanese Yen	1,50,00,000	2,50,00,000	3.20	2.40

(i) Determine the net exposure of each foreign currency in terms of Rupees.

(ii) Are any of the exposure positions offsetting to some extent?

(SM TYK – 10)

Solution:

(i) Net exposure of each foreign currency in ₹

Currency	Net inflow	Spread	Net exposure
\$	2,00,00,000	0.81	1,62,00,000
FFr	1,20,00,000	0.67	80,40,000
£	1,00,00,000	0.41	41,00,000
¥	-1,00,00,000	-0.80	80,00,000

(ii) In Japanese Yen, the exposure is payable, and the forward rate is quoted at a discount, effectively offsetting the position. Likewise, in the remaining currencies, the net exposures are in receivables, and the related currencies are at a premium, offsetting the positions in their respective currencies.



MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

An Indian exporting firm, Rohit and Bros. exported good worth of AUD 1 million to an importer in Sydney. Rohit and Bros. are worried about likely depreciation of AUD in near future as it is likely that the export sum will be received after 3 months. Today as such as there is no derivative contract is available in AUD to hedge itself from such depreciation.

The following data is given:

Spot rate	: ₹ 56.00/AUD
3 months interest rate : India	: 12 per cent per annum
	: Australia : 5 per cent per annum

From the information given above, choose the correct answer to the following questions:

- I.** hedging technique can be used Rohit and Bros. to hedge itself against the risk of depreciation of AUD.
- Forward Contract
 - Future Contract
 - Option Contract
 - Money Market Hedge
- II.** Suppose if Rohit and Bros. want to borrow some amount in AUD in such a manner that the receivable amount can be used to repay the amount borrowed along with interest. The

amount to be borrowed by Rohit and Bros. shall be approximately.....

- | | |
|-------------------|-------------------|
| (a) AUD 1 million | (b) AUD 9,87,654 |
| (c) ₹ 5,53,08,624 | (d) ₹ 5,69,67,882 |

- III.** Suppose if Rohit and Bros. borrows a designated amount in AUD for 3 months in such a manner that the receivable amount can be used to repay the amount borrowed along with interest and plan to invest same amount in Indian spot market. The same amount available for investment shall be approximately.....

- | | |
|-------------------|-------------------|
| (a) AUD 1 million | (b) AUD 9,87,654 |
| (c) ₹ 5,53,08,624 | (d) ₹ 5,69,67,882 |

- IV.** Suppose if Rohit and Bros. borrows a designated amount for 3 months in AUD in such a manner that the receivable amount can be used to repay the amount borrowed along with interest. Further he plans to invest same amount in Indian spot market. The amount obtained after 3 months of investment shall be approximately.....

- | | |
|-------------------|-------------------|
| (a) AUD 1 Million | (b) AUD 9,87,654 |
| (c) ₹ 5,53,08,624 | (d) ₹ 5,69,67,883 |

V. Suppose if Interest Rate Parity theory is held between INR and AUD, then forward rate between INR and AUD for 6 month should be.....

- (a) ₹ 51.54/AUD (b) ₹ 57.91/AUD
(c) ₹ 52.50/AUD (d) ₹ 59.73/AUD

Answer: Case Scenario – 01

- I. (d) Money Market Hedge**
II. (b) AUD 9,87,654
III. (c) ₹ 5,53,08,624
IV. (d) ₹ 5,69,67,883
V. (b) ₹ 57.91/AUD

Case Scenario – 02

On 1 October 2023 Mr. X an exporter enters into a forward contract with a BNP Bank to sell US\$ 1,00,000 on 31 December 2023 at ₹ 85.40/\$. However, due to the request of the importer, Mr. X received the amount on 28 November 2023. Mr. X requested the bank the take delivery of the remittance on 30 November 2023 i.e., before due date. The inter-banking rates on 28 November 2023 was as follows:

Spot ₹ 85.22/85.27

One Month Premium 10/15

Note: (1) Consider 365 days in a year.

(2) Prevailing Prime Lending Rate is 12%

Based on above case scenario, choose the most appropriate answer of the following:

- I.** The bank may accept the request of customer of delivery before due date of forward contract provided the customer is ready to bear the loss if any consisting of.....
- (a) Swap Difference
 (b) Interest on Outlay of Fund
 (c) Swap Difference Plus Interest on Outlay of Fund
 (d) Fixed Charges Plus Swap Difference and Interest on Outlay of Fund
- II.** In case of early delivery bank shall charge interest on outlay of fund at a rate not less than.....
- (a) 8% (b) 10%
 (c) 12% (d) 18%
- III.** Swap Difference for US\$ 1,00,000 is.....
- (a) ₹ 5,000 (b) ₹ 20,000
 (c) ₹ 18,000 (d) ₹ 8,000
- IV.** Interest on outlay of funds shall be approximately.....
- (a) ₹ 92 payable by X
 (b) ₹ 183 payable by X
 (c) ₹ 183 payable by Bank



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

(d) ₹ 122 payable by Bank

V. Net inflow to Mr. X is approximately.....

(a) ₹ 85,42,183 (b) ₹ 85,20,000

(c) ₹ 85,19,817 (d) ₹ 85,40,000

(MTP April – 2024)

Answer: Case Scenario – 02

I. (d) Fixed Charges Plus Swap Difference and Interest on Outlay of Fund

II. (c) 12%

III. (b) ₹ 20,000

IV. (b) ₹ 183 payable by X

V. (c) ₹ 85,19,817

Case Scenario – 03

A US parent company has subsidiaries in France, Germany, UK and Italy. The amounts due to and from the affiliates is converted into a common currency viz. US dollar and entered in the following matrix.

Inter Subsidiary Payments Matrix (US \$ Thousands)

Paying Affiliate						
		France	Germany	UK	Italy	Total
Receiving Affiliate	France	---	80	120	200	400
	Germany	120	---	80	160	360
	UK	160	120	---	140	420
	Italy	200	60	120	---	380

	Total	480	260	320	500	1560
--	-------	-----	-----	-----	-----	------

The treasurer of US Parent company is suggesting that by applying Multilateral Netting system the company can save a lot of transfer/exchange costs. The company's Board agreed with Treasurer's proposal.

From the above case scenario, choose the most appropriate answer of following MCQs.

I. Before applying Multilateral Netting it is necessary to apply.....

- (a) Unilateral Netting
- (b) Bilateral Netting
- (c) Multilateral Netting
- (d) Interest Rate Swapping

II. Through Multinational Netting these transfers will be reduced to

- (a) \$ 50,000 (b) \$ 100,000
- (c) \$ 150,000 (d) \$ 200,000

III. The Net Payment/ Net Receipts for France after netting off shall be.....

- (a) Net Receipt \$ 40,000
- (b) Net Payment \$ 80,000
- (c) Net Payment \$ 40,000
- (d) Net Receipt \$ 80,000



IV. The Net Payment/ Net Receipts for Italy after netting off shall be.....

- (a) Net Receipt \$ 60,000
- (b) Net Payment \$ 120,000
- (c) Net Payment \$ 60,000
- (d) Net Receipt \$ 120,000

V. Suppose if the transfer charges are 0.01% of the amount transferred then by applying multilateral netting techniques there will be reduction in overall cost of transfer by

- (a) US \$ 136
- (b) US \$ 156
- (c) US \$ 1,360
- (d) US \$ 1,560

(MTP April – 2024 & Aug – 2025, RTP Jan – 2026)

Answer: Case Scenario – 03

- I. (b) Bilateral Netting**
- II. (d) \$ 2,00,000**
- III. (b) Net Payment \$ 80,000**
- IV. (b) Net Payment \$ 1,20,000**
- V. (a) US \$ 136**

Case Scenario – 04

XYZ Ltd., a U.S. firm, will require £ 3,00,000 in 180 days and is evaluating different strategies to hedge against currency risk. The available market data is as follows:

Current Spot Rate: 1 £ = \$ 2.00

180-Day Forward Rate: 1 £ = \$ 1.96

Interest Rates:

• **UK (180 days):**

Deposit Rate: 4.5%

Borrowing Rate: 5%

• **US (180 days):**

Deposit Rate: 5%

Borrowing Rate: 5.5%

Option Contract:

Call option available on £, expiring in 180 days with:

Exercise price: \$ 1.97

Premium: \$ 0.04

XYZ Ltd. has forecasted the following probabilities for spot rates after 180 days:

Spot Rate	Probability
\$ 1.91	25%
\$ 1.95	60%
\$ 2.05	15%

From the information given above, choose the correct answer to the following questions:

I. The expected dollar amount neededif XYZ Ltd. does not hedge.

- (a) \$ 5,88,000
- (b) \$ 6,05,741
- (c) \$ 5,95,560
- (d) \$ 5,86,500



- II.** If XYZ Ltd. chooses the forward contract option, the expected dollar amount needed in 180 days, the total dollar amount required shall be.....
- (a) \$ 5,88,000
(b) \$ 6,05,741
(c) \$ 5,95,560
(d) \$ 5,86,500
- III.** Under the money market hedge, approximately how much will XYZ Ltd. need to repay after 180 days?
- (a) \$ 5,88,000
(b) \$ 6,05,741
(c) \$ 5,95,560
(d) \$ 5,86,500
- IV.** If XYZ Ltd. chooses the call option contract, after 180 days the total cost in dollar for £ 3,00,000 (inclusive of interest on Premium amount) shall be.....
- (a) \$ 5,88,000
(b) \$ 6,05,741
(c) \$ 5,95,560
(d) \$ 5,86,500
- V.** The expected spot rate per £ after 180 days shall be approximately.....
- (a) \$ 1.91
(b) \$ 1.94

- (c) \$ 1.96
(d) \$ 2.05

(MTP July – 2025)**Answer: Case Scenario – 04**

- I. (d) \$ 5,86,500**
II. (a) \$ 5,88,000
III. (b) \$ 6,05,741
IV. (c) \$ 5,95,560
V. (c) \$ 1.96

Case Scenario – 05

ABC Ltd., a UK firm, has a receivable \$ 20 Million due is 6 months. The company wants to cover full exposure. Following information are available:

Spot rate 1\$ = \$ 0.7720/£ 0.7840

6 months forward rate 1\$ = £ 0.7910/£ 0.8040

Interest rates are as follows:

	US	UK
6 months deposit rate	4.50% p.a.	5.50% p.a.
6 months borrowing rate	6.00% p.a.	7.50% p.a.

Following options on pound are available:

Option	Strike rate	Price
---------------	--------------------	--------------



FOREIGN EXCHANGE EXPOSURE & RISK MANAGEMENT

Call	£ 0.8100	£ 0.01
Put	£ 0.8100	£ 0.02

ABC Limited has forecasted the spot rates for 6 months as follows:

Future rates of 1\$	Probability
£ 0.7800	30%
£ 0.8100	50%
£ 0.8300	20%

From the information given above, choose the correct answer to the question no **I** to **IV**:

(EXAM MAY - 2025)

- I.** What will be the total expected value of option hedge in pounds, if the full exposure of 20 Million is covered ?
- (A) £ 16.125 million (B) £ 16.50 million
(C) £ 15.88 million (D) £ 15.70 million
- II.** If ABC Ltd. is using forward hedge strategy, what will be total proceed in pound?
- (A) £ 15.40 million (B) £ 15.88 million
(C) £ 16.125 million (D) £ 15.82 million
- III.** What will be expected spot rate at the end of 6 months and expected cost (proceed) if no hedge strategy is adopted by the company ?
- (A) £ 0.805/\$, £ 16.10 million
(B) £ 0.85/\$, £ 15.8 million

- (C) £ 0.7720/\$, £ 15.44 million
(D) £ 0.7910/\$, £ 15.82 million

IV. Identify which option gives the highest proceed.

- (A) Option hedge (B) Forward hedge
(C) Money market hedge (D) No hedge

Answer: Case Scenario – 05

- (C) £ 15.88 million
(D) £ 15.82 million
(A) £ 0.805/\$, £ 16.10 million
(D) No hedge

Case Scenario – 06

On 3rd April, 2025, Royal Bank quotes the following :

	Bid	Ask
Spot Exchange Rate (US \$ 1)	₹ 84.2525	₹ 85.5945
2 months' swap points	60	80
3 months' swap points	150	178

In a spot transaction, delivery is made after two days.
Assume spot date as 5th April, 2025 and 1 swap point

= 0.0001.

Calculation up to 4 decimal places.

From the information given above, choose the correct answer to the following question No. I to III :

(EXAM SEPTEMBER - 2025)



- I.** On an average rate, the annual rate of premium/discount of US \$ on Indian ₹ will be
- (a) Bid-0.0498 % and Ask — 0.0524 % (Premium)
 (b) Bid-0.0598 % and Ask — 0.0723 % (Discount)
 (c) Bid-0.0723 % and Ask — 0.0598 % (Premium)
 (d) Bid—0.0598 % and Ask - 0.0723 % (Premium)
- II.** The swap points for 2 months & 15 days (i.e., for 20th June, 2025) will be
- (a) Bid — 90 and Ask — 98
 (b) Bid — 105 and Ask — 129
 (c) Bid — 98 and Ask — 90
 (d) Bid — 129 and Ask — 105
- III.** Foreign exchange rate for 20th June, 2025 will be
- (a) Bid—₹ 83.2525 and Ask —₹ 84.5945
 (b) Bid —₹ 84.5945 and Ask —₹ 85.2525
 (c) Bid—₹ 84.2630 and Ask —₹ 85.6074
 (d) Bid —₹ 85.6074 and Ask —₹ 84.2630

Answer: Case Scenario – 06

- I. (d) Bid—0.0598 % and Ask - 0.0723 % (Premium)**
- II. (b) Bid — 105 and Ask — 129**

III. (c) Bid—₹ 84.2630 and Ask —₹ 85.6074

Case Scenario – 07

On 20.10.2024, the credit balance of an Indian bank in NOSTRO account with LMN Bank in London was £ 1,60,000 and the overbought position was £ 1,00,000. During the day, the following transactions have taken place.

Events	Time	Amount (£)
DD Purchased	11:08	50,000
Purchased a bill on London	11:50	150,000
Sold forward TT	13:15	100,000
Forward purchased contract	13:55	50,000
cancelled	14:45	85,000
Remitted by TT	15:00	40,000
Draft in London cancelled		

Based on the above scenario, choose the most appropriate answer for the following multiple-choice questions:

- I.** How much was the total amount of purchase commitments made during the day by the Indian Bank?
- (A) £ 2,00,000 (B) £ 1,50,000
 (C) £ 3,40,000 (D) £ 50,000
- II.** The final cash balance in the NOSTRO account at the end of 20.10.24 stands at
- (A) £ 85,000 (B) £ 75,000
 (C) £ 20,000 (D) £ 160,000



III. The transaction took place atshall affect both exchange & cash position of the bank with LMN Bank.

- | | |
|-----------|-----------|
| (A) 11:08 | (B) 11:50 |
| (C) 14:45 | (D) 15:00 |

IV. If at the end of day bank is required to maintain a credit balance of £ 20,000 in the NOSTRO account, then it.....

- (A) shall buy forward £ 15,000
 (B) shall sell spot TT £ 55,000
 (C) shall buy spot TT £ 55,000
 (D) shall sell forward £ 55,000

V. If bank takes required steps to maintain a credit balance of £ 20,000 in the Nostro account, then what additional step was required to achieve the overbought position of £ 65,000?

- (A) Buying forward £ 15,000
 (B) Selling forward £ 65,000
 (C) Buying forward £ 60,000
 (D) Selling forward £ 15,000

(MTP March: 2025)

Answer: Case Scenario – 07

I (D) £ 50,000

II (C) £ 20,000

III (C) 14:45

IV (B) shall sell spot TT £ 55,000

V (A) Buying forward £ 15,000

Case Scenario – 08

The Inter Banking Rates on 28th June, 2025 were as follows:

Spot US\$1 = ₹ 86.50/55

1 Month forward premium 5/8

On 1st May, 2025, Mr. M an exporter enters into a forward contract with DMP Bank to sell US \$ 2,50,000 on 31st July, 2025 at the rate US \$ 1 = ₹ 86.80. However, Mr. M received the amount on 28th June, 2025.

Mr. M requested the bank to take the delivery of the remittance on 30th June, 2025 i.e. before due date.

Note 1. – Consider 365 days in a year.

Note 2. – Prevailing Prime Lending Rate is 15% p.a.

From the information given above, choose the correct answer to the Question No. 4 to 6:

I. Swap loss is_____.

- | | |
|--------------|--------------|
| (A) ₹ 12,500 | (B) ₹ 32,500 |
| (C) ₹ 45,500 | (D) ₹ 25,000 |

II. Interest on outlay of funds shall be approximately_____.

- (A) ₹ 955 payable by Mr. M
 (B) ₹ 414 payable by Mr. M
 (C) ₹ 955 payable by DMP Bank



(B) ₹ 414 payable by DMP Bank

III. What is the net conversion rate per US \$ realized by Mr. M?

- (A) ₹ 86.6738 (B) ₹ 86.3662
(C) ₹ 86.3738 (D) ₹ 86.6662

(EXAM JANUARY – 2026)

Answer Case Scenario – 08

- (B) ₹ 32,500
(A) ₹ 955 payable by Mr. M
(D) ₹ 86.6662

Case Scenario – 09

A company is considering hedging its foreign exchange risk. It has made a purchase on 1st July, 2024 for which it has to make a payment of US\$ 60,000 on December 31, 2024. The present exchange rate is 1 US \$ = ₹ 83. It can purchase forward 1 \$ at ₹ 82. The company will have to make an upfront premium @ 2% of the forward amount purchased. The cost of funds to the company is 12% per annum.

The company hedges its foreign exchange risk using above-mentioned forward contract.

From the information given above, choose the correct answer to the following questions:

- I.** If exchange rate on 31st December, 2024 happens to be ₹ 86 per US \$ then due to forward position taken the company will make.....
- (a) a loss of ₹ 1,04,304
(b) a profit of ₹ 1,35,696
(c) a loss of ₹ 2,24,304
(d) a profit of ₹ 2,55,696
- II.** If exchange rate on 31st December, 2024 happens to be ₹ 80 per US \$ then due to forward position taken the company will make.....
- (a) a loss of ₹ 1,04,304
(b) a profit of ₹ 1,35,696
(c) a loss of ₹ 2,24,304
(d) a profit of ₹ 2,55,696
- III.** If exchange rate on 31st December, 2024 happens to be ₹ 88 per US \$ then due to forward position taken the company will make.....
- (a) a loss of ₹ 1,04,304
(b) a profit of ₹ 1,35,696
(c) a loss of ₹ 2,24,304
(d) a profit of ₹ 2,55,696

Answer Case Scenario – 09

- I. (b) a profit of ₹ 1,35,696**
- II. (c) a loss of ₹ 2,24,304**
- III. (d) a profit of ₹ 2,55,696**

Case Scenario – 10

An Indian exporting firm, Radhey & Sons exported dry fruits worth of AUD 1 million to an importer in Sydney. Radhey and Sons are worried about likely depreciation of AUD in near future as it is likely that the export sum will be received after 3 months. Today as such as there is no derivative contract is available in AUD to hedge itself from such depreciation.

The following data is given:

Spot rate : ₹ 64.00/AUD

3 months interest rates (per annum):

	Deposit	Borrowing
₹	10%	12%
AUD	4%	5%

From the information given above, choose the correct answer to the following questions:

- I.** In a money market hedge for a foreign currency receivable, the exporter should.....
- (a) Borrow in domestic currency and invest foreign currency
- (b) Borrow in foreign currency and invest domestic currency
- (c) Borrow and invest in domestic currency only

(d) Borrow and invest in foreign currency only

- II.** In case Radhey & Sons adopts Money Market hedge strategy, the interest rate to be used to compute amount borrowed shall be.....
- (a) 4% per annum
- (b) 5% per annum
- (c) 10% per annum
- (d) 12% per annum
- III.** If a forward contract in AUD were available, the money market hedge would.....
- (a) Always be superior
- (b) Always be inferior
- (c) Be compared with forward hedge for cost efficiency
- (d) Become invalid
- IV.** Interest Rate Parity theory assumes.....
- (a) No capital mobility
- (b) Presence of arbitrage opportunities
- (c) Efficient markets with no arbitrage opportunities
- (d) Fixed exchange rate system.

(RTP May – 2026)

Answer Case Scenario – 10

- I. (b) Borrow in foreign currency and invest domestic currency**
- II. (b) 5% per annum**
- III. (c) Be compared with forward hedge for cost efficiency**
- IV. (c) Efficient markets with no arbitrage opportunities**



Case Scenario – 11

ABC Inc., a U.S. firm, will require £ 6,00,000 in 6 months and is evaluating different strategies to hedge against currency risk. The available market data is as follows:

Current Spot Rate: 1 £ = \$ 2.50

6-months Forward Rate: 1 £ = \$ 2.45

Interest Rates:

- **UK (per annum):**
 - Deposit Rate: 9.00%
 - Borrowing Rate: 10.00%
- **US (per annum):**
 - Deposit Rate: 10%
 - Borrowing Rate: 11.00%

Option Contract:

- Call option available on £, expiring in 6 months with:
 - Exercise price: \$ 2.46
 - Premium: \$ 0.05

ABC Inc. has forecasted the following probabilities for spot rates after 6 months:

Spot Rate	Probability
\$ 2.39	25%
\$ 2.44	60%
\$ 2.56	15%

From the information given above, choose the correct answer to the following questions:

- Since ABC Inc. will require £ 600,000 in future, it is said to be.....
 - (a) Long in £
 - (b) Short in £
 - (c) Long in \$
 - (d) Neutral in currency
- Under a forward contract, the dollar cost of £ 600,000 is.....
 - (a) certain but flexible
 - (b) uncertain but capped
 - (c) certain and fixed
 - (d) nil
- Which hedging technique allows ABC Inc. to benefit if the Pound depreciates while protecting against appreciation?
 - (a) Forward contract
 - (b) Money market hedge
 - (c) Call option on £
 - (d) Futures contract
- If the spot rate after 6 months happens to be \$ 2.39/£, ABC Inc. under option hedge will.....
 - (a) exercise the option.
 - (b) let the option lapse.
 - (c) cancel the contract.
 - (d) incur loss equal to intrinsic value.
- Which of the following hedging methods involves borrowing and lending in two currencies simultaneously?
 - (a) Forward hedge
 - (b) Option hedge

- (c) Money market hedge (d) Futures hedge

Answer Case Scenario – 11

- I. (b) Short in £
II. (c) certain and fixed
III. (c) Call option on £
IV. (b) let the option lapse.
V. (c) Money market hedge



Student Notes



Student Notes



NORMAL PROBABILITY DISTRIBUTION TABLE

Normal Probability Distribution Table

Number of Standard Deviations from mean (Z)	Area to the Left of Right (One Tail)	Number of standard Deviations from Mean (Z)	Area to the lft or Right (One Tail)
0.00	0.5000	1.55	0.0606
0.05	0.4801	1.60	0.0548
0.10	0.4602	1.65	0.0495
0.15	0.4404	1.70	0.0446
0.20	0.4207	1.75	0.0401
0.25	0.4013	1.80	0.0359
0.30	0.3821	1.85	0.0322
0.35	0.3632	1.90	0.0287
0.40	0.3446	1.95	0.0256
0.45	0.3264	2.00	0.0228
0.50	0.3085	2.05	0.0202
0.55	0.2912	2.10	0.0179
0.60	0.2743	2.15	0.0158
0.65	0.2578	2.20	0.0139
0.70	0.2420	2.25	0.0122
0.75	0.2264	2.30	0.0107
0.80	0.2119	2.35	0.0094
0.85	0.1977	2.40	0.0082
0.90	0.1841	2.45	0.0071
0.95	0.1711	4.50	0.0062
1.00	0.1557	2.55	0.0054
1.05	0.1469	2.60	0.0047
1.10	0.3570	2.65	0.0040
1.15	0.1251	2.70	0.0035



NORMAL PROBABILITY DISTRIBUTION TABLE

1.20	0.1151	2.75	0.0030
1.25	0.1056	2.80	0.0026
1.30	0.0986	2.85	0.0022
1.35	0.0885	2.90	0.0019
1.40	0.0808	2.95	0.0016
1.45	0.0735	3.00	0.0013
1.50	0.0668		



Student Notes



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